Ambiguity, Global Financial Risk, and Emerging Market Business Cycles*

Brendan Epstein† Rahul Mukherjee‡ Musa Orak§ Alessandro Peri¶

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Abstract

This paper examines the impact of ambiguity (Knightian uncertainty) in U.S. financial markets on real economic activity in Emerging Market Economies (EMEs). Using a panel structural vector autoregression, we identify ambiguity shocks in U.S. equity markets and assess their transmission to EMEs. We find that an increase in ambiguity leads to a significant and persistent boom in EME GDP and investment. We attribute this result to investors' portfolio reallocations: heightened ambiguity prompts a shift out of U.S. assets into foreign bonds and equities, with no flight to safety, which eases global financial conditions for EMEs and boosts their activity; a portfolio model with ambiguity-averse investors delivers this reallocation, in contrast to the flight to safety that higher volatility produces. We also document a negative response of ambiguity to structurally identified global financial risk shocks, uncovering a key role of global financial risk in shaping investors' beliefs. Variance and historical decompositions underscore the role of ambiguity as a key driver of EME business cycles, comparable in importance to global financial risk and country spread shocks and accounting for about a third of the mid-2000s expansion in EME output.

Keywords: Emerging market economies, ambiguity, uncertainty, global financial conditions, international business cycles .

JEL Codes: F41, E44, D81, G15.

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†Department of Economics, University Massachusetts at Lowell. Falmouth Hall, Room 302. One University Avenue, Lowell, MA 01854, United States. E-mail: epsteinb@gmail.com.

‡University of Nottingham. B31 Sir Clive Granger Building, University Park, Nottingham, NG7 2RD, United Kingdom. E-mail: Rahul.Mukherjee@nottingham.ac.uk.

§Board of Governors of the Federal Reserve System, 20th St. and Constitution Ave. NW, Washington, D.C. 20551, United States. E-mail: musa.orak@frb.gov.

¶Department of Economics, University of Colorado Boulder, CO 80302, United States. E-mail: Alessandro.Peri@Colorado.edu.

1 Introduction

Assessing how real economic activity in Emerging Market Economies (EMEs) responds to external financial factors is a central question in the international finance literature, with implications for policymakers and investors alike. Prior research has examined the interplay between real economic activity in EMEs and financial factors, such as the U.S. real interest rate, country-specific interest rate spreads, and global financial conditions. Since Akıncı (2013), this literature has emphasized the key role of global financial risk in driving EME fluctuations. An important dimension of global financial conditions remains unexplored: the transmission of ambiguity, Knightian uncertainty about the probability distribution of financial returns (Knight, 1921), from U.S. equity markets to emerging market economies.[1] Ambiguity is conceptually distinct from risk and can therefore generate different portfolio-allocation and macroeconomic responses. We identify U.S. equity-market ambiguity shocks in a panel structural vector autoregression and trace their effects on EME GDP, investment, sovereign spreads, and cross-border portfolio flows.

In the last decades, U.S. listed companies have accounted for nearly 40% of the world's equity market capitalisation and over 60% of that within OECD countries (World Bank, 2025), highlighting a prominent role in global financial markets, and the S&P 500 is the reference risky asset on which both the VIX and our ambiguity measure are built. This paper investigates the extent to which fluctuations in Knightian uncertainty in U.S. equity markets—defined as ambiguity about the probability distribution of returns—have real effects on EMEs. To motivate this exploration, Panels (a) and (b) of Figure 1.1 show that our measure of Knightian uncertainty in the U.S. stock markets is positively correlated with aggregated EME GDP growth (with a pairwise coefficient of correlation of 0.36) and investment growth (with a pairwise coefficient of correlation of 0.32), suggesting that periods with rising ambi-

[1] In the context of decision-making, risk is associated with situations where uncertainty lies in the realization of outcomes whose probability distribution is known. On the other hand, ambiguity represents a distinct form of uncertainty where unique probabilities cannot be assigned to potential outcomes, so that their probability distribution is unknown or unknowable. While a large literature has documented meaningful real and financial effects of uncertainty shocks (e.g., Bloom, 2009; Jurado et al., 2015; Ludvigson et al., 2021; Cascaldi-Garcia et al., 2023), much less is known about how Knightian uncertainty, as distinct from risk, transmits internationally to emerging economies.

guity in U.S. equity markets coincide with increased EME activity. We interpret this pattern as reflecting a portfolio rebalancing mechanism. As ambiguity in U.S. equity markets rises, international financial markets are likely to respond, triggering portfolio reallocation that eases EME borrowing constraints, stimulating their economic activity. Panel (c) supports this interpretation, showing that ambiguity negatively comoves with EME country spreads (with a pairwise coefficient of correlation of -0.36), which in turn have strong and negative correlations with EME macroeconomic aggregates (Panel (d)), in line with the findings of Neumeyer and Perri (2005) and Uribe and Yue (2006).[2]

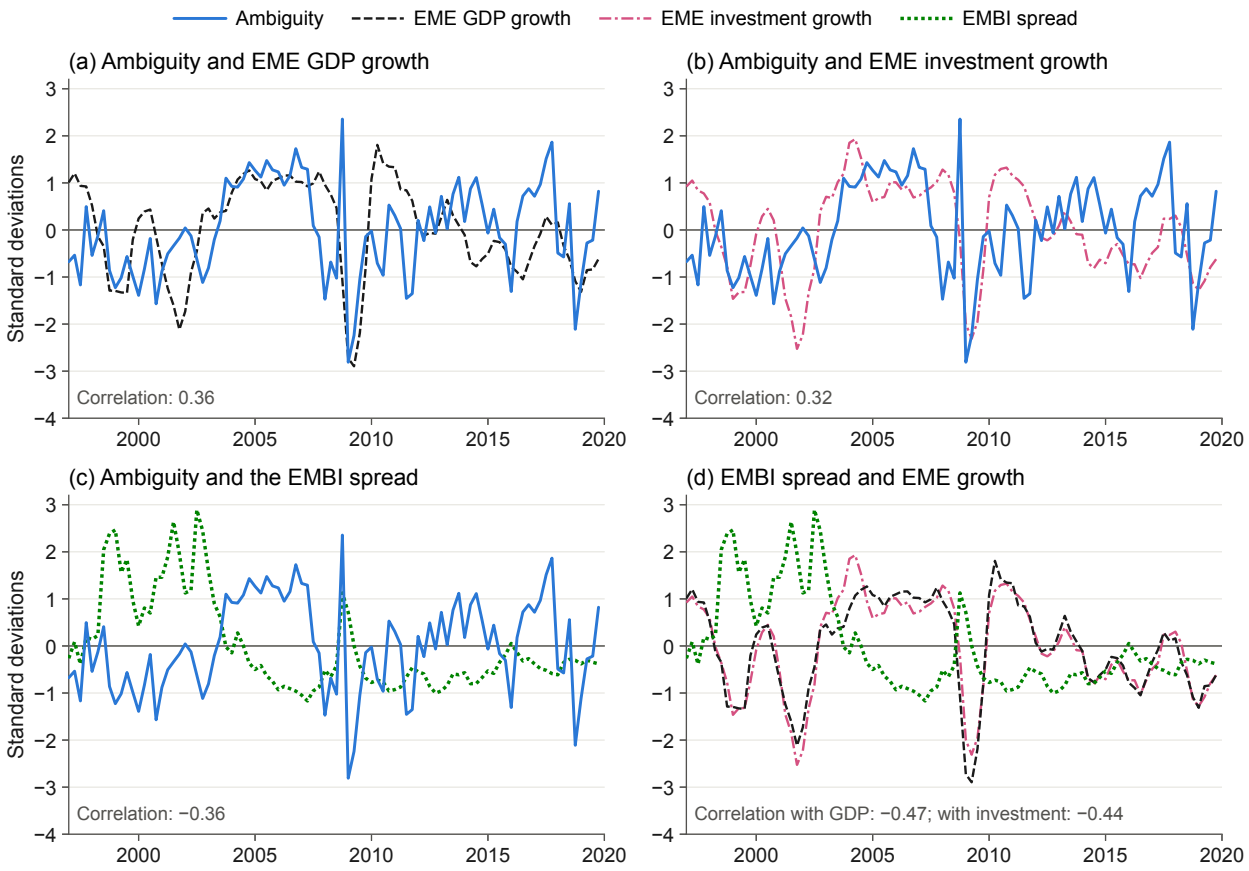


Figure 1.1: Ambiguity, EMBI spread, and EME business cycles

Note: The panels show standardized series. EME GDP and investment growth are year-over-year growth rates of real GDP and real gross fixed capital formation, averaged over Argentina, Brazil, Mexico, Peru, South Africa and Turkey (Peru's investment enters from 2008:Q1). Ambiguity is the log of the quarterly ambiguity index in deviation from a linear trend. The EMBI spread is averaged across countries after removing country means. Data start in 1997:Q1; each panel reports the correlation of the pair over 1997:Q1-2019:Q4.

This motivating evidence highlights the presence of a link between ambiguity in U.S.

[2]The pairwise coefficients of correlation between the EMBI spread and EME GDP growth and EME investment growth are -0.47 and -0.44, respectively.

equity markets and borrowing-constrained EMEs' activity. To rigorously assess this relationship, we empirically model the complex joint dynamics of U.S. equity markets' ambiguity, global financial risk, and EME business cycles within a panel structural vector autoregression framework in order to identify the real and financial effects of structurally identified U.S. ambiguity shocks on EMEs.

We quantify the degree of ambiguity in U.S. equity markets by adopting the ambiguity index developed by Izhakian (2020) and Brenner and Izhakian (2018). This index measures the degree of Knightian uncertainty associated with the probability distribution of U.S. equity returns, which it is derived within the expected utility with uncertain probabilities theory developed by Izhakian (2017).[3] An increase in the index means that the daily distribution of U.S. stock returns becomes less stable within the month, so that the probabilities of gains and losses are themselves more uncertain. We choose this index for three reasons. First, by construction it measures the degree of ambiguity separately from risk and from agents' attitudes towards risk and ambiguity. This property is key to our analysis, as it allows us to study shocks to the degree of ambiguity separately from changes in U.S. stock market risk, including those potentially correlated with global financial risk previously examined in the literature (Akıncı, 2013).[4] Second, this index has been shown to have important implications for U.S. financial variables that extend beyond those captured by traditional risk measures (e.g., VIX), such as expected stock returns (Brenner and Izhakian, 2018) and the pricing of credit default swaps, where higher ambiguity lowers spreads (Augustin and Izhakian, 2020). Third, it is independently constructed, facilitating replication and comparability with the literature.

Our empirical framework builds on Uribe and Yue (2006) and Akıncı (2013), and uses a 7-variable panel structural vector autoregression (PSVAR) model with: *(i)* four country-specific variables, which include three EME business cycle indicators (namely, GDP, invest-

[3] See Section 2 for more details.

[4] The ambiguity index is distinct by construction from conventional measures of risk, although the series are correlated in the data. The detrended index has a correlation of -0.38 with the BAA corporate spread, utilized by Akıncı (2013) to capture global financial risk, and of -0.68 with the log VIX in levels; the corresponding correlations in first differences are 0.01 and -0.28. Section 3.3.5 shows that the EME expansion also emerges when volatility is included in the model.

ment, and the trade balance) and one financial variable (the country spread); and *(ii)* three global financial variables given by the 3-month U.S. Treasury yield as a measure of the risk free rate, the BAA corporate bond spread as a measure of global financial risk, and the Izhakian (2020)'s ambiguity index as a measure of the degree of Knightian Uncertainty in U.S. stock markets. Our estimated PSVAR identifies four structural shocks: a U.S. risk-free interest rate shock, a global financial risk shock, a country spread shock, and a U.S. equity markets' ambiguity shock. Our identification scheme relies on few key restrictions, which extends assumptions previously made in the literature to our framework. First, we assume that the evolution of global financial variables are exogenous to country-specific developments in EMEs, extending the restrictions in Uribe and Yue (2006) and Akıncı (2013) to the U.S. ambiguity index. Following these studies, we impose a "Cholesky" structure to the contemporaneous relations among global financial variables by assuming that the risk free rate is the most exogenous variable, followed by global financial risk, while the ambiguity measure is the least exogenous of the three.[5] Specifically, we assume that the risk free rate affects both global financial risk and ambiguity contemporaneously, while global financial risk is assumed to affect only ambiguity contemporaneously. All the global financial variables of the PSVAR influence each other in lags. Last, we assume that EME financing conditions–proxied by country-specific EMBI+ sovereign bond spreads–react to contemporaneous shocks in own-country fundamentals (GDP, investment and trade balance, which are the other three variables entering the PSVAR) while the latter react to all financial variables only with a lag.

We estimate the parameters of the above-described PSVAR and assess the impact of the four identified shocks by computing impulse response functions, with confidence bands from a moving-block residual bootstrap (10,000 replications) that preserves the serial and cross-country dependence of the residuals. Moreover, we quantify the relevance of the four identified shocks in explaining movements across the variables in our model using variance and historical decompositions. Our analysis yields four main results.

[5]The ordering of the risk free rate and global financial risk follows earlier literature. Granger causality tests in the global block support placing ambiguity after global financial risk, and our results are the same under all six orderings of the global block (Section 4).

First, an identified unexpected increase in the degree of ambiguity in U.S. equity markets leads to a significant and persistent expansion in EME GDP and investment, which peak at about 0.4 and 1 percent above trend seven to eight quarters after the shock and remain above trend for five years. This stands in contrast to the negative transmission of U.S. uncertainty shocks measured by stock market volatility (Carrière-Swallow and Céspedes, 2013; Bhattarai et al., 2020).[6] U.S. ambiguity shocks, by contrast, are followed by a decline in U.S. investment, in line with earlier findings on both ambiguity and uncertainty shocks (e.g., Ilut and Schneider, 2014; Jurado et al., 2015). This contrast uncovers a previously undocumented asymmetry in business cycle responses to U.S. ambiguity shocks: while ambiguity shocks weigh on U.S. investment, their international transmission boosts EME activity, in sharp contrast with the impact of previously examined uncertainty (but not ambiguity) shocks. The boom is present in a larger sample of eleven EMEs and in ten of them taken individually, is specific to EMEs (small open advanced economies show no comparable response), and is obtained when the ambiguity shock is identified at monthly frequency, orthogonal within the month to stock market volatility, financial uncertainty, the excess bond premium and a broad set of risk-appetite, global-financial-cycle and monetary policy controls (Section 3.3.5).

Second, we identify international portfolio reallocation as the channel through which U.S. equity-market ambiguity stimulates EME activity, and we interpret it with a portfolio model in which an ambiguity-averse investor facing greater ambiguity about the mean of U.S. equity returns shifts toward correlated foreign risky assets, whereas a rise in common volatility produces a flight to safety. The financial-market responses reveal the shift toward foreign assets rather than a conventional flight to safety. The U.S. safe interest rate does not fall, the 10-year Treasury yield rises by 0.11 percentage points, the term premium rises by 0.06 to 0.08 percentage points, and the dollar weakens. U.S. residents raise their net purchases of foreign bonds, by 0.4 percent of quarterly U.S. GDP within a year and 1.3 percent within three years, and of foreign stocks; issuer-level data show that these are mostly

[6] Additional analysis in which we replace ambiguity with the VIX, with the financial, macro and real uncertainty indexes of Jurado et al. (2015) and with the economic uncertainty index of Bekaert et al. (2022) in our benchmark SVAR shows that these measures, all of which rise with volatility, are contractionary for EME GDP and investment (for the VIX, mildly so), in line with the existing literature; the ambiguity index is the only measure that moves against volatility.

European, advanced-economy and financial-centre bonds, with EME securities accounting for less than a tenth. This pattern differs from the flight to quality that Knightian uncertainty generates in Caballero and Krishnamurthy (2008). The BAA corporate spread narrows by about 9 basis points, largely reflecting the rise in Treasury yields.[7] Crucially, EMEs benefit through easier global financial conditions: their equity prices rise by 4 to 6 percent within a year, cross-border bank claims on them, mostly from euro-area banks, rise by 1.8 percent of their quarterly GDP after two years, their foreign exchange reserves accumulate, and their external costs of financing fall, with spreads about 11 basis points lower within a year and 30 basis points lower after two to three years. The joint dynamics of country spread and EME GDP and investment are consistent with the virtuous feedback loop documented in the literature (Uribe and Yue, 2006 and Akıncı, 2013): improved economic conditions reduce borrowing rates, which in turn help sustaining longer periods of above-trend EME GDP and investment. By and large, this analysis suggests that in response to greater ambiguity in the U.S. stock market, international investors shift investments away from U.S. assets and toward foreign assets, which eases the external financing conditions of EMEs and supports greater investment and GDP in EME countries.

Third, an identified positive shock to the BAA corporate spread leads to a sharp and significant decline in ambiguity, of about 8 percent one quarter after the shock; the decline is also present at monthly frequency, where it survives controlling for the VIX. Evidence from survey measures of expected stock returns and of the perceived probability of a crash points to a role for global financial risk in shaping investors' *beliefs* and, through this channel, the degree of ambiguity. Our analysis suggests that an unanticipated widening of corporate credit spreads provides a strong signal of a worsening outlook for real economic activity (e.g. Gertler and Lown, 1999; Gilchrist and Zakrajšek, 2012), stirring investors beliefs towards a narrower set of priors with greater weight on bad economic states, thereby dissipating Knightian uncertainty (ambiguity). In doing so, we suggest a further channel through which global financial risk shocks affect EMEs, by coordinating international investors' beliefs.

[7] Identified shocks to the BAA corporate spread, as in Akıncı (2013), are conceptually distinct from the endogenous response of the BAA spread to innovations in U.S. equity-market ambiguity.

Last, variance and historical decompositions reveal that ambiguity is a key driver of EME aggregate activity. Innovations to ambiguity are the largest external contributor to EME GDP fluctuations, accounting for about 6 percent of the variance at 20 quarters, versus roughly 2 percent each for global financial risk and the country spread (together with the U.S. interest rate, external shocks explain 10 percent of GDP variance). For investment, ambiguity and the country spread are the largest contributors (4.5 and 2.6 percent), while global financial risk adds about 1 percent. An expanding-window analysis shows that earlier samples (ending 2011) attribute far more variance to global financial risk and country-spread shocks (about 23 percent of GDP variance, ambiguity included). As more recent data are added, the contributions of global financial risk and the country spread recede while that of ambiguity remains stable, making ambiguity the leading external contributor to EME GDP fluctuations in the later, pre-pandemic portion of our sample. A historical decomposition translates these shares into episodes: ambiguity shocks account for 3.4 of the 9.4 percentage points by which average EME GDP rose above trend between 2003:Q2 and 2008:Q1, and for 15 percent of the variation of the cross-country average of EME GDP, whereas the 2008-09 collapse is a global financial risk event.

The rest of the paper proceeds as follows. Related literature is addressed in the following subsection. In section 2, we outline our methodology and data. Section 3 presents the results of our SVAR analysis using impulse response functions, sets out a portfolio interpretation of the mechanism and examines it with asset prices, capital flows and bank lending, EME financial variables, investors' beliefs, and alternative uncertainty measures, and closes with variance and historical decompositions. We then discuss a number of robustness analyses in section 4 before concluding with section 5.

Related Literature

Our paper contributes to three strands of research: the literature on the macro-financial implications of ambiguity, particularly for the U.S. economy; the work examining the role of global financial conditions in shaping EME business cycles; and the work on the international

transmission of uncertainty shocks.

On the financial implications of ambiguity, prior studies show that ambiguity influences asset pricing, portfolio allocation, and market dynamics (Epstein and Wang, 1994; Chen and Epstein, 2002; Epstein and Miao, 2003; Cao et al., 2005; Garlappi et al., 2007; Benigno and Nisticò, 2009; Ju and Miao, 2012; Drechsler, 2013; Brenner and Izhakian, 2018). These studies show how ambiguity helps explain puzzles such as limited market participation, equity home bias, and the equity premium. In portfolio problems, ambiguity aversion lowers the demand for the ambiguous asset (Garlappi et al., 2007; Maccheroni et al., 2013); the portfolio interpretation of Section 3.3.1 builds on this result and on the α-maxmin preferences of Ghirardato et al. (2004) to show when greater ambiguity about U.S. equity raises the demand for foreign risky assets. In open economies, ambiguity aversion helps account for deviations from uncovered interest parity (Ilut, 2012), flights to quality (Caballero and Krishnamurthy, 2008), and the premia that foreign lenders charge on emerging-market sovereign debt (Pouzo and Presno, 2016). From a macroeconomic perspective, increases in ambiguity have been found to generate economic contractions in closed-economy settings or advanced economies (Ilut and Schneider, 2014; Bianchi et al., 2018; Altug et al., 2020; Ilut and Saijo, 2021).[8] On the measurement side, Izhakian (2020) derives an index of the degree of ambiguity that is independent of risk and of attitudes toward ambiguity, and Brenner and Izhakian (2018) show that this index is priced in U.S. equities; using the same approach, Augustin and Izhakian (2020) find that higher ambiguity lowers credit default swap spreads, and Izhakian et al. (2022) that it raises corporate leverage. We build on these financial and macroeconomic mechanisms by studying whether, and how, ambiguity originating in U.S. financial markets transmits internationally to EME business cycles.

Most closely related on the macroeconomic effects of measured ambiguity is Asano et al. (2026). Using the same ambiguity index in a three-variable U.S. SVAR identified with event constraints, they find a temporary increase in industrial production after an ambiguity shock and a decline in ambiguity after a financial uncertainty shock. Their analysis estab-

[8] Ilut and Schneider (2023) survey the modeling of uncertainty as ambiguity in macroeconomics and finance.

lishes that measured ambiguity can have expansionary domestic effects. We complement this evidence by studying its international transmission to EME output and investment, sovereign spreads, and cross-border portfolio flows. The financial responses also differ: their ambiguity shock raises financial uncertainty and the VIX and lowers long-term government bond yields, whereas our estimated responses include lower financial uncertainty and higher Treasury yields. The different financial-market responses highlight the importance of shock identification. Section 3.3.5 therefore complements our quarterly analysis with a monthly identification that distinguishes ambiguity innovations from contemporaneous movements in volatility, financial uncertainty, and credit-market conditions.

In doing so, we contribute to the large empirical literature that explore the importance of *global financial conditions* for EME business cycles. Early work by Diaz-Alejandro (1985), Calvo et al. (1993), and Calvo (2002) stress the role of external financial factors, such as financial liberalization and global financial risk appetite, for EMEs. Subsequent research demonstrates U.S. monetary policy shocks have been important drivers of economic dynamics and fluctuations in EMEs (Canova, 2005; Mackowiak, 2007; Uribe and Yue, 2006; Fink and Schüler, 2015; Miranda-Agrippino and Rey, 2020; Degasperi et al., 2024). Several other studies show that unconventional monetary policy (Anaya et al., 2017; Chari et al., 2021) has strong effects on EME asset prices, capital flows, and output; while global financial risk (Akıncı, 2013; Rey, 2015; Caballero et al., 2019; Epstein et al., 2019) accounts for a meaningful share of EME credit spreads and business cycles.[9] Miranda-Agrippino and Rey (2022) review the evidence that a single global factor, driven by U.S. monetary policy and risk aversion, moves risky asset prices and capital flows worldwide, and Kalemli-Özcan (2019) and Kalemli-Özcan and Unsal (2023) show that U.S. shocks reach EME credit costs mainly through global investors' risk perceptions, more so where policy credibility is weak and dollar debt is high.

[9]Further work has refined these results. Horvath and Yang (2024) propose that the transmission of global financial risk shocks to EMEs occurs mainly through international capital flows, with countries experiencing larger capital inflows being more affected, while Horvath and Rothman (2021) identify the U.S. mortgage spread as the key driver of EME business cycles. Exposure also depends on who lends: EMEs that rely more on global mutual funds and ETFs are more sensitive to push factors (Cerutti et al., 2019; Converse et al., 2023).

A related literature studies the international transmission of uncertainty shocks and finds that they are contractionary abroad. Carrière-Swallow and Céspedes (2013) show that shocks to U.S. stock market volatility cause larger investment declines in emerging than in advanced economies, and Bhattarai et al. (2020) find that a U.S. uncertainty shock raises country spreads, reduces capital inflows, and lowers output in fifteen EMEs. Cesa-Bianchi et al. (2020), Bonciani and Ricci (2020), and Londono et al. (2025) reach similar conclusions using multicountry models, global realized-volatility factors, and real-uncertainty indices, and Akinci et al. (2022) trace the effect to a rise in global risk premia driven by U.S. financial intermediaries. At the same time, the sign of uncertainty effects depends on what is measured: Segal et al. (2015) show that uncertainty associated with positive growth news predicts higher investment, Berger et al. (2020) find that shocks to forward-looking uncertainty have no contractionary effect, and Bekaert et al. (2022) show that credit spreads load on economic uncertainty rather than on risk aversion (see Cascaldi-Garcia et al., 2023, for a review). We contribute to this literature by showing that ambiguity shocks raise EME activity and lower EME spreads, whereas shocks to the VIX and to the Jurado et al. (2015) uncertainty indices lower EME activity in the same model.

Our panel framework builds most directly on Uribe and Yue (2006) and Akıncı (2013), who show that U.S. interest rate and global financial risk shocks drive EME cycles. We introduce *U.S. financial ambiguity shocks* into this framework to study their international transmission. Our contribution is threefold. First, we document a medium-run expansion in EME GDP and investment following U.S. ambiguity shocks, whereas the U.S. uncertainty shocks studied by Bhattarai et al. (2020) and Carrière-Swallow and Céspedes (2013) lower EME activity; this difference holds when ambiguity and the VIX enter the same VAR. Second, we provide joint evidence on Treasury yields, the dollar, U.S. purchases of foreign securities by issuer, cross-border bank lending, EME asset prices and sovereign spreads that is consistent with a reallocation toward foreign assets without a flight to safety, which eases global financial conditions for EMEs, and we show in a portfolio model why ambiguity about U.S. equity returns, unlike a rise in common volatility, produces such a reallocation. Third, we quantify the interaction between BAA spread shocks and ambiguity and their

respective contributions to EME fluctuations: ambiguity's share of EME output variance remains stable as the shares of global financial risk and the country spread fall in samples extending beyond 2011, and a historical decomposition attributes about a third of the mid-2000s EME expansion to ambiguity shocks. A monthly identification of the ambiguity shock, orthogonal within the month to stock market volatility, financial uncertainty and the excess bond premium, delivers the same expansion, which is present in ten of eleven EMEs taken individually.

2 Data and Methodology

Our empirical model builds directly on Uribe and Yue (2006) and Akıncı (2013). In particular, we estimate the following PSVAR:

$$Ay_{i,t} = \alpha_i + \sum_{k=1}^{p} B_k y_{i,t-k} + \epsilon_{i,t}, \tag{2.1}$$

where i indexes countries and t indexes time, A is a square matrix summarizing the contemporaneous relationships between the endogenous variables, and

$$y_{i,t} = \left[gdp_{i,t}, inv_{i,t}, tby_{i,t}, R_t^{US}, GR_t, ambig_t, R_{i,t}\right]', \tag{2.2}$$

where, for EMEs, $gdp_{i,t}$ is real GDP, $inv_{i,t}$ is real investment, and $tby_{i,t}$ is the trade balance to GDP ratio. Also, R_t^{US} is the gross U.S. real interest rate (the 3-month real gross U.S. Treasury Bill rate deflated by a measure of expected inflation - in line with Akıncı, 2013), GR_t is the gross BAA corporate bond spread over the 10-year Treasury yield measured using data from Moody's (a shock to which, in line with earlier literature, is interpreted as an innovation to global financial risk), $ambig_t$ is ambiguity (in U.S. equity markets), and $R_{i,t}$ is the gross EME country spread (measured using EMBI spread data from J.P. Morgan).[10]

[10]The ordering of U.S. real interest rate and global financial risk in the foreign block is the same as in Akıncı (2013). Granger causality tests in the global block support placing ambiguity after global financial risk. They also find that global financial risk Granger-causes the U.S. real rate but not the reverse, so they do not support placing the U.S. real rate first (Online Appendix Table C.4).

In addition, α_i is a vector of country fixed effects, B_k are square matrices summarizing the effect of lagged variables of order p, and $\epsilon_{i,t}$ is the vector of innovations

$$\epsilon_{i,t} = \left[\epsilon_{i,t}^{gdp}, \epsilon_{i,t}^{inv}, \epsilon_{i,t}^{tby}, \epsilon_{t}^{R^{US}}, \epsilon_{i,t}^{GR}, \epsilon_{t}^{ambig}, \epsilon_{i,t}^{R}\right]'.$$

Except for $tby_{i,t}$ all data are in natural logarithms. Also, GDP, investment, the trade balance and ambiguity are in the form of deviations from a linear trend, while, as in Akıncı (2013), interest rates and spreads are not detrended. The log ambiguity index is trend-stationary: an augmented Dickey-Fuller test with a trend rejects a unit root (p-value 0.004), and a KPSS test rejects level stationarity but not trend stationarity. The index falls through the 1990s and rises from the early 2000s; the one break in its trend that is robust to the persistence of the series is a shift in level and slope in January 2001, the month in which the NYSE and AMEX completed decimal pricing, and break-search tests with critical values that account for this persistence select no further breaks at the 5 percent level (Online Appendix Table C.12). Section 4 shows that our results hold when trend terms enter every equation of the PSVAR instead, with a break in the trend of the index in 2001, and under quadratic, Hodrick-Prescott, rolling-mean and piecewise-linear trends (Online Appendix Table C.13).

Turning to our identifying assumptions, we build on Uribe and Yue (2006) and Akıncı (2013). First, we assume that the contemporaneous-effects matrix A is lower triangular with unit diagonal elements.[11] As such, regarding EME variables, these restrictions imply that: GDP is the most exogenous variable and does not contemporaneously react to any other variable in the system; investment reacts contemporaneously to GDP, exclusively; the trade balance reacts contemporaneously only to GDP and investment; and the country spread reacts contemporaneously to EME-specific variables and foreign financial variables (the gross U.S. real interest rate, global financial risk, and degree of ambiguity in U.S. equity markets).

As a baseline, following Akıncı (2013), we specify the model with two lags (the lag order selected by the Bayesian and Hannan-Quinn information criteria) and assume foreign

[11] We follow Akıncı (2013) in ordering the variables in the vector $y_{i,t}$ in equation (2.2), and Section 4 shows that our results are unchanged under all six orderings of the global block.

financial variables constitute an exogenous shock block: $A_{ij} = B_{1,ij} = B_{2,ij} = 0$ for $i = 4, 5, 6$ and $j = 1, 2, 3, 7$. This restriction implies that $\left[R_t^{US}, GR_t, ambig_t\right]'$ are not affected simultaneously or in lags by country-specific EME macroeconomic aggregates. Moreover, $A_{4j} = 0$ for all $j \neq 4$, $A_{5j} = 0$ for all $j \neq 4, 5$ and $A_{6j} = 0$ for all $j \neq 4, 5, 6$. This implies that: the real U.S. rate does not react contemporaneously to global financial risk or ambiguity; the BAA corporate bond spread reacts contemporaneously only to the real U.S. rate; and ambiguity reacts contemporaneously to both the real U.S. rate and the BAA corporate bond spread. Therefore, within the shock block, the U.S. rate is the most exogenous variable followed by global financial risk, while ambiguity is the least exogenous of the foreign financial variables. Since we do not impose any further restrictions on $\{B_k\}$, country-specific EME variables respond with lags to foreign financial variables.[12]

Our baseline PSVAR estimation uses pooled quarterly data from the same set of countries as Akıncı (2013) (Argentina, Brazil, Mexico, Peru, South Africa and Turkey) and, on net, our sample ranges from 1994:Q2 through 2019:Q4. Argentina enters as two panel units, before its 2001 default (1994:Q2-2001:Q3) and after its return to international bond markets (2016:Q2-2018:Q2), so the panel has seven units and 486 country-quarters (Table 2.1). This dataset allows for straightforward comparison of results with earlier literature and nets out potentially confounding dynamics owing to the pandemic period and its immediate aftermath. In addition, also in line with Akıncı (2013) and for similar reasons (time span considerably larger than country sample size), we use the least squares dummy variables estimator. The equations of the global block are estimated on the full 1994:Q2-2019:Q4 time series, and the country equations on the stacked panel.

Since relative to earlier work our analysis incorporates ambiguity, we discuss this measure in greater detail here, while additional information on the rest of our data can be found in Online Appendix A. We quantify the degree of ambiguity in U.S. equity markets ($ambig_t$) using the ambiguity index from Izhakian (2020). This index is theory-based and assumes the presence of a representative investor who: *(i)* aggregates the set of priors of all investors in the economy (henceforth, international investors); *(ii)* makes decisions according to the expected

[12]See Online Appendix B for a summary of matrix specifications resulting from these restrictions.

Table 2.1: Descriptive statistics

Variable	Units	Obs.	Mean	S.d.	Min.	Max.	AR(1)
A. Country variables (seven panel units, 1994Q2–2019Q4)							
GDP	%, within	486	–	4.43	−12.87	11.13	0.94
Investment	%, within	486	–	14.79	−46.66	40.49	0.95
Trade balance/GDP	% of GDP	486	−0.53	2.56	−8.41	9.07	0.90
EMBI spread	pp	486	3.62	2.61	0.67	17.32	0.92
B. Global variables (long sample, 1994Q2–2019Q4)							
U.S. real rate	pp	103	0.15	2.05	−2.97	3.76	0.92
Baa spread	pp	103	2.39	0.74	1.36	5.43	0.89
Ambiguity	%, dev. mean	103	–	28.77	−84.62	67.94	0.51
VIX	%, dev. mean	103	–	32.25	−52.06	117.33	0.79
JLN financial uncertainty (h=12)	%, dev. mean	103	–	5.62	−8.30	14.62	0.95
JLN macro uncertainty (h=12)	%, dev. mean	103	–	5.03	−7.54	21.38	0.94
JLN real uncertainty (h=12)	%, dev. mean	103	–	2.39	−3.73	10.54	0.88

C. Panel units

Unit	First	Last	Quarters	After lags	EMBI mean (pp)	GDP s.d. (%)	Inv. s.d. (%)
Argentina I	1994Q2	2001Q3	30	28	7.23	4.52	13.65
Argentina II	2016Q2	2018Q2	9	7	4.31	1.80	6.20
Brazil	1995Q1	2019Q4	100	98	4.78	5.33	14.03
Mexico	1994Q2	2019Q4	103	101	3.00	3.64	9.30
Peru	1997Q2	2019Q4	91	89	2.93	4.58	20.31
South Africa	2002Q2	2019Q4	71	69	2.00	4.24	13.79
Turkey	1999Q3	2019Q4	82	80	3.73	4.45	16.37
All			486	472	3.62		

Note: Panel A pools the 486 country-quarters of the benchmark panel; panel B uses the long global sample on which the block-exogenous global equations are estimated. GDP and investment are logs of real seasonally adjusted series and enter the PSVAR in deviation from a country-specific linear trend; they are shown as deviations from unit means (×100, percent). Ambiguity is the log of the square root of the Izhakian (2020) ambiguity measure for the S&P 500 (quarterly mean of the monthly index), linearly detrended; the VIX and the Jurado et al. (2015) indexes are logs, linearly detrended (used in Section 3.3.5 only); log series are shown as deviations from their sample means, in percent. Rates and spreads are annual, in percentage points: the EMBI spread is log(1 + spread), the BAA spread is log(1 + Baa minus 10-year Treasury yield), and the U.S. real rate is the 3-month Treasury bill rate minus expected inflation. AR(1) is the first-order autocorrelation (within units in panel A). Panel C gives the span of each panel unit.

utility with uncertain probabilities model (EUUP model henceforth) of Izhakian (2017) when confronted with uncertainty over outcomes and outcomes' probabilities. Specifically, in the EUUP model international investors respond to the two-tier uncertainty (over outcomes and outcomes' probabilities) by forming their perceived probabilities with respect to their payoff-relevant events and then use these certainty-equivalent probabilities (capacities) to assess expected stock market returns - gains or losses - with respect to the risk free interest rate (which is used as a reference point). Crucially for our analysis, the EUUP framework specializes the Choquet Expected Utility framework (Schmeidler, 1989) and Cumulative Prospect Theory (Wakker and Tversky, 1993) to derive a measure of the degree of ambiguity, which is independent of risk as well as individuals' attitudes towards ambiguity and risk. This empirically appealing separation is reached by assuming that investors' preferences for ambiguity (such as ambiguity aversion) do not depend on the stock return outcomes and apply exclusively to the uncertain probabilities associated with these outcomes.[13]

Operationally, the Izhakian (2020) index is constructed from intraday (5-minute frequency) data on the SPDR S&P 500 ETF (ticker SPY).[14] These high-frequency data are used to generate daily probability distributions of stock returns (outcomes), yielding an *endogenous* set of distinct (and equally likely) intra-month probability priors. Assuming daily stock returns are normally distributed, and after appropriately partitioning the stock return support, Izhakian (2020) measures the degree of ambiguity as the expected volatility of probabilities across all outcomes in the (discretized) stock returns support.[15] For our purposes, an increase in the ambiguity index measures an increase in the expected volatility of uncertain probabilities over stock return outcomes, that is, in how unstable the daily distribution

[13]In the EUUP model, agents preferences for ambiguity are formalized by agents' preferences for mean preserving spreads in uncertain probabilities, mirroring the canonical formalization of preferences for risk as agents' preferences for mean preserving spreads in lotteries' outcomes (Rothschild and Stiglitz, 1978). This assumption replaces the assumption that agents preferences for ambiguity are represented by agents' preferences for mean preserving spreads in certainty equivalent utilities (e.g., Klibanoff et al., 2005; Seo, 2009; Hayashi and Miao, 2011), which yields measures of ambiguity that are not independent of risk and preferences for risk. Fu et al. (2023) discuss the theoretical interpretation of the index; our empirical analysis uses its observed variation in financial markets and directly examines its relationship with conventional measures of risk and uncertainty.

[14]The SPDR S&P 500 ETF is a frequently traded fund designed to replicate the (relatively less liquid) S&P 500 index.

[15]See Izhakian (2020) for more details regarding the construction of the index.

of returns is within the month. Our variable $ambig_t$ is the log of the square root of this measure, averaged over the months of each quarter.

Inference. Confidence bands are computed with a recursive-design moving-block residual bootstrap (10,000 replications). Each replication draws blocks of eight consecutive quarters of dates, with wraparound (Künsch, 1989; Politis and Romano, 1992); at each drawn date every country receives its own structural residual vector and the global block receives its residual vector of that date, so that the serial dependence of the residuals within a block and their correlation across countries and with the global block at a given date are preserved. The replication regenerates the data from the estimated system starting at the observed initial conditions, re-estimates the PSVAR, and recomputes the impulse responses with shock sizes fixed at their point estimates. We report 68 and 95 percent intervals of Hall (1992), which reflect the bootstrap quantiles around the point estimate and thereby correct for the downward small-sample bias of least-squares estimates of persistent dynamics, and show responses over 20 quarters. The i.i.d. residual bootstrap with percentile bands of Uribe and Yue (2006) and Akıncı (2013), bands from a common simulated global path, a wild bootstrap by date, the bias-corrected bootstrap of Kilian (1998) and Bayesian credible sets give the same conclusions for the GDP and investment responses (Online Appendix C.3 and Table C.2).

3 Benchmark Analysis

We present our main findings from the estimated PSVAR using impulse response functions (IRFs) and forecast error variance decompositions (FEVDs). Section 3.1 reports EME responses to structurally identified shocks to the degree of ambiguity in U.S. equity markets. Section 3.2 does the same for shocks to global financial risk, the U.S. interest rate, and the country spread, Section 3.3 sets out a portfolio interpretation of the mechanism and inspects it, and Section 3.4 presents variance and historical decompositions. All IRF figures display responses to a positive, unanticipated, one standard deviation shock over 20 quarters, along

with 95% and 68% bootstrap confidence bands (Section 2), and Table 3.1 reports the peak responses.[16]

3.1 Impulse Response Functions: Ambiguity

Figure (3.1) shows IRFs to an unanticipated one standard deviation shock to the degree of ambiguity in U.S. equity markets, which raises the ambiguity index by 21.6 percent on impact. The joint responses point to portfolio reallocation toward foreign assets rather than a conventional flight to safety. In line with this, as shown in panel (e) of the figure, the BAA spread (labeled "global risk") declines by about 0.1 percentage points three quarters after the shock, a response significant at the 68 percent level for ten quarters. As in Akıncı (2013), an innovation to BAA spread is a proxy for a global financial risk shock. In contrast, the response of the BAA spread to shocks in other variables is intuitively in line with reflecting endogenous bond market dynamics. In the present case, Section 3.3.3 shows that the decline reflects higher Treasury yields rather than lower corporate yields.

Also in line with a shift toward risky foreign assets, EME investment exhibits a prominent and protracted increase that is significant at the 95 percent level from the sixth quarter to the end of the five-year horizon, peaking at roughly 1 percent above trend around eight quarters after the shock. In line with this, EME GDP follows similar dynamics, peaking at nearly 0.4 percent above trend seven quarters after the shock, with a response that is significant at the 95 percent level from the fourth quarter on. The expansion is a medium-run phenomenon: it builds up over two years and remains significantly above trend through the fifth year. The trade balance rises briefly above trend and then, consistent with foreign capital inflows, falls below trend from the second year on. Finally, the country spread ("EMBI spread") dynamics are dominated by a protracted decline below trend, of about 0.1 percentage points after four quarters (significant at the 68 percent level from the third to the fifteenth quarter). This response is consistent with the boom in EME GDP and investment taking place in tandem with EMEs being perceived as less risky in relative terms. Moreover, the joint response

[16] To place our findings in context, Online Appendix C.1 reviews key findings from Akıncı (2013)—the framework we build on—and compares them with our results.

of the EME real variables and the country spread reflect an important virtuous feedback loop that was previously emphasized by Uribe and Yue (2006) and Akıncı (2013). Indeed, stronger economic conditions lead to a lower country spread, which in turn helps keep EME GDP and investment above trend longer than would otherwise be the case.

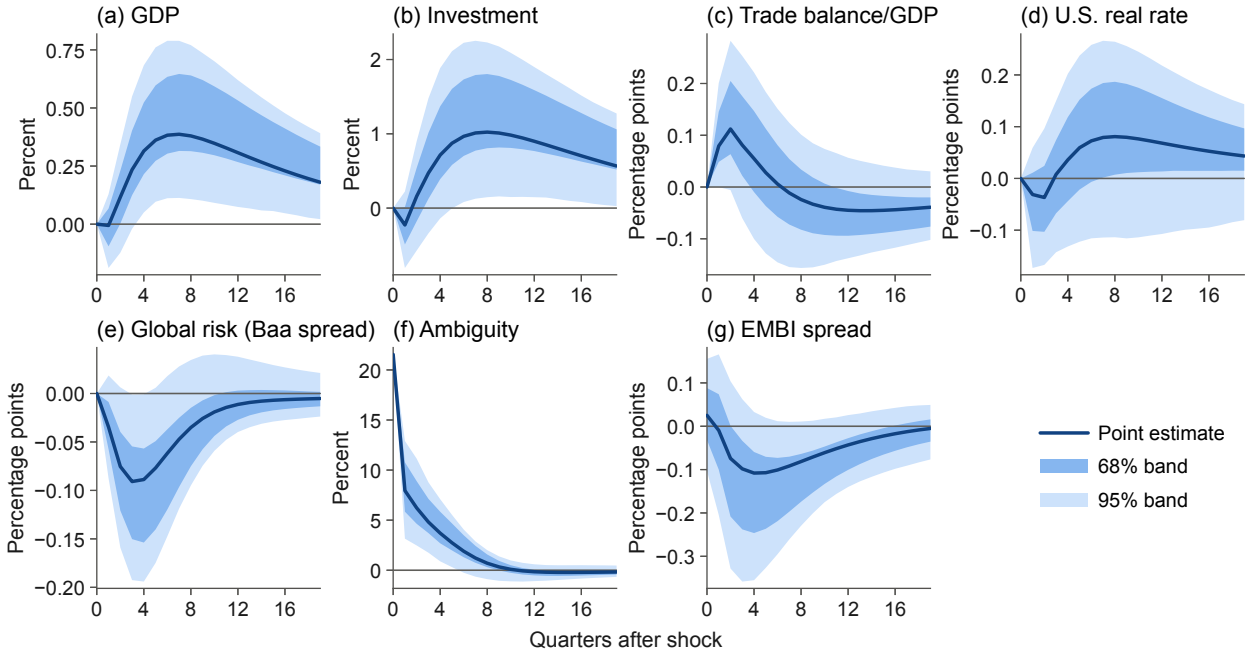


Figure 3.1: IRFs to one std. dev. shock to ambiguity

Note: Responses of the benchmark PSVAR to a one standard deviation ambiguity shock (21.6 percent on impact). Horizon 0 is the impact quarter. GDP, investment and ambiguity are in percent (log deviations from trend ×100); the trade balance/GDP ratio, the U.S. real interest rate, the BAA spread and the EMBI spread are in percentage points. Dark and light shaded areas: 68 and 95 percent Hall bands from 10,000 replications of the moving-block residual bootstrap (Section 2).

3.2 Additional Impulse Response Functions

As shown in Figure 3.2 and in Online Appendix Figures C.7 and C.8, the following results are in line with the 6-variable PSVAR from Akıncı (2013) and, therefore, unaffected by the inclusion of ambiguity in the analysis. Shocks to global financial risk and the country spread generate significant contractions in EMEs, while dynamics in the case of a U.S. interest rate shock are not significant, even if, in this case, also contractionary as in Uribe and Yue (2006).[17]

[17] Online Appendix C.1 complements the analysis presented here by running the Akıncı (2013) 6-variable PSVAR with our own benchmark data.

Table 3.1: Peak responses to the four structural shocks

Response of	Peak	h	68% interval	95% interval	68% range	95% range	Opp. sign, 68%
Shock: Ambiguity							
GDP	0.387	7	[0.32, 0.65]	[0.11, 0.79]	2–19	4–19	none
Investment	1.023	8	[0.81, 1.80]	[0.15, 2.23]	3–19	6–19	none
Trade balance/GDP	0.112	2	[0.06, 0.20]	[−0.01, 0.28]	1–3	none	11–19
U.S. real rate	0.081	8	[0.01, 0.19]	[−0.11, 0.26]	7–19	none	none
Baa spread	−0.091	3	[−0.15, −0.05]	[−0.19, 0.00]	1–10	3–4	none
Ambiguity (own)	21.552	0	[21.55, 21.55]	[21.55, 21.55]	0–8	0–5	13–19
EMBI spread	−0.108	4	[−0.25, −0.06]	[−0.36, 0.03]	3–15	none	none
Shock: Global risk (Baa spread)							
GDP	−0.352	2	[−0.55, −0.29]	[−0.72, −0.21]	1–6	1–4	none
Investment	−0.744	2	[−1.30, −0.52]	[−1.93, −0.22]	1–4	1–2	19
Trade balance/GDP	0.102	2	[0.06, 0.23]	[0.00, 0.34]	1–6	1	none
U.S. real rate	−0.463	4	[−0.57, −0.39]	[−0.65, −0.26]	1–19	1–15	none
Baa spread (own)	0.355	1	[0.34, 0.38]	[0.31, 0.41]	0–10	0–5	none
Ambiguity	−8.245	1	[−9.84, −5.78]	[−11.96, −2.39]	1–5	1–3	8–19
EMBI spread	0.356	0	[0.23, 0.42]	[0.08, 0.47]	0–3	0	8–19
Shock: U.S. real rate							
GDP	−0.089	2	[−0.18, 0.03]	[−0.29, 0.14]	none	none	none
Investment	−0.348	1	[−0.72, 0.01]	[−1.06, 0.29]	none	none	none
Trade balance/GDP	0.102	2	[0.03, 0.19]	[−0.03, 0.27]	1–10	none	none
U.S. real rate (own)	0.637	0	[0.64, 0.64]	[0.64, 0.64]	0–19	0–13	none
Baa spread	0.074	0	[0.01, 0.16]	[−0.05, 0.20]	0	none	8–12
Ambiguity	−3.444	1	[−5.94, −0.60]	[−8.59, 1.59]	1–19	none	none
EMBI spread	0.153	1	[0.07, 0.27]	[−0.03, 0.38]	0–19	0, 2–3, 10–19	none
Shock: Country spread (EMBI)							
GDP	−0.329	2	[−0.48, −0.25]	[−0.57, −0.10]	1–7	1–4	none
Investment	−1.100	2	[−1.57, −0.83]	[−1.84, −0.30]	1–9	1–4	none
Trade balance/GDP	0.186	2	[0.15, 0.27]	[0.07, 0.32]	1–10	1–4	none
U.S. real rate	zero by construction (block exogeneity)						
Baa spread	zero by construction (block exogeneity)						
Ambiguity	zero by construction (block exogeneity)						
EMBI spread (own)	0.859	0	[0.86, 0.86]	[0.86, 0.86]	0–19	0–14	none

Note: Peak (largest absolute) response over h = 0-19 of each variable to a one standard deviation shock in the benchmark PSVAR, its horizon (quarters; 0 = impact) and the 68 and 95 percent bootstrap intervals at that horizon. Ranges: horizons at which the 68 (95) percent band excludes zero with the sign of the peak; the last column gives horizons at which the 68 percent band excludes zero with the opposite sign. For the shocked variable the peak is the impact response, that is, the shock size (for the BAA spread, its peak one quarter after the shock). GDP, investment and ambiguity in percent; trade balance/GDP, rates and spreads in percentage points. Bands: 10,000 replications of the moving-block residual bootstrap, Hall intervals.

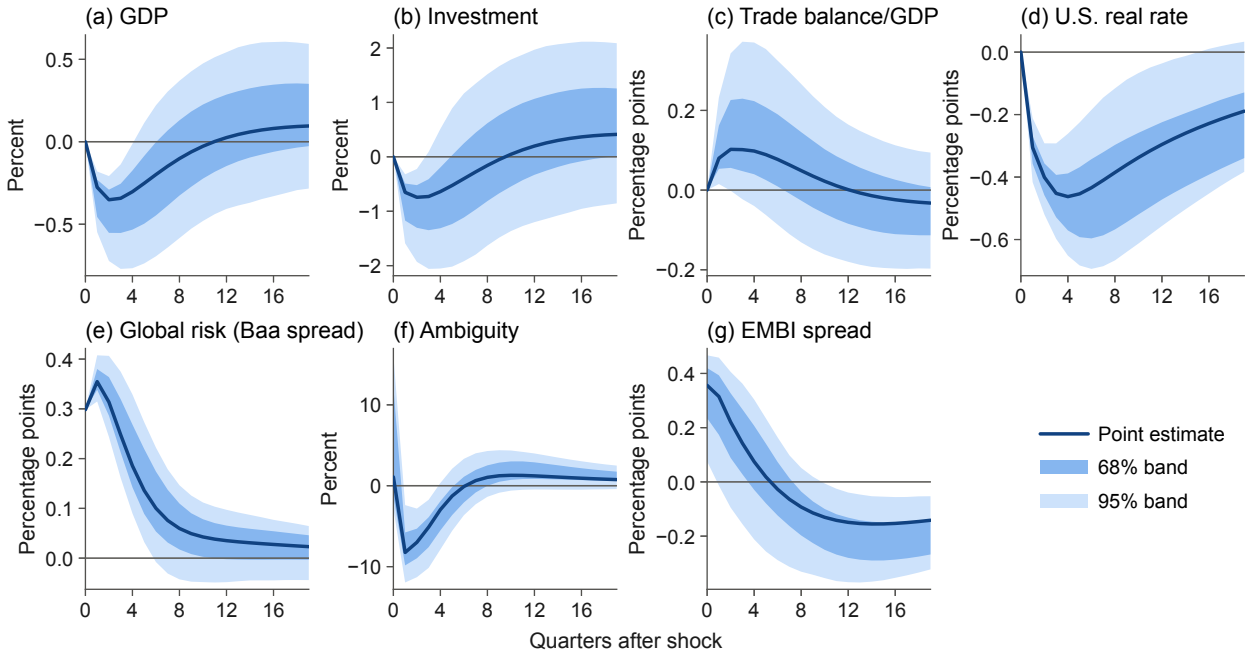


Figure 3.2: IRFs to one std. dev. shock to BAA spread

Note: Responses of the benchmark PSVAR to a one standard deviation global financial risk (BAA spread) shock (0.30 percentage points on impact). Units and bands as in Figure 3.1.

Figure 3.2 reveals an additional feature of global financial risk shocks: ambiguity falls sharply, by about 8 percent one quarter after the shock, with a response significant at the 95 percent level. This is in line with the shock to global financial risk signaling worldwide economic deterioration strongly enough so that uncertainty (in the Knightian sense) about conditions in U.S. equity market actually dissipates. Further in line with this, Section 3.3.4 presents suggestive evidence that global financial risk plays a central role in shaping investors *beliefs* about the underlying return generating process for U.S. equity returns and, through this channel, reduces the degree of ambiguity. Beyond these dynamics, the shock to global financial risk triggers a flight to safety that is reflected by an increase in the EMBI spread (0.36 percentage points on impact) and, consistent with foreign capital outflows, declines in EME GDP and investment (0.35 and 0.74 percent at the trough, two quarters after the shock) alongside an initial increase in the trade balance. Flight-to-safety dynamics are, in this case, further underscored by a steep decline in the U.S. real interest rate, of 0.46 percentage points after four quarters.[18]

[18]The decline in the U.S. real rate in response to a global financial risk shock is in line with other studies that build on Akıncı (2013), such as Epstein et al. (2019).

As anticipated, Online Appendix Figure C.7 shows that, in line with Uribe and Yue (2006), a shock to the U.S. interest rate is contractionary for EMEs but, in line with Akıncı (2013), these dynamics are not significant; the shock raises the country spread. Finally, and also as anticipated, Online Appendix Figure C.8 shows that a shock to the EME country spread is strongly and significantly contractionary, with EME GDP and investment dropping by about 0.3 and 1.1 percent below trend two quarters after the shock and the trade balance rising significantly above trend.

3.3 Evidence on the Transmission Mechanism

In Section 3.1 we document that an unexpected surge in the degree of ambiguity in U.S. equity markets has significant, protracted, and *counterintuitive* expansionary effects on EME GDP and investment. The point estimate dynamics of the EME trade balance further suggests a medium-term surge in foreign capital inflows into EMEs in response to heightened uncertainty in U.S. equity markets. Importantly, these patterns are accompanied by declines in the BAA corporate spread and country spreads. Taken together, these dynamics are consistent with a financial market channel in which international investors, faced with heightened ambiguity about U.S. equity returns, reallocate their portfolios toward foreign assets, easing the external financing conditions of EMEs.

We examine this transmission channel using U.S. real and financial outcomes, Treasury yields and credit spreads, international portfolio flows and bank lending, EME financial variables, survey measures of beliefs, and alternative uncertainty measures. We proceed in five steps. First, we set out the portfolio problem that organizes the evidence and gives the predictions we take to the data. Second, we examine whether a shock to the degree of ambiguity in U.S. equity markets generates "push" factors in the U.S. economy and stock market. Third, we use additional asset prices and data on capital flows to trace where the reallocated funds go, how they reach EMEs, and the extent to which the BAA corporate bond market amplifies the effect of ambiguity shocks for EMEs. Fourth, to shed light on the feedback effect between the corporate bond market and equity market ambiguity,

we provide suggestive evidence about the dynamics of investor beliefs. Finally, we test whether the effects of ambiguity in U.S. equity markets are distinct from those of other U.S.-centric measures of uncertainty commonly used in the literature and of measures of financial conditions.

3.3.1 A Portfolio Interpretation

A simple portfolio problem organizes the evidence in this section and tells us what to look for. A U.S. investor allocates wealth between a riskless asset, U.S. equity and a foreign risky asset. Given a vector of expected excess returns μ, excess returns are normal with a known covariance matrix Σ, where σ_U and σ_F denote the volatilities of the two risky returns and ρ their correlation, and the investor ranks portfolio shares $w = (w_U, w_F)'$ by the mean-variance criterion $w'\mu - \frac{\gamma}{2} w'\Sigma w$, with γ the coefficient of risk aversion. Ambiguity concerns the mean of U.S. equity returns: the investor does not know it and entertains every value in the interval $[m_U - a,\, m_U + a]$, where $a \geq 0$ is the degree of ambiguity, while the mean of the foreign asset, m_F, is known.[19] The investor evaluates a portfolio with the α-maxmin criterion of Ghirardato et al. (2004), a weighted average of its worst-case and best-case values over this interval with weight $\alpha \in [0, 1]$ on the worst case; $\alpha = 1$ is the maxmin portfolio problem of Garlappi et al. (2007), and $\alpha > 1/2$ is ambiguity aversion. Because the criterion is linear in the means, it reduces to

$$\max_{w} \; w'm - \kappa\,|w_U| - \tfrac{\gamma}{2}\, w'\Sigma w, \qquad \kappa \equiv (2\alpha - 1)\, a, \tag{3.1}$$

where $m = (m_U, m_F)'$. The degree of ambiguity a and the attitude α enter only through κ, which mirrors the separation between the degree of ambiguity that the index measures and attitudes toward it. Problem (3.1) has a closed-form solution (Online Appendix C.16), and two of its properties guide the empirical work.

Proposition 1 (Reallocation under ambiguity) *Let the investor be ambiguity averse*

[19] What matters is that ambiguity about the foreign asset moves less than ambiguity about U.S. equity; Online Appendix C.16 gives the condition.

($\alpha > 1/2$), hold U.S. equity, and let $\rho > 0$. An increase in ambiguity about U.S. equity returns lowers the U.S. equity share and raises the foreign risky share. Of each unit of U.S. equity sold, a fraction $\beta \equiv \rho\,\sigma_U/\sigma_F$ moves into the foreign risky asset and the remainder into the riskless asset.

Ambiguity about the U.S. mean acts like additional U.S.-specific variance that is uncorrelated with the foreign asset: it raises the cost of holding U.S. equity without raising the covariance between U.S. and foreign returns. Part of the risk-bearing capacity freed by selling U.S. equity therefore moves into the correlated foreign asset instead of into cash. The fraction β is the slope of a regression of U.S. on foreign returns; with annual volatilities of 17 and 28 percent and a correlation of 0.6, typical of U.S. and emerging-market equity, it is 0.36. The smooth ambiguity preferences of Klibanoff et al. (2005), in the mean-variance form of Maccheroni et al. (2013), give the same fraction. Each condition is essential: with $\rho = 0$ the foreign share does not move; with ambiguity seeking ($\alpha < 1/2$) every sign reverses; and once κ exceeds the premium on U.S. equity net of its foreign-asset hedge, the investor exits U.S. equity and further ambiguity has no effect.

Proposition 2 (Common volatility: flight to safety) *Let returns load on a common factor, so that $\Sigma = \sigma_G^2\, bb' + V$ with loadings $b > 0$ and diagonal V. An increase in the variance of the common factor, σ_G^2, lowers every risky share and raises the riskless share.*

A rise in common volatility raises the covariance of the two risky assets together with their variances. Neither becomes relatively more attractive, and the only escape is the riskless asset. Propositions 1 and 2 therefore separate two shocks that both raise uncertainty: U.S.-specific ambiguity pushes portfolios toward foreign risky assets, and common risk pushes them toward safety.[20] The effect reaches EMEs through the terms on which they borrow. With a spread schedule in the spirit of Uribe and Yue (2006), in which the EME spread falls with foreign demand for EME claims and with EME output while investment falls with the

[20] A rise in U.S.-specific volatility with ρ held fixed also raises the foreign share, by about two thirds as much per unit of U.S. equity sold as ambiguity does at the calibration above. The informative comparison is thus with common risk, which is how volatility and credit-spread shocks behave in the data: they raise EME spreads and lower EME equity prices (Section 3.2).

spread, the sign of the output response to U.S. ambiguity is the sign of the change in the foreign risky share, and the spread-output feedback amplifies it (Online Appendix C.16). The foreign asset need not be EME securities: what reaches EMEs is an easier external financing schedule, whoever the lender, which is how we read the evidence below.

The propositions yield the predictions that the rest of this section takes to the data: after an ambiguity shock, U.S. investors' exposure to U.S. assets falls without a rise in the demand for safe assets, the demand for foreign risky assets rises, EME spreads fall and EME activity expands; after a global financial risk shock, each of these responses reverses. The simplest version of the model leaves out three things that the evidence speaks to. First, in the two-asset problem the riskless share rises, so that if long Treasuries were the riskless asset their yields would fall; they rise in the data. Since about 2000 long Treasuries have hedged equity, with a negative stock-bond correlation, so that a cut in U.S. equity exposure lowers the hedging demand for Treasuries: with the Treasury bill as the riskless asset, the term premium rises after an ambiguity shock and falls after a common volatility shock (Online Appendix C.16). Second, in general equilibrium the U.S. equity sold must be absorbed; with an elastic supply of U.S. capital the reallocation survives and U.S. investment falls, as Section 3.3.2 documents. Third, a currency and intermediaries with limited risk-bearing capacity (Gabaix and Maggiori, 2015) are needed for the dollar: net U.S. purchases of foreign assets then depreciate it after an ambiguity shock, and a common risk shock appreciates it. The two-period model has no dynamics; we attribute the gradual two-to-three-year build-up of the EME expansion to the spread-output feedback and to investment adjustment.

3.3.2 U.S. Real and Financial Effects of U.S. Ambiguity Shocks.

We next examine how U.S. real activity and equity-market returns respond to ambiguity shocks. These responses help characterize the portfolio-reallocation mechanism and place the EME expansion alongside its domestic U.S. effects, including the temporary increase in U.S. industrial production that Asano et al. (2026) document after event-identified ambiguity shocks. We modify our benchmark specification by replacing the U.S. interest rate, in turn, with the expected returns on the SPDR S&P 500 ETF (SPY), U.S. GDP, and U.S.

investment, and, alternatively, by adding each of these variables to the benchmark global block, which keeps the U.S. interest rate.

Online Appendix Figure C.15 shows the impact of a shock to ambiguity after swapping out the U.S. interest rate for expected SPY returns.[21] Heightened ambiguity is followed by a decline in expected SPY returns one quarter after the shock, while realized U.S. equity prices do not fall (Section 3.3.3). The dynamics of all other variables remain close to the benchmark model.

Next, we show the impact of a shock to ambiguity after swapping out the U.S. rate, in turn, for U.S. real GDP (Online Appendix Figure C.17) and investment (Online Appendix Figure C.18).[22] The figures show that ambiguity shocks are followed by a decline in U.S. investment, while U.S. GDP barely moves. As before, the response of other variables is in line with the benchmark model, and the EME boom remains significant at the 95 percent level when U.S. GDP or investment is added to the benchmark model.

Together with the findings in Section 3.1, this analysis underscores a previously undocumented asymmetry in business cycle responses to U.S. ambiguity shocks: while ambiguity shocks weigh on U.S. investment, their international transmission stimulates EMEs' activity, in sharp contrast to the effects of previously examined uncertainty (but not ambiguity) shocks (e.g., Bhattarai et al., 2020). The muted response of U.S. output is in line with the temporary increase in U.S. industrial production that Asano et al. (2026) document after event-identified ambiguity shocks.

3.3.3 No Flight to Safety: Prices and Quantities

We first ask where the decline in the BAA spread comes from. To this end, we replace the BAA spread in the benchmark model by its two components, the spread of Aaa corporate yields over the 10-year Treasury yield and the Baa-Aaa default spread, and we add, one at a time, U.S. asset prices as an eighth variable ordered after ambiguity: the 10-year and 2-year

[21] The ordering of the shock block now assumes that global financial risk is the most exogenous variable. Then, since the construction of the ambiguity variable is based off the behavior of the stock market, in the shock block ambiguity goes last, preceded by expected SPY returns.

[22] For consistency with the analysis of expected SPY returns from above, we incorporate U.S. GDP and investment as the second variable in the shock block, before ambiguity.

Treasury yields, the 10-year term premium of Adrian et al. (2013), credit spreads, a financial conditions index and equity returns. Panels (a) to (d) of Figure 3.3 show the main results. About two thirds of the decline in the BAA spread is a fall in the Aaa-Treasury spread (0.06 percentage points after four quarters), while the Baa-Aaa default spread falls by only 0.03 percentage points. At the same time, the 10-year Treasury yield rises by 0.11 percentage points after a year, the 2-year yield by 0.14 percentage points, and the term premium by 0.06 to 0.08 percentage points in the first three quarters.[23] In local projections, Baa yields themselves do not fall. The compression of the BAA spread is therefore mostly a Treasury leg: Treasury yields rise faster than corporate yields, which is the opposite of a flight to safety. As a validation, a global financial risk shock produces the textbook flight to safety in the same systems, with falling Treasury yields and a rising excess bond premium (Online Appendix Figure C.34).

Next, we trace quantities with local projections (Jordà, 2005) of portfolio flows and EME outcomes on the identified ambiguity shock, that is, the structural residual of the ambiguity equation of the benchmark model. Table 3.2 and panels (e) to (h) of Figure 3.3 report the results. U.S. residents raise their net purchases of foreign bonds by 0.42 percent of quarterly U.S. GDP in the year after the shock and by 1.28 percent over three years, and their net purchases of foreign stocks by 0.47 percent within a year; inflows into world equity funds also rise. We find no sign that investors sell U.S. equities: U.S. equity prices, domestic equity fund flows and foreign purchases of U.S. stocks do not fall. Issuer-level data on U.S. cross-border holdings (Bertaut and Judson, 2014) show which foreign bonds are bought. Of the roughly 12 billion dollars of foreign bonds that U.S. residents purchase in the twelve months after the monthly identified ambiguity shock of Section 3.3.5, about half are issued in Europe and a quarter in Caribbean financial centres, and a third in the twelve small open advanced economies of Section 4 (the groups overlap); bonds issued by eighteen EMEs account for 7 percent, and U.S. purchases of EME stocks do not rise (Online Appendix Table C.8 and

[23] The rise in the 10-year yield is significant at the 95 percent level from the second quarter to the end of the horizon, and that of the term premium in the second quarter. In local projections, the rise in the term premium is significant at the 90 percent level in the first four quarters, as is the rise in the 10-year yield after four to six quarters.

Figure C.27). The reallocation is therefore a move into foreign assets at large, and EMEs benefit from the easier global financial conditions that follow rather than from direct U.S. purchases of their securities. On the receiving side, EME dollar equity prices rise by 4 to 6 percent within a year, cross-border bank claims on eighteen EMEs rise by 1.8 percent of their quarterly GDP after two years, 85 percent of them from euro-area banks, their foreign exchange reserves rise by about 2 percent, and their EMBI spreads fall with a lag, by 0.30 percentage points after ten quarters (Online Appendix Figures C.28 and C.31). The six EMEs in our sample also attract non-resident portfolio equity inflows of 0.56 percent of their quarterly GDP within a year over the estimation windows of the PSVAR, and of 0.3 to 0.5 percent over the countries' full data spans and in the eleven-country panel (Online Appendix Table C.9), and their international debt issuance rises after two years (Online Appendix C.8). A global financial risk shock produces the mirror image: a sudden stop in portfolio and bank inflows, falling EME equity prices and depreciating EME currencies.

The EME side of the reallocation is visible in a broader set of EME financial variables, which we add one at a time to the country block of the benchmark model and also study with local projections (Online Appendix Figure C.31 and Table C.10). After an ambiguity shock, EME equity prices in dollars rise by 4.4 percent within a year (significant at the 95 percent level) and real foreign exchange reserves by about 2 percent, a rise that persists through the fifth year (significant at the 68 percent level throughout and at the 95 percent level from the tenth quarter); central banks cut policy rates briefly, by 0.3 to 0.4 percentage points in the first year; consumption rises with a lag; and EME currencies do not appreciate for about three years, consistent with central banks absorbing part of the inflows in reserves. A global financial risk shock does the opposite: EME currencies depreciate by about 2 percent within a quarter, equity prices fall by 5 percent and reserves do not move. The dollar tells the same story from the U.S. side: the broad dollar index falls by about 1.2 percent in the year after an ambiguity shock and strengthens on impact after a global financial risk shock (Online Appendix Figure C.32), the pattern expected when U.S. investors move into foreign assets rather than into dollar safe assets. The EME output expansion remains significant at the 95 percent level in all 22 of these augmented systems.

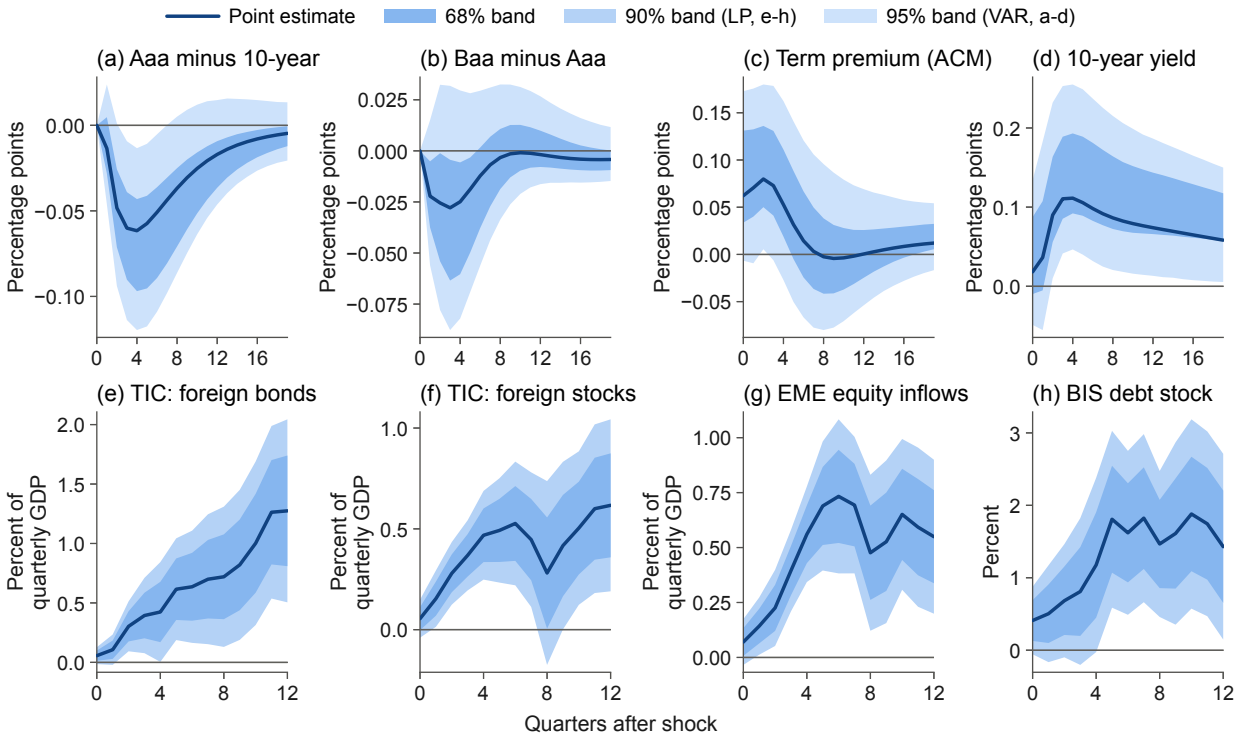


Figure 3.3: Financial-market prices and quantities after an ambiguity shock

Note: Responses to a one standard deviation ambiguity shock. Panels (a)-(d): impulse responses (percentage points) from the benchmark model with the BAA spread replaced by its two components, the Aaa-10-year Treasury spread and the Baa-Aaa spread (a, b), or with an eighth variable ordered after ambiguity, the Adrian et al. (2013) 10-year term premium (c) or the 10-year Treasury yield (d); 68 and 95 percent bands from 10,000 bootstrap replications. Panels (e)-(h): local projections on the benchmark ambiguity shock (the structural residual of the ambiguity equation, standardized), with 68 and 90 percent bands from Newey and West (1987) (e, f) or Driscoll and Kraay (1998) (g, h) standard errors: cumulative U.S. net purchases of foreign bonds and of foreign stocks (TIC), in percent of quarterly U.S. GDP; cumulative non-resident portfolio equity inflows into the six sample countries (IMF balance of payments), in percent of quarterly GDP; and the stock of international debt securities issued by residents (BIS), $100 \times$ log. Horizons: 0-19 (VAR), 0-12 (local projections).

Table 3.2: Quantities after an ambiguity shock

	Benchmark ambiguity shock		VIX-orthogonal shock	
	$h = 4$	$h = 8$	$h = 4$	$h = 8$
A. U.S. investors (time-series LPs, percent of quarterly U.S. GDP, cumulative)				
TIC: U.S. net purchases of foreign bonds	0.42^{**}	0.72^{**}	0.32^{*}	0.70^{*}
	(0.26)	(0.36)	(0.28)	(0.49)
TIC: U.S. net purchases of foreign stocks	0.47^{**}	0.28^{*}	0.10	-0.02
	(0.13)	(0.28)	(0.17)	(0.25)
ICI: world equity fund flows	0.44^{**}	0.24^{*}	0.37^{**}	0.06
	(0.10)	(0.23)	(0.12)	(0.23)
ICI: bond fund flows (2007–2019)	-0.34	-0.69^{*}	0.22	0.06
	(0.40)	(0.61)	(0.30)	(0.40)
Z.1: mutual funds, corporate and foreign bonds	0.34^{**}	0.10	0.43^{**}	0.04
	(0.16)	(0.24)	(0.19)	(0.30)
B. EMEs (panel LPs, six countries; percent of quarterly GDP unless stated)				
BIS international debt, net issuance (cumulative)	0.02	0.56^{**}	-0.47^{*}	-0.35
	(0.26)	(0.20)	(0.32)	(0.37)
BIS international debt stock (log $\times$ 100)	1.18^{*}	1.47^{**}	0.76^{*}	0.76
	(0.73)	(0.61)	(0.69)	(0.81)
IMF BoP portfolio debt inflows (cumulative)	0.19	-0.16	0.15	-0.42^{*}
	(0.30)	(0.46)	(0.32)	(0.39)
IMF BoP portfolio equity inflows (cumulative)	0.56^{**}	0.48^{**}	0.24^{*}	0.05
	(0.13)	(0.22)	(0.21)	(0.25)
MSCI country index, USD (log $\times$ 100)	6.28^{**}	4.22^{*}	7.39^{**}	7.77^{**}
	(2.55)	(3.31)	(1.97)	(3.06)

Note: Local-projection coefficients (standard errors in parentheses) of each outcome at horizons $h = 4$ and 8 quarters on a one standard deviation shock: the benchmark ambiguity shock (structural residual of the ambiguity equation of the benchmark PSVAR, 1994:Q4-2019:Q4) and the VIX-orthogonal ambiguity shock (from the eight-variable model with the log VIX ordered before ambiguity). Flows are cumulated over quarters 0 to h. Panel A: U.S. time series, controls two lags of the outcome and of the U.S. real rate, BAA spread and ambiguity; Newey-West standard errors (Bartlett, lag $h+1$). Panel B: panel of the six benchmark countries (seven units), country fixed effects, controls add two lags of GDP, investment, trade balance/GDP and the EMBI spread; Driscoll-Kraay standard errors. Quarter dummies for non-seasonally-adjusted flows. **, *: 90 and 68 percent normal intervals exclude zero. Outcomes stop in 2019:Q4; ICI flows cover 2007-2019.

The portfolio-reallocation mechanism also appears when ambiguity shocks are identified separately from contemporaneous changes in the VIX. We construct a VIX-orthogonal shock from the eight-variable model that orders the VIX before ambiguity (Section 3.3.5). Under this identification, EME output and sovereign spreads continue to respond strongly: EME GDP rises by 0.45 percent after four quarters in the local projections, while EME equity prices, Treasury yields, term premia, and U.S. purchases of foreign bonds increase. Corporate-spread compression is substantially weaker, as are EME debt issuance and portfolio equity inflows (Table 3.2). These findings distinguish the EME expansion from the contemporaneous decline in volatility that accompanies the benchmark ambiguity shock.

Finally, we explore the extent to which the Baa corporate bond market serves as a transmission channel of the shock to ambiguity. To this end, we generate IRFs from the benchmark model (without reestimating it) but assuming that the Baa spread is simply an AR2 process not depending on the global financial variables, the U.S. interest and ambiguity. As such, in the Baa equation we set the R^{US}_{t-j} coefficients equal to zero for $j = 0, 1, 2$ and the $ambig_{t-j}$ coefficients equal to zero for $j = 1, 2$ (note that in the benchmark case the ambiguity coefficient is already zero for $j = 0$ in the case of the Baa equation), while keeping everything else the same as in the benchmark model. Overall, as shown in Online Appendix Figure C.44, the results of this exercise are in essence the same as in our baseline analysis in the case of the Baa shock. The ambiguity-driven expansion persists when the BAA spread is restricted not to respond endogenously to global financial variables. EME GDP still peaks 0.32 percent above trend, compared with 0.39 percent in the benchmark. Thus, the endogenous BAA-spread response amplifies the expansion by roughly one fifth but is not necessary for it. Together with the asset-price evidence, this pattern points to the broader compression of spreads relative to Treasury yields as an amplification channel.

3.3.4 Global Financial Risk, Investors Beliefs, and Ambiguity.

The decline in ambiguity after a global financial risk shock suggests a connection between credit-market conditions and investors' *beliefs* about equity returns. We investigate this connection using a three-variable VAR for the BAA spread, the ambiguity index, and the

dispersion of expected one-year S&P 500 returns reported in the CFO survey (Online Appendix C.5). This evidence extends the analysis of ambiguity to the belief-formation channel of global financial risk shocks. We measure dispersion as the difference between the average 90th and the average 10th percentiles of the expected S&P 500 return over the next year reported by respondents to the CFO survey of Duke University and Federal Reserve Banks of Richmond and Atlanta.[24] Online Appendix Table C.6 shows that global financial risk Granger causes investors' beliefs (p-value 0.006), which in turn Granger cause ambiguity (p-value 0.002). Crucially, panel (b) of Online Appendix Figure C.14 shows that an unanticipated increase in BAA spreads widens the range of expected returns across CFO respondents, by up to 0.7 percentage points (significant at the 95 percent level for four quarters). In turn, panel (f) shows that a widening of the range of expected returns reduces ambiguity, by 6 to 8 percent one quarter later depending on the ordering. The same pattern holds, at the 5 percent level, when we measure dispersion by the cross-respondent standard deviation of expected returns in the CFO survey files for 2004-2019. Monthly data tell the same story with more observations. In a monthly VAR for the BAA spread and ambiguity, a global financial risk shock lowers ambiguity by 4 to 6 percent after five to six months, and by 3 to 5 percent when the VIX is also in the system (Online Appendix Figure C.12). Among monthly survey measures of investors' beliefs, the one closest to the probability of a bad state, the share of individual investors in the Yale survey who attach a low probability to a crash (Goetzmann et al., 2017), completes the chain: a global financial risk shock lowers this crash confidence by 0.8 percentage points over six months, a fall in crash confidence is followed by 3 to 4 percent lower ambiguity one to three months later, both significant at the 95 percent level, and crash confidence Granger-causes ambiguity (p-value 0.03) while not responding to the identified ambiguity shock (Online Appendix Table C.5).

These results point to a belief-coordination channel: a widening of BAA spreads can provide information about a worsening economic outlook (e.g. Gertler and Lown, 1999;

[24] On the CFO survey, see Ben-David et al. (2013). Giglio et al. (2021) show that the average expectations in the CFO survey are quite similar to the average beliefs they derived from a newly-designed survey administered to a large panel of wealthy retail investors over 2017-2019, and hence can be used as a proxy for investors' beliefs over 1-year ahead market returns.

Gilchrist and Zakrajšek, 2012), shifting probability assessments toward adverse states and narrowing the set of priors even as the dispersion of expected returns increases.We illustrate this mechanism with a simple example adopted from Brenner and Izhakian (2018). Suppose stock returns follow a discrete binomial distribution, with payoff $b = -5$ in the bad state and $g = 5$ in the good state. Assume a representative investor holds the following equally likely beliefs: $(P_1(b), P_1(g)) = (.5, .5)$, $(P_2(b), P_2(g)) = (.4, .6)$, $(P_3(b), P_3(g)) = (.34, .66)$ and $(P_4(b), P_4(g)) = (.22, .78)$, corresponding to expected returns 0, 1, 1.6 and 2.8, respectively. Now, consider an unexpected increase in the BAA spread that signals a worsening in economic conditions, captured by a lowering of the payoff in the bad state to $b = -30$, while keeping $g = 5$.[25] Crucially, assume further that an increase in global financial risk provides key information regarding the worsening of the economic outlook (e.g. Gertler and Lown, 1999; Gilchrist and Zakrajšek, 2012), spurring the representative investor to update its priors towards a narrower set of distributions that put more weights on bad economic outcomes, represented by the following smaller set of equally likely beliefs: $(P_1(b), P_1(g)) = (.5, .5)$ and $(P_2(b), P_2(g)) = (.4, .6)$. In this setting, an increase in the BAA spread produces an increase in the dispersion of expected returns, from $2.8 = 2.8 - 0$ to $3.5 = -9 - (-12.5)$, while simultaneously reducing the degree of ambiguity, from 0.10 to 0.05, as a result of the alignment of investors' beliefs about the bad outcome. This analysis contributes to the international finance literature—which highlights the role of global financial risks for EMEs (Akıncı, 2013)—by suggesting a further channel through which unexpected shocks to BAA spreads act as a coordination mechanism of international investors' beliefs, with consequences for EMEs.

3.3.5 Ambiguity and Conventional Measures of Uncertainty

To assess the extent to which accounting for ambiguity can yield unique insights relative to other commonly used measures of U.S.-centric uncertainty, we first estimate the benchmark model by swapping out ambiguity for, each in turn, the Chicago Board Options Exchange

[25] Among others, Gilchrist and Zakrajšek (2012) discusses a mechanism through which an increase in corporate credit spreads (bond market) may affect outcomes in the stock market. In particular, a widening of corporate credit spreads may signal a worsening in credit intermediary balance sheets, which, in turn, reduces the supply of credit, raises the cost of firms' external finance, and thereby curtails production and investment, ultimately worsening future profitability.

Volatility Index (VIX) and the financial, macro, and real uncertainty indexes of Jurado et al. (2015) (FU, MU and RU). As shown in Online Appendix Figures C.20 to C.24, in contrast to the effect of ambiguity, a shock to any one of these variables leads to a contraction in economic activity in EMEs (mildly so for the VIX) along with an increase in the country spread and increases in the Baa spread, and a shock to financial uncertainty also leads to a sharp and protracted decrease in the U.S. interest rate. These results suggest that shocks to any one of these uncertainty variables decrease risk appetite, with asset allocations shifting towards U.S. bonds, and that these U.S.-centric measures capture aspects of uncertainty similar to the Baa or global risk shock.

We next identify ambiguity shocks jointly with conventional uncertainty measures to separate their contributions. Because the ambiguity index is correlated with the VIX (-0.68) and financial uncertainty (-0.66) in levels, we add each measure to the benchmark global block, placing it alternatively before and after ambiguity. Table 3.3 and Figure 3.4 report the results. The boom survives the VIX, macro uncertainty and real uncertainty in either order, at the 95 percent level. With the VIX ordered first, the GDP response peaks at 0.32 percent, and with macro or real uncertainty in the system it is larger than in the benchmark (0.44 to 0.63 percent). Shocks to each of these measures, by contrast, are contractionary for EMEs (panels (c) and (d) of Figure 3.4). The same holds for the economic uncertainty index of Bekaert et al. (2022), whose shocks lower EME GDP in the first quarters after the shock (by 0.15 to 0.35 percent, significant at the 95 percent level), whereas shocks to the dispersion of professional forecasts, the proxy for ambiguity used by Ilut and Schneider (2014), have no effect on EME GDP; the ambiguity boom survives either measure in the system (Table 3.3 and Online Appendix Figure C.19). Every one of these measures rises with the VIX, and none is positively correlated with the ambiguity index, which is the only measure that moves against volatility. With the VIX ordered first, the decline in the BAA spread shrinks to about 0.04 percentage points and loses significance and the decline in the EMBI spread arrives later and is significant at the 68 percent level only, while GDP and investment remain significant at the 95 percent level; the spread compression is the part of the ambiguity shock that comes with a lower VIX, and the boom does not need it, as the

monthly identification below confirms.

The closest overlap arises with financial uncertainty and the excess bond premium, which incorporate information from related equity and credit markets. When financial uncertainty enters after ambiguity, EME GDP peaks 0.33 percent above trend and is significant at the 95 percent level; when it enters first, the peak is 0.19 percent and is significant at the 68 percent level. Adding the excess bond premium of Gilchrist and Zakrajšek (2012) to the quarterly VAR also substantially attenuates the output and investment responses. By contrast, conditioning on the components of the BAA spread, Treasury yields, term premia, or the Chicago Fed financial conditions index leaves the expansion statistically significant, with GDP peaks between 0.37 and 0.48 percent (Online Appendix Figure C.35). These differences motivate the monthly identification below, which separates ambiguity innovations from volatility, uncertainty, and credit-market movements within the month.

Monthly identification provides a complementary way to disentangle ambiguity innovations from contemporaneous changes in volatility, financial uncertainty, and credit-market conditions. We estimate a monthly VAR for 1993:M2-2019:M12 with two lags (selected by the BIC) in which ambiguity is ordered last, after the U.S. real rate, the BAA spread, the VIX, financial uncertainty at the one-month horizon, the excess bond premium and the NASDAQ stock return, so that the ambiguity shock is orthogonal within the month to all six. Summed within the quarter, this shock has a correlation of 0.69 with our benchmark ambiguity shock. Panels (a) to (c) of Figure 3.5 show that, in the monthly VAR, the shock is followed by a small and short-lived rise in the VIX and by a fall in financial uncertainty, while the excess bond premium and the BAA spread do not move. The decline in the VIX and the compression of the BAA spread that accompany our quarterly ambiguity shock are therefore co-movements within the month rather than responses to the shock.

The EME expansion is fully present under this identification. In panel local projections for the six countries (panels (d) to (f) of Figure 3.5), EME GDP rises by 0.5 percent and investment by 1.9 percent eleven quarters after the monthly shock, significant at the 90 percent level between the ninth and the twelfth quarter, and the EMBI spread falls by 0.34 percentage points, significant at the 90 percent level between the eighth and the eleventh

quarter. The response to the monthly shock builds up gradually, and GDP is still about 0.75 percent above trend after sixteen quarters. Adding two lags of the excess bond premium, the VIX and financial uncertainty to the controls leaves the GDP response at 0.57 percent. The aggregated monthly shock is also a strong instrument for the ambiguity innovation of the benchmark PSVAR: it explains half of its variance, with a first-stage F statistic of about 100, and used as an external instrument it yields GDP and investment responses of the same shape as the benchmark, with peaks of 0.3 and 0.9 percent when, as in the benchmark, the country variables do not respond within the quarter (Online Appendix C.14 and Table C.22). The boom in EMEs therefore does not hinge on how within-quarter movements in volatility, financial uncertainty and credit market conditions are attributed: an ambiguity shock that is orthogonal to all of them within the month produces it with a similar magnitude. Our monthly estimates share with the event-identified ambiguity shocks of Asano et al. (2026) a short-lived rise in the VIX after an ambiguity shock and an expansionary effect on activity; we extend their U.S. evidence to the international transmission of ambiguity shocks.

The identified shock is not a risk-appetite shock. Its largest positive realizations are crash months, October 2008 (7.3 standard deviations), October 1997, July 2002 and August 1998, when the within-month distribution of returns was most unstable, and its largest negative realizations are moderate sell-offs such as December 2018 and January 2016 (Online Appendix Figure C.56 and Table C.19). Months with large positive shocks are risk-on months, with rising stock prices and a falling VIX, no more often than the average month (47 against 46 percent); the contemporaneous stock return and VIX change explain 2 percent of the variance of the shock, and fifteen candidate drivers, from policy and geopolitical uncertainty indices to monetary policy surprises and survey sentiment, explain 14 percent.

The identification is also unaffected by a broad set of controls for risk appetite, the global financial cycle and monetary policy (Online Appendix Table C.21 and Figure C.57). Adding to the monthly VAR, ordered before ambiguity, any one of the risk aversion and uncertainty indices of Bekaert et al. (2022), the global factor of Miranda-Agrippino and Rey (2022), the monetary policy and central bank information shocks of Bauer and Swanson (2023) and Jarociński and Karadi (2020), the variance risk premium, realized variance and

skewness, the policy uncertainty and equity market volatility indices of Baker et al. (2016), the geopolitical risk index of Caldara and Iacoviello (2022) or commodity prices leaves the shock nearly unchanged (correlations of 0.94 to 1.00 with the baseline shock and first-stage F statistics of 80 to 101) and the EME GDP response between 0.45 and 0.59 percent, significant at the 90 percent level; a principal component of all twelve controls gives 0.48 percent. The shock is not predictable from lagged EME growth, investment, spreads or trade balances, and extending the monthly sample to 2021:M9 with controls for the pandemic months gives a GDP response of 0.55 percent. Risk aversion rises after the shock, so the shock is not a disguised increase in risk appetite. When the Bekaert et al. (2022) risk aversion index and the variance risk premium, both of which move closely with the VIX, are ordered before ambiguity together with financial uncertainty, the excess bond premium and returns, the response falls to 0.2 to 0.3 percent; this is a question of how the component of ambiguity shared with these near-volatility measures is attributed within the month, since the same controls ordered after ambiguity or entered as lags leave the response at 0.4 to 0.7 percent.

Taken together, ambiguity shocks are distinct from the uncertainty shocks studied in the literature. Shocks to stock market volatility and to macro, real, financial and economic uncertainty, every measure that rises with volatility, are contractionary for EMEs; ambiguity shocks are expansionary, and the expansion holds whether ambiguity is conditioned on these measures at quarterly frequency or purged of them, together with credit market conditions, at monthly frequency. The compression of the BAA spread, by contrast, is tied to the decline in volatility that accompanies ambiguity shocks.

3.4 Variance and Historical Decompositions

Figure 3.6 presents forecast error variance decompositions (FEVDs) from our benchmark model (Eq. (2.1)) at different horizons for the following EME variables: GDP, investment, trade balance and country spreads. Each panel corresponds to one of the variables and reports the percentage contributions of the ambiguity shocks (solid lines), global financial risk shocks (dashed lines), country spread shocks (dash-dotted lines), and the U.S. real

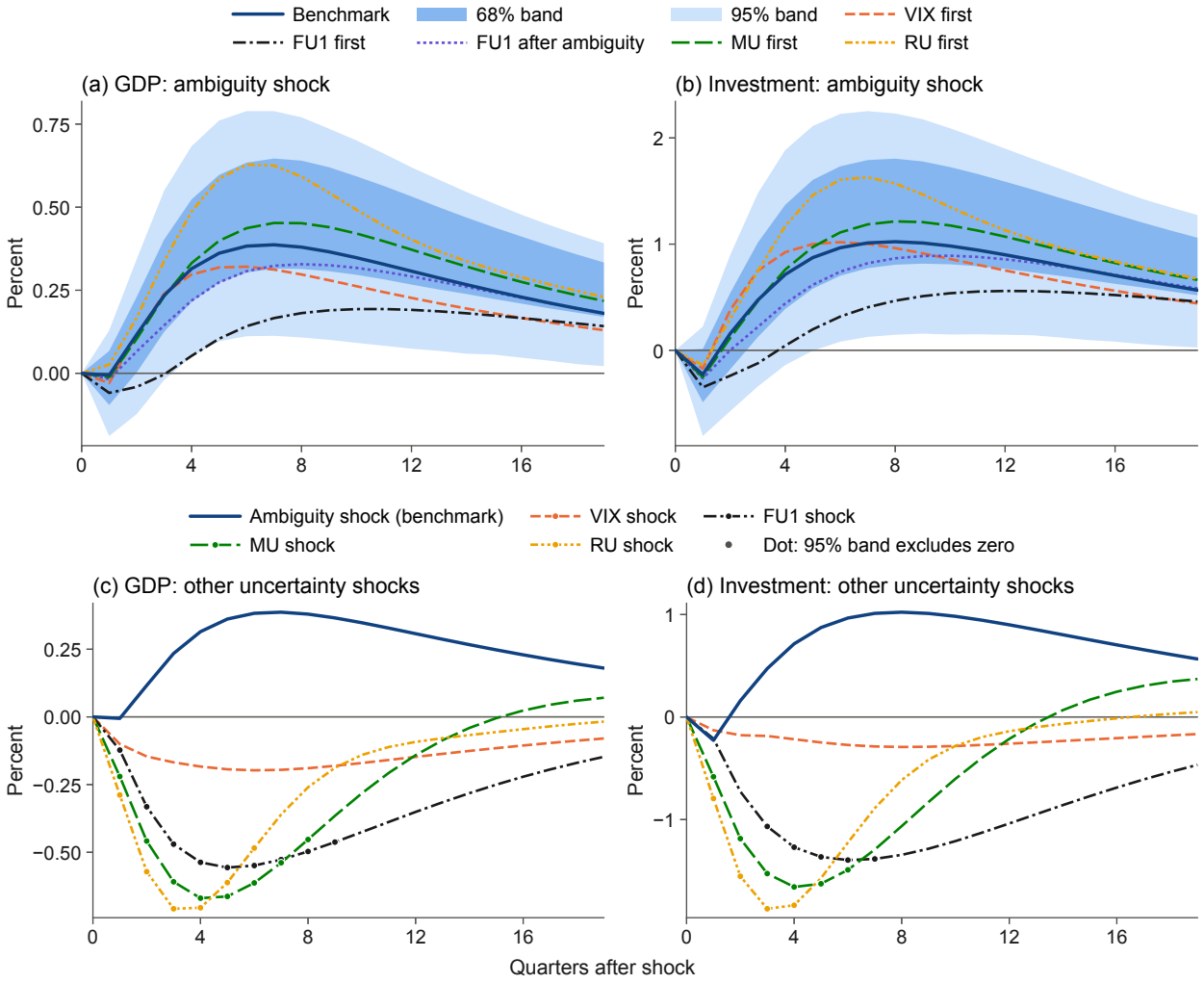


Figure 3.4: Ambiguity shocks versus other uncertainty shocks

Note: Panels (a) and (b): responses of EME GDP and investment (percent) to a one standard deviation ambiguity shock. Solid line and shaded areas: benchmark PSVAR, point estimate with 68 and 95 percent bands. Other lines: point estimates from eight-variable models that add one uncertainty measure to the global block, ordered before ambiguity (VIX; Jurado et al. (2015) financial uncertainty at the one-month horizon, FU1; macro uncertainty, MU; real uncertainty, RU) or after it (FU1). Panels (c) and (d): responses of GDP and investment to a one standard deviation shock to each uncertainty measure in the same eight-variable models (measure ordered before ambiguity); dots mark horizons at which the 95 percent band excludes zero. The benchmark ambiguity-shock response is repeated for comparison.

Table 3.3: Ambiguity versus other uncertainty measures

	GDP		Investment		Baa	EMBI	FEVD		x shock: GDP
Control x, order	Peak (h)	68% / 95%	Peak (h)	68% / 95%	trough (68%)	trough (68%)	amb.	x	peak (h), 95%
Benchmark	0.39 (7)	2–19 / 4–19	1.02 (8)	3–19 / 6–19	−0.09 (1–10)	−0.11 (3–15)	6.1	–	–
VIX, before	0.32 (6)	2–19 / 4–17	1.02 (6)	2–19 / 3–19	−0.03 (none)	−0.04 (7–19)	4.1	1.8	−0.20 (6), none
VIX, after	0.38 (6)	2–19 / 3–15	1.00 (6)	2–19 / 3–11	−0.08 (1–12)	−0.11 (2–16)	5.8	0.1	−0.10 (1), none
FU1, before	0.19 (11)	6–19 / none	0.56 (12)	6–19 / none	−0.02 (none)	−0.05 (9–19)	1.6	10.3	−0.56 (5), 1–9
FU1, after	0.33 (8)	3–19 / 5–15	0.89 (10)	4–19 / none	−0.07 (3–12)	−0.08 (6–19)	4.4	7.5	−0.50 (5), 1–7
FU12, before	0.15 (13)	8–19 / none	0.47 (14)	8–19 / none	−0.01 (none)	−0.06 (9–19)	1.0	12.5	−0.60 (5), 1–11
FU12, after	0.27 (9)	4–19 / none	0.77 (11)	5–19 / none	−0.05 (6–10)	−0.08 (7–19)	3.1	10.4	−0.57 (5), 1–9
MU12, before	0.45 (7)	2–19 / 3–19	1.21 (8)	3–19 / 4–19	−0.10 (1–13)	−0.12 (2–13)	7.3	9.1	−0.67 (4), 1–8
MU12, after	0.44 (8)	2–19 / 3–19	1.19 (8)	3–19 / 5–19	−0.10 (1–12)	−0.12 (3–14)	7.0	9.3	−0.68 (4), 1–9
RU12, before	0.63 (6)	2–19 / 2–19	1.63 (7)	2–19 / 3–19	−0.15 (1–19 [16])	−0.16 (3–10)	11.7	8.1	−0.71 (3), 1–6
RU12, after	0.57 (7)	3–19 / 4–19	1.50 (7)	3–19 / 4–19	−0.13 (1–19 [15])	−0.16 (3–11)	9.7	10.1	−0.76 (4), 1–7
VIX+FU12, before	0.27 (6)	3–19 / none	0.95 (6)	3–19 / 5–7	−0.01 (none)	−0.06 (6–19)	2.7	11.8	−0.59 (6), 2–10
BEX, before	0.35 (8)	3–19 / 5–19	0.92 (9)	4–19 / 7–19	−0.08 (1–9)	−0.07 (5–19)	5.0	2.6	0.24 (12), none
BEX, after	0.33 (8)	3–19 / 5–19	0.86 (9)	4–19 / 8–16	−0.09 (1–10)	−0.08 (4–19)	4.5	3.1	0.26 (12), none
SPF, before	0.36 (7)	2–19 / 4–19	0.98 (8)	3–19 / 6–19	−0.07 (1–7)	−0.07 (4–16)	5.7	0.2	−0.09 (7), none
SPF, after	0.36 (7)	2–19 / 4–19	0.98 (8)	3–19 / 6–19	−0.08 (1–9)	−0.07 (3–16)	5.8	0.1	−0.07 (7), none

Note: Responses to a one standard deviation ambiguity shock when an alternative uncertainty measure x is added to the global block of the benchmark model, ordered before or after ambiguity (eight variables; the last row adds both the VIX and FU12 before ambiguity). FU1 and FU12: Jurado et al. (2015) financial uncertainty at the 1- and 12-month horizons; MU12 and RU12: macro and real uncertainty, 12-month horizon. Peak: largest response over $h = 0$-19 (percent) and its horizon; 68%/95%: horizons at which the band excludes zero with the sign of the peak. BAA and EMBI: trough (percentage points) and the horizons at which the 68 percent band excludes zero. FEVD: shares of the ambiguity and x shocks in the 20-step-ahead GDP forecast error variance, percent. Last column: peak GDP response to a one standard deviation x shock (percent), horizon, and the horizons at which its 95 percent band excludes zero. BEX: the Bekaert et al. (2022) economic uncertainty index; SPF: the interquartile range of one-quarter-ahead real GDP growth forecasts in the Survey of Professional Forecasters. Bands: 10,000 replications of the moving-block residual bootstrap, Hall intervals.

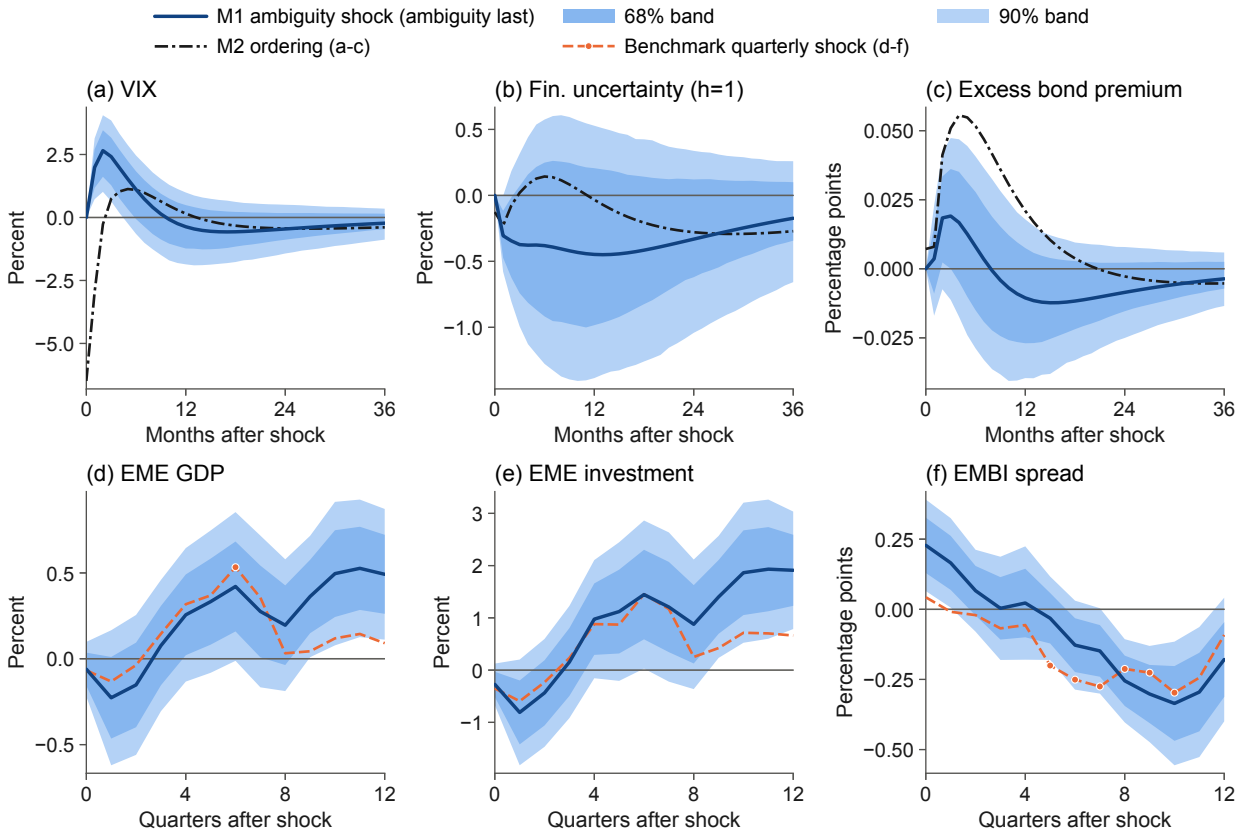


Figure 3.5: The ambiguity shock identified at monthly frequency

Note: Panels (a)-(c): responses of the log VIX, the log of the Jurado et al. (2015) financial uncertainty index at the one-month horizon (percent) and the Gilchrist and Zakrajšek (2012) excess bond premium (percentage points) to a one standard deviation ambiguity shock in a monthly VAR (1993:M2-2019:M12, two lags) ordered [U.S. real rate, BAA spread, VIX, financial uncertainty, excess bond premium, NASDAQ return, ambiguity], so that the shock is orthogonal within the month to all six variables; 68 and 90 percent bands from 2,000 replications of a recursive-design wild bootstrap. Dash-dot line: point estimate with ambiguity ordered before the VIX, financial uncertainty, the excess bond premium and the return. Panels (d)-(f): panel local projections of EME GDP, investment (percent) and the EMBI spread (percentage points) on the monthly shock summed within the quarter and standardized; six countries, unit fixed effects, controls two lags of the outcome, of the country variables and of the U.S. real rate, BAA spread and ambiguity; 68 and 90 percent bands from Driscoll and Kraay (1998) standard errors. Dashed line: the same projections on the benchmark quarterly ambiguity shock (dots: 90 percent band excludes zero). Horizons in months in panels (a)-(c) and in quarters in panels (d)-(f).

interest rate shock (dotted lines) to the variance of the k-quarter-ahead forecast errors. Following Uribe and Yue (2006), we associate business-cycle fluctuations with the variance of the forecast error at a horizon of about five years (20 quarters), which falls in the middle of the window of 6 to 32 quarters that typically defines business cycles (Stock and Watson, 1999). Table 3.4 reports the shares at 4, 8 and 20 quarters.

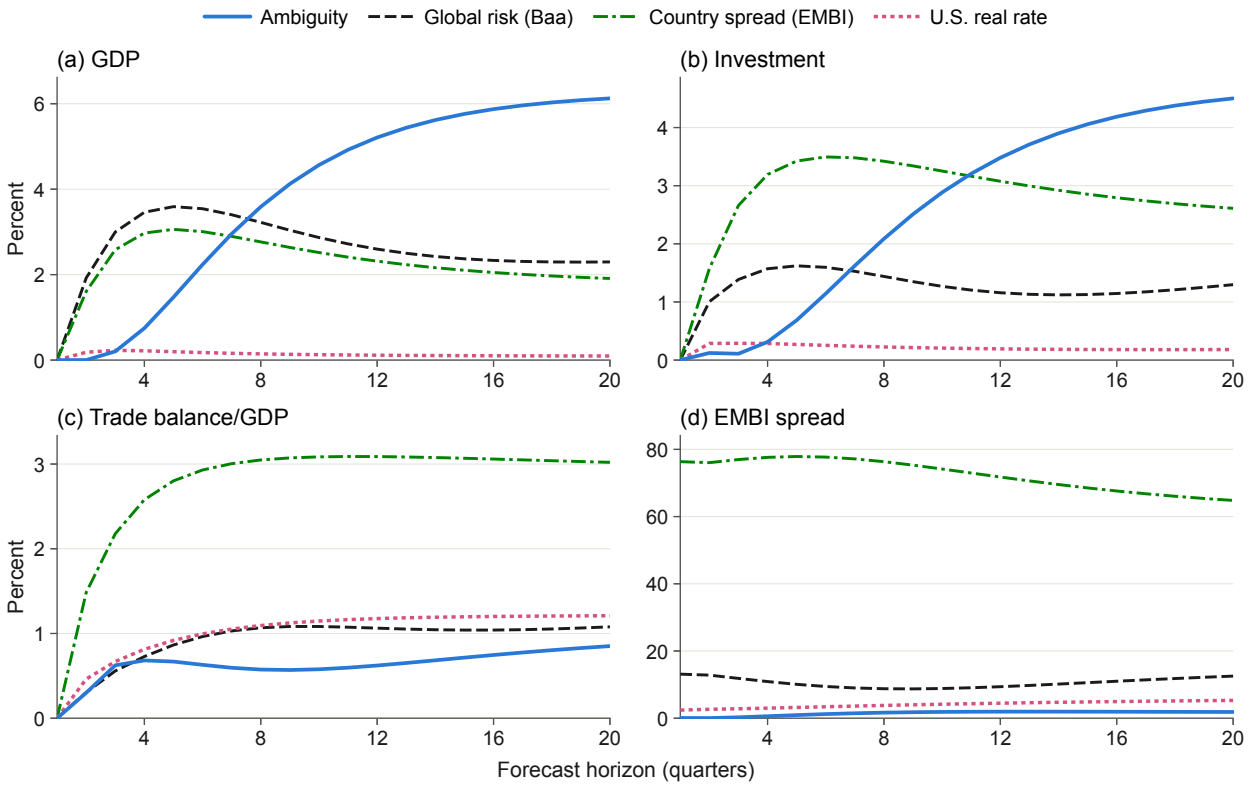


Figure 3.6: Forecast Error Variance Decomposition

Note: This chart shows the fraction of the variance of the forecasting error accounted for by ambiguity (the solid lines), the BAA spread (the dashed lines), the country spreads (the dash-dotted lines), and the U.S. real interest rate (the dotted lines) at each horizon, in percent. Point estimates.

Panel (a) shows that innovations to ambiguity are the largest external contributor to the movements in the EME domestic economic activity at 20 quarters, accounting for 6 percent of the variations in emerging markets GDP. The global financial risk (the BAA spread) and the country spread are less important, with each explaining about 2 percent of the fluctuations in domestic economic activity in the EMEs. Overall, these three factors explain about 10 percent of GDP fluctuations in emerging market economies. Shocks to the U.S. interest rate, on the other hand, do not have a material impact on the EME GDP. The ambiguity share

builds up with the horizon (0.7 percent at four quarters and 3.6 percent at eight), whereas the global financial risk share is largest at short horizons (3.5 percent at four quarters).

Similarly, Panel (b) shows that shocks to ambiguity, global financial risk, and the country spread jointly explain about 8 percent of movements in emerging market investment. At the 20-quarter horizon, ambiguity explains 4.5 percent of investment forecast error variance, compared with 2.6 percent for country-spread shocks and 1.3 percent for global financial risk.

Panel (c) shows that country spread shocks are the main driver of EME trade balance fluctuations, accounting for as much as the combined contribution of ambiguity, global financial risk, and U.S. interest rate shocks—about 3 percent. Panel (d) concludes by showing that global financial risk shocks are the most important driver of fluctuations in the country spread, though much less so than the country spreads' own innovations.

The contribution of ambiguity is particularly clear when we compare nested specifications on the same data. At the 20-quarter horizon, the six-variable specification of Akıncı (2013), estimated on our sample, attributes 5.9 percent of EME GDP forecast error variance to the U.S. interest rate, global financial risk, and country-spread shocks. Adding ambiguity raises the combined contribution of external and country-spread shocks to 10.4 percent (Table 3.4), with ambiguity itself accounting for 6.1 percent. The substantially larger contributions reported in earlier studies (Uribe and Yue, 2006; Akıncı, 2013) reflect, in part, differences in the estimation samples, which we examine next.

To investigate whether these differences stem from the sample period used in these earlier studies, we generate 20-quarters ahead FEVDs for sample periods ending at different dates, beginning with the first quarter of 2011 and expanding one quarter at a time. Figure 3.7 presents the results from this exercise, where each year shows the average FEVDs of the respective variable obtained by running the estimation through each quarter of that year.

The expanding-window results also reveal why the relative contributions differ across studies: in samples ending around 2011, global financial risk and country-spread shocks account for a substantially larger share of EME fluctuations. In this earlier window, the BAA spread, country spreads, the U.S. interest rate, and ambiguity jointly account for 23

Table 3.4: Forecast error variance decomposition

Variable	h	Ambiguity	Baa spread	U.S. real rate	EMBI spread	Sum
A. Seven-variable model (benchmark)						
GDP	4	0.7	3.5	0.2	3.0	7.4
	8	3.6	3.2	0.1	2.8	9.7
	20	6.1	2.3	0.1	1.9	10.4
Investment	4	0.3	1.6	0.3	3.2	5.4
	8	2.1	1.4	0.2	3.4	7.2
	20	4.5	1.3	0.2	2.6	8.6
Trade balance/GDP	4	0.7	0.7	0.8	2.6	4.8
	8	0.6	1.1	1.1	3.0	5.8
	20	0.9	1.1	1.2	3.0	6.2
EMBI spread	4	0.6	10.9	3.0	77.6	92.1
	8	1.7	8.8	3.8	76.3	90.6
	20	1.9	12.5	5.3	64.8	84.5
B. Six-variable model (Akinci 2013 specification)						
GDP	4	–	4.4	0.8	3.2	8.4
	8	–	4.4	0.7	3.2	8.4
	20	–	3.0	0.5	2.4	5.9
Investment	4	–	2.1	0.8	3.4	6.3
	8	–	2.1	0.8	3.9	6.8
	20	–	1.4	0.6	3.2	5.2
Trade balance/GDP	4	–	0.7	0.8	2.6	4.0
	8	–	1.0	1.3	3.0	5.3
	20	–	1.0	1.5	3.0	5.5
EMBI spread	4	–	11.7	4.1	76.4	92.2
	8	–	9.5	5.2	75.8	90.5
	20	–	12.1	6.3	65.3	83.6

Note: Share of the h-step-ahead forecast error variance of each EME variable explained by the ambiguity, BAA spread, U.S. real rate and EMBI shocks, in percent; $h = 1$ is the impact quarter. Sum: the four shares together (for the EMBI spread this includes its own shock). The six-variable model is the Akıncı (2013) specification without ambiguity, estimated on our data. Sample 1994:Q2-2019:Q4. Point estimates.

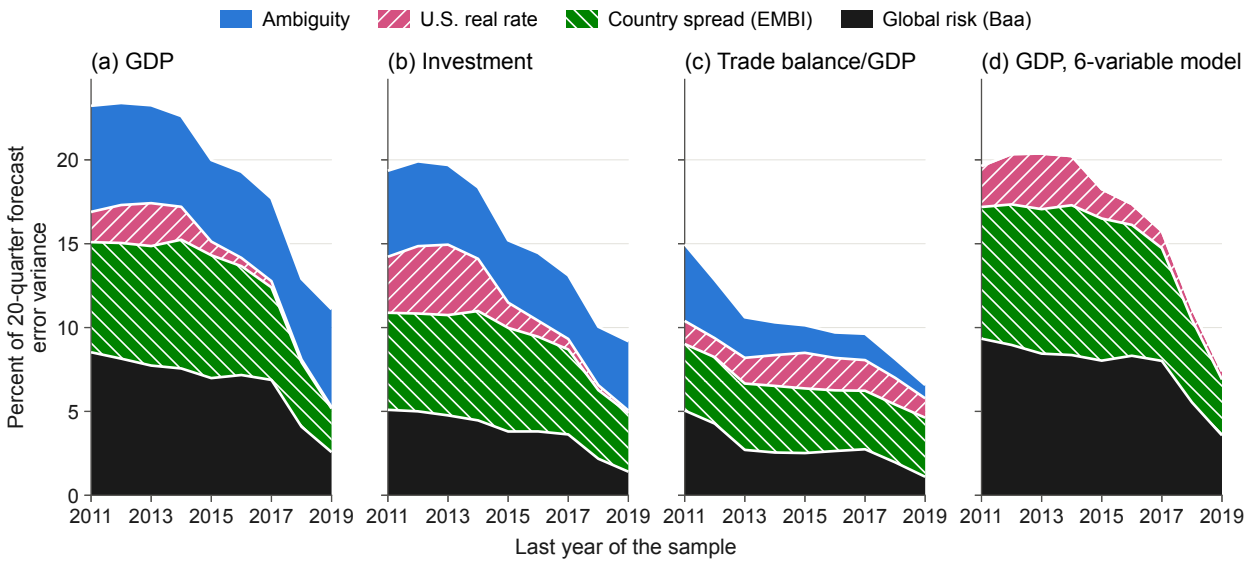


Figure 3.7: 20-Quarter Ahead Forecast Error Variance Decomposition
(for different sample period ends)

Note: This chart shows 20-quarters ahead forecast error variance decomposition for an expanding window of the sample period, in percent. Each region represents the fraction of the variance of the forecasting error accounted for by the BAA spread, the country spreads, the U.S. real interest rate, and ambiguity. Each year shows the averages obtained by running the estimation through each quarter of that year. Panel (d) shows GDP in the 6-variable model of Akıncı (2013), without ambiguity, estimated on the same samples. Point estimates.

percent of the movements in EME domestic activity, with the BAA spread and country spreads playing the most important roles (panel (a)). Similarly, these four factors explain nearly 20 percent of fluctuations in emerging markets' investment (panel (b)) and 15 percent of the fluctuations in the trade balance (panel (c)). Extending the sample through 2019 changes the composition of external drivers: the shares attributed to global financial risk and country-spread shocks decline, particularly in windows ending in 2018 and 2019, while the ambiguity share remains between 4.8 and 6.3 percent. Ambiguity consequently becomes the largest external contributor to EME GDP forecast error variance in the later pre-pandemic windows. The six-variable specification displays a comparable decline in the total external contribution, from about 20 to 7 percent (Figure 3.7, panel (d)).

Variance shares average over the sample; a historical decomposition shows when the shocks mattered. Because the structural shocks are identified, average EME GDP relative to trend in each quarter can be written exactly as the sum of a deterministic component and the cumulated contributions of the identified shocks. Figure 3.8 reports this decomposition for the cross-country average of EME GDP. Ambiguity shocks account for 3.4 of the 9.4 percentage-point rise of average EME GDP relative to trend between 2003:Q2 and 2008:Q1, more than three times the contribution of global financial risk shocks (0.9), and their cumulative contribution peaks at 2.8 percent of trend GDP in 2008:Q1 (95 percent band 0.5 to 5.6; positive at the 95 percent level from 2007:Q4 to 2008:Q4). The collapse of 2008:Q1-2009:Q2 (8.2 points) is a global financial risk event: global risk shocks account for 2.8 points of the decline and ambiguity shocks for 0.7, as the support from the 2003-07 ambiguity shocks fades. Domestic shocks dominate the 2010-13 recovery and the 2015-19 slowdown. Over the whole sample, ambiguity shocks account for 6.2 percent of the in-sample variation of EME GDP, in line with the FEVD, and for 15 percent of the variation of the cross-country average, because country-specific shocks average out across countries while the global shocks do not. The attribution is the same under all six orderings of the global block, with indicators for the crisis quarters, and with the ambiguity shock identified at monthly frequency as in Section 3.3.5 (Online Appendix Table C.23 and Figure C.67). The two external shocks also differ in their cumulative impact: a one standard deviation ambiguity shock

adds 1.3 percent-years to EME GDP over five years, against a loss of 0.4 percent-years after a one standard deviation global financial risk shock, whose effect reverses within the horizon.

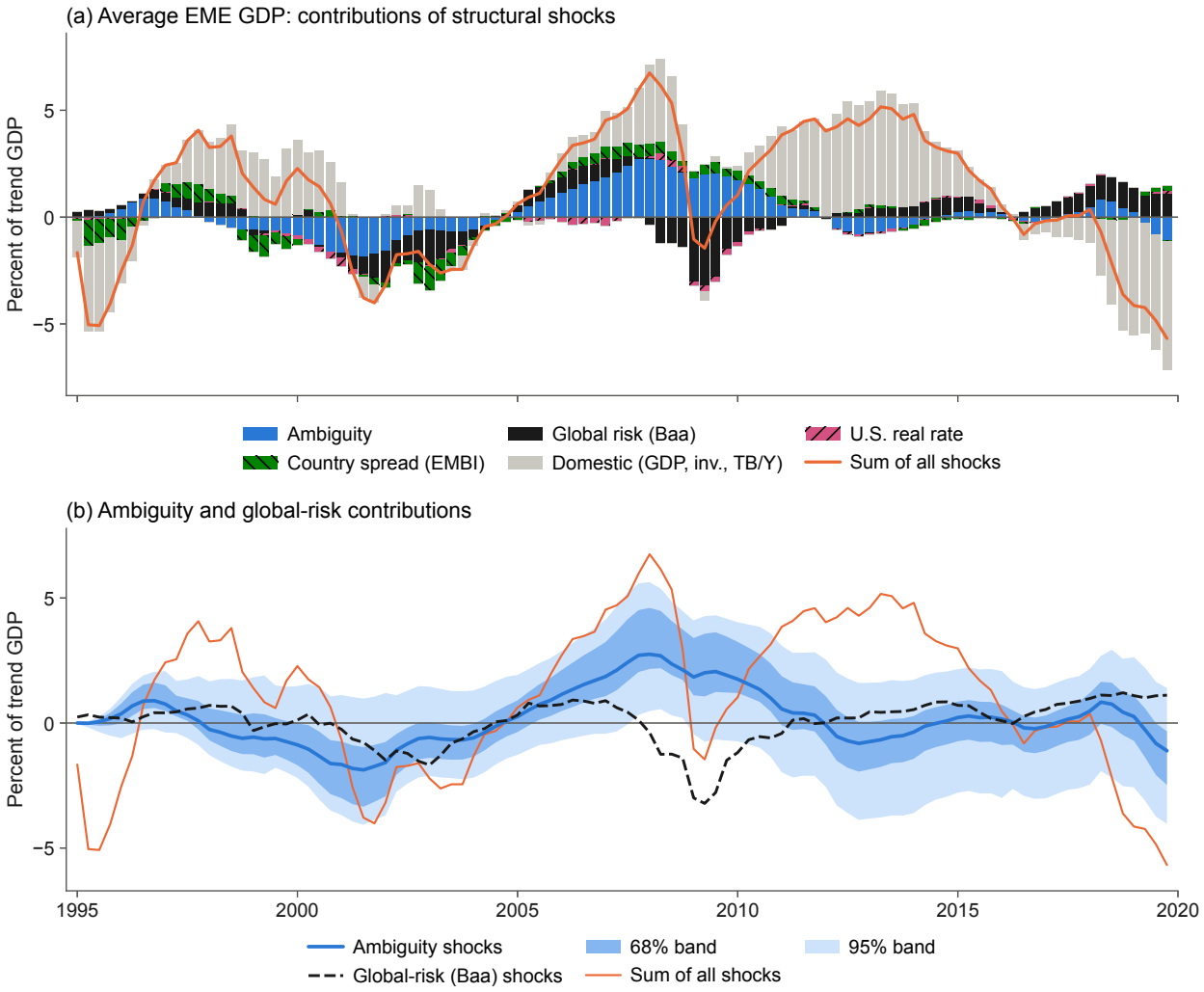


Figure 3.8: Historical decomposition of average EME GDP

Note: Panel (a): contributions of the identified shocks to the cross-country average of EME GDP (percent deviation from trend), 1995:Q1-2019:Q4, from the benchmark PSVAR; the line is the sum of all shock contributions (actual GDP minus the deterministic component). Domestic: the GDP, investment and trade balance shocks. Panel (b): the cumulated contributions of ambiguity shocks (with 68 and 95 percent Hall bands from 10,000 replications of the moving-block bootstrap) and of global financial risk shocks, and the sum of all shocks.

4 Robustness and Additional Analysis

This section performs a battery of robustness checks including: alternative filters and trend treatments, lag orders, exogenous block specifications, samples of countries and periods, and

a comparison with advanced economies. Online Appendix Tables C.14–C.16 summarize the responses to an ambiguity shock in every specification.

4.1 Alternative Filters

The expansion following ambiguity shocks is persistent and is robust to trend specifications that retain lower-frequency movements. With the Christiano-Fitzgerald filter spanning 2 to 250 quarters, EME GDP peaks 0.37 percent above trend; with an HP smoothing parameter of 10^5, the peak is 0.31 percent. In both cases the expansion is significant at the 95 percent level (Online Appendix Figure C.41). The Baxter-King filter with the same nominal band substantially attenuates the response because its finite window removes much of the relevant low-frequency variation. Turning to global financial risk shocks, the HP and Christiano-Fitzgerald filters yield results close to our baseline analysis, while the contraction of EME GDP and investment is deeper with the Baxter-King filter.

4.2 Detrending and Frequency

The frequency content of the response explains this sensitivity. In the benchmark model, 87.5 percent of the variance of the ambiguity-driven component of EME GDP lies at periods longer than 32 quarters, compared with 58 percent for the global-risk-driven component. Conventional business-cycle filters designed to isolate fluctuations between 6 and 32 quarters therefore remove most of the estimated ambiguity response by construction. The expansion also persists when deterministic trend terms enter every PSVAR equation instead of linearly detrending the data: EME GDP peaks 0.29 percent above trend, significantly at the 95 percent level around the peak (Online Appendix C.9 and Table C.16).

The output effect is thus a medium-run effect: it is identified from the multi-year co-movement of ambiguity and EME activity, in particular the rise of the index during the 2000s and the EME expansion of 2003-08 that Figure 3.8 attributes in part to it. Any detrending that removes multi-year movements removes the effect by construction, whether a business-cycle band-pass filter or piecewise trends with level shifts in 2000 and 2007, which carve the

EME boom out of the sample. Break-search tests with critical values that account for the persistence of the index do not select those breaks at the 5 percent level, and the one break they do select, in 2001, leaves the response intact (GDP peak 0.32 percent, significant at the 95 percent level). Quadratic, Hodrick-Prescott and rolling-mean trends and continuous piecewise-linear trends give GDP peaks of 0.31 to 0.42 percent, all significant at the 95 percent level, and the identified shock itself is the same under every trend treatment, with correlations of 0.95 to 1.00 with the benchmark shock (Online Appendix Tables C.12 and C.13).

4.3 Alternative Lag Orders

The Bayesian and Hannan-Quinn information criteria suggest that the optimal lag order of our benchmark PSVAR in Equation (2.1) is 2, in line with Akıncı (2013), while the Akaike criterion selects 4. We perform robustness analysis by re-estimating the model with 1, 3 and 4 lags. Online Appendix Figures C.42 and C.43 report the results.

In the case of one lag, a shock to global financial risk yields results in line with our benchmark analysis, with a somewhat deeper and more persistent contraction. Nonetheless, in contrast to our baseline results, a shock to the U.S. interest rate results in a deep and protracted decline in EME GDP and investment that is significant at the 68 percent level. In addition, a shock to the country spread is contractionary at the 95 percent level from the fifth quarter to the end of the horizon, with a peak effect on GDP that arrives only after ten quarters. The one-lag specification also produces a smaller and more delayed output expansion after an ambiguity shock, consistent with its more restrictive representation of propagation dynamics. The information criteria favor a two-lag baseline, and the results with three and four lags reinforce the main finding: EME GDP peaks 0.53 and 0.61 percent above trend, respectively, with responses significant at the 95 percent level from the fourth quarter to the end of the horizon.

4.4 Alternative Exogenous Block Specifications

4.4.1 Endogenous Global Financial Variables

In the baseline model, global financial variables—U.S. interest rate, global financial risk, and ambiguity in U.S. equity markets—are treated as exogenous to all country-specific variables. We relax this block-exogeneity assumption by allowing all variables, including the global financial variables, to interact endogenously (while preserving their baseline ordering). Online Appendix Figure C.49 and Table C.14 show that the results remain consistent with our benchmark specification (EME GDP peaks at 0.37 percent, significant at the 95 percent level from the third quarter to the end of the horizon).

4.4.2 Ambiguity: Exogenous to Other Global Financial Variables

To assess the extent to which the endogenous dependence of ambiguity from U.S. interest rate and global financial risk affect the results, we examine the case in which ambiguity does not depend on these variables. To do so, we set the R^{US}_{t-j} and GR_{t-j} coefficients equal to zero for $j = 0, 1, 2$ in the ambiguity equation, while keeping everything else the same as in the benchmark model. Online Appendix Figure C.46 shows that the ambiguity-driven expansion remains close to the benchmark even when ambiguity is restricted not to respond to the U.S. real rate or BAA spread, confirming that the main impulse responses do not depend on these feedback terms.

4.4.3 U.S. Interest Rate: Exogenous to Other Global Financial Variables

To assess the extent to which the endogenous dependence of the U.S. interest rate from global financial risk and the degree of ambiguity affect the results, we examine the case in which the U.S. interest rate does not depend on these variables. To do so, we set the $ambig_{t-j}$ and GR_{t-j} coefficients equal to zero for $j = 1, 2$ in the interest rate equation (note that in the benchmark case these coefficients are already zero for $j = 0$), while keeping everything else the same as in the benchmark model. Online Appendix Figure C.45 shows that the IRFs are in essence the same as in our baseline analysis in the case of the ambiguity shock.

In response to a global financial risk shock, the contraction in EME GDP and investment is somewhat deeper than in the benchmark, and the point estimates show a rebound after about three years in both cases, which is significant at the 68 percent level only in the last quarters of the horizon.

4.4.4 Country Spread: Exogenous to Global Financial Variables

To examine the extent to which the interaction of the exogenous block variables with the country spread have an impact on results, we examine the case in which the country spread does not depend on these variables. As such, in the country spread equation we set $ambig_{t-j}$, GR_{t-j} and R^{US}_{t-j} equal to zero for $j = 0, 1, 2$ while keeping everything else the same as in the benchmark model. Regarding IRFs, recall that in the baseline case a shock to Baa leads to a jump in the country spread and a significant decline in EME GDP. As shown in Online Appendix Figure C.47, in the present case the country spread no longer jumps, and GDP still declines significantly, although by about a third less than in the benchmark (0.23 against 0.35 percent at the trough). Given a shock to ambiguity, the expansion in EME GDP and investment is similar to the benchmark analysis, and the later decline in the country spread, driven by the improvement in EME economic conditions, is smaller but lasts longer.

4.4.5 Country Spread: Exogenous to other Country-Specific Real Variables

To assess the extent to which the endogenous dependence of the country spreads from country-specific variables—GDP, investment, trade balance— affect the results, we examine the case in which the country spread does not depend on these variables. To do so, we set the coefficients on $gdp_{i,t-j}$, $inv_{i,t-j}$ and $tby_{i,t-j}$ equal to zero for $j = 0, 1, 2$ in the country spread equation, while keeping everything else the same as in the benchmark model. The expansion in EME GDP and investment persists when country spreads are restricted not to respond to domestic activity (Online Appendix Figure C.48). Its magnitude is moderately smaller and the decline in the country spread is shorter-lived, significant at the 68 percent level from the second through the eighth quarter. This attenuation is consistent with the amplification mechanism in the benchmark (Uribe and Yue, 2006; Akıncı, 2013): stronger

domestic conditions reduce borrowing costs, which in turn sustain the output and investment expansion. The EME responses to global financial risk shocks are nearly unchanged.

4.4.6 Alternative Exogenous Block Orderings

We examine the robustness of our main results by ordering the exogenous block across all possible five remaining permutations (relative to the baseline specification) and re-estimating the model. Online Appendix Figure C.49 and Table C.14 show that our baseline results are robust to these alternative specifications: EME GDP peaks at 0.37 to 0.39 percent and is significant at the 95 percent level in all orderings, including the two orderings that are consistent with the Granger causality tests, which place global financial risk first.

4.5 Alternative Samples and Measures

The expansion is stable across leave-one-country-out estimates: the GDP and investment responses are significant at the 95 percent level in all six exclusions. Estimated country by country, the GDP response is positive over quarters 4 to 12 in ten of the eleven EMEs of the larger sample below (Mexico is the exception) and the EMBI spread falls in ten of eleven, while a global financial risk shock lowers GDP and raises the spread in all eleven; the mean-group estimate of Pesaran and Smith (1995), which averages the country responses, equals the pooled estimate (GDP 0.30 against 0.29 percent at eight quarters, with a 95 percent interval based on the dispersion of the country estimates that excludes zero from the sixth quarter), so pooling is not what produces the result (Online Appendix Figure C.50 and Table C.17). The response is larger in the higher-spread, more volatile economies (Turkey, South Africa, Chile, Ecuador) than in Brazil, Mexico or Malaysia, consistent with a mechanism that operates through external financing conditions; precision- or size-weighted averages of the country responses are accordingly smaller, 0.1 to 0.2 percent. Beyond the level of spreads, no measure of exposure to foreign investors, from portfolio liabilities and U.S. holdings to dollar credit and capital-account openness, systematically amplifies the output response across the eleven EMEs, as expected when the channel is global financial

conditions rather than bilateral U.S. purchases; in the cross-section, economies with larger current-account deficits respond more to both shocks (Online Appendix C.13). Estimates on samples ending in 2011:Q3 or beginning in 2003:Q1 also deliver significant GDP expansions; shorter subsamples retain positive but less precisely estimated responses. All nine monthly-to-quarterly aggregations of the ambiguity index that we consider keep the GDP response significant at the 95 percent level, as do the Baa spread measured against the 20-year Treasury yield, an alternative construction of the U.S. real rate, and the trade balance without detrending (Online Appendix Tables C.14-C.16).

4.6 Larger sample of countries

For comparability, our paper focuses on the set of emerging economies considered in Akıncı (2013): Argentina, Brazil, Mexico, Peru, South Africa and Turkey. Online Appendix Figure C.53 shows that our results hold when we expand the sample to 11 countries, adding Chile, Colombia, Ecuador, Malaysia, and the Philippines to the baseline set. In the larger sample, EME GDP peaks at 0.29 percent eight quarters after the shock, significant at the 95 percent level from the fifth quarter to the end of the horizon, and investment at 0.79 percent, significant at the 95 percent level between the sixth and the eighteenth quarter; ambiguity accounts for 4.4 percent of GDP variance at 20 quarters. These estimates are more precise than the six-country baseline, with confidence bands about a fifth narrower; the moderately smaller peaks reflect primarily the inclusion of the Philippines and Ecuador, and the result does not hinge on any single country.

4.7 Advanced Economies and the Pandemic

To check whether the response to ambiguity shocks is specific to EMEs, we estimate the benchmark model on twelve small open advanced economies (Australia, Austria, Belgium, Canada, Denmark, Finland, Israel, the Netherlands, New Zealand, Norway, Sweden, and Switzerland), with the 10-year sovereign yield in place of the country spread (Online Appendix Figure C.54). The advanced-economy estimates reveal a clear contrast with EMEs.

Following an ambiguity shock, the GDP response in these economies is modest (a peak of 0.11 percent) and is not statistically distinguishable from zero at the 95 percent level; ambiguity explains about 2 percent of their GDP forecast error variance. Global financial risk shocks, by contrast, produce a large and statistically significant contraction in these economies (GDP falls by 0.48 percent, and global financial risk accounts for 37 percent of GDP variance). As in the United States, the 10-year yield rises after an ambiguity shock. The pronounced expansion following U.S. ambiguity shocks is thus a distinctive feature of the EME sample. Finally, extending the sample to 2021:Q3, the last quarter for which the ambiguity index is available, and controlling separately for the exceptional quarters of 2020 preserves the expansionary response: EME GDP peaks 0.28 percent above trend, with significance at the 95 percent level from the seventh quarter to the end of the horizon (Online Appendix Figure C.55). The pre-pandemic benchmark avoids the unusually large and concentrated output movements associated with the onset of the pandemic.

5 Conclusion

This paper provides new empirical evidence on the role of Knightian uncertainty (ambiguity) in U.S. equity markets as a driver of business cycles in EMEs. While prior literature has emphasized the impact of global risk and U.S. interest rates on EMEs, we isolate and quantify the distinct influence of ambiguity associated with investors' beliefs over U.S. stock market returns in shaping macroeconomic and financial dynamics in EMEs. Using a panel SVAR framework, we identify ambiguity shocks in U.S. equity markets and analyze their spillover effects on EME output, investment, trade balance and country risk premia, together with their effects on U.S. asset prices and cross-border portfolio flows.

In contrast to previously examined uncertainty shocks, which typically depress economic activity, we show that ambiguity shocks in U.S. financial markets are followed by economic expansions in EMEs. This novel and counterintuitive result is explained by a portfolio reallocation channel without a flight to safety, the response that a simple portfolio model predicts for ambiguity-averse investors who face greater ambiguity about U.S. equity returns:

after an ambiguity shock, U.S. Treasury yields and term premia rise, the dollar weakens, U.S. investors increase their purchases of foreign bonds and stocks, and global financial conditions ease for EMEs, whose equity prices rise, which receive more cross-border bank credit and accumulate reserves, and which pay lower country spreads, which fuels investment and GDP growth. The expansion is a medium-run phenomenon that builds over two years and lasts about five. It is specific to EMEs, holds in a larger sample of countries and in ten of eleven EMEs taken individually, and is obtained when the ambiguity shock is identified at monthly frequency, orthogonal to stock market volatility, financial uncertainty, credit market conditions and a broad set of risk-appetite and monetary policy controls. Our results also reveal a virtuous feedback loop: improved domestic conditions reinforce lower financing costs, which, in turn, prolong the expansionary effects of ambiguity shocks. Moreover, our analysis quantifies the interplay between ambiguity in U.S. equity markets and global financial risk—the key driver of EME business cycles identified in the literature. We find that increases in global financial risk tend to reduce ambiguity, consistent with risk shocks providing investors with clear signals of a worsening economic outlook, which dissipates Knightian uncertainty. Last, variance and historical decompositions confirm that ambiguity is at least as important as global financial risk and country spread shocks in explaining fluctuations in EME business cycles, that it accounts for about a third of the rise of EME output above trend between 2003 and 2008, and that its relative importance increases across the later pre-pandemic estimation windows as the contributions of global financial risk and country-spread shocks decline.

Our findings establish U.S. equity-market ambiguity as a distinct source of international financial transmission, with effects that differ sharply from those of conventional uncertainty shocks. They highlight how shifts in investors' beliefs and global portfolio allocation can ease external financing conditions and stimulate activity in emerging markets, while changes in global financial risk can themselves reshape the degree of ambiguity. Accounting for this interaction adds an important dimension to the analysis of EME macroeconomic fluctuations.

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Online Appendix

A Data Details

A.1 Common Variables Across Countries

US real GDP: span: 1993:Q4-2022:Q4; source: U.S. Bureau of Economic Analysis, Real Gross Domestic Product [GDPC1], retrieved from FRED, Federal Reserve Bank of St. Louis (https://fred.stlouisfed.org/series/GDPC1). **US real investment**: source: U.S. Bureau of Economic Analysis, Real Gross Private Domestic Investment [GPDIC1], retrieved from FRED. **BAA corporate spread**: span: 1993:Q4-2022:Q4; source: Federal Reserve Bank of St. Louis, Moody's Seasoned Baa Corporate Bond Yield Relative to Yield on 10-Year Treasury Constant Maturity [BAA10Y], retrieved from FRED, Federal Reserve Bank of St. Louis (https://fred.stlouisfed.org/series/BAA10Y); quarterly mean, entered as log(1+spread/100). **Ambiguity index**: span: 1993:Q4-2021:Q3 ; source: Izhakian (2020). We use the log of the square root of the monthly ambiguity measure, averaged over the months of each quarter and linearly detrended. **U.S. nominal interest rate (3-month US Treasury bill rate)**: span: 1993:Q4-2022:Q4; source: Board of Governors of the Federal Reserve System (US), 3-Month Treasury Bill Secondary Market Rate, Discount Basis [TB3MS], retrieved from FRED, Federal Reserve Bank of St. Louis (https://fred.stlouisfed.org/series/TB3MS). **U.S. price level**: source: U.S. Bureau of Labor Statistics, Consumer Price Index for All Urban Consumers: All Items in U.S. City Average [CPIAUCSL], retrieved from FRED. **U.S. gross real interest rate**: calculated as the ratio of the U.S. gross nominal interest rate to U.S. gross expected inflation, where gross expected inflation is annualized. Following Akıncı (2013), U.S. (annualized) gross expected inflation is calculated by regressing the inverse of the (annualized) gross inflation in period $t+1$ (obtained using the U.S. CPI) on a constant and two lags. **VIX**: span: 1993:Q4-2022:Q4; source: retrieved from FRED, Federal Reserve Bank of St. Louis (https://fred.stlouisfed.org/series/vixcls); log of the quarterly mean, linearly detrended. **Financial, macro, and real uncertainty**: the financial (FU), macroeconomic (MU) and real (RU) uncertainty indexes of Jurado et al. (2015) at the 1-,

3- and 12-month horizons, from Sydney Ludvigson's website (August 2026 vintage); logs of quarterly means, linearly detrended. The analysis that replaces ambiguity with these measures uses the 12-month indexes; the horse race in Section 3.3.5 uses the 1-month financial uncertainty index (FU1) and reports the 12-month index (FU12) alongside. **CFO survey**: mean 10th and 90th percentiles and cross-respondent standard deviation of the expected one-year S&P 500 return, Duke University and Federal Reserve Banks of Richmond and Atlanta CFO Survey (Ben-David et al., 2013), 2001:Q4-2019:Q4.

Additional U.S. and global series (Sections 3.3.3, 3.3.4 and 3.3.5 and Online Appendix C): daily S&P 500 and VIX histories from Cboe (realized variance, skewness and kurtosis of daily returns; the variance risk premium as the squared VIX minus realized variance; the VVIX from 2006); the variance risk premium series of Hao Zhou (Bollerslev et al., 2009); the risk aversion and economic uncertainty indices of Bekaert et al. (2022); the global factor in risky asset prices of Miranda-Agrippino and Rey (2022) (2024 update); the monetary policy surprises of Bauer and Swanson (2023) and the policy and central bank information shocks of Jarociński and Karadi (2020) (updated series from the authors); the news-based economic policy uncertainty and equity market volatility indices of Baker et al. (2016); the geopolitical risk index of Caldara and Iacoviello (2022); IMF primary commodity price indices; the Federal Reserve Board's broad and emerging-market nominal dollar indices (FRED); the Survey of Professional Forecasters dispersion series of the Federal Reserve Bank of Philadelphia; the Yale School of Management stock market confidence indices (Goetzmann et al., 2017); the University of Michigan Surveys of Consumers probability of a stock price increase (mean and binned dispersion); and the American Association of Individual Investors sentiment survey. The Yale and AAII series are used under their terms of use and are not redistributed.

Additional cross-border and EME series: U.S. cross-border holdings of and net purchases of foreign bonds and stocks by issuing country, monthly, from the issuer-based estimates of Bertaut and Judson (2014) and the TIC SLT data (Board of Governors and U.S. Treasury); BIS locational banking statistics (cross-border claims on each country, all currencies and U.S. dollars) and consolidated banking statistics by lender nationality (foreign

claims, immediate counterparty basis); IMF balance of payments (BPM6) portfolio equity, portfolio debt, other investment and reserve flows; BIS central bank policy rates, broad nominal and real effective exchange rate indices and credit to the private non-financial sector; IMF International Financial Statistics reserves excluding gold and consumer prices; MSCI country equity indices in dollars; the external wealth of nations estimates of Lane and Milesi-Ferretti for nominal GDP and external positions; IMF coordinated portfolio investment survey holdings; the Chinn and Ito capital-account openness index; and OECD quarterly national accounts for the additional countries. Table A.1 lists the series of the benchmark analysis. Table A.1 lists all series, including the asset prices and capital flows used in Section 3.3.3.

A.2 Country Specific Variables

In all countries, the trade balance is the ratio of nominal net exports to nominal GDP. Argentina enters the benchmark panel as two units, 1994:Q2-2001:Q3 and 2016:Q2-2018:Q2, which exclude the default period.

- ***Argentina***. **Real GDP**: span: 1993:Q4-2022:Q4; source: The National Institute of Statistics and Censuses of Argentina (INDEC). **Real investment**: span: 1993:Q4-2022:Q4; source: The National Institute of Statistics and Censuses of Argentina (INDEC). **Price level**: span: 2012:Q3-2022:Q4; source: The National Institute of Statistics and Censuses of Argentina (INDEC). **Real exports**: span: 1993:Q4-2022:Q4; source: The National Institute of Statistics and Censuses of Argentina (INDEC). **Real imports**: span: 1993:Q4-2022:Q4; source: The National Institute of Statistics and Censuses of Argentina (INDEC).[26] **Trade balance**: ratio of nominal net exports to nominal GDP. **Country spread**: span: 1993:Q4 - 2024:Q1; source: The J.P. Morgan Emerging Markets Bond Index Plus (EMBI+).

- ***Brazil***. **Real GDP**: span: 1994:Q1-2022:Q4; source: Brazilian Institute of Geography and Statistics. **Real investment**: span: 1994:Q1-2022:Q4; source: Brazilian

[26] In all cases for data from national statistics agency please see https://www.indec.gob.ar/ and https://www.indec.gob.ar/indec/web/Institucional-Indec-InformacionDeArchivo-5

Institute of Geography and Statistics. **Price level**: span: 1994:Q1-2022:Q4; source: Brazilian Institute of Geography and Statistics and Uribe and Yue (2006). **Real exports**: span: 1994:Q1-2022:Q4; source: Brazilian Institute of Geography and Statistics. **Real imports**: span: 1994:Q1-2022:Q4; source: Brazilian Institute of Geography and Statistics. **Trade balance**: ratio of nominal net exports to nominal GDP. **Country spread**: span: 1994:Q2 - 2024:Q1; source: The J.P. Morgan Emerging Markets Bond Index Plus (EMBI+).

- ***Mexico***. **Real GDP**: span: 1994:Q1-2022:Q4; source: National Statistical and Geographic Information System of Mexico (INEGI). **Real investment**: span: 1994:Q1-2022:Q4; source: National Statistical and Geographic Information System of Mexico (INEGI). **Price level**: span: 1994:Q1-2022:Q4; source: National Statistical and Geographic Information System of Mexico (INEGI). **Real exports**: span: 1994:Q1-2022:Q4; source: National Statistical and Geographic Information System of Mexico (INEGI). **Real imports**: span: 1994:Q1-2022:Q4; source: National Statistical and Geographic Information System of Mexico (INEGI). **Trade balance**: ratio of nominal net exports to nominal GDP. **Country spread**: span: 1993:Q4 - 2024:Q1; source: The J.P. Morgan Emerging Markets Bond Index Plus (EMBI+).

- ***Peru***. **Real GDP**: span: 1994:Q1-2022:Q4; source: Central Reserve Bank of Peru. **Real investment**: span: 1994:Q1-2022:Q4; source: Central Reserve Bank of Peru. **Price level**: Central Reserve Bank of Peru. **Real exports**: span: 1994:Q1-2022:Q4; source: Central Reserve Bank of Peru. **Real imports**: span: 1994:Q1-2022:Q4; source: Central Reserve Bank of Peru. **Trade balance**: ratio of nominal net exports to nominal GDP. **Country spread**: span: 1997:Q2 - 2024:Q1; source: The J.P. Morgan Emerging Markets Bond Index Plus (EMBI+) and Uribe and Yue (2006).

- ***South Africa***. **Real GDP**: span: 1994:Q1-2022:Q4; source: Statistics South Africa (Stats SA). **Real investment**: span: 1994:Q1-2022:Q4; source: South African Reserve Bank. **Price level**: span: 1994:Q1-2022:Q4; source: Statistics South Africa (Stats SA). **Real exports**: span: 1994:Q1-2022:Q4; source: South African Reserve Bank.

Real imports: span: 1994:Q1-2022:Q4; source: South African Reserve Bank. **Trade balance**: ratio of nominal net exports to nominal GDP. **Country spread**: span: 1994:Q4 - 2021:Q1; source: The J.P. Morgan Emerging Markets Bond Index Global (EMBIG); the benchmark panel uses the series from 2002:Q2.

- ***Turkey***. **Real GDP**: span: 1993:Q4-2022:Q4; source: Turkish Statistical Institute (TURKSTAT). **Real investment**: span: 1998:Q1-2022:Q4; sources: Turkish Statistical Institute (TURKSTAT) and Organisation for Economic Co-operation and Development (OECD) VPVOBARSA. **Price level**: span: 1993:Q4-2022:Q4; source: Turkish Statistical Institute (TURKSTAT). **Real exports**: span: 1998:Q1-2022:Q4; sources: Turkish Statistical Institute (TURKSTAT) and OECD CARSA. **Real imports**: span: 1998:Q1-2022:Q4; sources: Turkish Statistical Institute (TURKSTAT) and OECD CARSA. **Trade balance**: ratio of nominal net exports to nominal GDP. **Country spread**: span: 1996:Q3 - 2021:Q1; source: The J.P. Morgan Emerging Markets Bond Index Plus (EMBI+).

A.3 Additional Countries

The larger sample of Section 4 adds the following countries, each entering from the first quarter in which all four country variables are available (Chile 1999:Q2, Colombia 2005:Q1, Ecuador 1995:Q1-2014:Q4, Malaysia 2010:Q1, the Philippines 1994:Q2).

- ***Chile***. **Real GDP**: span: 1996:Q1-2022:Q4; source: Central Bank of Chile. **Real investment**: span: 1996:Q1-2022:Q4; source: Central Bank of Chile. **Price level**: span: 1996:Q1-2022:Q4; source: National Statistics Institute of Chile. **Real exports**: span: 1996:Q1-2022:Q4; source: Central Bank of Chile. **Real imports**: span: 1996:Q1-2022:Q4; source: Central Bank of Chile. **Trade balance**: ratio of nominal net exports to nominal GDP. **Country spread**: span: 1999:Q2 - 2024:Q1; source: The J.P. Morgan Emerging Markets Bond Global Index (EMBIG).

- ***Colombia***. **Real GDP**: span: 2000:Q1-2022:Q4; source: National Administrative Department of Statistics Colombia (DANE). **Real investment**: span: 2005:Q1-

2022:Q4; source: National Administrative Department of Statistics Colombia (DANE). **Price level**: span: 2005:Q1-2022:Q4; source: National Administrative Department of Statistics Colombia (DANE). **Real exports**: span: 2005:Q1-2022:Q4; source: National Administrative Department of Statistics Colombia (DANE). **Real imports**: span: 2005:Q1-2022:Q4; source: National Administrative Department of Statistics Colombia (DANE). **Trade balance**: ratio of nominal net exports to nominal GDP. **Country spread**: span: 1997:Q2 - 2024:Q1; source: The J.P. Morgan Emerging Markets Bond Global Index (EMBIG).

- ***Ecuador***. **Real GDP**: span: 2000:Q1-2022:Q4; source: Central Bank of Ecuador. **Real investment**: span: 2000:Q1-2022:Q4; source: Central Bank of Ecuador. **Price level**: span: 1994:Q1-2022:Q4; source: Central Bank of Ecuador. **Real exports**: span: 2000:Q1-2022:Q4. source: Central Bank of Ecuador and Uribe and Yue (2006); **Real imports**: span: 2000:Q1-2022:Q4; source: Central Bank of Ecuador and Uribe and Yue (2006). **Trade balance**: ratio of nominal net exports to nominal GDP. **Country spread**: span: 1995:Q1 - 2014:Q4; source: The J.P. Morgan Emerging Markets Bond Index Plus (EMBI+).

- ***Malaysia***. **Real GDP**: span: 1994:Q1-2022:Q4; source: Department of Statistics Malaysia (DOSM). **Real investment**: span: 2010:Q1-2022:Q4; source: Department of Statistics Malaysia (DOSM) **Price level**: span: 2014:Q1-2022:Q4; source: Department of Statistics Malaysia (DOSM). **Real exports**: span: 2010:Q1-2022:Q4; source: Department of Statistics Malaysia (DOSM). **Real imports**: span: 2010:Q1-2022:Q4; source: Department of Statistics Malaysia. **Trade balance**: ratio of nominal net exports to nominal GDP. **Country spread**: span: 1996:Q4 - 2024:Q1; source: The J.P. Morgan Emerging Markets Bond Global Index (EMBIG).

- ***Philippines***. **Real GDP**: span: 1994:Q1-2022:Q4; source: Philippine Statistics Authority (PSA). **Real investment**: span: 1994:Q1-2022:Q4; source: Philippine Statistics Authority (PSA). **Price level**: span: 1994:Q1-2022:Q4; source: Philippine Statistics Authority (PSA). **Real exports**: span: 1994:Q1-2022:Q4; source: Philippine

Statistics Authority (PSA). **Real imports**: span: 1994:Q1-2022:Q4; source: Philippine Statistics Authority (PSA). **Trade balance**: ratio of nominal net exports to nominal GDP. **Country spread**: span: 1993:Q4 - 2021:Q1; source: The J.P. Morgan Emerging Markets Bond Index Plus (EMBI+).

Advanced economies. For Australia, Austria, Belgium, Canada, Denmark, Finland, Israel, the Netherlands, New Zealand, Norway, Sweden, and Switzerland, real GDP, real investment and the real trade balance (ratio of real net exports to real GDP) come from the same national accounts compilation (IMF IFS via Haver Analytics), and the 10-year government bond yield replaces the country spread.

B Matrix Specifications

In this subsection, we present a summary of the matrix forms from the restrictions we imposed on the model specification in equation 2.1. More specifically, we estimate the following structural panel vector autoregression model with two lags:

$$Ay_{i,t} = \alpha_i + B_1 y_{i,t-1} + B_2 y_{i,t-2} + \epsilon_{i,t}, \tag{B.1}$$

where the order of variables is as follows:

$$y_{i,t} = \begin{bmatrix} gdp_{i,t} \\ inv_{i,t} \\ tby_{i,t} \\ R_t^{US} \\ GR_t \\ ambig_t \\ R_{i,t} \end{bmatrix}.$$

After imposing all the restrictions discussed in the paper, the matrix summarizing the contemporaneous relationships, A, becomes:

Table A.1: Data sources

Variable	Definition and transformation	Source	Span
A. Benchmark VAR			
GDP	Log real GDP, s.a.; linear trend removed by country	National accounts (IMF IFS / Haver; authors' compilation); rebuilt from OECD QNA, CEPALSTAT, IMF	1994Q2–2019Q4
Investment	Log real gross fixed capital formation, s.a.; detrended by country	As GDP	1994Q2–2019Q4
Trade balance/GDP	Nominal (exports − imports)/GDP, s.a.; detrended by country	As GDP	1994Q2–2019Q4
EMBI spread	log(1 + J.P. Morgan EMBI country spread), quarterly mean	J.P. Morgan via Haver/Bloomberg (proprietary)	1994Q2–2019Q4
U.S. real rate	3-month T-bill rate minus expected inflation from a 2-lag CPI regression	FRED TB3MS, CPIAUCSL	1994Q2–2019Q4
Baa spread	log(1 + Moody's Baa yield minus 10-year Treasury), quarterly mean	FRED BAA10Y	1994Q2–2019Q4
Ambiguity	Log of the square root of the Izhakian (2020) ambiguity index for the S&P 500 (SPDR), quarterly mean of the monthly index, linearly detrended	Izhakian (2020), provided by the author	1994Q2–2019Q4
B. Horse race and robustness			
VIX	Log CBOE VIX, quarterly mean, detrended	FRED VIXCLS	1994Q2–2019Q4
JLN uncertainty	Log financial, macro and real uncertainty, h = 1, 3, 12 months, quarterly mean, detrended	Jurado, Ludvigson and Ng (2015), Ludvigson's website (Aug. 2026 vintage)	1994Q2–2019Q4
U.S. GDP, investment	Log real GDP and gross private domestic investment, detrended	FRED GDPC1, GPDIC1	1994Q2–2019Q4
Expected / realized SPY return	Series from the authors' dataset	Authors' compilation	1994Q2–2019Q4
C. Mechanism			
Corporate spreads, yields	Aaa and Baa minus 10-year Treasury, Baa minus Aaa, Baa minus 20-year, 2- and 10-year Treasury yields; quarterly means	FRED AAA10Y, BAA10Y, AAA, BAA, DGS2, DGS10, DGS20	1990Q1–2021Q4
Credit spread, EBP	Gilchrist-Zakrajšek credit spread and excess bond premium	Federal Reserve Board	1990Q1–2021Q4
Term premia	ACM and Kim-Wright 10-year term premia	New York Fed; FRED THREEFYTP10	1990Q1–2021Q4
Financial conditions, equity	Chicago Fed NFCI; NASDAQ Composite and MSCI USA log returns	FRED NFCI, NASDAQCOM; MSCI	1990Q1– (MSCI 1997Q1–)
U.S. portfolio flows	Net purchases of foreign bonds and stocks (TIC form S); mutual fund net purchases (Z.1 F.122); fund flows (ICI)	U.S. Treasury TIC; Federal Reserve Z.1; ICI	1990Q1–2021Q4 (ICI 2007–2020)
EME debt and portfolio flows	International debt securities, net issues and stocks; BoP portfolio debt and equity liabilities; scaled by GDP	BIS debt securities statistics; IMF BoP (BPM6); Lane and Milesi-Ferretti EWN (GDP)	1990Q1–2021Q4 (varies)
EME equity prices, REER	MSCI country price indices (USD); BIS real effective exchange rates	MSCI; BIS	1997Q1–2021Q4
CFO beliefs	Average respondent's 90th minus 10th percentile of the expected 1-year S&P 500 return	Duke-Richmond Fed CFO Survey (public file)	2001Q4–2019Q4

Note: Benchmark country series differ in span by unit (Table 2.1); Argentina enters as two units (1994:Q2-2001:Q3 and 2016:Q2-2018:Q2). The BAA series is the spread against the 10-year Treasury yield. Mechanism series stop in 2019:Q4 in the estimation.

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ A_{21} & 1 & 0 & 0 & 0 & 0 & 0 \\ A_{31} & A_{32} & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & A_{54} & 1 & 0 & 0 \\ 0 & 0 & 0 & A_{64} & A_{65} & 1 & 0 \\ A_{71} & A_{72} & A_{73} & A_{74} & A_{75} & A_{76} & 1 \end{bmatrix}$$

Following two matrices summarize the restrictions put on the first (B_1) and the second (B_2) lags, respectively:

$$B_1 = \begin{bmatrix} B_{1,11} & B_{1,12} & B_{1,13} & B_{1,14} & B_{1,15} & B_{1,16} & B_{1,17} \\ B_{1,21} & B_{1,22} & B_{1,23} & B_{1,24} & B_{1,25} & B_{1,26} & B_{1,27} \\ B_{1,31} & B_{1,32} & B_{1,33} & B_{1,34} & B_{1,35} & B_{1,36} & B_{1,37} \\ 0 & 0 & 0 & B_{1,44} & B_{1,45} & B_{1,46} & 0 \\ 0 & 0 & 0 & B_{1,54} & B_{1,55} & B_{1,56} & 0 \\ 0 & 0 & 0 & B_{1,64} & B_{1,65} & B_{1,66} & 0 \\ B_{1,71} & B_{1,72} & B_{1,73} & B_{1,74} & B_{1,75} & B_{1,76} & B_{1,77} \end{bmatrix}$$

$$B_2 = \begin{bmatrix} B_{2,11} & B_{2,12} & B_{2,13} & B_{2,14} & B_{2,15} & B_{2,16} & B_{2,17} \\ B_{2,21} & B_{2,22} & B_{2,23} & B_{2,24} & B_{2,25} & B_{2,26} & B_{2,27} \\ B_{2,31} & B_{2,32} & B_{2,33} & B_{2,34} & B_{2,35} & B_{2,36} & B_{2,37} \\ 0 & 0 & 0 & B_{2,44} & B_{2,45} & B_{2,46} & 0 \\ 0 & 0 & 0 & B_{2,54} & B_{2,55} & B_{2,56} & 0 \\ 0 & 0 & 0 & B_{2,64} & B_{2,65} & B_{2,66} & 0 \\ B_{2,71} & B_{2,72} & B_{2,73} & B_{2,74} & B_{2,75} & B_{2,76} & B_{2,77} \end{bmatrix}$$

C Robustness and Additional Analysis

C.1 Comparison with the results of Akinci (2013)

To assess the hypothesis that global financial risk plays an important role in driving country spreads and aggregate fluctuations in EMEs, Akıncı (2013) sets up a 6-variable structural panel VAR for a large set of EME countries with the following variables: EME real GDP, EME real investment, EME trade balance, U.S. rate, global financial risk (measured by the BAA corporate spread), and the country spread (measured by the EMBI). With the variables in this order, the VAR uses the typical Sims restrictions and in addition imposes that U.S. rate and global financial risk have an impact on all other variables but not the other way around. Recall that we build on this modeling strategy for our own analysis. For her benchmark analysis, Akinci uses quarterly data for Argentina, Brazil, Mexico, Peru, South Africa and Turkey from 1994 through 2011, included. The model is estimated using the least square dummy variable (LSDV) estimator. Recall also that our own benchmark analysis uses quarterly data for the same set of countries as Akinci but extends her sample through the pre-pandemic period: the fourth quarter of 2019.

Our estimation code reproduces Akinci's results on her own data (Figure C.4): at 20 quarters, global financial risk, U.S. rate and country spread shocks account for 16.8, 6.1 and 19.4 percent of the forecast error variance of EME GDP, against about 18, 6 and 18 percent in Akıncı (2013). For the purposes of comparison to our benchmark 7-variable PSVAR, we present results from using Akinci's 6-variable model with our own benchmark data. On net, results are consistent with Akinci's findings. Figures C.1 to C.3 show IRF results from an unanticipated one standard deviation shock to: the country spread (Figure C.1), the U.S. rate (Figure C.2) and global financial risk (Figure C.3).

As shown in Figure C.1, a shock to the country spread results in a contraction of EME GDP and investment and an expansion of the EME trade balance. These results are intuitive, as a shock to the country spread is in line with borrowing rates rising exogenously. As shown in Figure C.2, a shock to the U.S. rate yields similar results for GDP, investment, and the trade balance, although of less significance (only at the 68 percent level, for the first two

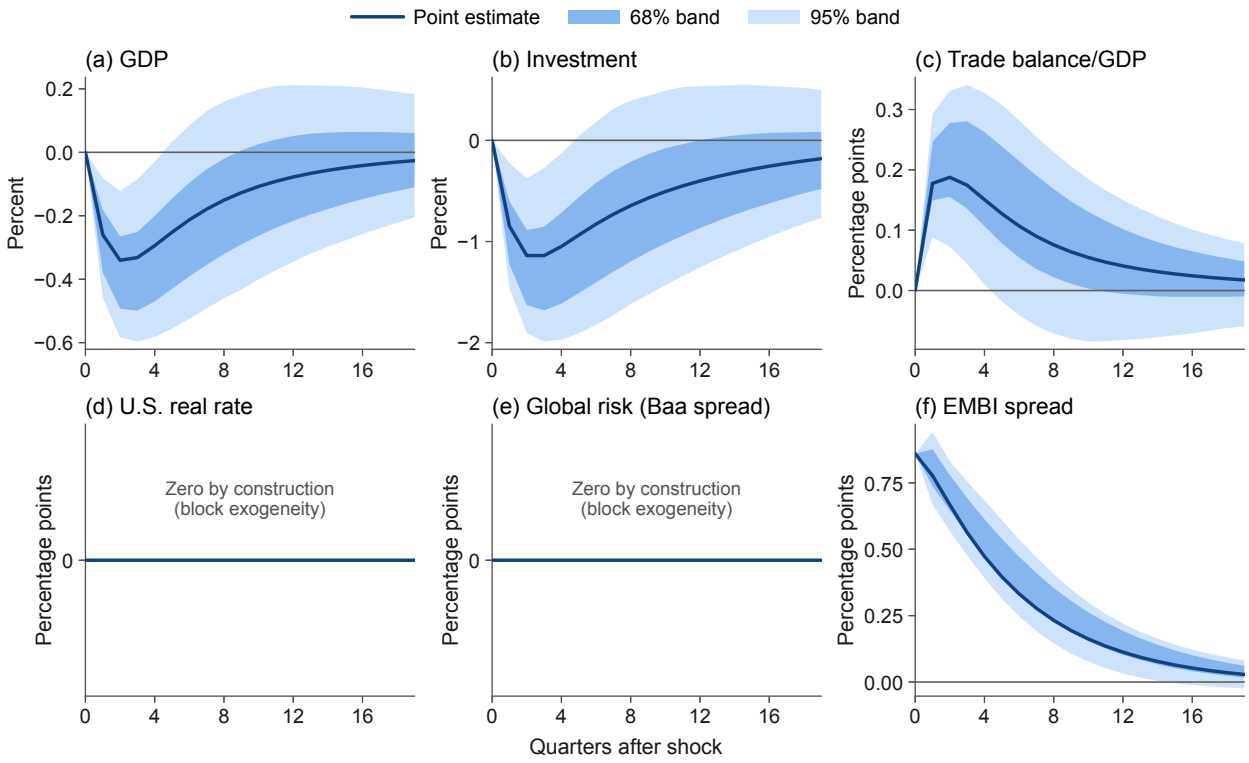


Figure C.1: IRFs to one std. dev. shock to country spread

Note: Estimated using our data through 2019:Q4 and the Akıncı (2013) specification with six variables. Shaded areas: 68 and 95 percent bands from 10,000 bootstrap replications.

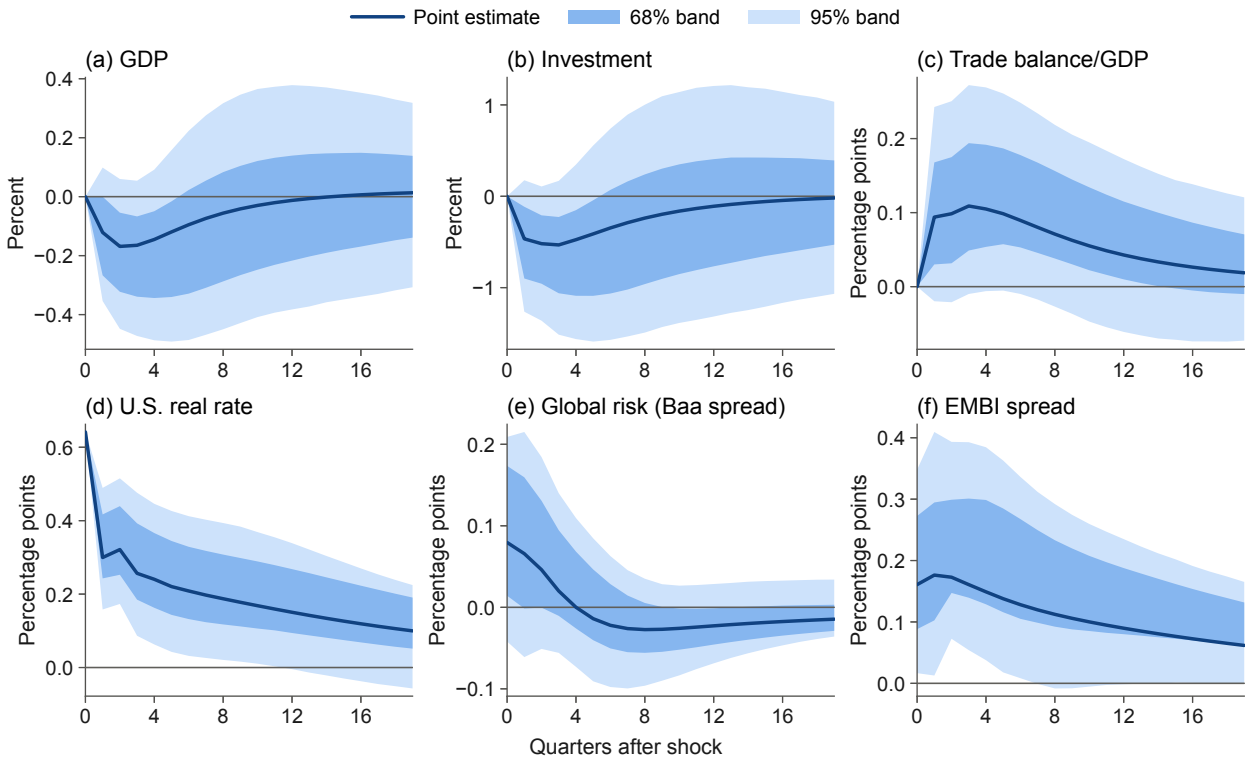


Figure C.2: IRFs to one std. dev. shock to U.S. interest rate

Note: Estimated using our data through 2019:Q4 and the Akıncı (2013) specification with six variables. Shaded areas: 68 and 95 percent bands from 10,000 bootstrap replications.

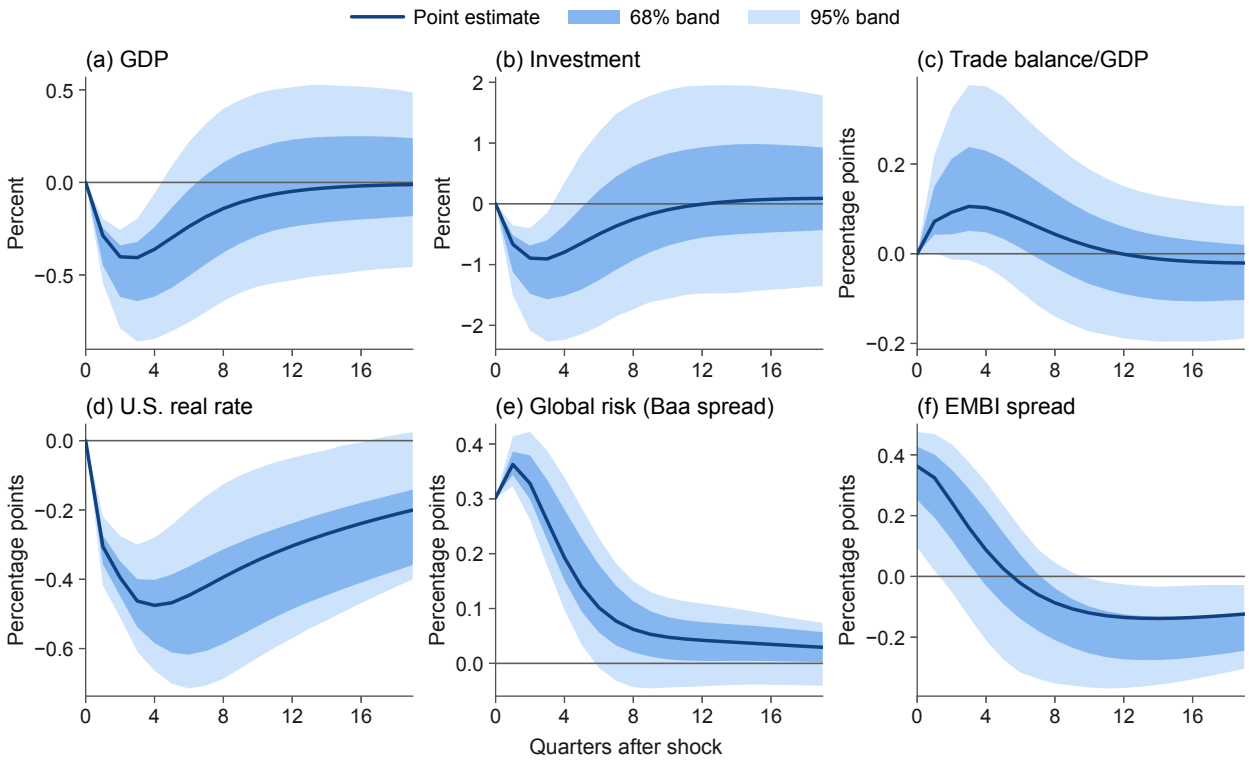


Figure C.3: IRFs to one std. dev. shock to BAA spread

Note: Estimated using our data through 2019:Q4 and the Akıncı (2013) specification with six variables. Shaded areas: 68 and 95 percent bands from 10,000 bootstrap replications.

to three quarters). This shock also raises the country spread, while the response of global financial risk is small and not significant. Intuitively, the shock to the U.S. rate can be interpreted as resulting in a rotation away from relatively riskier bonds, EME country bonds in particular, to the relatively safer U.S. sovereign bonds whose interest rate is now higher. As such, with lower relative demand for EME country bonds, their yields increase. Ultimately, the ensuing higher borrowing costs for EMEs drive similar dynamics in country variables as in the case of the shock to the country spread. (In the baseline analysis in Akıncı (2013), a U.S. interest rate shock also triggers an insignificant decrease in the BAA corporate spread.)

Finally, Figure C.3 shows the impact of a shock to the BAA spread, which can be interpreted as an exogenous shock to global financial risk. Against this backdrop there is a flight to safety to U.S. sovereign bonds, which drives a substantial and significant decline in the U.S. interest rate. The retreat to safer assets lowers demand for EME country bonds, which increases the country spread and, therefore, EME borrowing costs resulting in EME GDP and investment decreasing and doing so by a greater amount, both quantitatively and in terms of significance, compared to the case of the country interest rate and U.S. rate shocks. Restricting the sample to Akinci's end date (2011:Q3) magnifies the size and significance of these results: in the 6-variable model on our data, EME GDP falls by 0.58 percent after a global financial risk shock (significant at the 95 percent level for five quarters), and global financial risk accounts for 9.4 percent of GDP variance at 20 quarters, against 3.0 percent on the full sample (Table C.1). Figures C.5 and C.6 show our 7-variable model on the same shorter sample; the expansion after an ambiguity shock is of similar size (EME GDP peaks at 0.36 percent) but arrives later and is significant at the 95 percent level only between the ninth and the eleventh quarter.

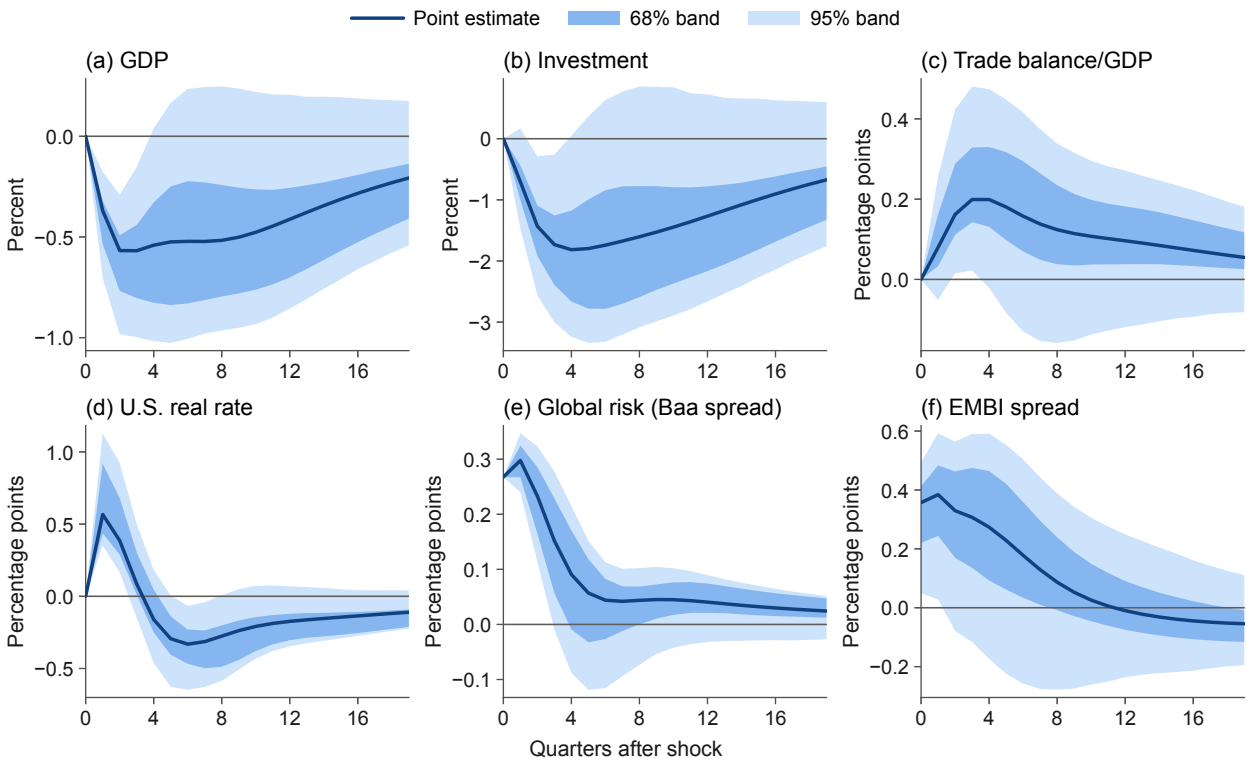


Figure C.4: IRFs to one std. dev. shock to BAA spread: Akinci (2013) model and data

Note: The Akıncı (2013) 6-variable specification estimated with our code on Akinci's own data (panel through 2011:Q3, long global sample from 1987:Q3). Rates and spreads in annualized percentage points, as in Akıncı (2013). Shaded areas: 68 and 95 percent bands from 10,000 bootstrap replications.

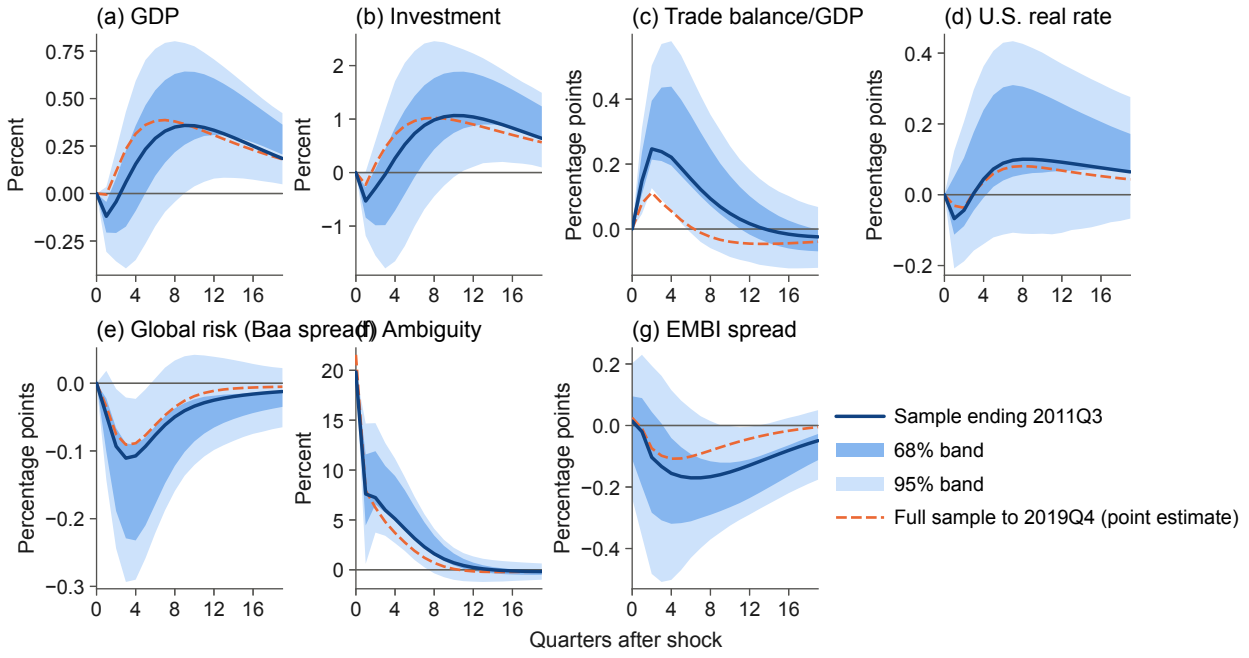


Figure C.5: IRFs to one std. dev. shock to ambiguity through 2011:Q3

Note: Benchmark 7-variable model estimated on samples ending in 2011:Q3 (panel and global block). Solid line and shaded areas: point estimate with 68 and 95 percent bands; dashed line: full-sample benchmark.

Table C.1: Forecast error variance decomposition, samples ending in 2011:Q3

Variable	h	Ambiguity	Baa spread	U.S. real rate	EMBI spread	Sum
A. Seven-variable model, sample ending 2011Q3						
GDP	4	0.2	6.5	0.8	4.9	12.4
	8	1.8	9.4	0.8	6.6	18.7
	20	6.5	8.6	1.3	6.8	23.1
Investment	4	0.3	3.6	1.3	4.1	9.4
	8	1.2	5.2	1.6	5.7	13.7
	20	5.3	5.0	3.0	5.9	19.2
Trade balance/GDP	4	3.4	1.1	1.3	3.0	8.8
	8	4.8	2.5	1.3	3.8	12.4
	20	4.7	4.6	1.3	4.1	14.8
EMBI spread	4	0.8	10.5	1.6	75.7	88.6
	8	2.9	8.4	2.0	72.0	85.3
	20	5.0	11.4	4.4	58.4	79.2
B. Six-variable model, sample ending 2011Q3						
GDP	4	–	7.3	1.8	5.1	14.2
	8	–	10.3	2.3	7.0	19.6
	20	–	9.4	2.5	8.0	19.9
Investment	4	–	4.1	2.2	4.2	10.4
	8	–	5.6	3.1	5.9	14.6
	20	–	4.9	4.3	6.9	16.1
Trade balance/GDP	4	–	0.7	0.6	2.3	3.7
	8	–	1.9	0.7	2.7	5.2
	20	–	3.2	0.7	2.7	6.6
EMBI spread	4	–	11.6	3.2	74.3	89.1
	8	–	9.5	4.1	71.9	85.5
	20	–	10.9	5.6	60.4	76.9

Note: Share of the h-step-ahead forecast error variance of each EME variable explained by the ambiguity, BAA spread, U.S. real rate and EMBI shocks, in percent; $h = 1$ is the impact quarter. Sum: the four shares together (for the EMBI spread this includes its own shock). The 6-variable model is the Akıncı (2013) specification without ambiguity, estimated on our data. Panel and long global samples end in 2011:Q3. Point estimates.

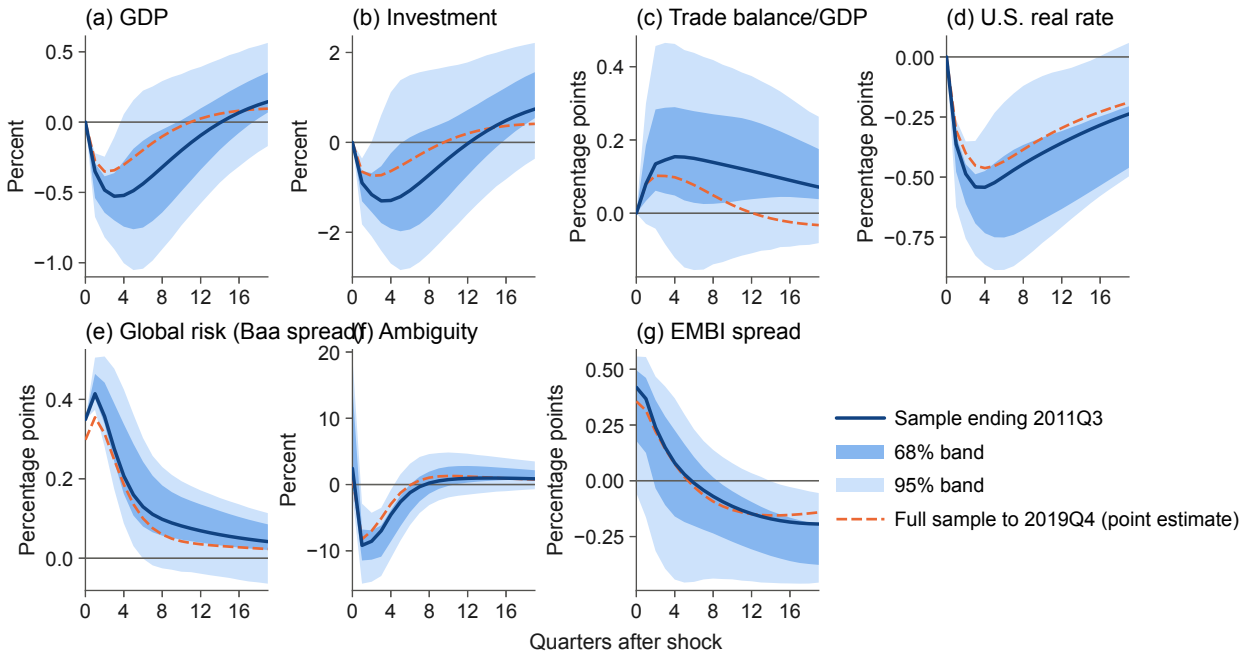


Figure C.6: IRFs to one std. dev. shock to BAA spread through 2011:Q3

Note: Benchmark 7-variable model estimated on samples ending in 2011:Q3 (panel and global block). Solid line and shaded areas: point estimate with 68 and 95 percent bands; dashed line: full-sample benchmark.

C.2 Additional impulse responses

Figures C.7 and C.8 show the responses of the benchmark model to U.S. interest rate and country spread shocks. A U.S. interest rate shock (0.64 percentage points) lowers EME investment by 0.35 percent one quarter after the shock (not significant) and raises the trade balance and the country spread (the latter by 0.15 percentage points, significant at the 95 percent level on impact and in the second and third quarters); the response of EME GDP is not significant, and the BAA spread rises on impact (significant at the 68 percent level only there). Ambiguity falls by 3.4 percent one quarter after the shock, a small move relative to the standard deviation of the ambiguity shock (21.6 percent). A country spread shock (0.86 percentage points) lowers EME GDP and investment by 0.33 and 1.10 percent two quarters after the shock and raises the trade balance by 0.19 percentage points, all significant at the 95 percent level between the first and the fourth quarter.

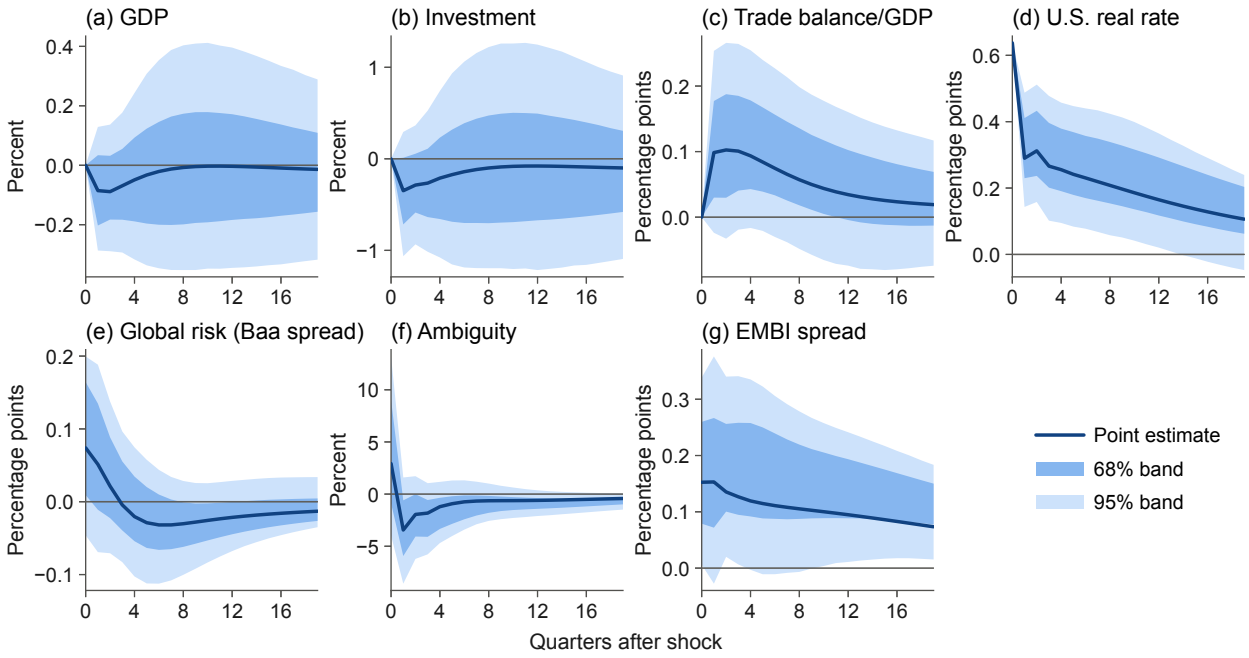


Figure C.7: IRFs to one std. dev. shock to U.S. interest rate

Note: Responses of the benchmark PSVAR to a one standard deviation U.S. real interest rate shock (0.64 percentage points on impact). Units and bands as in Figure 3.1.

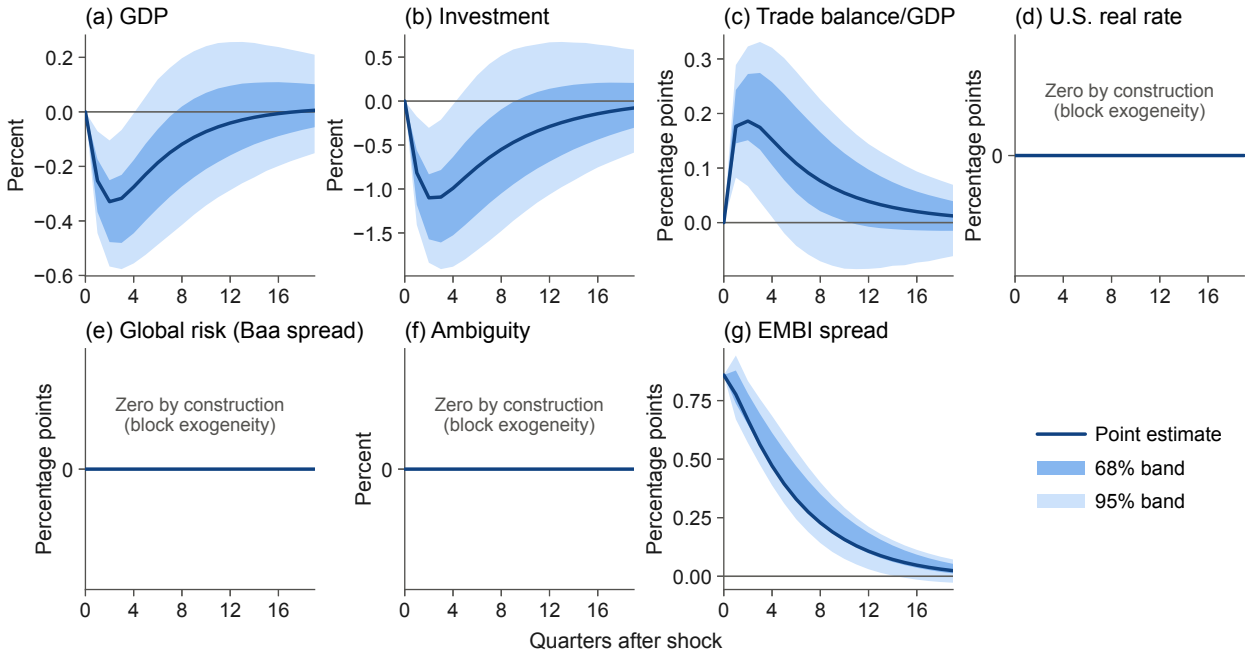


Figure C.8: IRFs to one std. dev. shock to country spread

Note: Responses of the benchmark PSVAR to a one standard deviation country spread (EMBI) shock (0.86 percentage points on impact). The global variables do not respond, by construction (block exogeneity). Units and bands as in Figure 3.1.

C.3 Inference

The benchmark bands come from a recursive-design moving-block residual bootstrap with Hall intervals (Section 2). Blocks of eight consecutive dates are drawn with wraparound (Künsch, 1989; Politis and Romano, 1992); every country receives its own structural residual vector of each drawn date and the global block its residual vector of that date, so that serial dependence within a block and the same-date dependence across countries and with the global block are preserved. A country that is not observed at a drawn date receives a draw from its own residuals; this affects 17 percent of the country-quarters in the benchmark, almost all of them in the two short Argentine samples. Table C.2 and Figures C.9 and C.10 compare this scheme with the i.i.d. residual bootstrap of Uribe and Yue (2006) and Akıncı (2013), which resamples each country's structural residual vectors and, separately, the residuals of the global block, and with three further alternatives. A design that feeds a single simulated path of the global block to all countries respects the fact that the global block is common to them. A wild bootstrap with one Rademacher sign per date, shared by all countries and by the global block, preserves the same-date correlation of the residuals across countries, which matters because the EMBI residuals are strongly correlated across countries (mean pairwise correlation 0.44; Pesaran, 2015 CD statistic 9.5; Table C.4). Both give nearly the same significance ranges as the i.i.d. bootstrap; the date-wild bands are the widest of the three, by roughly 5 to 10 percent. Given the seven panel units, country-cluster wild bootstrap inference is uninformative because it admits only $2^7 = 128$ distinct sign patterns (Cameron et al., 2008); the date-based wild bootstrap instead preserves the relevant cross-country dependence.

Two features of the bootstrap distribution decide the choice of the benchmark. First, block resampling widens the bands: the 95 percent bands of the GDP and investment responses are about 1.2 times as wide as under i.i.d. resampling. Second, under every scheme the bootstrap distribution sits below the point estimate at longer horizons, which reflects the usual downward bias of least-squares estimates of persistent dynamics in short samples; at the GDP peak the median of the block bootstrap distribution is 0.28 percent against a point

estimate of 0.39, and 28 percent of the draws lie above the point. Percentile bands inherit this bias and are conservative at long horizons: with block resampling, percentile bands of the GDP response exclude zero at the 90 but not at the 95 percent level. The intervals of Hall (1992) reflect the bootstrap quantiles around the point estimate and correct for the bias; with them, the GDP and investment responses to an ambiguity shock are significant at the 95 percent level from the fourth and sixth quarter to the end of the horizon under block resampling, and from the third and fifth quarter under i.i.d. resampling. The bias-corrected bootstrap of Kilian (1998) points the same way: it raises the GDP response at eight quarters from 0.38 to 0.49 percent (0.58 under block resampling) and gives 95 percent significance over quarters 4 to 16 (4 to 18), but it also raises the largest root of the system from 0.94 to 0.99 and makes the upper bounds very wide at long horizons, so we report it only as a check. Bayesian credible sets with a flat prior or a Minnesota prior give 95 percent significance for GDP over quarters 4 to 17 and 6 to 19. Under every rule, the decline in the BAA spread after an ambiguity shock is significant at the 68 percent level but not, or only at isolated horizons, at the 95 percent level, and the EMBI decline at the 68 percent level only.

The FEVD shares in the main text are point estimates. Table C.3 and Figure C.11 add their 68 percent bootstrap bands, computed as percentiles of the moving-block bootstrap distribution (Hall intervals are not range-respecting and can be negative for a share). Because shares are bounded below by zero and sampling noise adds variance to every shock's contribution, the bootstrap distribution of a small share sits above its point estimate (for the BAA share of GDP at 20 quarters, the point estimate is 2.3 percent and the bootstrap median 4.8 percent), so a point estimate can lie at or below its lower band, as several of the U.S. real rate shares do. The bands of the ambiguity, BAA and EMBI shares overlap at all but the shortest horizons.

Table C.2: Inference: significance ranges under alternative bootstrap schemes

	GDP		Investment		Baa spread		EMBI spread	
Scheme	68%	95%	68%	95%	68%	95%	68%	95%
Shock: Ambiguity								
Block bootstrap (block 8), Hall (benchmark)	2–19	4–19	3–19	6–19	1–10	3–4	3–15	none
Block bootstrap (block 8), percentile	2–19	none	4–19	none	1–6	none	none	none
Block bootstrap (block 8), Kilian bias-corrected	2–19	4–18	3–19	7–17	1–12	2–5	3–13	none
Residual bootstrap, percentile	2–19	4–13	3–19	6–12	1–7	none	2–9	none
Residual bootstrap, Hall	2–19	3–19	3–19	5–19	1–19	none	3–14	none
Residual bootstrap, Kilian bias-corrected	2–19	4–16	3–19	6–15	1–8	none	3–13	none
Common global path, percentile	2–19	4–12	3–19	6–12	1–7	none	3–9	none
Wild bootstrap by date, percentile	2–19	5–13	3–19	7–13	1–7	2	2–9	none
Bayesian, flat prior	2–19	4–17	3–19	6–15	1–8	none	3–10	none
Bayesian, Minnesota prior ($\lambda = 0.5$)	2–19	4–18	3–19	6–17	1–8	none	3–12	none
Bayesian, Minnesota prior ($\lambda = 0.2$)	3–19	6–19	4–19	8–19	none	none	5–15	none
Bayesian, hierarchical prior (pooled)	2–14	none	3–14	none	1–8	none	none	none
Panel local projections (Driscoll-Kraay)	4–7, 16–19	none	4–7, 15–19	19	3–6	none	5–11, 13–19	5–7, 16–18
Country-cluster wild (shared code; rejected)	2–19	2–19	2–19	2–19	1–8	none	2–14	3–9
Shock: Global risk (Baa spread)								
Block bootstrap (block 8), Hall (benchmark)	1–6	1–4	1–4	1–2	0–10	0–5	0–3	0
Block bootstrap (block 8), percentile	1–5	1	1–3	none	0–19	0–4	0–3	0–2
Block bootstrap (block 8), Kilian bias-corrected	1–8	1–4	1–6	1–2	0–10	0–6	0–4	0–2
Residual bootstrap, percentile	1–6	1–4	1–5	1–2	0–6	0–4	0–3	0–2
Residual bootstrap, Hall	1–5	1–3	1–4	1–2	0–19	0–6	0–4	0–2
Residual bootstrap, Kilian bias-corrected	1–6	1–3	1–5	1–2	0–19	0–5	0–4	0–3
Common global path, percentile	1–5	1–3	1–5	1–2	0–6	0–4	0–3	0–2
Wild bootstrap by date, percentile	1–4	1–3	1–3	1	0–5	0–4	0–2	0–1
Bayesian, flat prior	1–6	1–4	1–5	1–2	0–15	0–5	0–3	0–2
Bayesian, Minnesota prior ($\lambda = 0.5$)	1–6	1–4	1–5	1–2	0–15	0–7	0–3	0–2
Bayesian, Minnesota prior ($\lambda = 0.2$)	1–7	1–4	1–5	1–2	0–14	0–9	0–4	0–3
Bayesian, hierarchical prior (pooled)	1–5	1–2	1–4	1	0–15	0–5	0–3	0–2
Panel local projections (Driscoll-Kraay)	0–8	0–6	0–6	4	0–8	0–4, 6–8	0–3	0–1
Country-cluster wild (shared code; rejected)	1–8	1–6	1–7	1–5	0–6	0–4	0–4	0–3

Note: Horizons ($h = 0$-19) at which the 68 and 95 percent bands of the responses to a one standard deviation

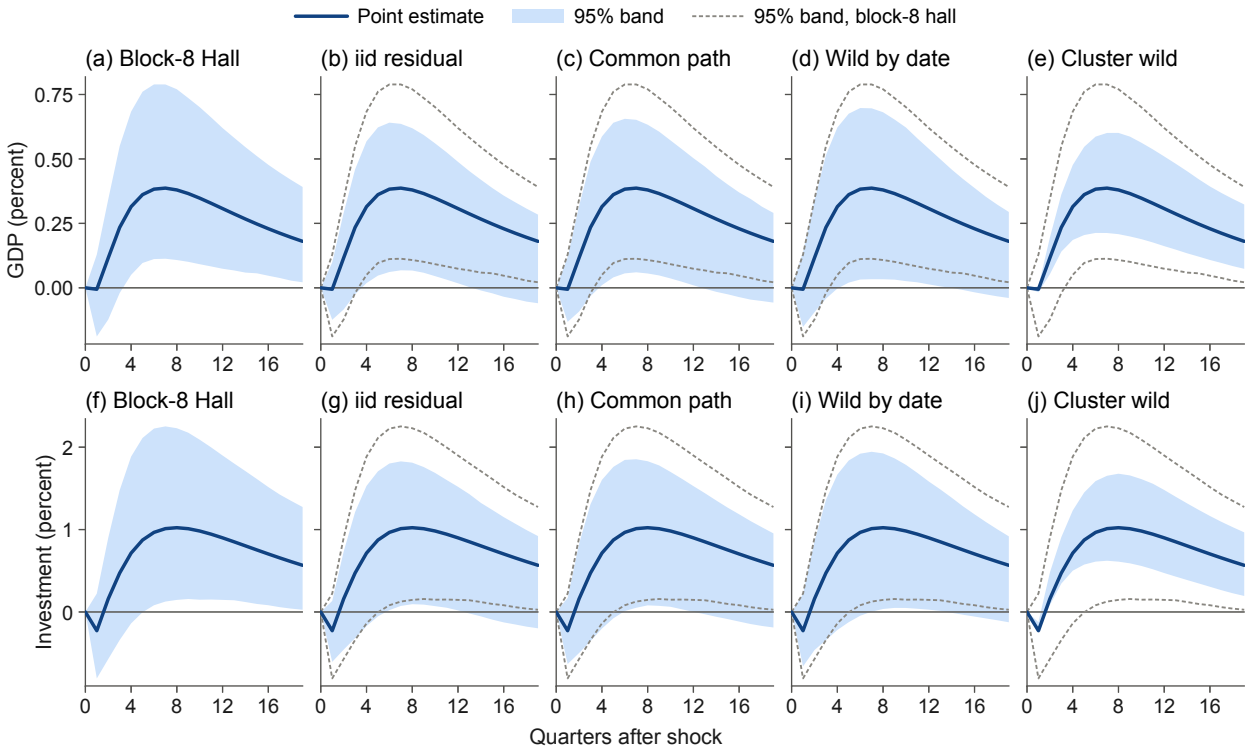


Figure C.9: Ambiguity-shock responses under alternative bootstrap schemes

Note: Responses of EME GDP and investment (percent) to a one standard deviation ambiguity shock in the benchmark model, with 95 percent bands under alternative bootstrap schemes (10,000 replications each): the i.i.d. residual bootstrap with percentile bands; a common path of the global block for all units; a wild bootstrap with one Rademacher sign per date; the country-cluster wild bootstrap (one sign per panel unit); and the benchmark moving-block bootstrap with Hall intervals. Dashed gray lines repeat the i.i.d. percentile band.

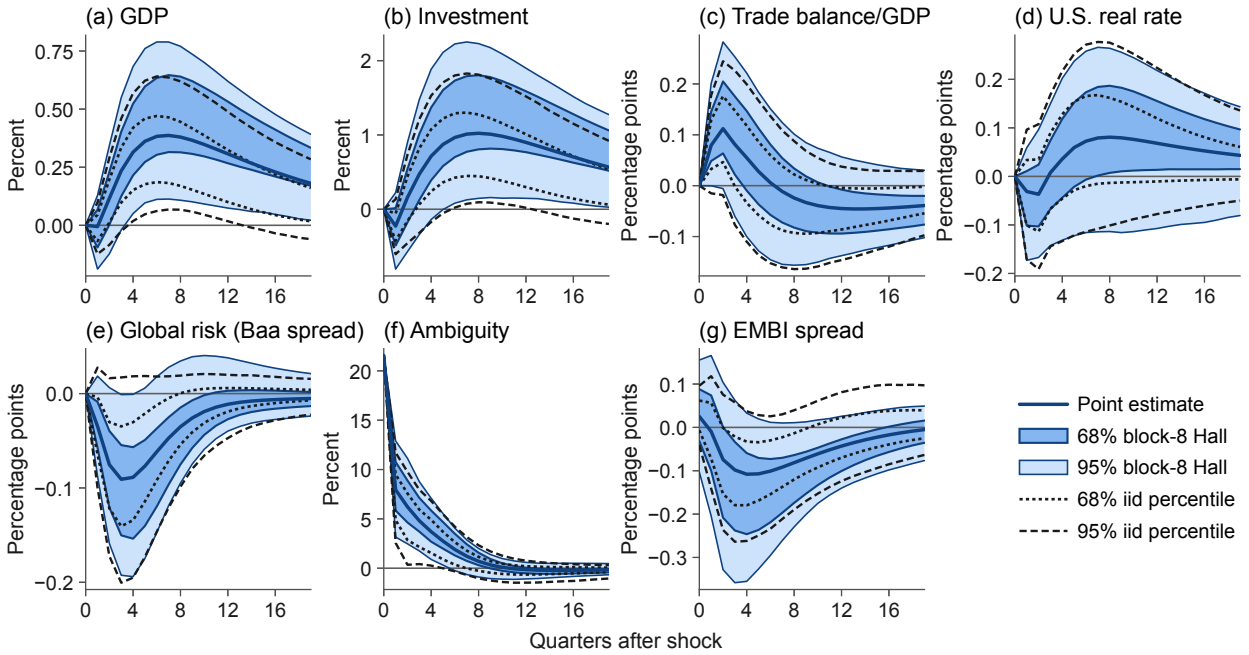


Figure C.10: Benchmark bands and i.i.d. percentile bands

Note: Responses of the benchmark PSVAR to a one standard deviation ambiguity shock. Solid line and shaded areas: point estimate with the benchmark 68 and 95 percent Hall bands from 10,000 replications of the moving-block residual bootstrap. Dashed lines: 68 and 95 percent percentile bands from 10,000 replications of the i.i.d. residual bootstrap.

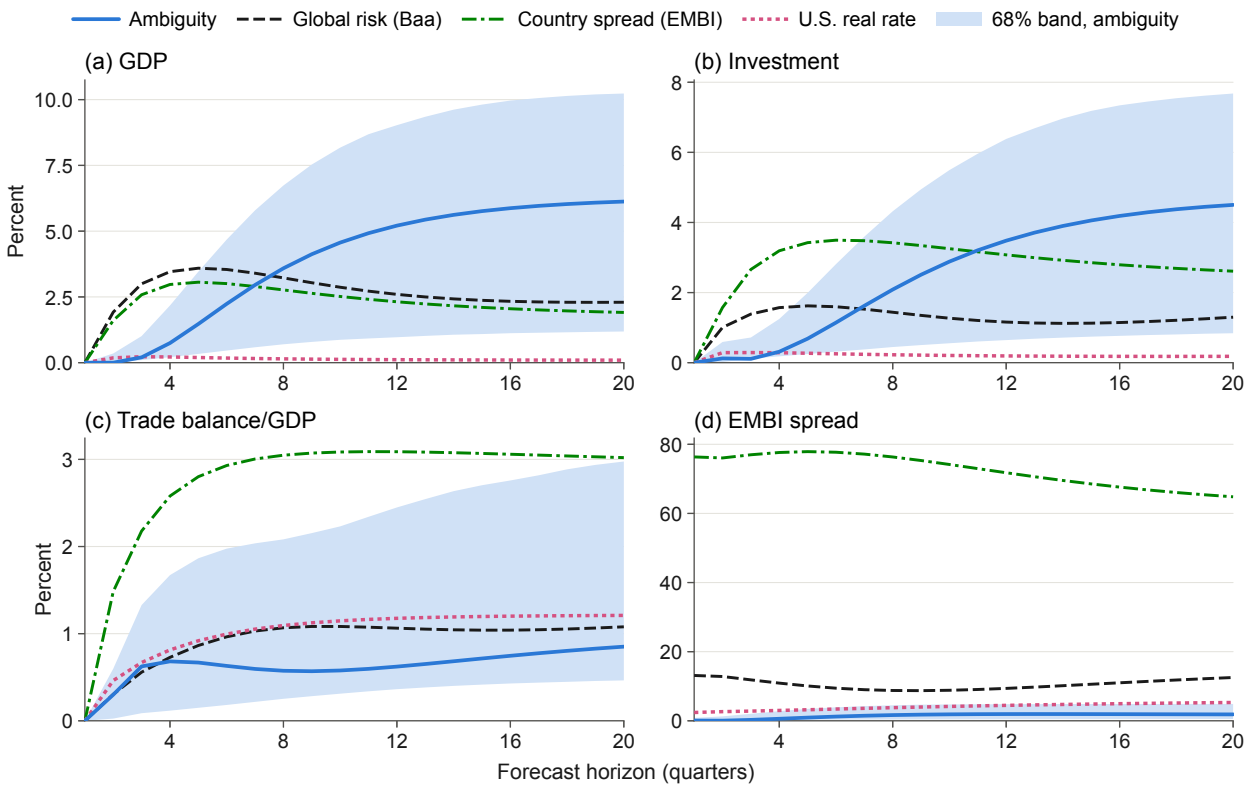


Figure C.11: Forecast error variance decomposition with the band of the ambiguity share

Note: As Figure 3.6; the shaded area is the 68 percent bootstrap band of the ambiguity share.

Table C.3: Forecast error variance decomposition with bootstrap bands

Variable	h	Ambiguity	Baa spread	U.S. real rate	EMBI spread	Sum
A. Seven-variable model (benchmark)						
GDP	4	0.7 [0.2, 2.2]	3.5 [0.8, 5.6]	0.2 [0.1, 1.4]	3.0 [1.0, 5.2]	7.4 [5.1, 11.7]
	8	3.6 [0.7, 6.7]	3.2 [1.3, 7.6]	0.1 [0.2, 2.3]	2.8 [1.0, 6.0]	9.7 [7.3, 18.8]
	20	6.1 [1.2, 10.2]	2.3 [2.1, 10.2]	0.1 [0.4, 4.8]	1.9 [1.0, 6.0]	10.4 [9.7, 26.8]
Investment	4	0.3 [0.2, 1.2]	1.6 [0.4, 3.3]	0.3 [0.1, 1.5]	3.2 [1.2, 5.6]	5.4 [3.7, 9.8]
	8	2.1 [0.4, 4.3]	1.4 [0.7, 5.1]	0.2 [0.2, 2.3]	3.4 [1.2, 7.3]	7.2 [5.6, 16.3]
	20	4.5 [0.8, 7.7]	1.3 [1.3, 8.7]	0.2 [0.4, 4.9]	2.6 [1.2, 7.5]	8.6 [8.2, 25.0]
Trade balance/GDP	4	0.7 [0.1, 1.7]	0.7 [0.1, 1.8]	0.8 [0.1, 2.5]	2.6 [0.9, 3.8]	4.8 [3.1, 7.8]
	8	0.6 [0.3, 2.1]	1.1 [0.4, 3.3]	1.1 [0.3, 3.1]	3.0 [0.9, 5.2]	5.8 [4.4, 11.3]
	20	0.9 [0.5, 3.0]	1.1 [0.8, 5.2]	1.2 [0.5, 3.8]	3.0 [1.0, 6.0]	6.2 [6.0, 15.2]
EMBI spread	4	0.6 [0.2, 2.8]	10.9 [6.7, 21.4]	3.0 [0.6, 5.5]	77.6 [63.2, 81.7]	92.1 [85.7, 95.9]
	8	1.7 [0.4, 4.5]	8.8 [6.4, 19.5]	3.8 [1.0, 6.3]	76.3 [59.7, 79.0]	90.6 [83.2, 93.0]
	20	1.9 [0.5, 4.9]	12.5 [7.8, 21.9]	5.3 [1.4, 7.6]	64.8 [51.6, 72.1]	84.5 [77.9, 89.5]
B. Six-variable model (Akinci 2013 specification)						
GDP	4	–	4.4 [1.2, 6.8]	0.8 [0.1, 2.4]	3.2 [1.1, 5.4]	8.4 [4.6, 12.5]
	8	–	4.4 [1.7, 9.6]	0.7 [0.3, 3.0]	3.2 [1.1, 6.6]	8.4 [5.7, 16.6]
	20	–	3.0 [2.4, 13.5]	0.5 [0.6, 5.3]	2.4 [1.1, 6.8]	5.9 [7.0, 22.9]
Investment	4	–	2.1 [0.5, 4.0]	0.8 [0.2, 2.3]	3.4 [1.3, 5.9]	6.3 [3.5, 10.4]
	8	–	2.1 [0.9, 6.3]	0.8 [0.3, 3.0]	3.9 [1.3, 7.8]	6.8 [4.8, 15.2]
	20	–	1.4 [1.5, 10.7]	0.6 [0.6, 5.5]	3.2 [1.3, 8.4]	5.2 [6.2, 22.1]
Trade balance/GDP	4	–	0.7 [0.1, 1.7]	0.8 [0.1, 2.3]	2.6 [0.9, 3.7]	4.0 [2.3, 6.4]
	8	–	1.0 [0.4, 3.2]	1.3 [0.3, 3.2]	3.0 [0.9, 5.0]	5.3 [3.2, 9.7]
	20	–	1.0 [0.8, 5.0]	1.5 [0.6, 3.9]	3.0 [1.0, 5.9]	5.5 [4.4, 12.9]
EMBI spread	4	–	11.7 [7.4, 21.9]	4.1 [0.8, 6.7]	76.4 [63.8, 81.3]	92.2 [85.5, 95.9]
	8	–	9.5 [6.8, 20.4]	5.2 [1.2, 7.5]	75.8 [60.2, 79.2]	90.5 [82.6, 92.6]
	20	–	12.1 [7.7, 22.0]	6.3 [1.6, 8.6]	65.3 [52.4, 72.9]	83.6 [76.3, 88.6]

Note: Shares as in Table 3.4, in percent, with the 68 percent bootstrap band in brackets (10,000 replications of the moving-block residual bootstrap, percentile intervals). The bands are percentiles of the bootstrap distribution of each share, so a point estimate can lie outside its band.

C.4 Lag order, Granger causality and residual diagnostics

Table C.4 reports the diagnostics behind our specification choices. For the panel system, the Bayesian and Hannan-Quinn criteria select two lags and the Akaike criterion four. Granger causality tests in the global block find one-way causality from global financial risk to the U.S. real rate and from global financial risk to ambiguity; the recursive orderings consistent with these tests are (global financial risk, U.S. rate, ambiguity) and (global financial risk, ambiguity, U.S. rate), which give the same results as the benchmark (Section 4). The residuals of the global block show no autocorrelation, while the EMBI residuals are strongly correlated across countries at the same date.

Table C.4: Lag order, Granger causality and residual diagnostics

A. Lag-order criteria (bold: selected)

	Full system			Country block			Global block			Largest
p	AIC	BIC	HQ	AIC	BIC	HQ	AIC	BIC	HQ	root
1	−58.672	−57.951	−58.388	−34.402	**−33.816**	−34.171	−24.110	**−23.717**	−23.951	0.919
2	−59.249	**−58.195**	**−58.834**	−34.635	−33.797	**−34.305**	**−24.326**	−23.697	**−24.072**	0.939
3	−59.346	−57.959	−58.800	−34.647	−33.557	−34.218	−24.258	−23.393	−23.908	0.942
4	**−59.489**	−57.768	−58.812	**−34.649**	−33.307	−34.121	−24.267	−23.166	−23.822	0.951

B. Granger causality in the global block (p-values)

Cause → affected	p=1	p=2	p=3	p=4
Baa → R_{US}	0.000	0.000	0.000	0.000
Baa → Ambiguity	0.000	0.000	0.000	0.000
R_{US} → Baa	0.722	0.195	0.185	0.061
R_{US} → Ambiguity	0.000	0.136	0.432	0.217
Ambiguity → R_{US}	0.024	0.554	0.767	0.627
Ambiguity → Baa	0.641	0.224	0.037	0.115
Number of consistent orderings	2	2	3	2

C. Cross-sectional dependence of residuals

Equation	Mean corr.	CD	p
GDP	0.08	1.72	0.086
Investment	−0.02	−0.91	0.363
Trade balance/GDP	−0.03	−1.04	0.300
EMBI spread	0.44	9.53	0.000

Note: Panel A: information criteria of the full panel system, the country block and the long global block for lag orders 1-4 on a common sample (the first four observations of every unit and of the long block serve as presample); bold marks the minimum. Largest root: modulus of the largest eigenvalue of the benchmark companion matrix estimated with p lags. Panel B: p-values of F tests that x does not Granger-cause y in a VAR(p) with a constant in the long global block (1994:Q2-2019:Q4). Panel C: average same-date correlation of the country-equation residuals across units and the Pesaran (2015) CD statistic for unbalanced panels.

C.5 Global risk, ambiguity, and dispersion of beliefs

In this subsection, we explore how an unexpected increase in global financial risk (proxied by an increase in BAA spread) leads to a decline in the degree of ambiguity in U.S. equity markets. Our proposed mechanism is that global financial risk influences investors' beliefs about expected market returns, which in turn affect the degree of ambiguity. We examine this belief-coordination channel with a three-variable VAR with two lags, a constant and a linear trend, including: *(i)* global financial risk (proxied by the BAA spread); *(ii)* a measure of dispersion of investors' beliefs, proxied by the difference between the average 90th and the average 10th percentile of expectations for the return of the S&P 500 over the next year from the CFO survey of Duke University and Federal Reserve Banks of Richmond and Atlanta; and *(iii)* U.S. equity markets' ambiguity. The sample is 2001:Q4-2019:Q4, with the missing value for 2019:Q1 interpolated. Table C.6 and Figure C.14 present the results suggesting that an increase in the BAA spread leads to rise in dispersion of beliefs, which in turn reduces ambiguity. Global financial risk Granger-causes dispersion (p-value 0.006), and dispersion and global financial risk Granger-cause ambiguity (p-values 0.002 and 0.001). A global financial risk shock raises dispersion by up to 0.7 percentage points (significant at the 95 percent level for four quarters), and a dispersion shock lowers ambiguity by 6 to 8 percent one quarter later, depending on the ordering. Panel B of Table C.6 repeats the tests on the CFO survey files for 2004-2019, which include the cross-respondent standard deviation of expected returns: with this measure of disagreement, global financial risk Granger-causes dispersion (p-value 0.046), dispersion Granger-causes ambiguity (p-value 0.008), and a dispersion shock lowers ambiguity by about 7 percent one quarter later.

Monthly data allow a sharper look at both links of the chain. Figure C.12 shows that in a monthly VAR for the BAA spread and ambiguity (two lags, 1993:M2-2019:M12) a global financial risk shock lowers ambiguity by 4 to 6 percent after five to six months, and by 3 to 5 percent with the log VIX ordered second, so the decline is not a volatility response in disguise. Table C.5 and Figure C.13 replace the CFO dispersion measure by nine monthly measures of investors' beliefs: the institutional and individual one-year, crash and valuation

confidence indices of the Yale survey (Goetzmann et al., 2017), the mean and cross-household standard deviation of the probability of a stock price increase in the Michigan Surveys of Consumers, and the bull-bear spread of the American Association of Individual Investors. The first link is broad: a global financial risk shock lowers the Michigan mean probability by 1.3 percentage points, the bull-bear spread by 3.4 points and crash confidence by 0.8 to 1 point, all significant at the 95 percent level, while valuation confidence rises, as prices fall. The second link is specific to the measure closest to the probability of the bad state: a fall in individual crash confidence, that is a rise in the share of investors who see a crash as likely, is followed by 3 to 4 percent lower ambiguity one to three months later in both orderings and with the VIX in the system, and it is the only measure that Granger-causes ambiguity at the 5 percent level (p-value 0.03); shocks to the broad optimism measures are instead followed by lower ambiguity after one to three months, and the Michigan dispersion measure, a coarse binned approximation of household disagreement, moves ambiguity in the opposite direction of the CFO measure. None of the belief measures responds to the identified ambiguity shock.

Table C.5: Global financial risk, investors' beliefs and ambiguity: monthly survey measures

	Baa → belief		Belief → amb., A		Belief → amb., B		Granger p			
Measure	Peak (h)	95%	Peak (h)	95%	Peak (h)	95%	B→b	b→A	B→A	Chain
Monthly survey measures, 1993M2–2019M12 (windows by measure; h in months, 0–12)										
Shiller crash confidence, individual	−0.76 (4)	−1–6	4.09 (2)	+1–12	3.33 (3)	+2–12	0.065	0.030	0.152	holds (95%)
Shiller crash confidence, institutional	−0.99 (6)	−2–8	2.55 (1)	none	1.30 (1)	none	0.185	0.567	0.041	partial
Shiller one-year confidence, individual	−0.45 (2)	−1	1.93 (4)	none	1.53 (6)	none	0.128	0.396	0.014	partial
Shiller one-year confidence, institutional	0.68 (5)	none	1.19 (3)	none	0.75 (4)	none	0.239	0.664	0.012	fails (opp. sign)
Shiller valuation confidence, individual	1.19 (12)	+5–7	1.66 (7)	none	1.67 (7)	none	0.150	0.277	0.005	fails (opp. sign)
Shiller valuation confidence, institutional	0.95 (6)	none	2.70 (4)	+3–4	2.51 (5)	+3	0.283	0.079	0.003	fails (opp. sign)
Michigan mean P(stock prices up)	−1.30 (5)	−0–12	2.62 (1)	none	−2.08 (2)	none	0.002	0.356	0.004	fails (opp. sign)
Michigan s.d. of P(stock prices up) (disp.)	0.06 (4)	+4–7	3.76 (2)	+4–8	3.75 (2)	+4–9	0.096	0.100	0.001	fails (opp. sign)
AAII bull minus bear	−3.40 (0)	−0	3.44 (1)	+1	−1.89 (3)	none	0.025	0.263	0.006	fails (opp. sign)
CFO survey (quarterly; h in quarters, 0–4)										
CFO range p90–p10 (paper) (disp.)	0.68 (1)	+1–4	−6.29 (1)	−1	−7.50 (1)	−1	0.006	0.002	0.001	holds (95%)
CFO s.d., additional data 2004–2019 (disp.)	0.62 (0)	+0–4	−7.43 (1)	−1	−7.56 (1)	−1	0.046	0.008	0.021	holds (95%)
CFO mean, additional data 2004–2019	−0.49 (1)	−0–1	−1.84 (1)	none	−5.47 (1)	−1	0.002	0.026	<0.001	fails (opp. sign)

Note: Three-variable monthly VARs in the BAA spread, one belief measure and log ambiguity, two lags, constant and linear trend, 1993:M2-2019:M12 or the measure's sample. Baa → belief: peak response of the measure to a one standard deviation global financial risk shock and the horizons at which the 95 percent band excludes zero. Belief → ambiguity: peak lagged response of ambiguity to a one standard deviation innovation in the measure, with the measure ordered before (A) or after (B) ambiguity. GC: p-values of Granger causality tests. Chain: whether both links hold with the signs of the belief-coordination channel at the 95 percent level. Yale indices are six-month moving averages of the monthly survey values; the CFO rows reproduce the quarterly results of Table C.6. Percentile bands from 10,000 replications of a residual bootstrap.

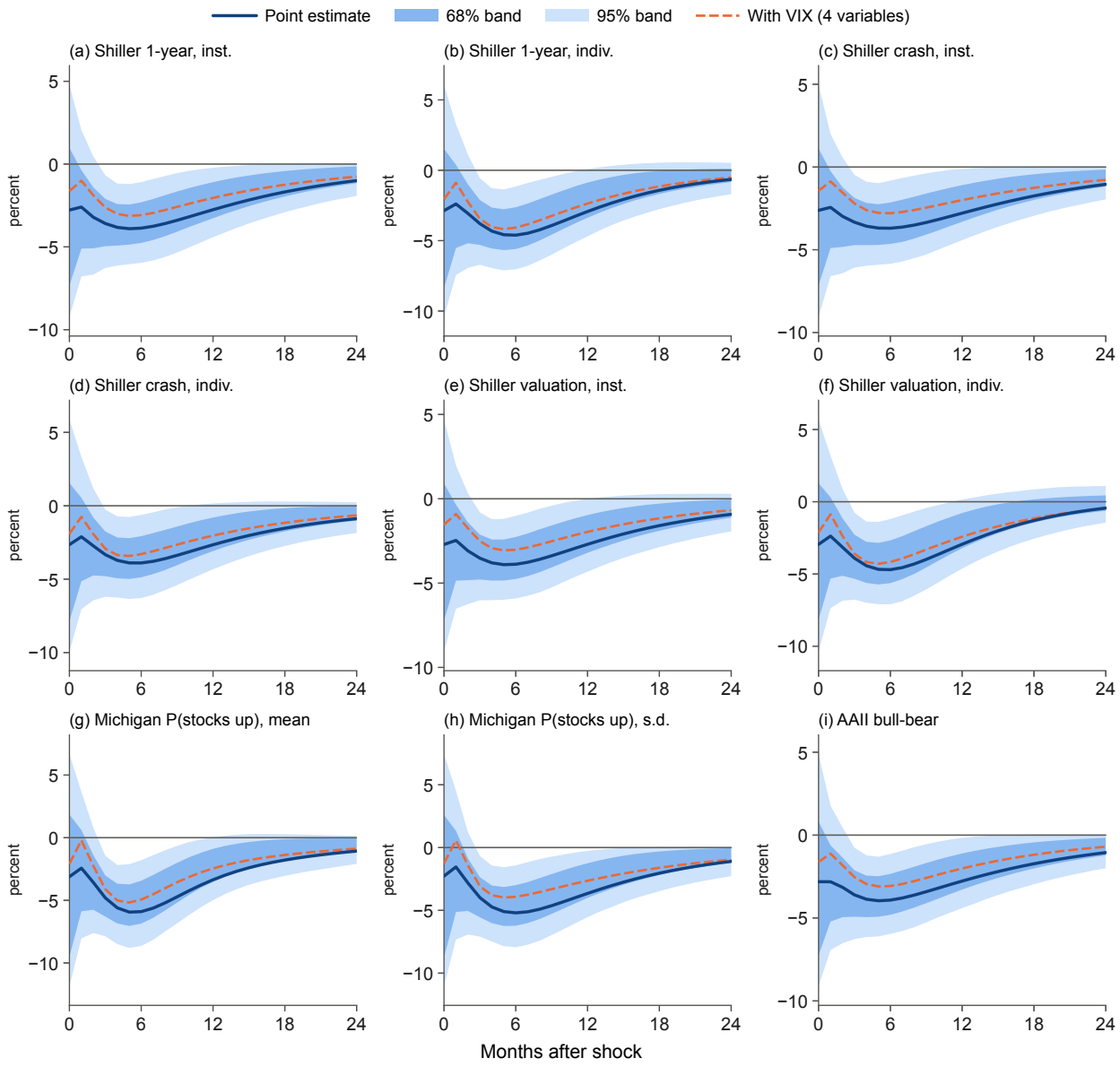


Figure C.12: Ambiguity after a global financial risk shock at monthly frequency

Note: Response of log ambiguity (percent) to a one standard deviation BAA spread shock in monthly VARs with two lags, a constant and a linear trend, 1993:M2-2019:M12: the two-variable system and the system with the log VIX ordered between the BAA spread and ambiguity; 68 and 95 percent bands from 10,000 bootstrap replications.

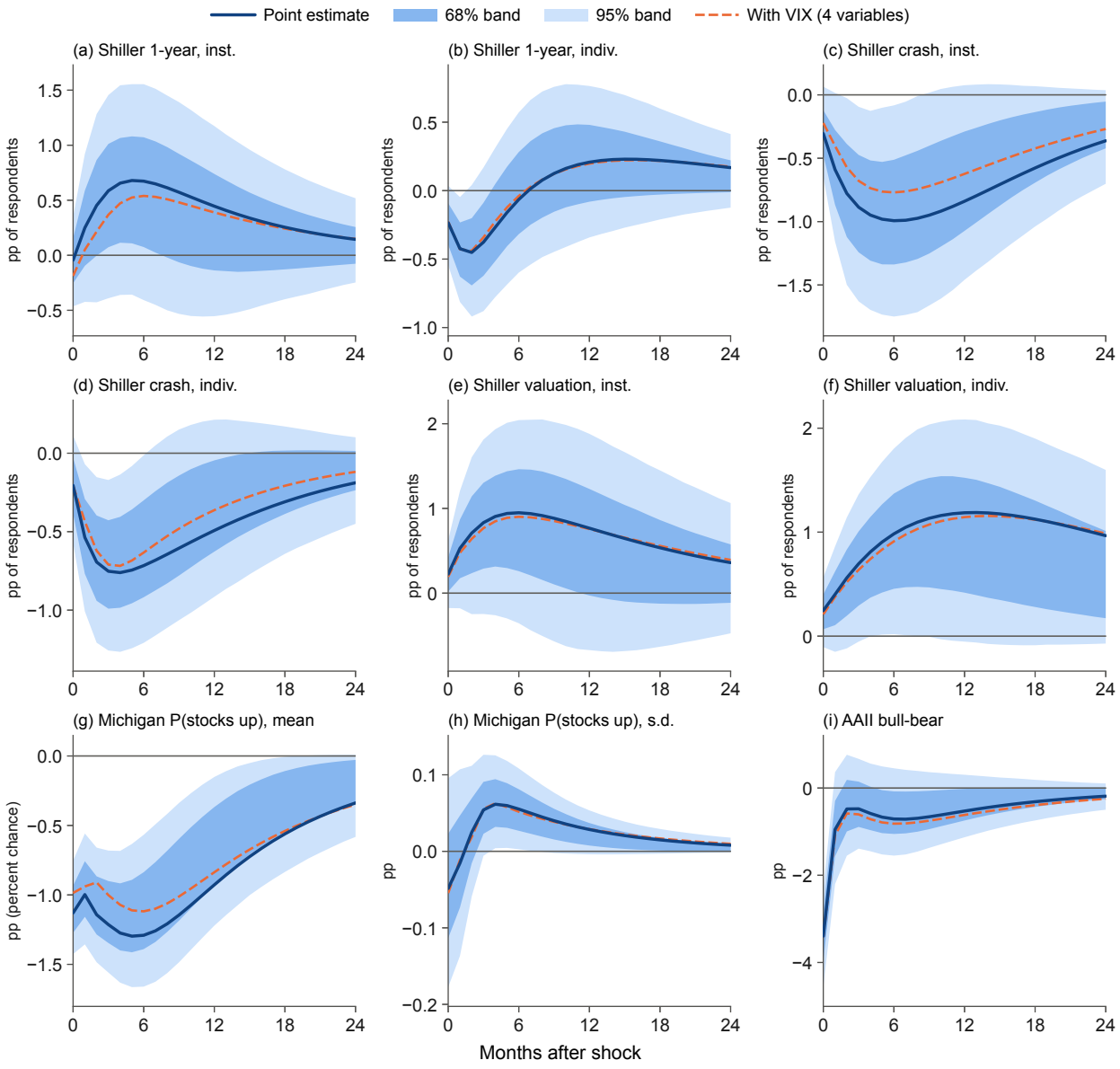


Figure C.13: Responses of survey measures of beliefs to a global financial risk shock

Note: Responses of nine monthly belief measures to a one standard deviation BAA spread shock in the three-variable monthly VARs of Table C.5 (solid line, 68 and 95 percent bands from 10,000 bootstrap replications) and in the four-variable systems that add the log VIX (dashed line). Yale confidence indices in percentage points of respondents; Michigan mean probability and standard deviation and the AAII bull-bear spread in percentage points.

Table C.6: Granger Causality Wald Tests

A. Public CFO Survey file, 2001Q4–2019Q4

Dependent	Regressor	Baseline	No 2019Q1	To 2018Q4	Detrended	No trend
Dispersion	Baa spread	0.006***	0.004***	0.004***	0.006***	0.008***
	Ambiguity	0.336	0.451	0.463	0.337	0.288
Baa spread	Dispersion	0.228	0.139	0.157	0.232	0.205
	Ambiguity	0.050*	0.012**	0.010**	0.051*	0.062*
Ambiguity	Dispersion	0.002***	0.002***	0.001***	0.002***	0.001***
	Baa spread	0.001***	0.001***	0.001***	0.001***	0.001***

B. Additional CFO Survey Data, 2004Q1–2019Q4

		To 2019Q4 (2019Q1 interpolated)			To 2018Q4			
Dependent	Regressor	S.d.	Mean	Public	S.d.	Range	Mean	Public
Dispersion	Baa spread	0.046**	0.002***	0.027**	0.069*	0.021**	0.003***	0.020**
	Ambiguity	0.694	0.003***	0.229	0.543	0.248	0.002***	0.255
Baa spread	Dispersion	0.694	0.276	0.175	0.839	0.069*	0.276	0.069*
	Ambiguity	0.229	0.216	0.032**	0.091*	0.004***	0.080*	0.004***
Ambiguity	Dispersion	0.008***	0.026**	0.002***	0.006***	0.000***	0.046**	0.000***
	Baa spread	0.021**	0.000***	0.010**	0.037**	0.022**	0.000***	0.023**

Note: p-values of Wald (χ^2) tests that the regressor does not Granger-cause the dependent variable in a VAR(2) in the BAA spread, belief dispersion and ambiguity with a constant and a linear trend. Panel A uses the public CFO survey from 2001:Q4 through 2019:Q4, with the missing 2019:Q1 observation interpolated (Baseline); dispersion is the average respondent's 90th minus 10th percentile of the expected one-year S&P 500 return. The remaining columns vary the treatment of the missing quarter (No 2019Q1: observations using the interpolated quarter dropped; To 2018Q4: sample ends before the gap), and the trend specification (Detrended: each variable detrended before estimation instead of a trend in the VAR; No trend: no trend term). Panel B: the same VAR on additional CFO survey data for 2004-2019 (estimation sample from 2004:Q3). Dispersion is measured by the cross-respondent standard deviation of expected one-year returns (S.d.); by the 90th-10th percentile range (Range), which matches the public range; by the mean expected return (Mean), as a placebo; or by the public range on the same sample (Public). ***, **, *: significant at 1, 5 and 10 percent.

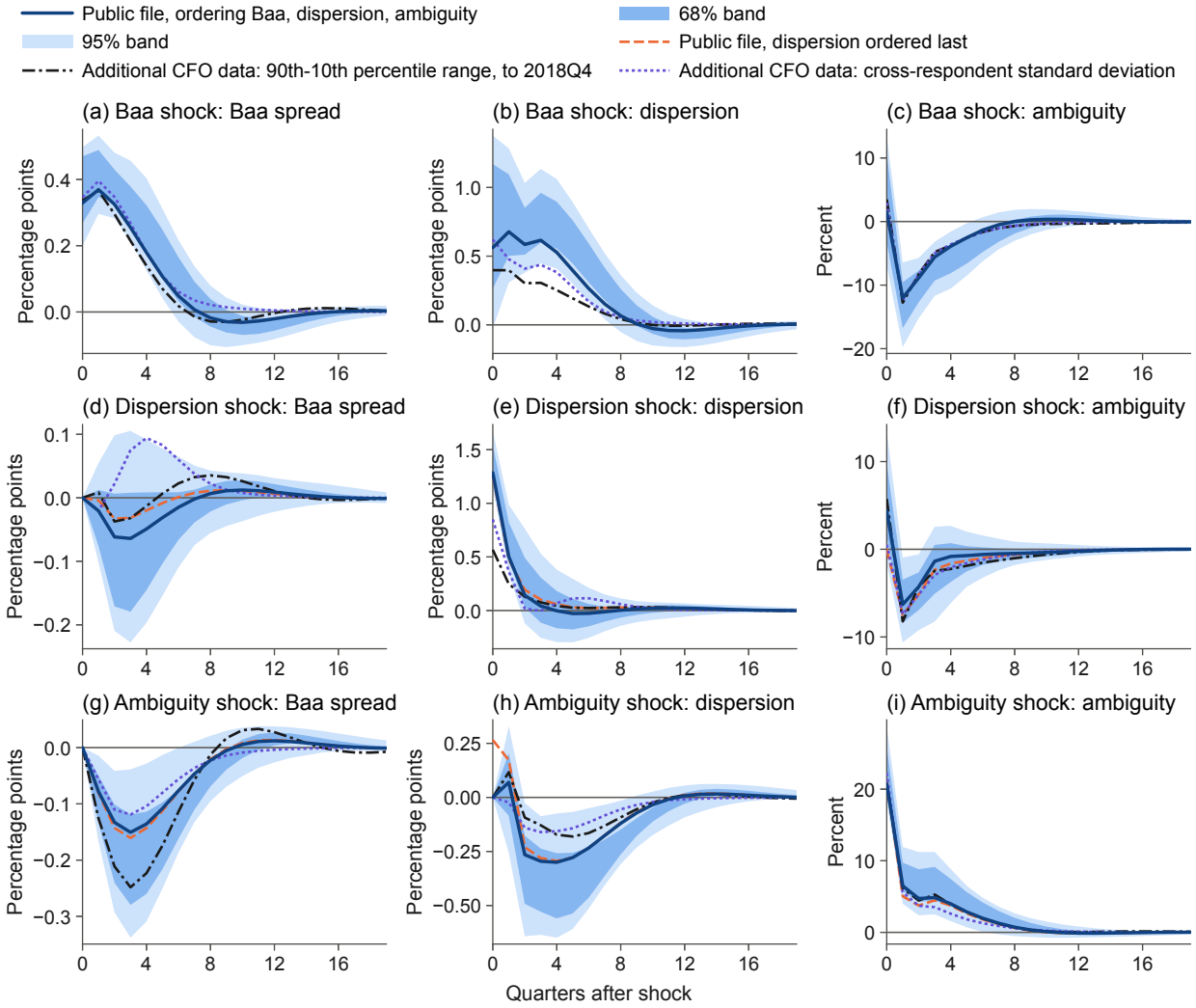


Figure C.14: Impulse Response Functions from 3-Variable VAR

Note: Responses to one standard deviation Cholesky shocks in the VAR(2) with a constant and a linear trend in the BAA spread (percentage points), the dispersion of CFO expectations of one-year S&P 500 returns (percentage points) and log ambiguity (×100), 2001:Q4-2019:Q4. Rows: shocks to the BAA spread, dispersion and ambiguity. Solid line and shaded areas: public survey file, ordering BAA spread, dispersion, ambiguity, with 68 and 95 percent bands (10,000 bootstrap replications); dashed line: dispersion ordered last. Other lines: point estimates on the survey files for 2004-2019, with dispersion measured by the 90th-10th percentile range (to 2018:Q4) or by the cross-respondent standard deviation of expected returns.

C.6 U.S. variables and expected returns

Figures C.15 to C.18 show the systems discussed in Section 3.3, in which the U.S. interest rate is replaced by expected or realized SPY returns, U.S. GDP or U.S. investment, together with the point estimates of eight-variable models that keep the U.S. interest rate and add each variable before or after ambiguity. With the U.S. interest rate kept, EME GDP peaks at 0.35 to 0.49 percent and is significant at the 95 percent level when U.S. GDP or investment is added, U.S. GDP responds at the 68 percent level only (a rise after two years with U.S. GDP ordered before ambiguity, a dip in the first quarter with it ordered after), and U.S. investment falls by 0.5 to 0.8 percent two quarters after the shock (significant at the 68 percent level). With SPY returns in the system, the EME point estimates are close to the benchmark (GDP peaks at 0.29 to 0.37 percent) and significant at the 95 percent level in all six systems.

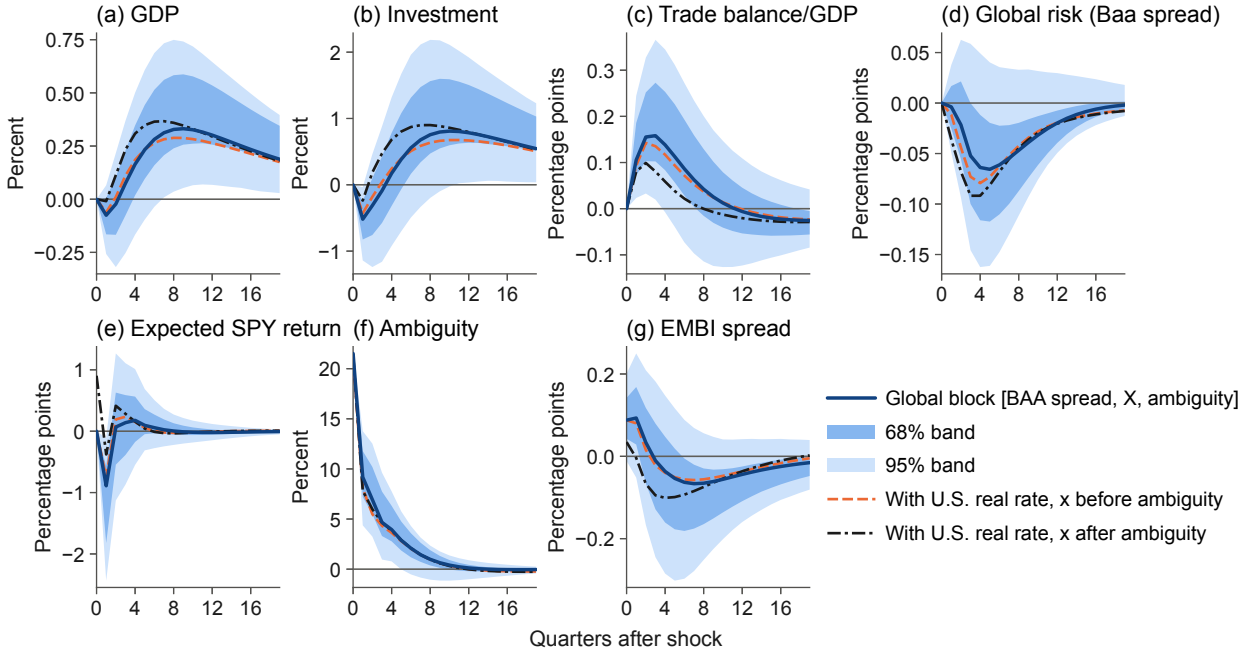


Figure C.15: IRFs to one std. dev. shock to ambiguity with U.S. interest rate replaced by expected SPY returns

Note: Global block ordered BAA spread, expected SPY return, ambiguity. Solid line and shaded areas: point estimate with 68 and 95 percent bands (10,000 bootstrap replications). Dashed lines: point estimates of eight-variable models that keep the U.S. real rate and add the expected SPY return before or after ambiguity.

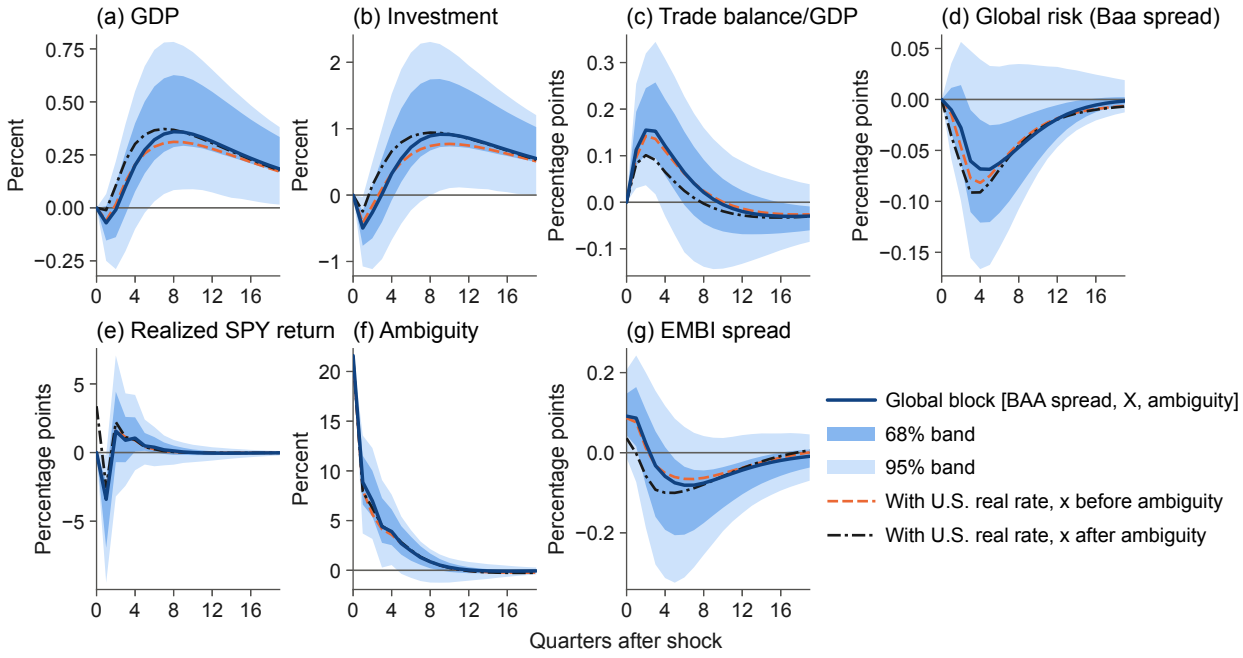


Figure C.16: IRFs to one std. dev. shock to ambiguity with U.S. interest rate replaced by realized SPY returns

Note: As Figure C.15, with the realized SPY return.

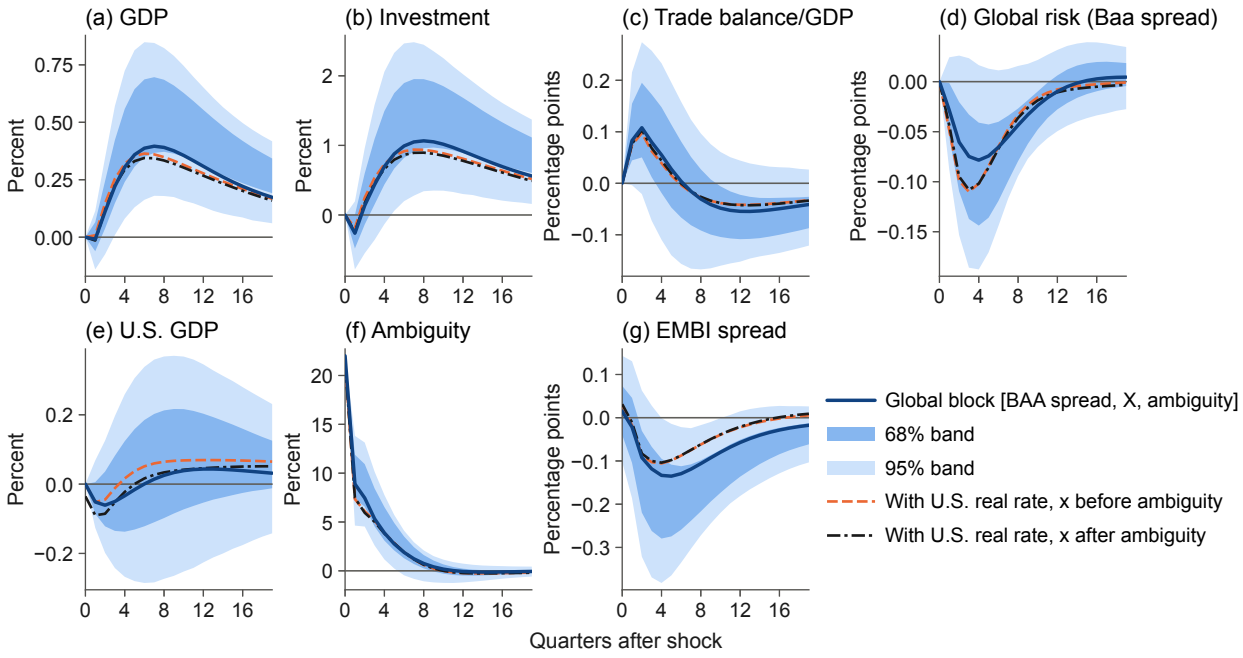


Figure C.17: IRFs to one std. dev. shock to ambiguity with U.S. interest rate replaced by U.S. GDP

Note: As Figure C.15, with U.S. GDP.

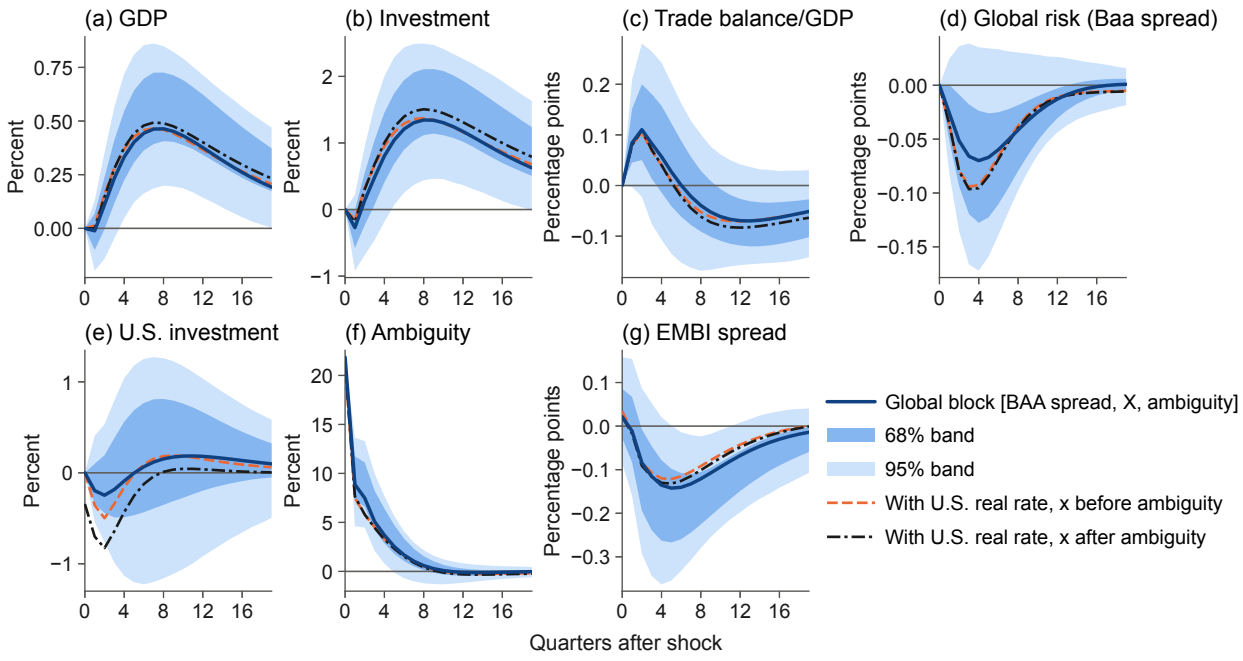


Figure C.18: IRFs to one std. dev. shock to ambiguity with U.S. interest rate replaced by U.S. investment

Note: As Figure C.15, with U.S. investment.

C.7 Ambiguity and other uncertainty measures

Figures C.20 to C.24 replace ambiguity in the benchmark model with the VIX and with the Jurado et al. (2015) financial, macro and real uncertainty indexes. Shocks to financial, macro and real uncertainty lower EME GDP by 0.56 to 0.86 percent (significant at the 95 percent level) and raise the BAA spread by 0.22 to 0.26 percentage points; a VIX shock lowers EME GDP by 0.18 percent (significant at the 68 percent level only) and raises the BAA spread by 0.09 percentage points. Every one of these shocks raises the EMBI spread on impact.

Figure C.19 extends the comparison to the other proxies for uncertainty and ambiguity in use, each added to the global block of the benchmark model before or after ambiguity, or swapped for it: the interquartile range of professional forecasters' real GDP growth and corporate profit forecasts at the one- and four-quarter horizons, the proxy for ambiguity of Ilut and Schneider (2014); the economic uncertainty and risk aversion indices of Bekaert et al. (2022); the variance risk premium (Bollerslev et al., 2009), from the Cboe data and from Hao Zhou's series; the VVIX; the Michigan dispersion of stock price expectations; and the Yale crash confidence index. None of these measures is positively correlated with the ambiguity index (quarterly detrended correlations between -0.12 and -0.47), and every one of them is positively correlated with the VIX (0.27 to 0.90). The ambiguity boom survives each of them at the 95 percent level in every ordering, with GDP peaks of 0.31 to 0.52 percent, and any proxy ordered first absorbs at most 13 percent of the variance of the ambiguity innovation (Table C.7). Shocks to the economic uncertainty index lower EME GDP by 0.15 to 0.35 percent in the first quarters after the shock (significant at the 95 percent level in every arrangement) and raise the EMBI spread by 0.3 percentage points on impact; shocks to forecast dispersion have no significant effect on EME GDP. A shock to the variance risk premium is expansionary (GDP 0.43 percent at three quarters), but it is a different object: its innovations are falls in realized variance (monthly correlation of the changes -0.81), the response is front-loaded, and the EMBI spread does not respond.

In levels, ambiguity has a correlation of -0.68 with the VIX and of -0.66 to -0.67 with

financial uncertainty at the 1-, 3- and 12-month horizons; in first differences the correlations are -0.28 and -0.19. Its correlations with macro and real uncertainty are -0.26 and -0.16 in levels. Figure C.25 shows the financial uncertainty case in detail. The horizon and vintage of the index matter little. Generalized impulse responses (Pesaran and Shin, 1998) coincide with the responses obtained when ambiguity is ordered first in the global block. Under this specification, the GDP response peaks at 0.32 percent with the 12-month financial uncertainty index and is significant at the 95 percent level from the ninth quarter on; with the one-month index, the response is significant at the 95 percent level from the seventh quarter on. Replacing ambiguity with its residual from a regression on current and lagged financial uncertainty reproduces the result with financial uncertainty ordered first (GDP peaks at 0.10 to 0.12 percent), while the residual from a regression on the VIX keeps a GDP response of 0.32 percent that is significant at the 95 percent level. Figure C.26 shows that the early declines in the BAA and EMBI spreads after an ambiguity shock disappear when the VIX is ordered before ambiguity, or when the VIX is not allowed to respond to ambiguity; a later decline in the EMBI spread, from the seventh quarter on, remains significant at the 68 percent level, and the GDP response remains significant at the 95 percent level.

Table C.7: Ambiguity versus other proxies for uncertainty and ambiguity

	GDP		Investment		Baa	EMBI	FEVD		x shock: GDP
Control x, order	Peak (h)	68% / 95%	Peak (h)	68% / 95%	trough (68%)	trough (68%)	amb.	x	peak (h), 95%
SPF GDP 4q, before	0.35 (6)	2–19 / 4–18	0.84 (7)	3–19 / none	−0.09 (1–11)	−0.10 (3–15)	4.8	1.0	0.17 (5), none
SPF GDP 4q, after	0.37 (6)	2–19 / 4–19	0.92 (7)	3–19 / none	−0.09 (1–11)	−0.11 (2–16)	5.3	0.5	0.13 (4), none
SPF profits 1q, before	0.40 (7)	2–19 / 4–19	1.09 (8)	3–19 / 5–19	−0.09 (1–10)	−0.11 (3–16)	6.6	0.2	−0.10 (1), none
SPF profits 1q, after	0.40 (7)	3–19 / 4–19	1.11 (8)	3–19 / 5–19	−0.09 (1–10)	−0.11 (3–17)	6.6	0.1	−0.10 (1), none
SPF profits 4q, before	0.36 (7)	2–19 / 4–19	0.94 (8)	3–19 / 6–19	−0.09 (1–11)	−0.11 (2–15)	5.3	1.3	0.17 (8), none
SPF profits 4q, after	0.38 (7)	2–19 / 3–19	0.99 (8)	3–19 / 5–19	−0.09 (1–11)	−0.11 (2–15)	5.8	0.8	0.13 (8), none
BEX RA, before	0.52 (6)	2–19 / 3–19	1.46 (7)	2–19 / 3–19	−0.12 (1–14)	−0.12 (3–11)	10.3	0.1	−0.13 (1), none
BEX RA, after	0.51 (6)	2–19 / 3–19	1.37 (7)	2–19 / 3–19	−0.12 (1–14)	−0.14 (2–12)	9.9	0.5	0.13 (6), none
VRP (Cboe), before	0.42 (4)	1–19 / 2–19	1.03 (7)	2–19 / 2–18	−0.10 (1–8)	−0.08 (4–16)	7.8	3.2	0.44 (2), 1–4
VRP (Cboe), after	0.31 (7)	2–19 / 3–16	0.85 (9)	3–19 / none	−0.08 (1–8)	−0.09 (3–17)	5.0	6.1	0.53 (3), 1–10
VRP (Zhou), before	0.41 (5)	2–19 / 3–19	1.01 (6)	2–19 / 3–16	−0.10 (2–9)	−0.07 (5–13)	6.9	0.6	0.20 (2), none
VRP (Zhou), after	0.36 (6)	2–19 / 3–19	0.90 (8)	3–19 / 8–19	−0.10 (1–10)	−0.11 (2–16)	5.8	1.7	0.29 (3), none

Note: Columns as in Table 3.3. SPF: interquartile range of Survey of Professional Forecasters forecasts of real GDP growth or corporate profits at the one- and four-quarter horizons; BEX: Bekaert et al. (2022) risk aversion index; VRP: variance risk premium from Cboe data (squared VIX minus realized variance) and from Hao Zhou's series. Proxies linearly detrended, 1994:Q2-2019:Q4. Bands: 10,000 replications of the moving-block residual bootstrap, Hall intervals.

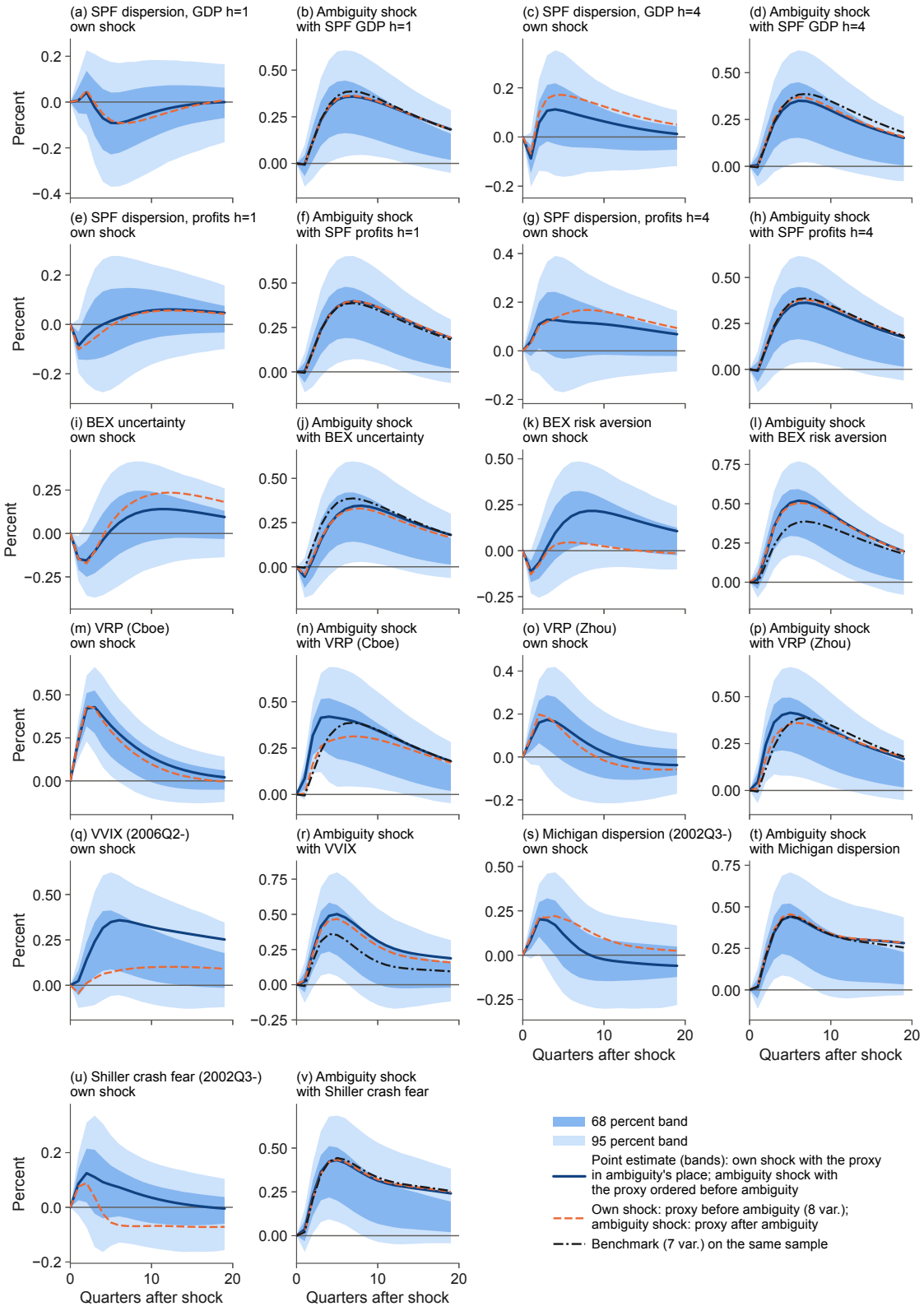


Figure C.19: Ambiguity and other proxies for uncertainty and ambiguity

Note: EME GDP responses (percent). Odd columns: response to a one standard deviation shock to the proxy, with the proxy in ambiguity's place in the benchmark model (solid line, 68 and 95 percent bands) or ordered before ambiguity in an eight-variable model (dashed line). Even columns: response to a one standard deviation ambiguity shock with the proxy ordered before (solid, with bands) or after (dashed) ambiguity, and the benchmark model on the same sample (dash-dot). SPF: interquartile range of Survey of Professional Forecasters real GDP growth and profit forecasts at horizons h; BEX: Bekaert et al. (2022) indices; VRP: variance risk premium from Cboe data and from Hao Zhou's series; Michigan: cross-household standard deviation of the probability of a stock price increase; Shiller: Yale individual crash confidence index. 10,000 bootstrap replications, moving-block Hall bands.

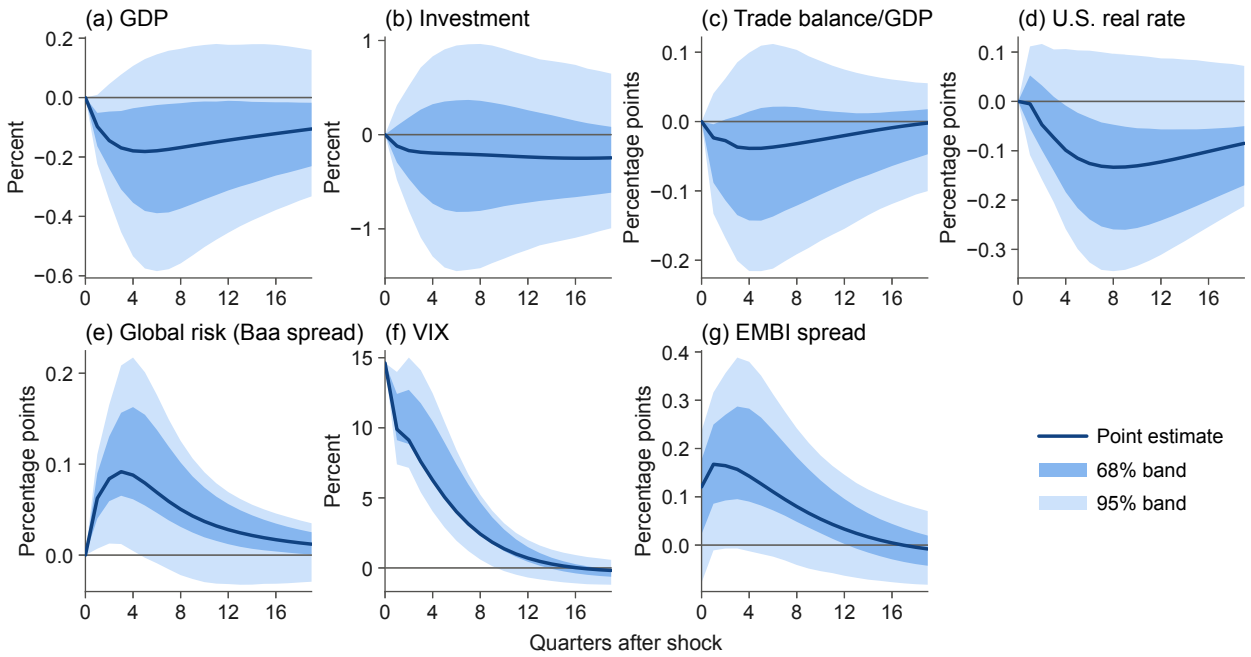


Figure C.20: IRFs to one std. dev. shock to the VIX, with the VIX replacing ambiguity

Note: Benchmark model with ambiguity replaced by the log VIX (linearly detrended, same position in the global block); shock of 14.6 percent on impact. Shaded areas: 68 and 95 percent bands, 10,000 bootstrap replications.

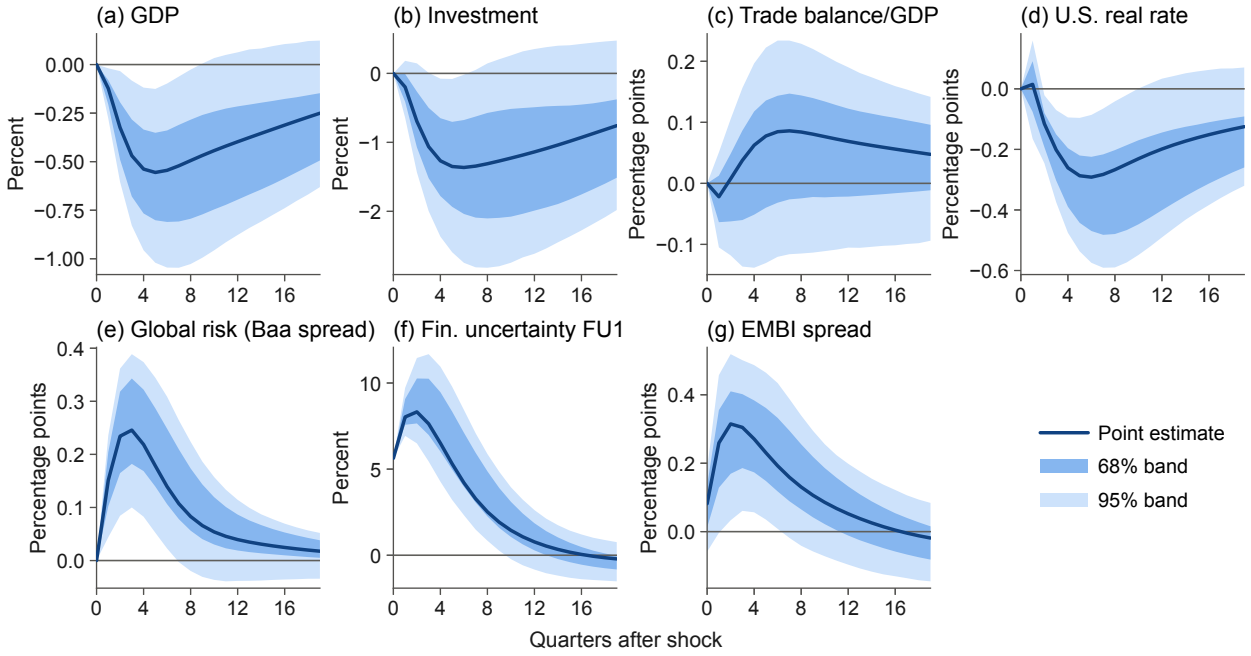


Figure C.21: IRFs to one std. dev. shock to financial uncertainty (1-month), replacing ambiguity

Note: Benchmark model with ambiguity replaced by the log of the Jurado et al. (2015) 1-month financial uncertainty index (linearly detrended); shock of 5.6 percent on impact. Shaded areas: 68 and 95 percent bands.

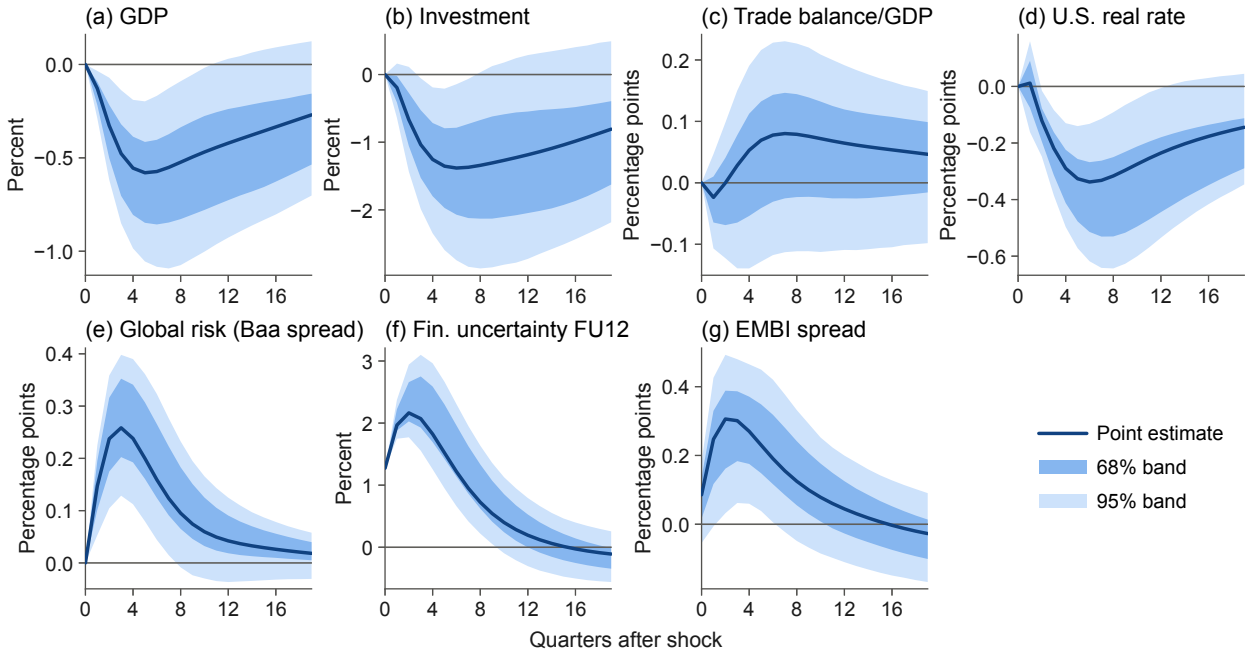


Figure C.22: IRFs to one std. dev. shock to financial uncertainty (12-month), replacing ambiguity

Note: Benchmark model with ambiguity replaced by the log of the Jurado et al. (2015) 12-month financial uncertainty index (linearly detrended); shock of 1.3 percent on impact. Shaded areas: 68 and 95 percent bands.

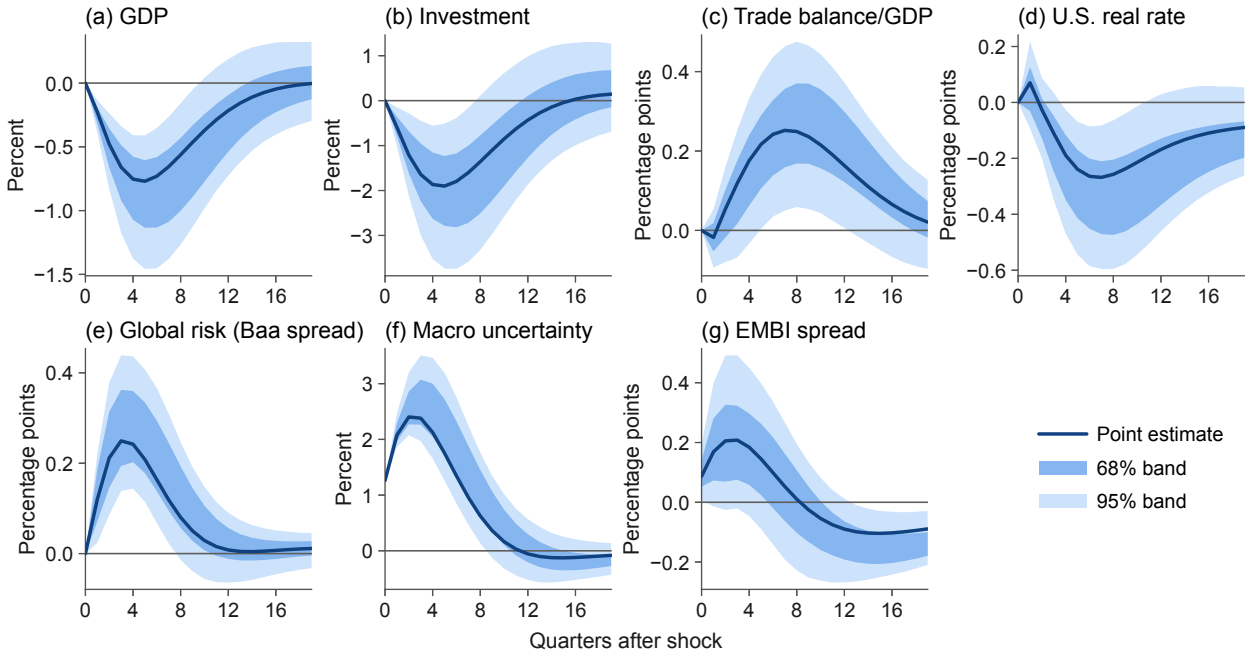


Figure C.23: IRFs to one std. dev. shock to macro uncertainty, replacing ambiguity

Note: Benchmark model with ambiguity replaced by the log of the Jurado et al. (2015) 12-month macro uncertainty index (linearly detrended); shock of 1.3 percent on impact. Shaded areas: 68 and 95 percent bands.

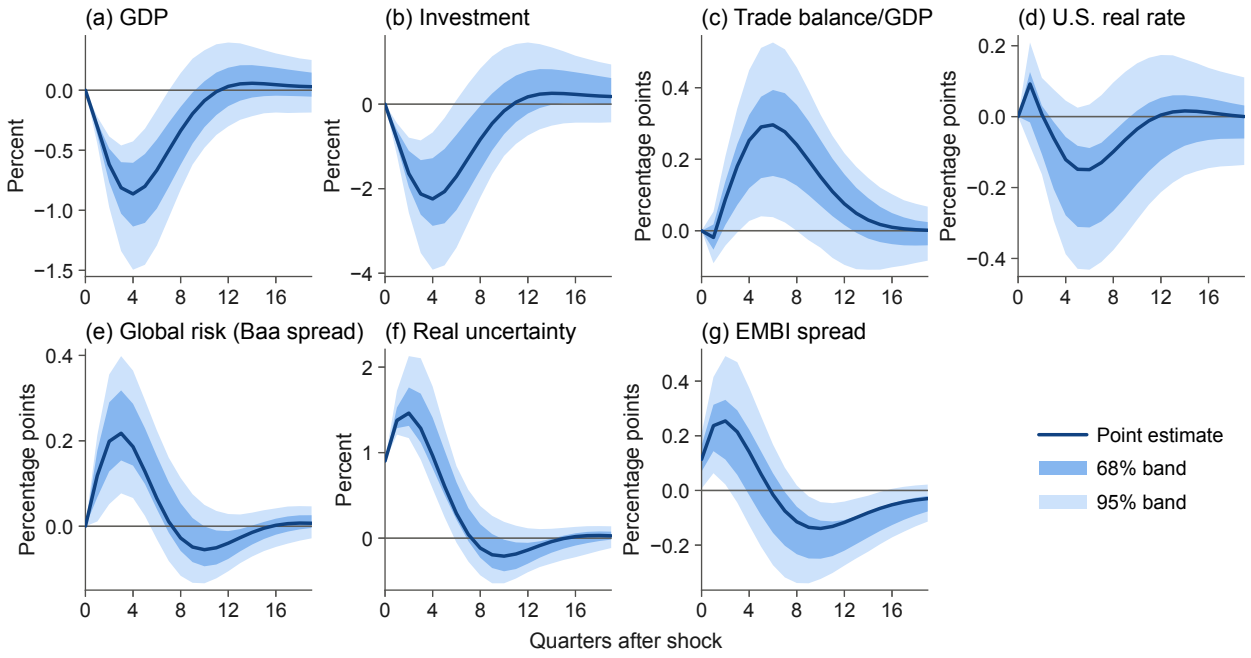


Figure C.24: IRFs to one std. dev. shock to real uncertainty, replacing ambiguity

Note: Benchmark model with ambiguity replaced by the log of the Jurado et al. (2015) 12-month real uncertainty index (linearly detrended); shock of 0.9 percent on impact. Shaded areas: 68 and 95 percent bands.

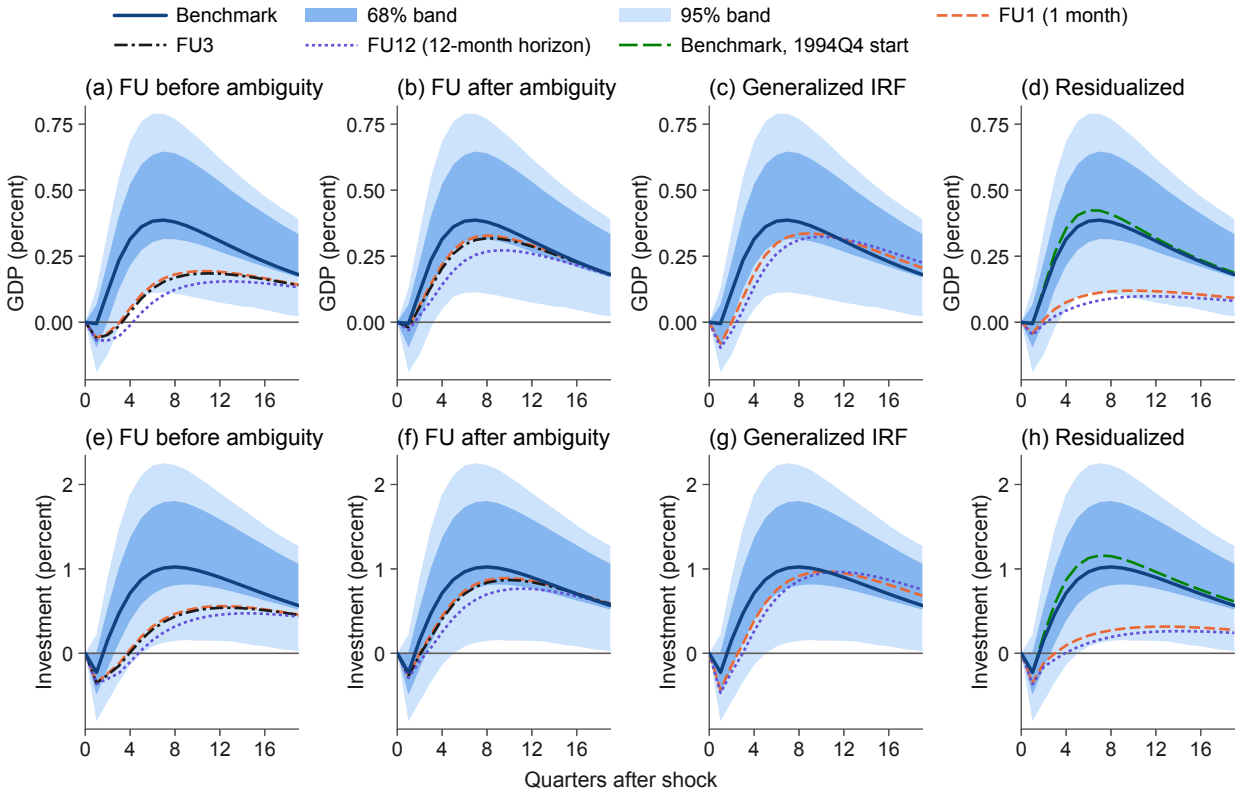


Figure C.25: Ambiguity and financial uncertainty

Note: Responses of EME GDP (top row) and investment (bottom row) to a one standard deviation ambiguity shock. Shaded areas and solid line: benchmark with 68 and 95 percent bands. Other lines: point estimates with Jurado et al. (2015) financial uncertainty at the 1-, 3- and 12-month horizons (FU1, FU3, FU12) ordered before ambiguity (first column) or after it (second column); generalized impulse responses to the ambiguity innovation in the eight-variable systems, which equal the responses with ambiguity ordered first (third column); and the benchmark model with ambiguity replaced by its residual from a regression on financial uncertainty (current and two lags) and two own lags, estimated from 1994:Q4 (fourth column, with the benchmark on the same sample).

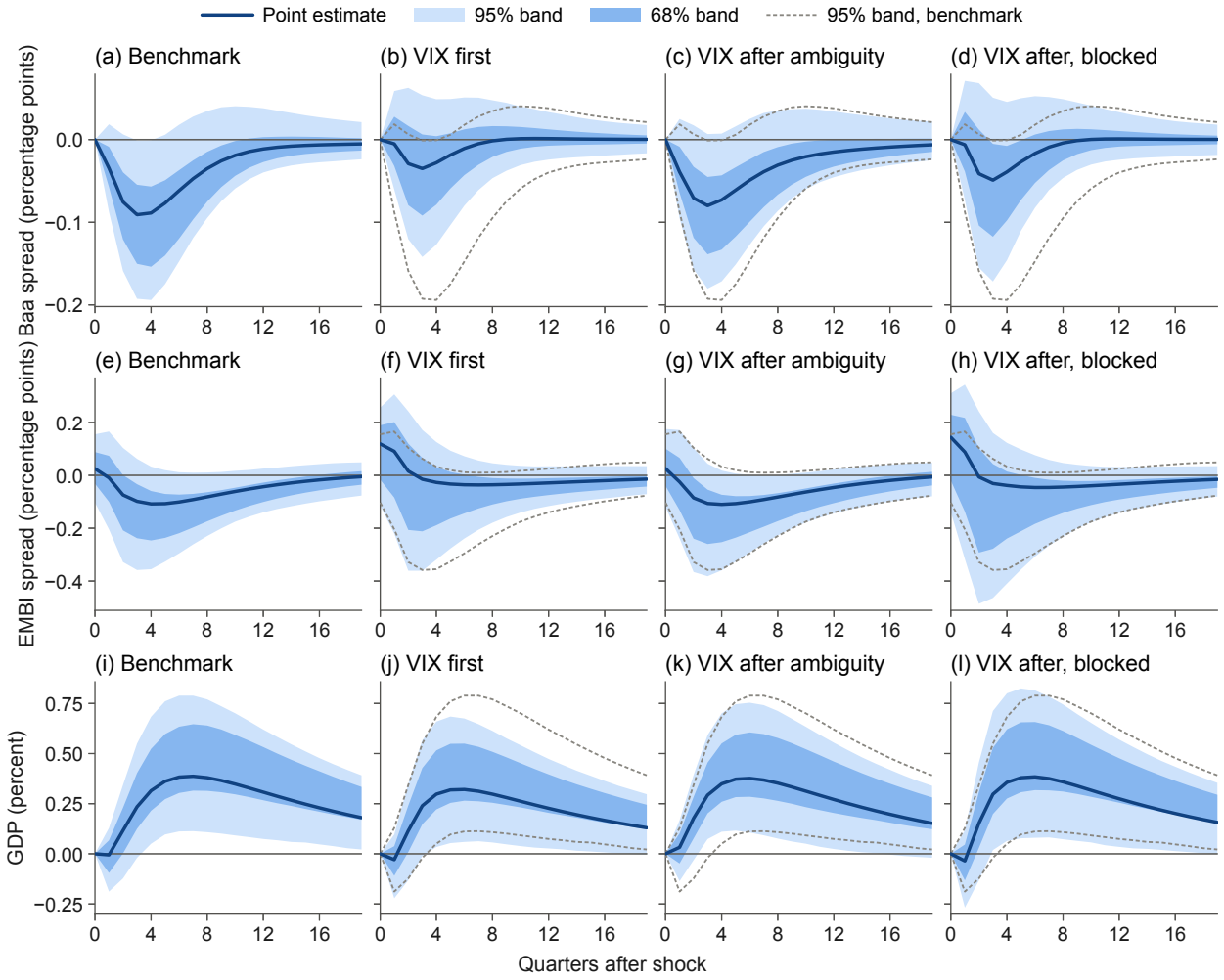


Figure C.26: Spread responses with and without the VIX

Note: Responses of the BAA spread, the EMBI spread (percentage points) and EME GDP (percent) to a one standard deviation ambiguity shock, each column with its own point estimate and 68 and 95 percent bands: the benchmark; the eight-variable model with the log VIX ordered before ambiguity; with the VIX after ambiguity; and the latter with the VIX's response to ambiguity set to zero (coefficients not re-estimated). Dashed gray lines repeat the benchmark 95 percent band.

C.8 Mechanism: additional results

Figure C.33 shows the responses of U.S. asset prices to an ambiguity shock in eight-variable models that add each price to the benchmark global block, ordered after or before ambiguity, and Figure C.34 shows the same prices after a global financial risk shock. The global financial risk shock produces a textbook flight to safety: on impact, the Gilchrist-Zakrajšek credit spread rises by 0.35 percentage points and the excess bond premium by 0.23, the 10-year and 2-year Treasury yields fall by 0.15 and 0.24 percentage points, and U.S. equity returns fall by 5.3 percent, all significant at the 95 percent level (the 10-year yield in the first two quarters after impact). Figure C.35 shows the EME responses to an ambiguity shock when the excess bond premium or the credit spread of Gilchrist and Zakrajšek (2012) enters the model. With the excess bond premium in the model, excluding it from the country equations restores the GDP response (0.38 percent), which shows that the loss comes from the country equations rather than from a change in the identified shock.

Figures C.36 to C.38 provide additional evidence on financial-market and cross-border responses. U.S. bond mutual funds do not experience a sustained inflow after an ambiguity shock, while their portfolios shift temporarily toward corporate and foreign bonds. Foreign purchases of U.S. Treasuries do not rise in the first year, whereas foreign purchases of U.S. equities increase on impact. In the six benchmark EMEs, international debt issuance increases after two years, primarily among governments and financial firms, although this debt-issuance response does not extend to the twelve-country sample. The estimated GDP response is sensitive to controlling for lagged excess bond premia, while the responses of sovereign spreads, international debt issuance, EME equity inflows, and equity prices remain. We find no statistically significant difference in the GDP responses between countries with high and low initial portfolio debt liabilities.

Which foreign assets, and who lends to EMEs. The issuer-level estimates of U.S. cross-border positions of Bertaut and Judson (2014), which combine the Treasury's annual surveys with the monthly TIC flows and are available by issuing country from 1994 at monthly frequency, show where the foreign bonds and stocks that U.S. residents buy after

an ambiguity shock are issued. Figure C.27 and Table C.8 report cumulative net purchases by issuer from local projections on the monthly identified shock of Section 3.3.5. In the twelve months after the shock U.S. residents buy about 12 billion dollars of foreign bonds (close to the 15 billion implied by the aggregate TIC response), of which European issuers account for about half, issuers in the twelve small open advanced economies for a third, Caribbean financial centres for a quarter and the eighteen EMEs for which issuer data exist for 7 percent, an amount that is not significantly different from zero; the largest single issuers are the Cayman Islands, the United Kingdom and Australia. Relative to U.S. holdings, the purchases are largest in Ireland, Singapore, Poland, Australia and Luxembourg. U.S. purchases of EME stocks do not rise and turn negative after eighteen months, whereas purchases of stocks issued in Caribbean centres and Germany rise. Part of the financial-centre purchases may finance EME borrowers through offshore subsidiaries and investment funds (Coppola et al., 2021); even if every Cayman bond purchase were EME-related, U.S. purchases of EME-related bonds would be about the advanced-economy figure. The EME asset-price response is a repricing rather than the result of U.S. portfolio quantities: EME equity prices in dollars are flat for six months and rise by 3 percent after a year and 5 to 6 percent after two, and the valuation gain on existing U.S. holdings of EME equities within a year, 9 to 13 billion dollars, is an order of magnitude larger than U.S. net purchases of EME securities.

The financing that reaches EMEs is bank credit and reserve accumulation. Figure C.28 shows that BIS cross-border bank claims on eighteen EMEs rise by 1.8 percent of their quarterly GDP after two years and 2.7 percent after three (about 40 billion dollars at two years, significant at the 90 percent level from the seventh quarter), in dollars and in other currencies roughly in proportion to their shares. By lender nationality, euro-area banks account for about 85 percent of the response and U.K. banks for most of the remainder; U.S. banks contribute little and late and Japanese banks nothing. EMEs with an above-median dollar share of their cross-border bank claims receive less credit after the shock, not more, so the expansion of credit does not run through a dollar-funding channel of the type studied by Bruno and Shin (2015). In the balance of payments, the EMEs' portfolio debt liabilities

and bank inflows rise with a lag and their reserve assets rise by 2.6 percent of quarterly GDP after three years (eleven EMEs), while the small open advanced economies' portfolio debt liabilities rise by 1.3 percent of GDP within a year and 3.4 percent after three (Figures C.29 and C.30). A global financial risk shock produces a sudden stop in every category: portfolio equity, portfolio debt and bank inflows into eleven EMEs fall by 0.4, 1.1 and 1.4 percent of quarterly GDP within a year.

The portfolio equity inflow into the six benchmark EMEs reported in Section 3.3.3 (0.56 percent of quarterly GDP within a year) is estimated over the estimation windows of the PSVAR. Table C.9 traces it across alternative samples: data vintage, scaling, the shock series, controls, quarter dummies and the treatment of Argentina move it by at most 0.03; using each country's full data span lowers it to 0.31 percent, almost entirely because South Africa's 1994-2002 quarters, which include the 1999-2001 migration of several primary listings to London, enter the sample (0.47 percent with South Africa from 2002:Q2); in the eleven-country panel it is 0.23 to 0.33 percent. All variants are positive, and all but the smallest significant at the 90 percent level.

EME financial variables and the dollar. Figure C.31 adds EME financial variables one at a time to the country block of the benchmark model, ordered after the EMBI spread, and estimates panel local projections of the same variables on the benchmark and the monthly ambiguity shocks: the bilateral exchange rate against the dollar, the real effective exchange rate, the policy rate, real foreign exchange reserves, bank credit to the private sector relative to GDP, and dollar equity prices. After an ambiguity shock, equity prices rise by 4.4 percent after four quarters (95 percent) and reserves by 1.8 to 1.9 percent (68 percent throughout, 95 percent in the first quarter and from the tenth quarter on), the policy rate falls by 0.4 percentage points in the second quarter (95 percent in quarters 1 to 3), the currencies do not move for about three years and appreciate afterwards in the local projections, and bank credit relative to GDP does not respond. A global financial risk shock depreciates the currencies by 2.4 percent and the real effective exchange rate by 1.5 percent within a quarter (95 percent), lowers equity prices by 5 percent on impact and leaves reserves unchanged.

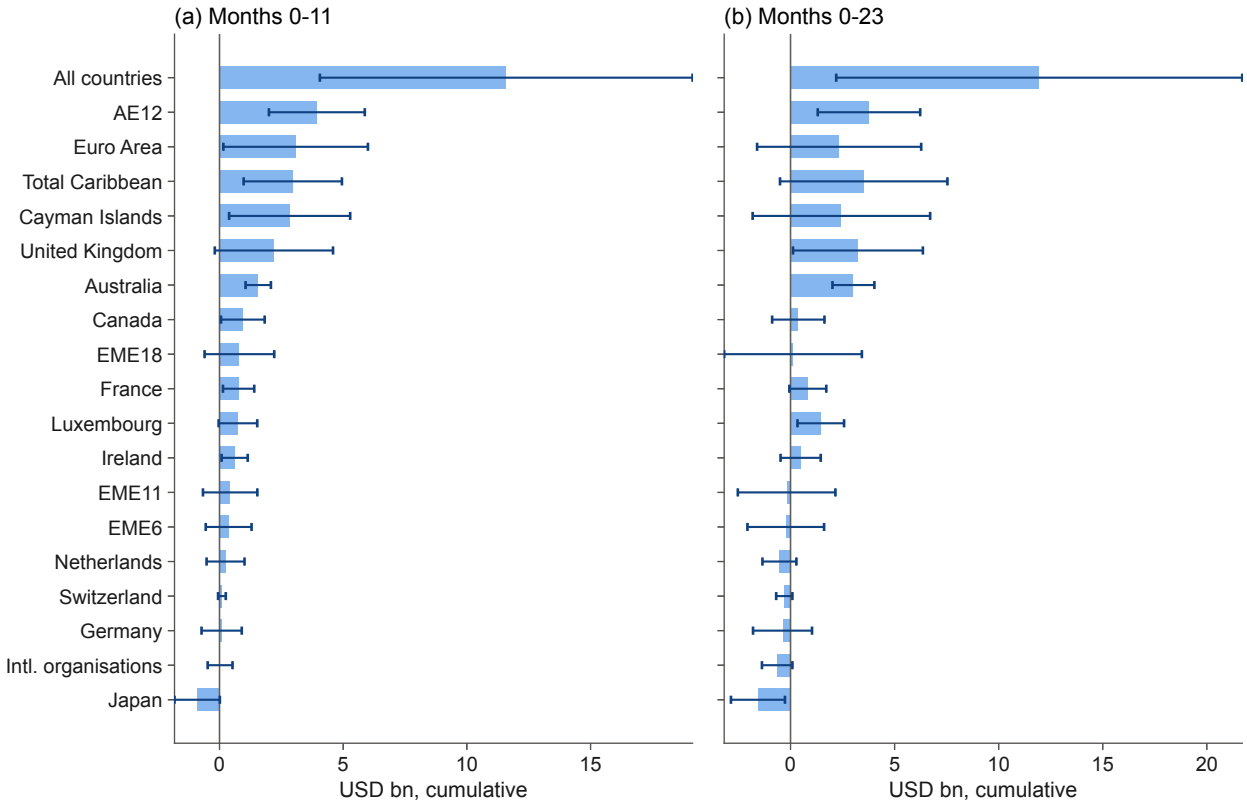


Figure C.27: U.S. net purchases of foreign bonds after an ambiguity shock, by issuer

Note: Cumulative U.S. net purchases of bonds issued in each country or group, in billions of dollars, over months 0-11 (panel a) and 0-23 (panel b) after a one standard deviation monthly ambiguity shock (M1), from local projections of the monthly issuer-based flows of Bertaut and Judson (2014) in percent of U.S. quarterly GDP, converted at the 1994-2019 average; three lags of the flow and of the shock, month dummies, Newey and West (1987) 90 percent bands. AE12: the twelve small open advanced economies; EME6, EME11, EME18: the benchmark, larger-sample and all available EMEs. Groups overlap.

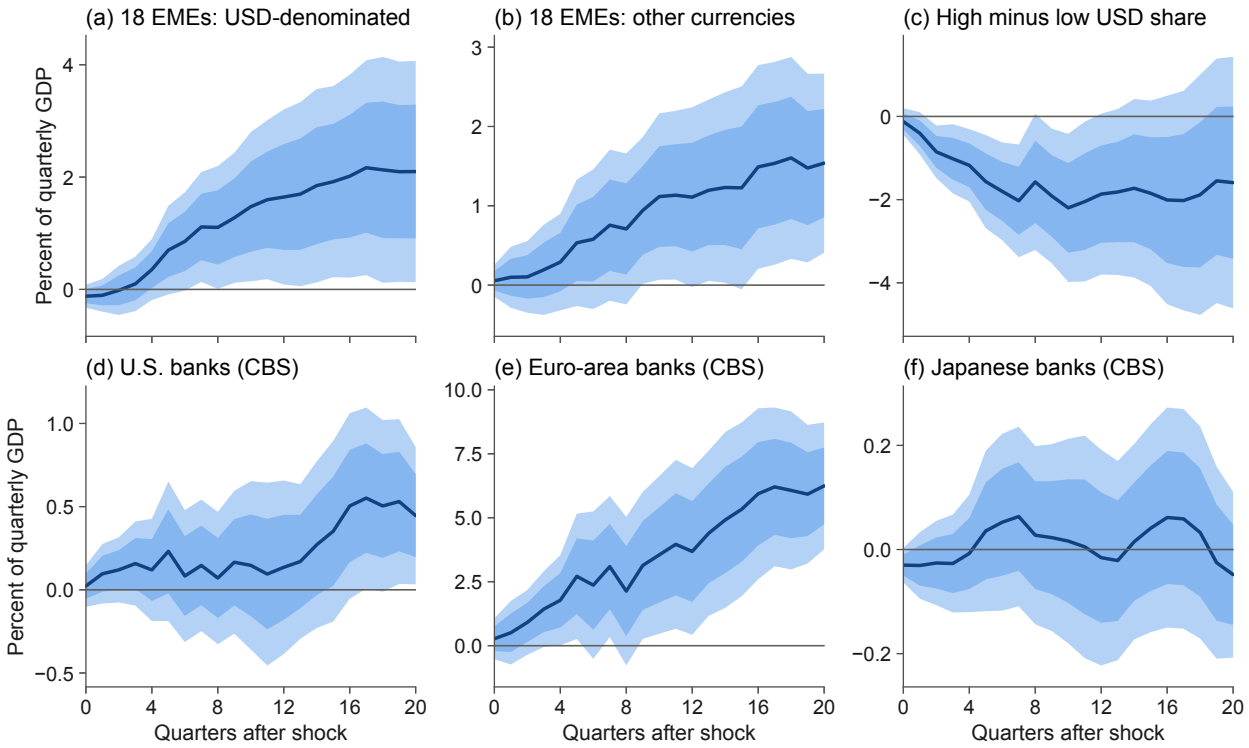


Figure C.28: Cross-border bank claims on EMEs after an ambiguity shock

Note: Panel local projections of BIS cross-border bank claims on eighteen EMEs (cumulative, percent of quarterly GDP) on the benchmark ambiguity shock. Panels (a) and (b): locational statistics, dollar-denominated and other-currency claims. Panel (c): difference between EMEs with above- and below-median dollar shares of their claims. Panels (d)-(f): consolidated statistics by lender nationality. Unit fixed effects, two lags of the outcome, of the country variables and of the global variables; Driscoll and Kraay (1998) 68 and 90 percent bands.

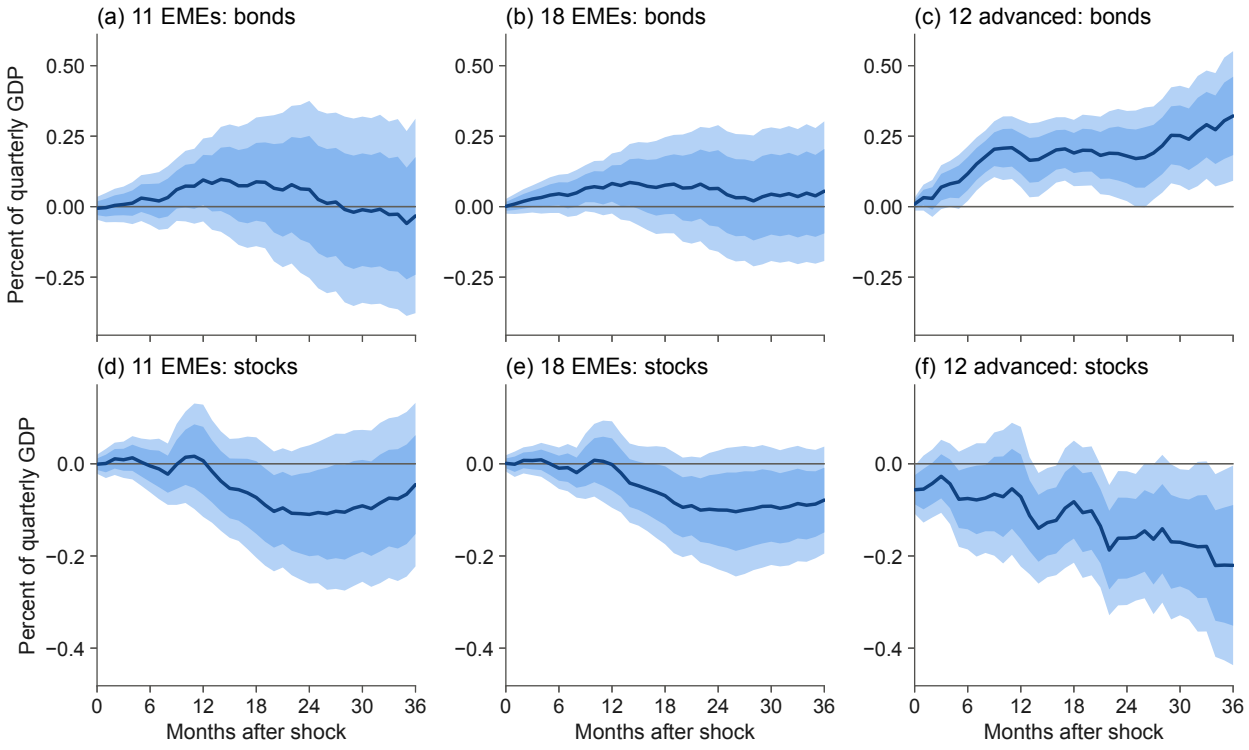


Figure C.29: U.S. purchases of EME and advanced-economy securities after an ambiguity shock

Note: Local projections of U.S. net purchases of bonds and stocks issued in the benchmark EMEs and in the twelve small open advanced economies (issuer-based monthly estimates of Bertaut and Judson (2014), cumulative, percent of U.S. quarterly GDP) on the monthly ambiguity shock (M1); 68 and 90 percent Newey and West (1987) bands.

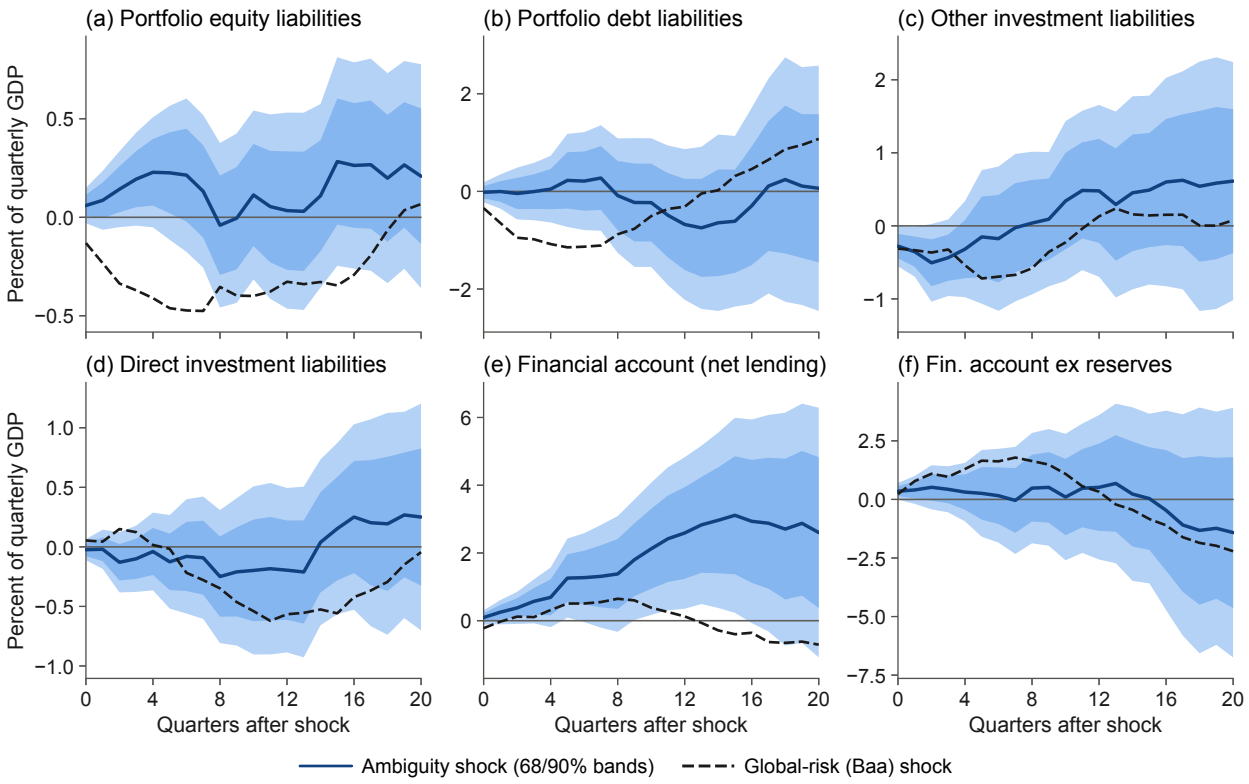


Figure C.30: Balance-of-payments inflows after ambiguity and global financial risk shocks

Note: Panel local projections of IMF balance-of-payments liabilities and reserve flows (cumulative, percent of quarterly GDP) on the benchmark ambiguity shock and on the global financial risk shock, for the EME and advanced-economy panels; Driscoll and Kraay (1998) 68 and 90 percent bands.

Table C.8: U.S. purchases of foreign bonds by issuer and bank claims on EMEs by lender

A. U.S. net purchases of foreign bonds by issuer, cumulative after the monthly (M1) shock

	Percent of U.S. quarterly GDP		USD bn			
Issuer	Months 0–11	90% band	Months 0–11	90% band	90% months	0–23, USD bn
All countries	0.337	[0.118, 0.556]	11.6	[4.1, 19.1]	+3–24	11.9
Total Europe	0.166	[0.024, 0.308]	5.7	[0.8, 10.6]	+3–12, 19, 21–23	5.6
Euro area	0.089	[0.005, 0.174]	3.1	[0.2, 6.0]	+0–1, 3–11	2.3
12 small advanced economies (sum)	0.114	[0.058, 0.171]	3.9	[2.0, 5.9]	+4–24	3.8
Total Caribbean	0.086	[0.028, 0.144]	3.0	[1.0, 4.9]	+0–14, 19–20	3.5
18 EMEs (sum)	0.023	[−0.018, 0.064]	0.8	[−0.6, 2.2]	none	0.1
11 EMEs (sum)	0.013	[−0.019, 0.044]	0.4	[−0.7, 1.5]	none	−0.2
6 benchmark EMEs (sum)	0.011	[−0.016, 0.038]	0.4	[−0.6, 1.3]	none	−0.2
Total Asia	−0.030	[−0.072, 0.013]	−1.0	[−2.5, 0.4]	none	−1.7
Japan	−0.026	[−0.053, 0.000]	−0.9	[−1.8, 0.0]	−3–4, 8, 10, 12, 18–24	−1.6

B. BIS consolidated foreign claims on the 18 EMEs by lender nationality, cumulative, percent of quarterly GDP

Lender	$h = 8$	90% band	$h = 12$	90% band	90% horizons (0–20)
Benchmark quarterly ambiguity shock					
All lender nationalities	2.61	[−1.81, 7.02]	4.27	[−0.81, 9.34]	+13–20
Euro-area banks	2.13	[−0.77, 5.04]	3.68	[0.43, 6.94]	+4–5, 7, 9–20
U.K. banks	0.49	[−0.35, 1.32]	0.57	[−0.40, 1.54]	+14–20
U.S. banks	0.07	[−0.33, 0.47]	0.14	[−0.39, 0.66]	+17, 19–20
Japanese banks	0.03	[−0.14, 0.20]	−0.02	[−0.22, 0.19]	none
Monthly (M1) shock, summed to quarters					
All lender nationalities	3.67	[0.31, 7.04]	6.58	[1.38, 11.78]	+5, 7–20
Euro-area banks	2.03	[−0.35, 4.40]	3.58	[0.19, 6.97]	+5, 7, 9–20
U.K. banks	1.33	[0.75, 1.91]	1.98	[1.03, 2.94]	+3–20
U.S. banks	0.46	[0.11, 0.80]	0.85	[0.51, 1.19]	+7–20
Japanese banks	0.00	[−0.13, 0.12]	0.04	[−0.09, 0.17]	+16–20

Note: Panel A: cumulative U.S. net purchases of bonds by issuer group twelve months after a one standard deviation monthly ambiguity shock, in billions of dollars and in percent of the total, with 90 percent bands, from the local projections of Figure C.27. Panel B: cumulative change in consolidated cross-border claims on eighteen EMEs by lender nationality, percent of quarterly GDP, at the horizons shown, from the local projections of Figure C.28.

Table C.9: Portfolio equity inflows into EMEs: alternative samples

Variant	$h = 4$		$h = 8$		Obs. (units), $h = 4$
A. One change at a time from the paper's specification (T3.2)					
Paper (T3.2): VAR windows, seven units	0.56**	(0.13)	0.48**	(0.22)	444 (7)
Drop country-block controls	0.54**	(0.15)	0.43**	(0.25)	444 (7)
Drop global controls	0.57**	(0.19)	0.47**	(0.26)	444 (7)
Own lags only	0.55**	(0.19)	0.48**	(0.28)	444 (7)
Add two shock lags	0.53**	(0.15)	0.46**	(0.24)	440 (7)
J9 controls (own lags and two shock lags)	0.55**	(0.17)	0.43**	(0.25)	440 (7)
No quarter dummies	0.56**	(0.13)	0.44**	(0.22)	444 (7)
Argentina I and II as one unit	0.57**	(0.13)	0.48**	(0.22)	444 (6)
Full country windows, 1993–2019	0.31**	(0.17)	0.16	(0.24)	582 (6)
B. Full country windows: one window change at a time					
Only South Africa cut to its VAR start (2002Q2)	0.47**	(0.12)	0.47**	(0.20)	550 (6)
Only Brazil cut to its VAR start (1995Q1)	0.31**	(0.17)	0.16	(0.24)	579 (6)
Only Peru cut to its VAR start (1997Q2)	0.31**	(0.17)	0.13	(0.26)	570 (6)
Only Turkey cut to its VAR start (1999Q3)	0.32**	(0.17)	0.15	(0.25)	561 (6)
Argentina default gap removed	0.30*	(0.19)	0.14	(0.26)	518 (6)
VAR start dates	0.50**	(0.12)	0.47**	(0.21)	510 (6)
VAR start dates, Argentina gap removed	0.51**	(0.13)	0.46**	(0.22)	446 (6)
VAR start dates, J9 controls	0.51**	(0.15)	0.48**	(0.23)	510 (6)
C. Country sets					
J9 design, six EMEs (full windows, J9 controls)	0.33**	(0.19)	0.22	(0.26)	570 (6)
J9 design, 11 EMEs	0.23*	(0.17)	−0.04	(0.25)	1006 (11)
J9 design, 18 EMEs	0.15*	(0.14)	−0.18*	(0.18)	1650 (18)
J9 design, 11 EMEs, South Africa from 2002Q2	0.33**	(0.16)	0.12	(0.26)	976 (11)
J9 design, 18 EMEs, South Africa from 2002Q2	0.21*	(0.13)	−0.10	(0.18)	1620 (18)
Full windows, globals and own lags, 11 EMEs	0.22*	(0.15)	−0.07	(0.22)	1022 (11)
Full windows, globals and own lags, 18 EMEs	0.14*	(0.12)	−0.15	(0.17)	1678 (18)
VAR windows for the six, full windows for five more	0.33**	(0.15)	0.04	(0.23)	884 (12)

Note: Cumulative non-resident portfolio equity inflows (IMF balance of payments, percent of quarterly GDP) four and eight quarters after a one standard deviation benchmark ambiguity shock, from panel local projections with unit fixed effects and Driscoll and Kraay (1998) standard errors (in parentheses), across samples and designs; the first row is the estimate of Table 3.2. **, *: 90 and 68 percent intervals exclude zero.

The GDP response to the ambiguity shock in each of the 22 augmented systems (six and eleven EMEs) is reported in Table C.10. Figure C.32 shows the broad dollar index: it falls by 0.8 percent after a year in the monthly VAR and local projections and by 1.2 percent in quarterly local projections on the benchmark shock (90 percent at quarters 3 to 5), and it strengthens on impact after a global financial risk shock, by 0.7 percent against the broad index and 1.1 percent against EME currencies.

Table C.10: The EME expansion with EME financial variables in the model

		EME GDP		Added variable		
Added variable	Units	Peak (h)	95% horizons	Peak (h)	95% horizons	Unit
Benchmark panel (six countries)						
Exchange rate vs USD	6	0.38 (6)	3–19	0.73 (4)	none	%
NEER	7	0.35 (8)	5–19	−0.41 (7)	none	%
REER	7	0.39 (7)	4–19	−0.30 (0)	none	%
Policy rate	7	0.42 (7)	4–19	−0.40 (2)	1–3	pp
Money-market rate	6	0.40 (6)	3–14	−0.60 (2)	1–2	pp
Real reserves	7	0.43 (7)	3–19	1.91 (8)	1, 10–19	%
Reserves/GDP	7	0.37 (7)	4–19	−0.05 (4)	none	pp
Bank credit/GDP	6	0.41 (6)	4–19	0.26 (10)	none	pp
Equity (real, USD)	6	0.63 (6)	3–19	4.42 (4)	2–7	%
Equity (real, local)	6	0.58 (7)	3–19	3.19 (4)	2–6	%
Consumption	7	0.38 (7)	4–19	0.31 (9)	8–19	%
Eleven EMEs						
Exchange rate vs USD	11	0.31 (7)	4–19	0.82 (5)	15–19	%
NEER	12	0.27 (9)	6–19	0.32 (17)	none	%
REER	12	0.29 (8)	5–19	−0.20 (0)	none	%
Policy rate	11	0.33 (8)	4–19	−0.31 (2)	1–3	pp
Money-market rate	10	0.32 (7)	4–19	−0.43 (2)	2–3	pp
Real reserves	12	0.30 (8)	4–18	1.00 (14)	none	%
Reserves/GDP	12	0.29 (8)	5–19	−0.11 (5)	none	pp
Bank credit/GDP	9	0.35 (7)	4–19	0.16 (14)	none	pp
Equity (real, USD)	8	0.56 (7)	3–19	4.55 (4)	2–7	%
Equity (real, local)	8	0.52 (8)	3–19	3.25 (4)	2–7	%
Consumption	12	0.29 (8)	5–19	0.23 (10)	none	%

Note: Eight-variable PSVARs that add one EME financial variable to the country block of the benchmark model, ordered after the EMBI spread, for the six benchmark countries and the eleven EMEs. GDP: peak response to a one standard deviation ambiguity shock (percent), horizon and horizons at which the 95 percent band excludes zero. Added variable: peak response and significance ranges. Bands: 10,000 replications of the moving-block residual bootstrap, Hall intervals.

Table C.11 reports average changes after the ten largest positive and the ten largest negative ambiguity shocks. Over four quarters, the BAA spread falls by 0.71 percentage points more, the 10-year Treasury yield rises by 0.48 percentage points more, and EME GDP rises by 1.5 percent more after positive than after negative shocks. The contrast is driven

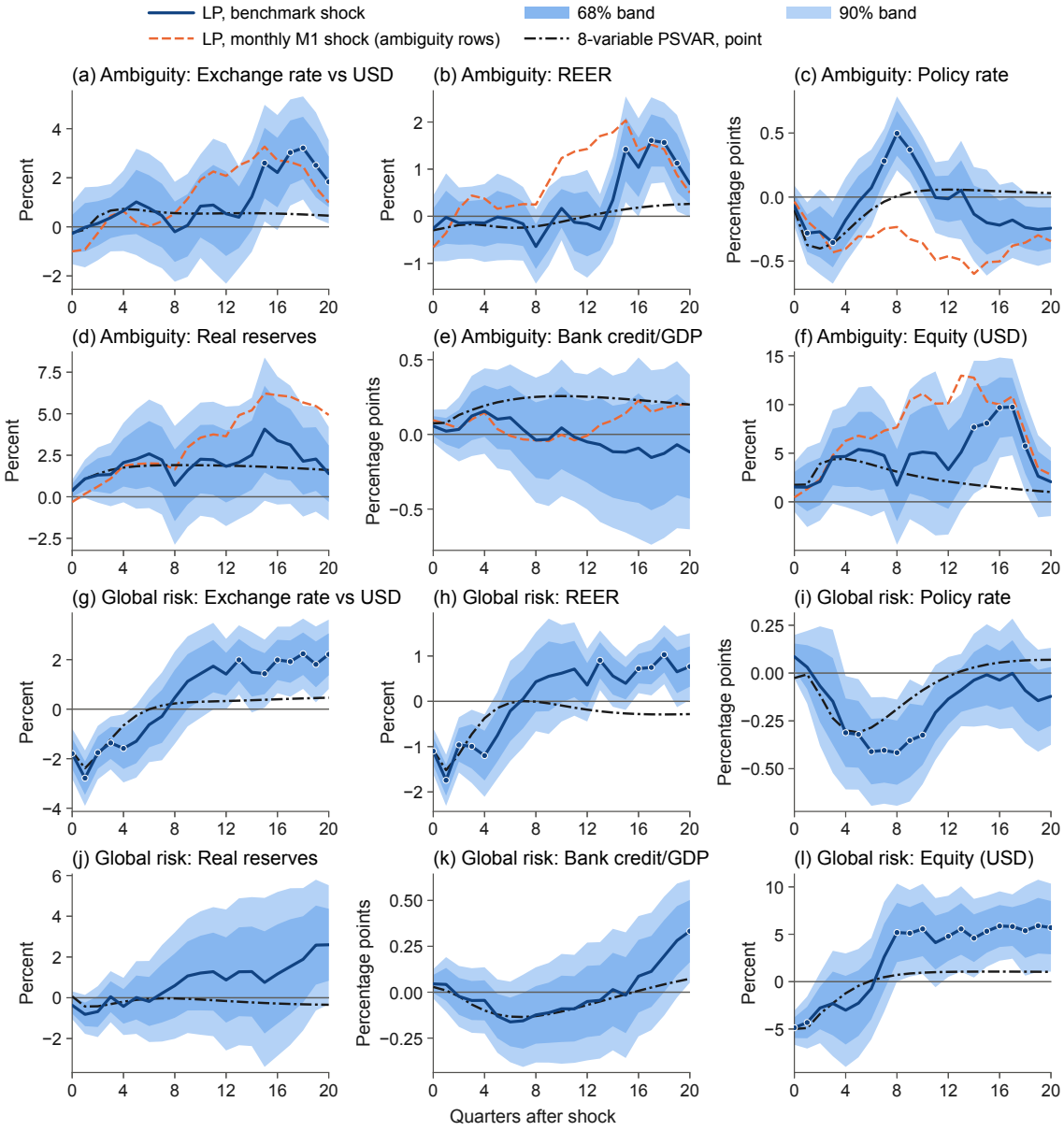


Figure C.31: EME financial variables after ambiguity and global financial risk shocks

Note: Rows 1-2: responses to a one standard deviation ambiguity shock; rows 3-4: to a global financial risk shock. Solid line and shaded areas: panel local projections on the benchmark shock with unit fixed effects, two lags of the outcome, of the country variables and of the global variables, Driscoll and Kraay (1998) 68 and 90 percent bands; dots mark horizons at which the 90 percent band excludes zero. Dashed line: local projections on the monthly ambiguity shock (M1). Dash-dot line: point estimate of the eight-variable PSVAR with the variable added to the country block. Exchange rates: 100 × log, increase = appreciation (BIS); policy rate in percentage points (BIS); real reserves: 100 × log of total reserves minus gold deflated by the U.S. CPI (IMF); bank credit to the private non-financial sector in percent of GDP (BIS); equity: 100 × log of the MSCI dollar index deflated by the U.S. CPI. Six benchmark countries, 1994:Q2-2019:Q4.

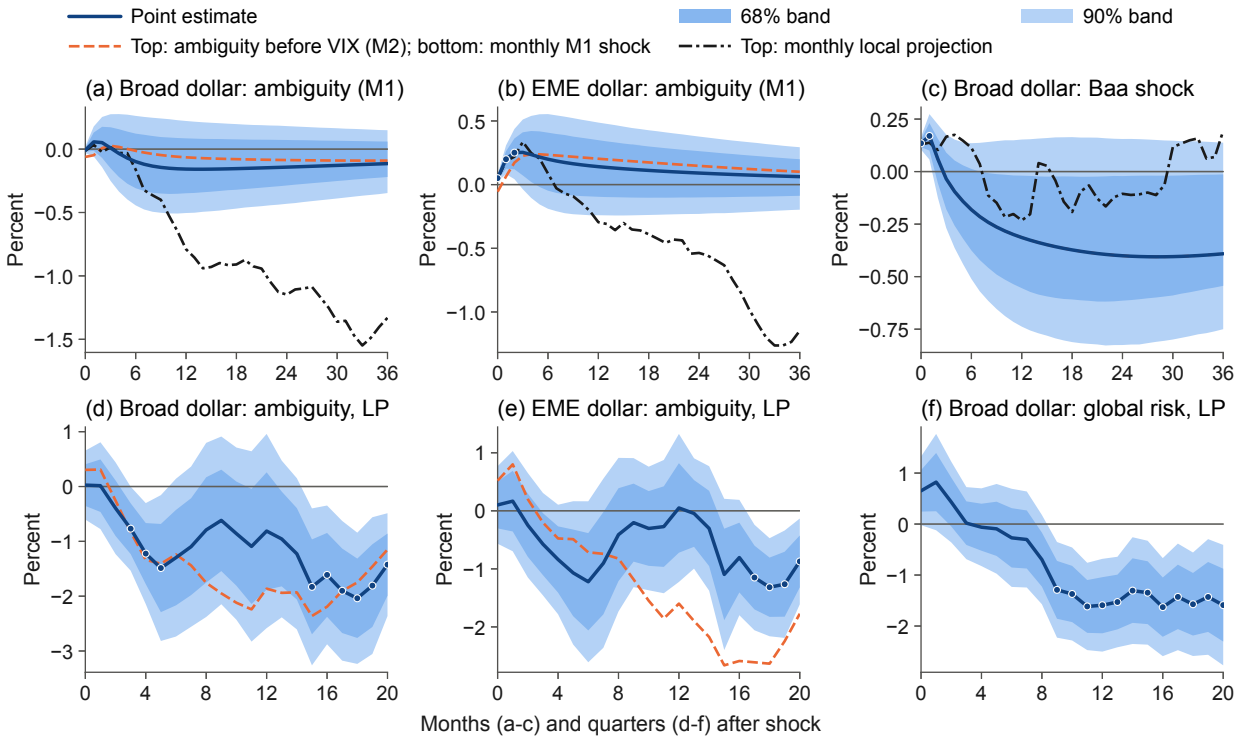


Figure C.32: The dollar after an ambiguity shock and after a global financial risk shock

Note: Increase = dollar appreciation. Panels (a)-(c): monthly VAR in the U.S. real rate, the BAA spread, the log VIX, financial uncertainty, the excess bond premium, the NASDAQ return, ambiguity and the log broad or EME dollar index, 1993:M2-2019:M12, two lags; responses (percent) to a one standard deviation ambiguity or BAA spread shock, 68 and 90 percent wild-bootstrap bands; dash-dot: monthly local projection on the same shock. Panels (d)-(f): local projections of the quarterly log dollar index on the benchmark ambiguity and global financial risk shocks, two lags of the index and of the shock, Newey and West (1987) 68 and 90 percent bands.

by the negative shocks, which coincide with stress episodes (2008:Q1, 2011:Q3, 2018:Q4), and the largest positive shock occurs in 2008:Q4; we therefore read the event averages as descriptive.

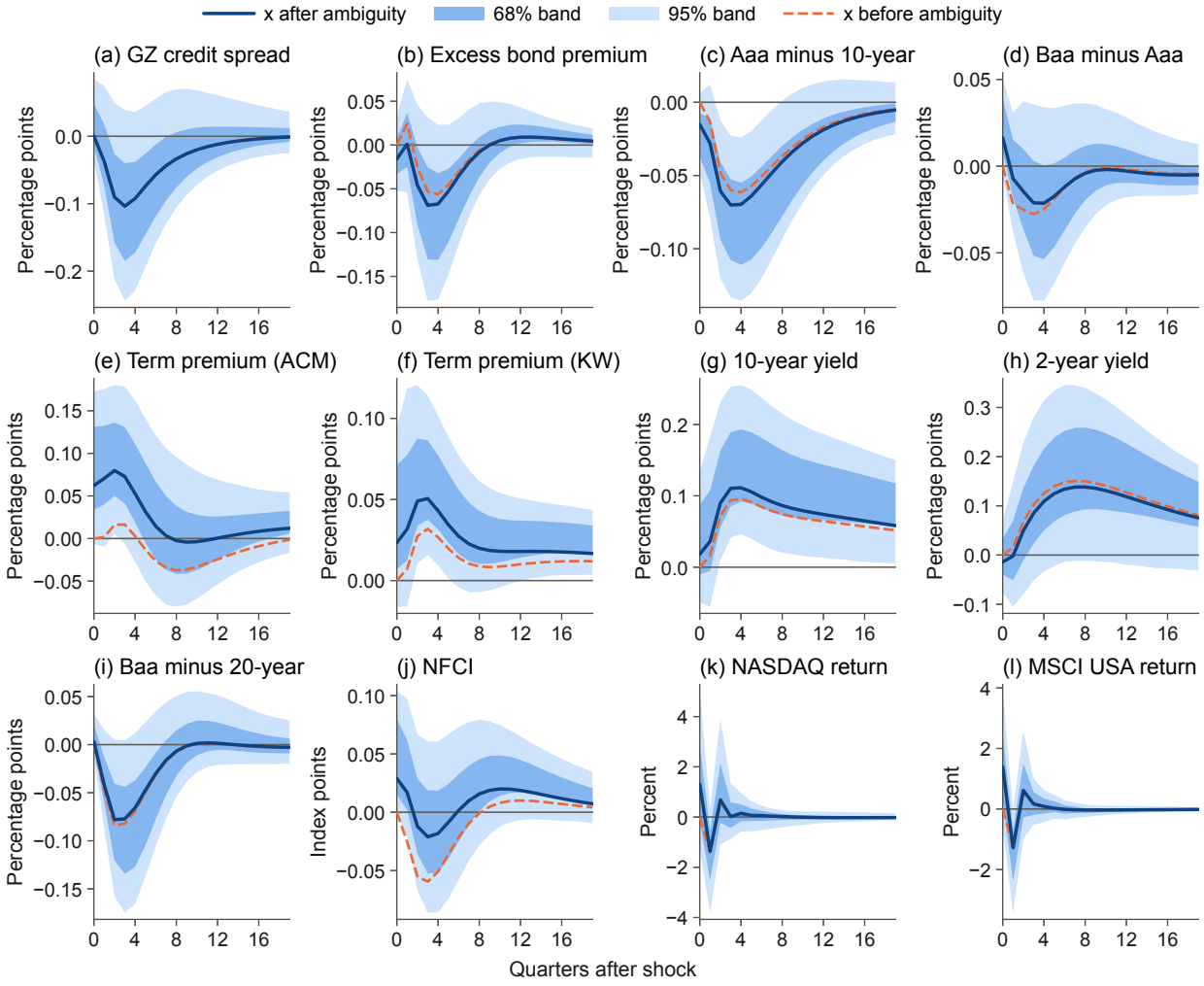


Figure C.33: U.S. asset prices after an ambiguity shock

Note: Responses of U.S. asset prices to a one standard deviation ambiguity shock in eight-variable models that add one U.S. financial variable to the benchmark global block. Solid line and shaded areas: variable ordered after ambiguity, point estimate with 68 and 95 percent bands; dashed line: variable ordered before ambiguity. Spreads and yields enter as $100 \times \log(1 + x/100)$ (percentage points to first order), term premia in percentage points (Adrian et al., 2013; Kim and Wright, 2005), the NFCI in index points, equity as quarterly log returns $\times 100$ (MSCI USA from 1997:Q2). 10,000 bootstrap replications.

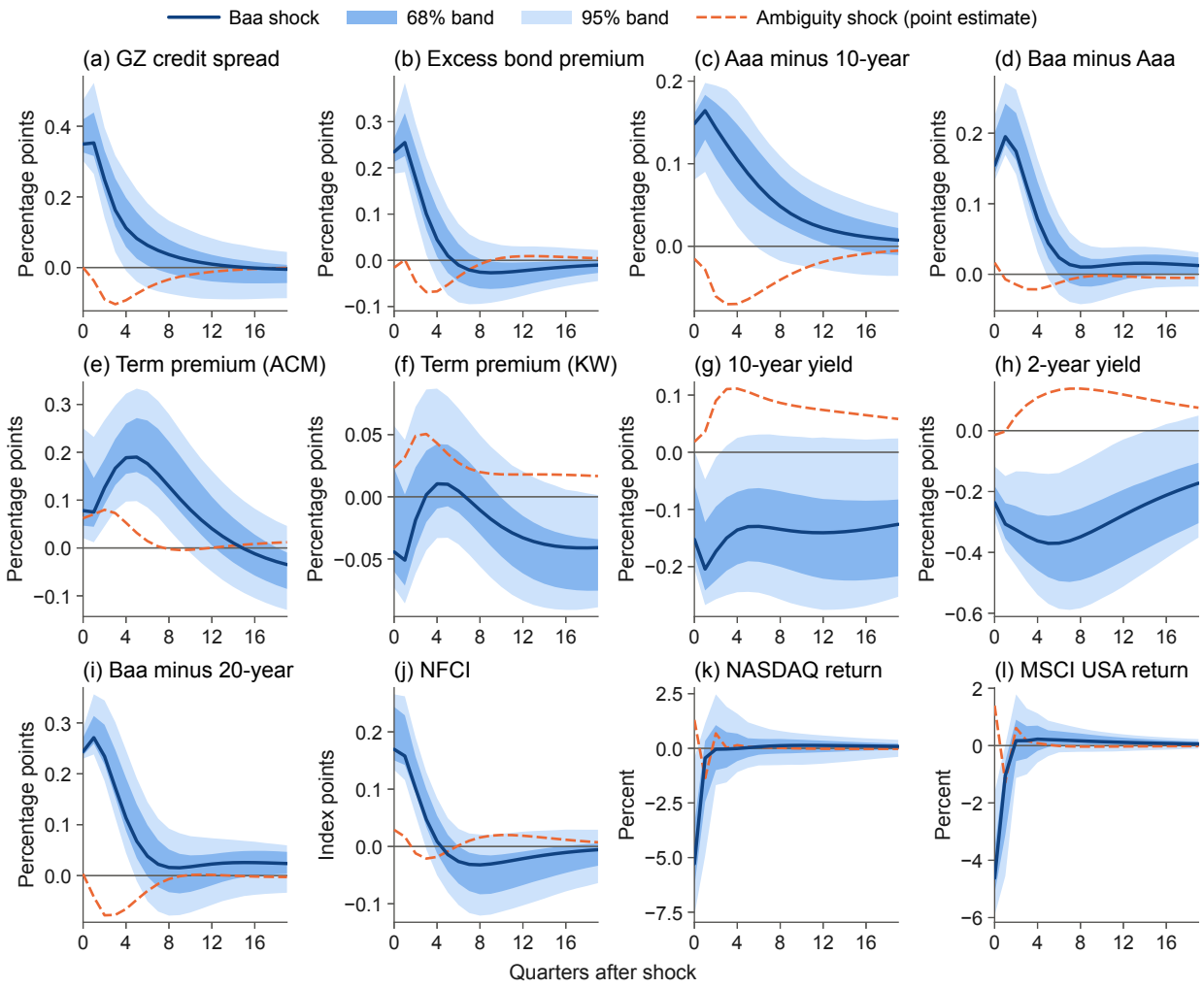


Figure C.34: U.S. asset prices after a global financial risk shock

Note: Responses of U.S. asset prices to a one standard deviation global financial risk (BAA spread) shock in the same eight-variable models (variable after ambiguity). Solid line and shaded areas: point estimate with 68 and 95 percent bands; dashed line: the response to the ambiguity shock in the same model. Units as in Figure C.33.

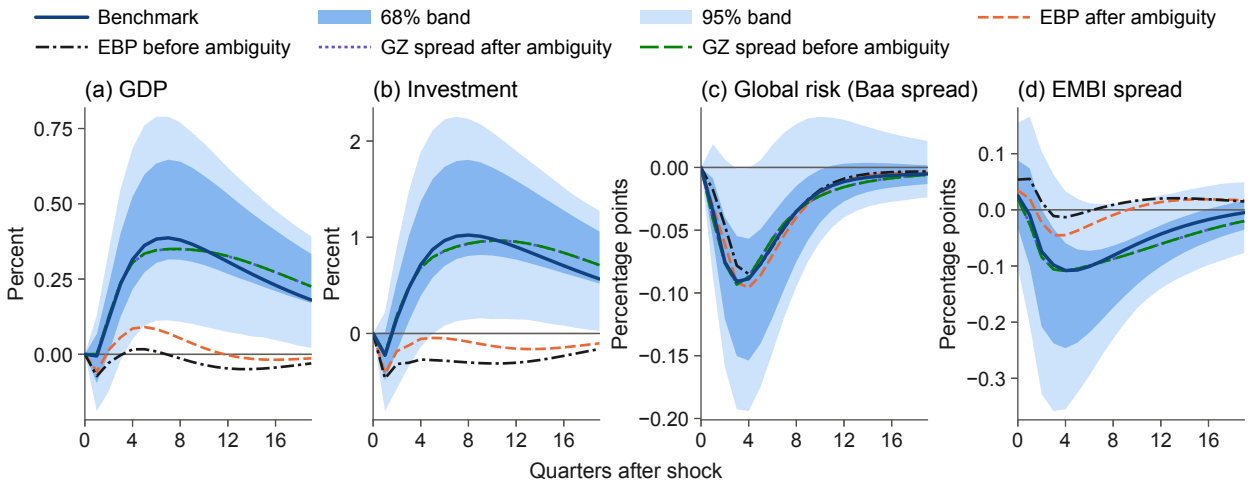


Figure C.35: Ambiguity shock with the excess bond premium or the credit spread in the model

Note: Responses of EME GDP, investment (percent), the BAA spread and the EMBI spread (percentage points) to a one standard deviation ambiguity shock. Solid line and shaded areas: benchmark with 68 and 95 percent bands. Other lines: point estimates of eight-variable models that add the Gilchrist and Zakrajšek (2012) excess bond premium (EBP) or credit spread (GZ) to the global block, after or before ambiguity.

Table C.11: Large ambiguity shocks: event averages

	Benchmark ambiguity shock					VIX-orth.
Change in	Diff. 1q	Pos. 4q	Neg. 4q	Diff. 4q	ex 2008Q4	Diff. 4q
Baa − 10y Treasury (pp)	−0.04	−0.14	0.57	−0.71	−0.70	−0.62
GZ spread (pp)	−0.12	−0.14	0.73	−0.88	−0.80	−0.80
EBP (pp)	−0.12	−0.17	0.43	−0.60	−0.48	−0.47
ACM term premium (pp)	0.12	−0.05	−0.24	0.19	0.10	0.06
10y Treasury yield (pp)	0.01	−0.11	−0.59	0.48	0.51	0.46
VIX (index)	−3.85	2.00	5.88	−3.88	−3.70	−2.68
Z.1 bond funds[a]	1.89	1.67	−3.38	5.05	5.01	2.08
Z.1 equity funds[a]	0.06	−0.29	−1.86	1.57	1.30	1.86
TIC foreign bonds[a]	0.21	0.23	−1.29	1.52	1.31	0.73
EMBI, EME mean (pp)	0.15	−0.05	0.85	−0.90	−0.90	−0.99
GDP, EME mean (%)	−0.19	−0.25	−1.74	1.49	2.16	1.97

Note: Average changes after the ten largest positive and the ten largest negative benchmark ambiguity shocks: $x_{t-1+k} - x_{t-1}$ for $k = 1$ and 4 quarters. [a] Abnormal cumulative flows relative to the previous four-quarter average, percent of GDP. EME series are within-unit changes averaged across units. Diff.: positive-shock mean minus negative-shock mean; ex 2008Q4: the 4-quarter difference without the 2008:Q4 shock. Positive shocks: 2008:Q4, 2010:Q4, 2016:Q2, 2012:Q1, 2017:Q3, 2019:Q4, 2006:Q4, 1995:Q4, 1997:Q4, 2017:Q4. Negative shocks: 2018:Q4, 2018:Q1, 2011:Q3, 2000:Q4, 2008:Q1, 2018:Q2, 2016:Q1, 2010:Q2, 2000:Q1, 1997:Q3. VIX-orth.: the same contrast for the VIX-orthogonal shock. No inference.

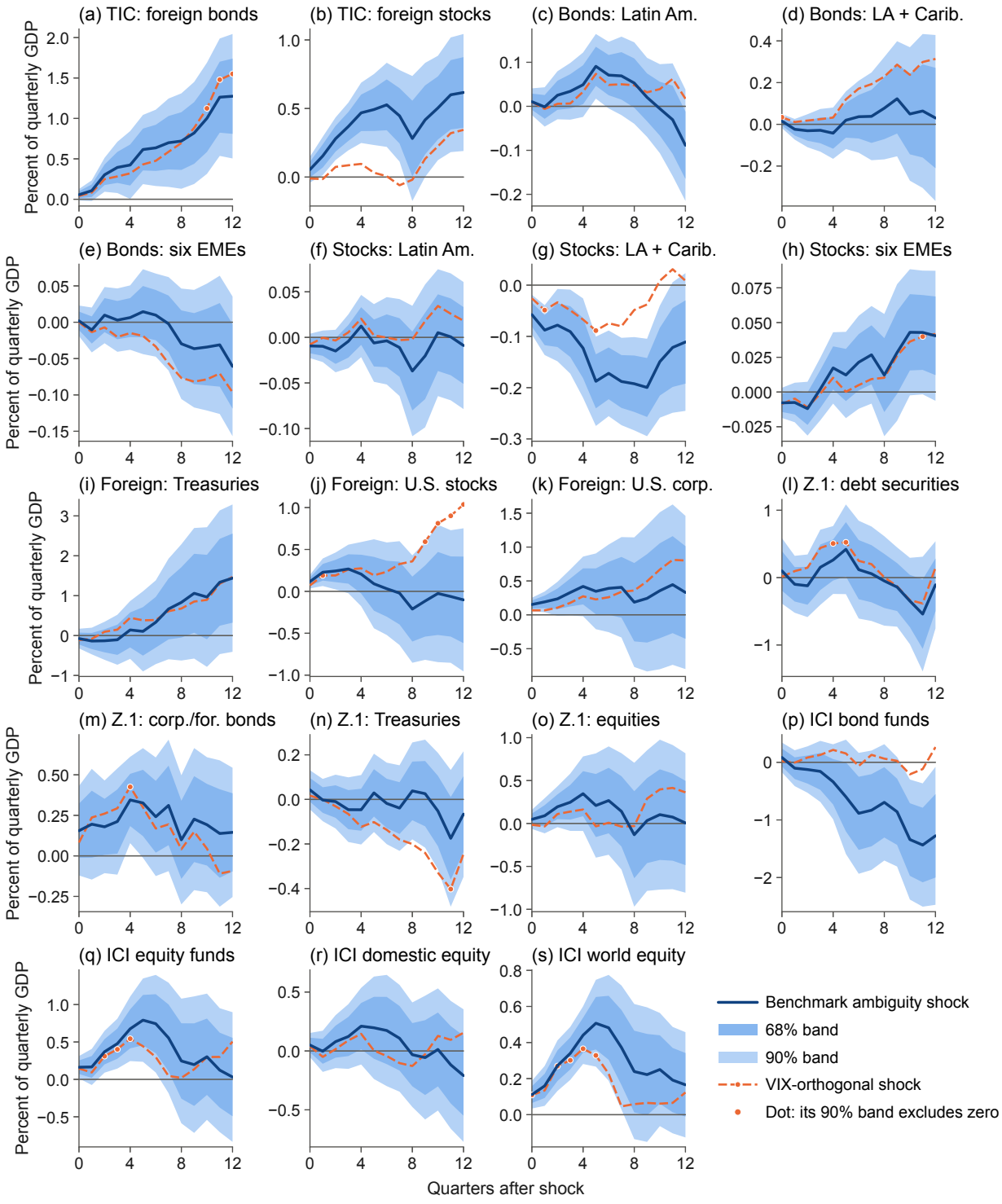


Figure C.36: U.S. portfolio flows after an ambiguity shock (local projections)

Note: Local-projection responses of U.S. and foreign portfolio flows (cumulative over quarters 0 to h, percent of quarterly U.S. GDP) to a one standard deviation ambiguity shock. Solid line and shaded areas: benchmark shock, coefficient with 68 and 90 percent bands; dashed line: the VIX-orthogonal shock, with dots where its 90 percent band excludes zero. Controls: two lags of the outcome and of the U.S. real rate, BAA spread and ambiguity; Newey and West (1987) standard errors. Horizons 0-12.

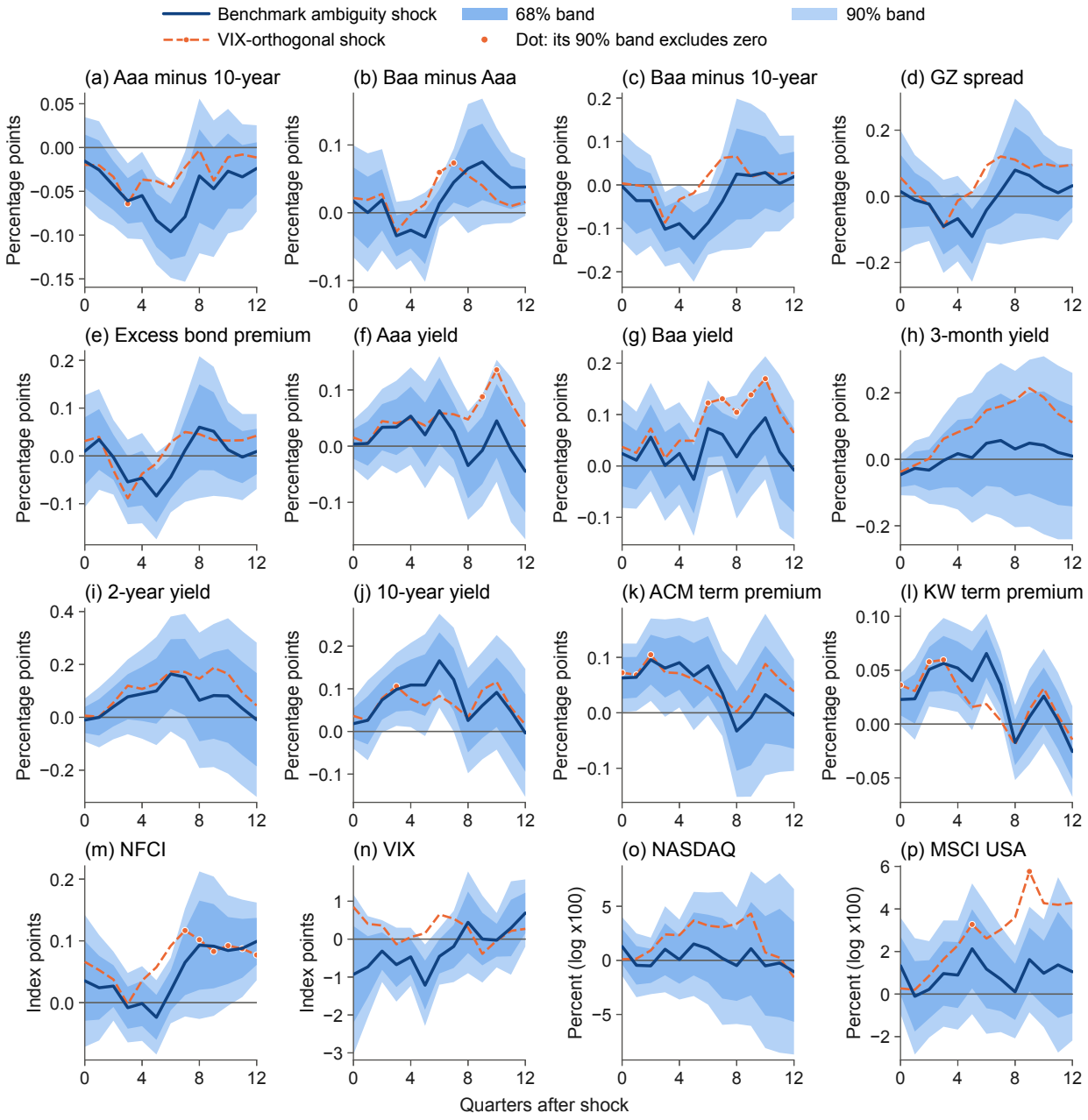


Figure C.37: U.S. asset prices after an ambiguity shock (local projections)

Note: Local-projection responses of U.S. asset prices (spreads, yields and term premia in percentage points; equity indexes 100 × log) to a one standard deviation ambiguity shock. Lines, bands and controls as in Figure C.36.

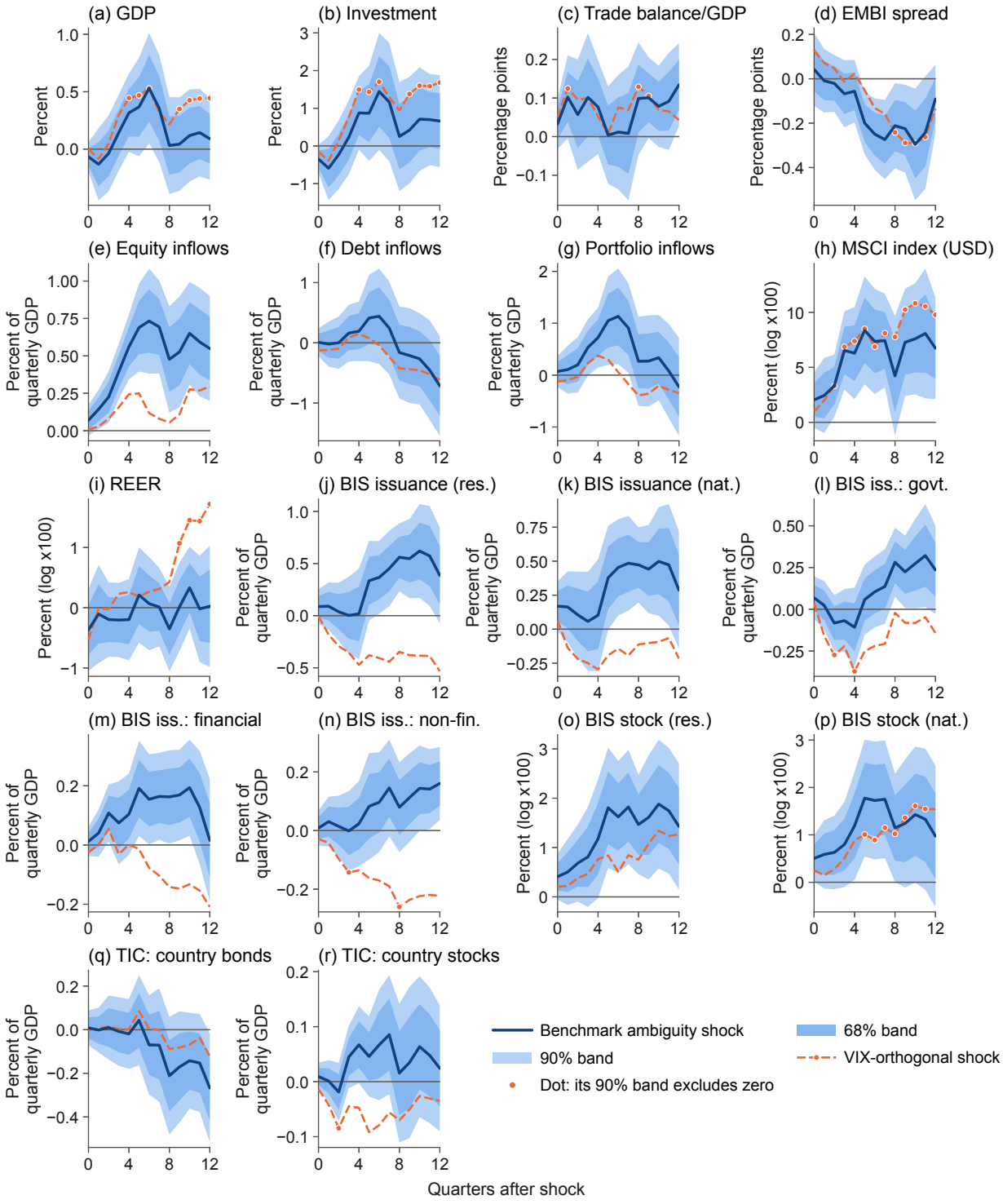


Figure C.38: EME quantities and prices after an ambiguity shock (local projections)

Note: Local-projection responses of EME outcomes (flows cumulative over quarters 0 to h in percent of quarterly GDP; stocks and prices 100 × log; GDP, investment, trade balance and EMBI spread as in the PSVAR) to a one standard deviation ambiguity shock, panel of the seven benchmark units with unit fixed effects. Controls add two lags of the country variables; Driscoll and Kraay (1998) standard errors. Lines and bands as in Figure C.36.

C.9 Alternative filters, detrending and frequency

We assess the implications of relaxing the assumption of the benchmark linear trend by allowing for curvature on the margin using the HP filter with smoothing parameter 10^5 and the Baxter-King (BK) and Christiano-Fitzgerald (CF) filters with lower band equal to 2 and upper band equal to 250. These parameter choices allow for some, but not excessive, curvature in the trend component relative to its linear counterpart for the HP and CF filters. The BK filter, with a window of 12 quarters, cannot pass periods anywhere near 250 quarters, acts as a high-pass filter, and loses 12 quarters at each end of the sample. Figure C.41 shows the GDP and investment responses to ambiguity and global financial risk shocks across these filters and across filters with standard business-cycle settings (HP with smoothing parameter 1600 and band-pass filters with bands of 6 and 32 quarters).

The filters are applied to GDP, investment, the trade balance and ambiguity. Filtering ambiguity alone leaves the results unchanged (EME GDP peaks at 0.39 percent with the HP filter with smoothing parameter 10^5); the loss of the boom under business-cycle settings comes from filtering the country variables. In the benchmark model, 87.5 percent of the variance of the component of EME GDP driven by ambiguity shocks lies at periods longer than 32 quarters, against 58 percent for the component driven by global financial risk shocks, and the HP filter with smoothing parameter 1600 passes only 13 percent of the ambiguity-driven variance against 42 percent of the global-risk-driven variance. The identified ambiguity shock itself is close to white noise, so the low-frequency character of the response comes from its propagation.

For the log ambiguity index, an augmented Dickey-Fuller test with a trend rejects a unit root (p-value 0.004), a KPSS test rejects level stationarity but not trend stationarity, and the trend slope is 1.07 percent per quarter (Newey-West t-statistic 7.95). These tests support treating the log ambiguity index as trend-stationary. Consistently treating trends across the system also preserves the expansionary response: with deterministic trend terms included in every PSVAR equation, the GDP response peaks 0.29 percent above trend and is significant at the 95 percent level around the peak. The specification that retains the trend

in ambiguity but detrends the country variables yields a substantially smaller response, reflecting its inconsistent treatment of trending series. First-differencing changes the shock from a transitory innovation around trend to a permanent shift in the level and generates a slightly negative GDP response (Table C.16).

The trend of the index. The linear trend is an approximation: the monthly log index falls through the 1990s (1.6 percent per quarter), rises steeply from 2000 to 2007 (3.9 percent per quarter), more slowly to about 2012 and is flat afterwards. Table C.12 tests for breaks. Because the index is persistent, the usual critical values are not valid, and we use heteroskedasticity- and autocorrelation-consistent and dynamic versions of the Chow test and a parametric bootstrap with autoregressive errors for the Bai and Perron (1998) statistics. A shift in level and slope in January 2001, when the NYSE and AMEX completed decimal pricing, is significant in every version (p-values below 0.001); September 2008 is not a break and January 2012 at most a weak one. The Bai and Perron (1998) procedures select no break at the 5 percent level and two breaks, in April 2000 and August 2007, at the 10 percent level. The trend is absent from every daily-close moment of returns, which do not trend over 1993-2019, and residualizing the index on realized variance, skewness, kurtosis and the VIX leaves its slope unchanged, which locates the trend in the intraday construction of the index rather than in the behaviour of risk.

Table C.13 re-estimates the benchmark model with the index detrended in fourteen ways. The identified ambiguity shock is nearly invariant to the choice: its correlation with the benchmark shock is 0.95 to 1.00 under every smooth or single-break treatment and 0.84 to 0.88 under segmented trends with level shifts, and seven to ten of the ten largest shocks of each sign coincide. The dynamics depend on how much low-frequency variation the treatment keeps: quadratic, Hodrick-Prescott and rolling-mean trends, continuous piecewise-linear trends with kinks in 2000 and 2007, and single breaks in 2001, 2008 or 2012 give GDP peaks of 0.31 to 0.42 percent, significant at the 95 percent level, whereas segmented trends with free level shifts at the break dates selected at the 10 percent level reduce the peak to 0.12 to 0.20 percent and remove its significance, as does the raw index with no detrending.

The last column shows why: the transmission per unit of ambiguity persistence, the GDP peak per 100 log-point-quarters of the cumulative ambiguity response, is stable at 0.6 to 0.8 across treatments, and the segmented trends halve the persistence of the ambiguity path. Applied to EME output as well, the same segmented trends remove a third of the within-country variation of GDP levels while leaving four-quarter growth untouched, that is, they remove the 2003-07 expansion and the 2008-12 adjustment themselves; under that treatment even the monthly identified shock loses its GDP effect.

Measurement of the index and volatility. Figure C.39 relates the monthly log index to the moments of daily returns. Realized variance and the VIX account for about half of the detrended variation of the index, with an elasticity of the index to realized variance of -0.46, half of the -1 that a mechanical rescaling of volatility would imply; realized skewness and kurtosis add little. Figure C.40 re-estimates the benchmark model with the index replaced by its residual from monthly regressions on realized variance, skewness and kurtosis, on the VIX and realized variance, or on all five moments (current and three lags). The EME output expansion survives every residualization, with GDP peaks of 0.32 to 0.46 percent, significant at the 95 percent level for the VIX and five-moment residualizations and at the 68 percent level for the realized-moment residualization; the early decline in the BAA spread vanishes, and the EMBI spread rises slightly on impact before falling after six to eleven quarters, as it does after the monthly identified shock. The volatility component of the benchmark shock accounts for the early spread compression, not for the expansion.

Table C.12: Breaks in the trend of the ambiguity index

A. Chow tests at candidate dates (p-values)

	Level and slope			Slope only		
Break date(s)	OLS	HAC	Dynamic	OLS	HAC	Dynamic
Sample 1993M2–2019M12						
2001M1	<0.001	<0.001	<0.001	<0.001	0.053	0.137
2008M9	0.015	0.334	0.648	0.135	0.556	0.570
2012M1	0.002	0.095	0.303	0.172	0.547	0.591
All three	<0.001	<0.001	<0.001	<0.001	0.020	0.309
Sample 1993M2–2021M9						
2001M1	<0.001	<0.001	<0.001	0.005	0.187	0.373
2008M9	0.393	0.852	0.889	0.786	0.914	0.853
2012M1	<0.001	0.013	0.083	0.403	0.710	0.667
All three	<0.001	<0.001	<0.001	<0.001	0.036	0.378

B. Bai-Perron tests (level and slope; AR(2) bootstrap critical values and p-values)

Test	Statistic	95% c.v.	p	Break dates
Sample 1993M2–2019M12				
supF(1|0)	30.82	40.08	0.145	2003M4
supF(2|0)	38.63	33.91	0.019	2000M4, 2007M8
supF(3|0)	31.78	28.75	0.025	2000M4, 2007M8, 2012M8
supF(2|1)	34.68	23.34	0.006	2003M4 ; added 2007M8
supF(3|2)	12.48	17.22	0.163	2000M4, 2007M8 ; added 2012M8
UDmax	38.63	41.45	0.067	m = 2
Sample 1993M2–2021M9				
supF(1|0)	32.33	42.61	0.155	2003M5
supF(2|0)	32.21	35.56	0.093	2003M9, 2012M1
supF(3|0)	31.21	30.59	0.044	2000M4, 2007M8, 2012M8
supF(2|1)	24.32	23.51	0.043	2003M5 ; added 2012M1
supF(3|2)	10.03	16.79	0.266	2003M9, 2012M1 ; added 1998M9
UDmax	32.33	43.15	0.188	m = 1

C. Number of breaks selected

Sample	Sequential, 5%	UDmax, then sequential, 5%	BIC	LWZ
1993M2–2019M12	0	0	3	2
1993M2–2021M9	0	0	3	3

D. Trend slopes by subperiod, 1993M2–2019M12 (percent per quarter)

Segmentation	Segment	Months	Slope	HAC t
Linear trend	1993M2–2019M12	323	0.95	7.0
Break at 2001M1	1993M2–2000M12	95	−1.55	−7.3
	2001M1–2019M12	228	0.96	4.2
Breaks at 2001M1, 2008M9, 2012M1	1993M2–2000M12	95	−1.55	−7.3
	2001M1–2008M8	92	1.94	1.9
	2008M9–2011M12	40	0.85	0.4
	2012M1–2019M12	96	0.42	0.8
Bai-Perron, one break	1993M2–2003M3	122	−0.66	−2.0
	2003M4–2019M12	201	0.68	2.7
Bai-Perron, two breaks	1993M2–2000M3	86	−1.81	−9.2
	2000M4–2007M7	88	3.87	13.2
	2007M8–2019M12	149	1.62	4.4
Bai-Perron, three breaks	1993M2–2000M3	86	−1.81	−9.2
	2000M4–2007M7	88	3.87	13.2
	2007M8–2012M7	60	1.62	2.9
	2012M8–2019M12	89	0.15	0.3

Note: Monthly log index, 1993:M2-2019:M12. Panel A: p-values of Chow tests for a break in level and slope, or in slope only, at the dates shown: static OLS, heteroskedasticity- and autocorrelation-consistent (12 lags), and dynamic (two lags of the index, HAC). Panel B: Bai and Perron (1998) supF statistics with 15 percent trimming and up to three breaks, with p-values from a parametric bootstrap with AR(2) errors (1,999 draws), and the break dates selected. Panel C: trend slopes by subperiod, percent per quarter, with HAC t-statistics.

Table C.13: Responses to an ambiguity shock under alternative trend treatments of the index

Detrending of the index	Corr.	GDP peak (h)	95% horizons	Inv. peak	Baa	EMBI	Cum. amb.	Multiplier [95%]
Paper's series (linear, 1994Q2–2019Q4)	1.00	0.39 (7)	4–19	1.02	−0.09	−0.11	49.6	0.78 [0.24, 1.60]
Linear OLS	1.00	0.40 (7)	4–19	1.06	−0.09	−0.10	49.9	0.80 [0.28, 1.61]
Quadratic	1.00	0.41 (7)	4–19	1.10	−0.09	−0.10	50.3	0.81 [0.30, 1.62]
Level and slope break, 2003Q2	0.95	0.20 (7)	none	0.34	−0.07	−0.01	41.8	0.48 [−0.15, 1.45]
Kink, 2003Q2	1.00	0.40 (7)	4–19	1.07	−0.09	−0.10	50.1	0.80 [0.28, 1.62]
Level and slope breaks, 2000Q2, 2007Q3	0.88	0.17 (4)	none	0.38	0.00	−0.02	26.2	0.64 [−0.21, 1.91]
Kinks, 2000Q2, 2007Q3	0.98	0.31 (7)	4–14	0.70	−0.09	−0.05	46.1	0.68 [0.11, 1.58]
Level and slope breaks, 2000Q2, 2007Q3, 2012Q3	0.84	0.12 (4)	none	0.30	0.00	−0.03	20.5	0.57 [−0.32, 1.93]
Kinks, 2000Q2, 2007Q3, 2012Q3	0.98	0.33 (6)	4–16	0.73	−0.08	−0.04	44.0	0.76 [0.16, 1.72]
Level and slope break, 2001Q1	0.96	0.32 (6)	3–15	0.77	−0.07	−0.09	39.6	0.82 [0.33, 1.83]
Level and slope break, 2008Q3	0.99	0.42 (7)	3–19	1.14	−0.09	−0.10	43.3	0.97 [0.45, 1.85]
Level and slope break, 2012Q1	0.99	0.42 (7)	4–19	1.18	−0.08	−0.11	50.3	0.83 [0.33, 1.65]
HP filter, $\lambda = 10^5$	1.00	0.39 (7)	4–19	1.04	−0.09	−0.10	48.4	0.81 [0.30, 1.64]
10-year trailing rolling mean	0.98	0.37 (9)	6–19	0.99	−0.05	−0.13	65.3	0.56 [−0.07, 1.26]
10-year centered rolling mean	0.99	0.35 (6)	4–19	0.87	−0.10	−0.11	45.5	0.78 [0.21, 1.68]
None (raw log index)	0.95	0.11 (13)	none	0.31	−0.06	−0.19	96.4	0.12 [−0.48, 0.51]

Note: Benchmark model with the quarterly log index detrended as indicated (fitted on 1994:Q2-2019:Q4). Corr.: correlation of the identified ambiguity shock with the benchmark shock. GDP, investment: peak response (percent), horizon, and horizons at which the 95 percent band excludes zero; BAA, EMBI: trough (percentage points). Cum. ambiguity: cumulative response of ambiguity over 20 quarters (log points × quarters). Multiplier: GDP peak per 100 log-point-quarters of the cumulative ambiguity response. Bands: 10,000 replications of the moving-block residual bootstrap, Hall intervals.

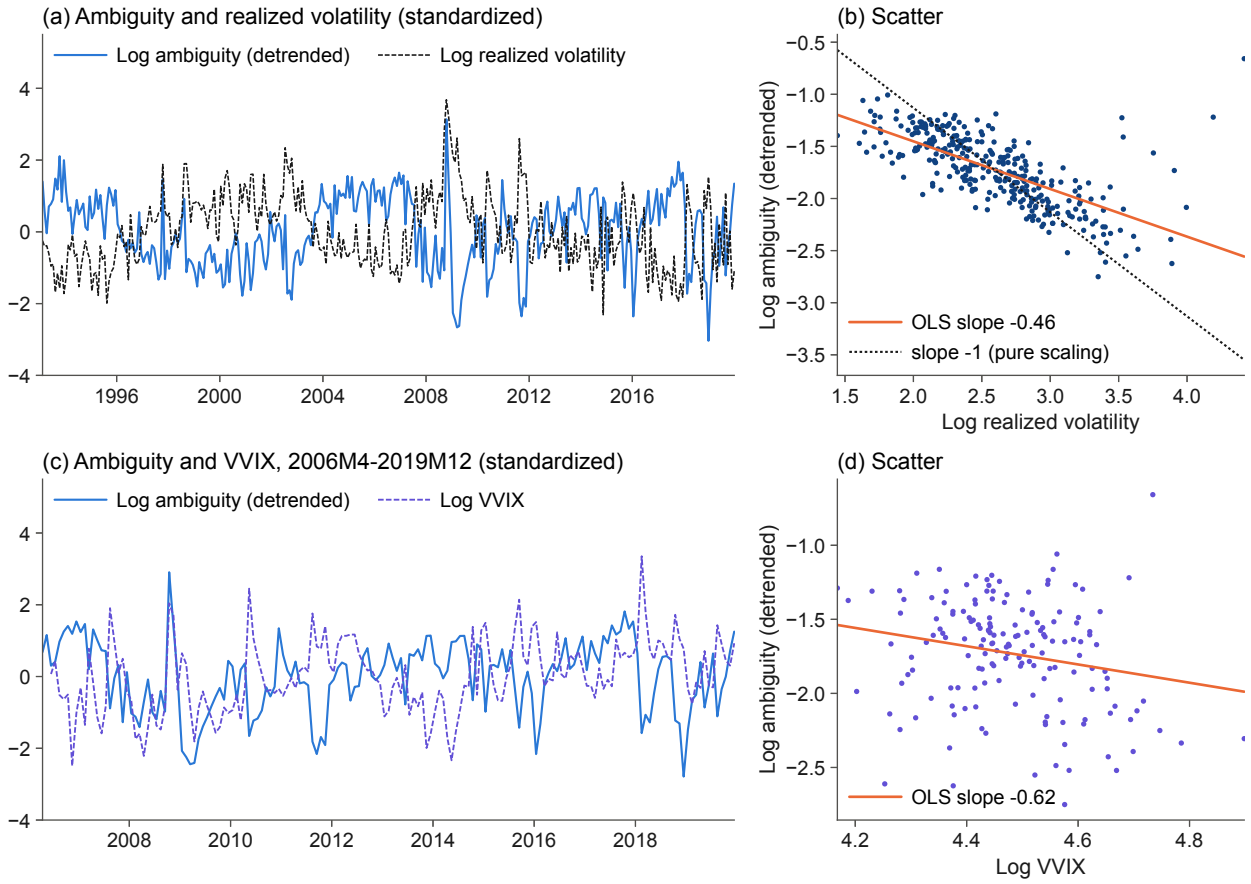


Figure C.39: The ambiguity index and the moments of daily returns

Note: Monthly log ambiguity index (detrended) against realized variance, the VIX, realized skewness and kurtosis of daily S&P 500 returns, 1993:M2-2019:M12, with the fitted values of the regressions described in the text.

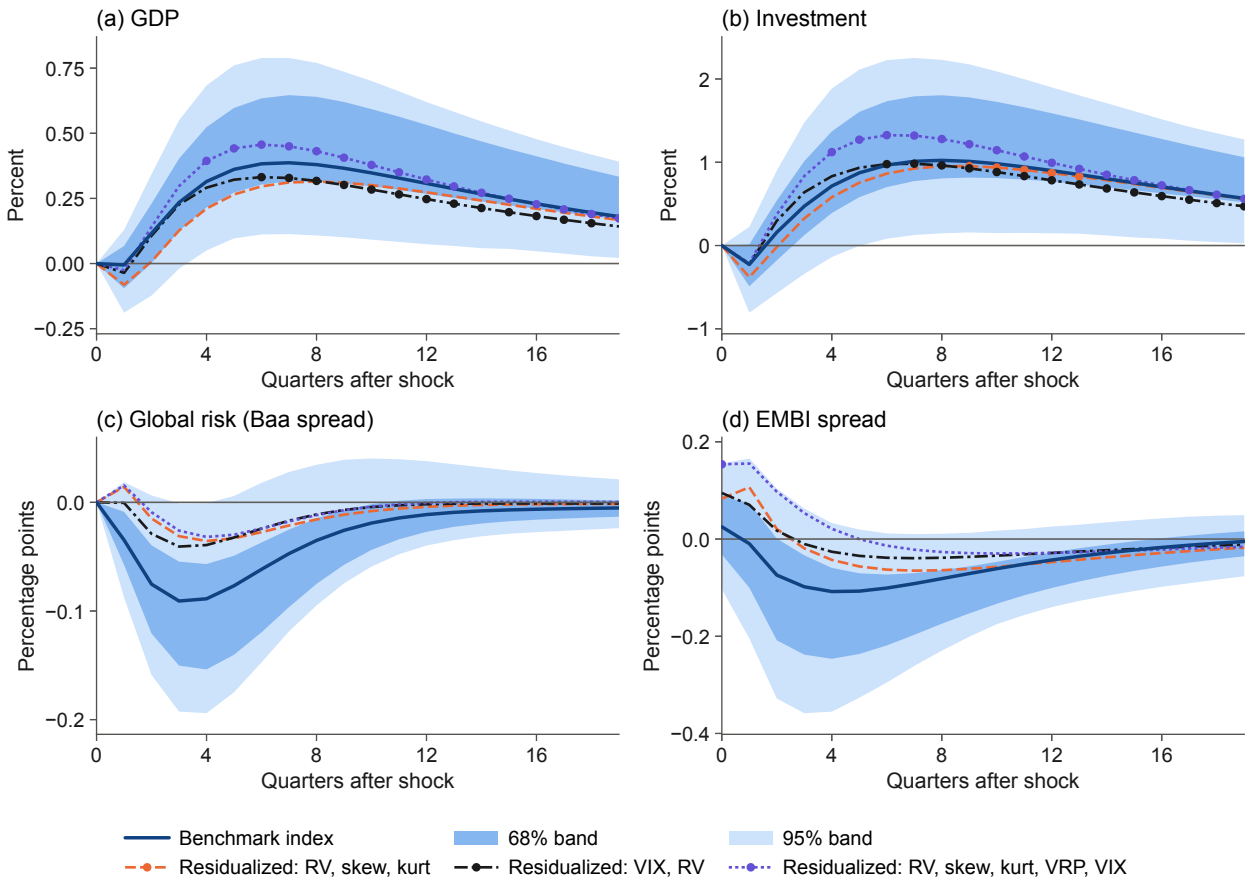


Figure C.40: Responses to an ambiguity shock with the index purged of volatility and higher moments

Note: Responses of EME GDP, investment (percent), the BAA spread and the EMBI spread (percentage points) to a one standard deviation ambiguity shock in the benchmark model (solid line, 68 and 95 percent bands) and in the same model with the index replaced by its residual from monthly regressions on realized variance, skewness and kurtosis; on the VIX and realized variance; or on all five moments (current values and three lags), aggregated to quarters and linearly detrended. 10,000 replications of the moving-block residual bootstrap, Hall intervals.

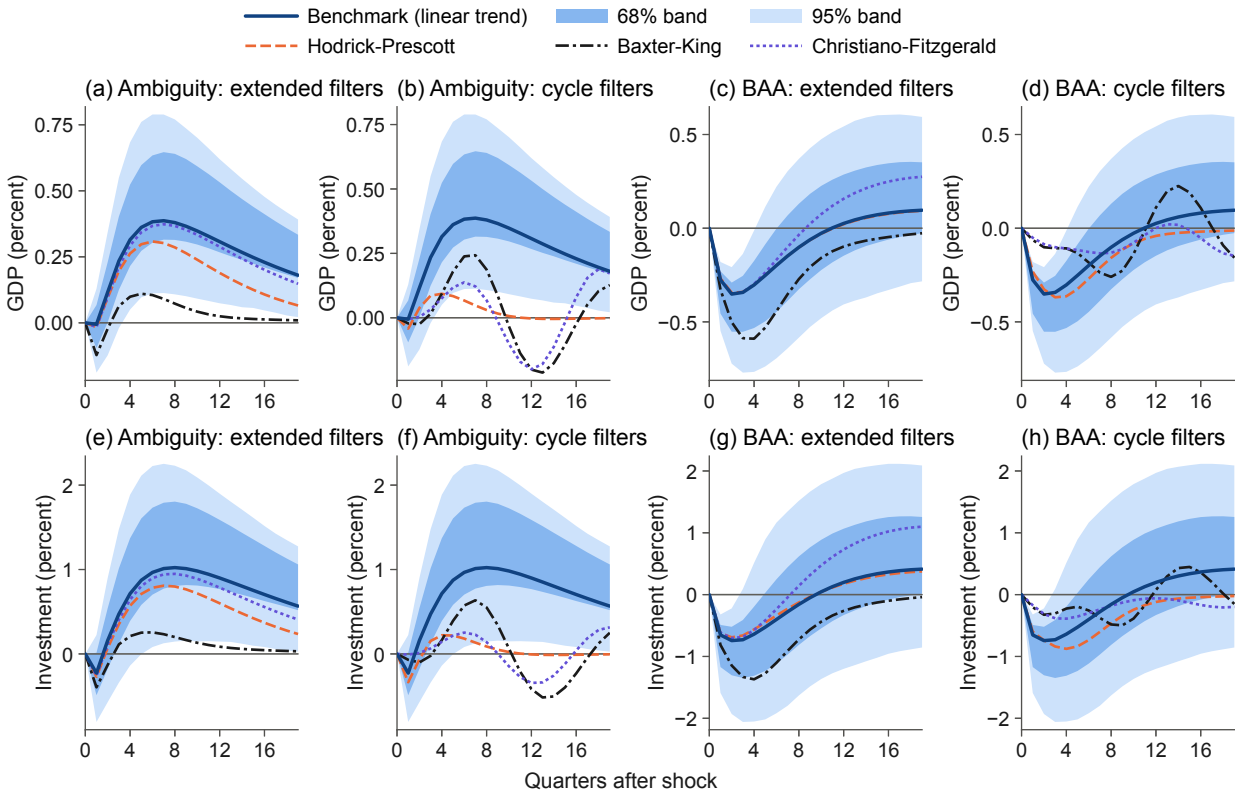


Figure C.41: Alternative filters

Note: Responses of EME GDP (top row) and investment (bottom row) to one standard deviation ambiguity and global financial risk shocks when GDP, investment, the trade balance and ambiguity are filtered instead of linearly detrended. Columns 1 and 3: HP with smoothing parameter 10^5 and band-pass filters keeping periods of 2 to 250 quarters; columns 2 and 4: standard business-cycle settings (HP 1600; periods of 6 to 32 quarters). The BK filter uses a 12-quarter window. Solid line and shaded areas: benchmark with 68 and 95 percent bands; other lines: point estimates.

C.10 Alternative lag orders

The figures below show results from estimation with, alternatively, 1, 3 and 4 lags.

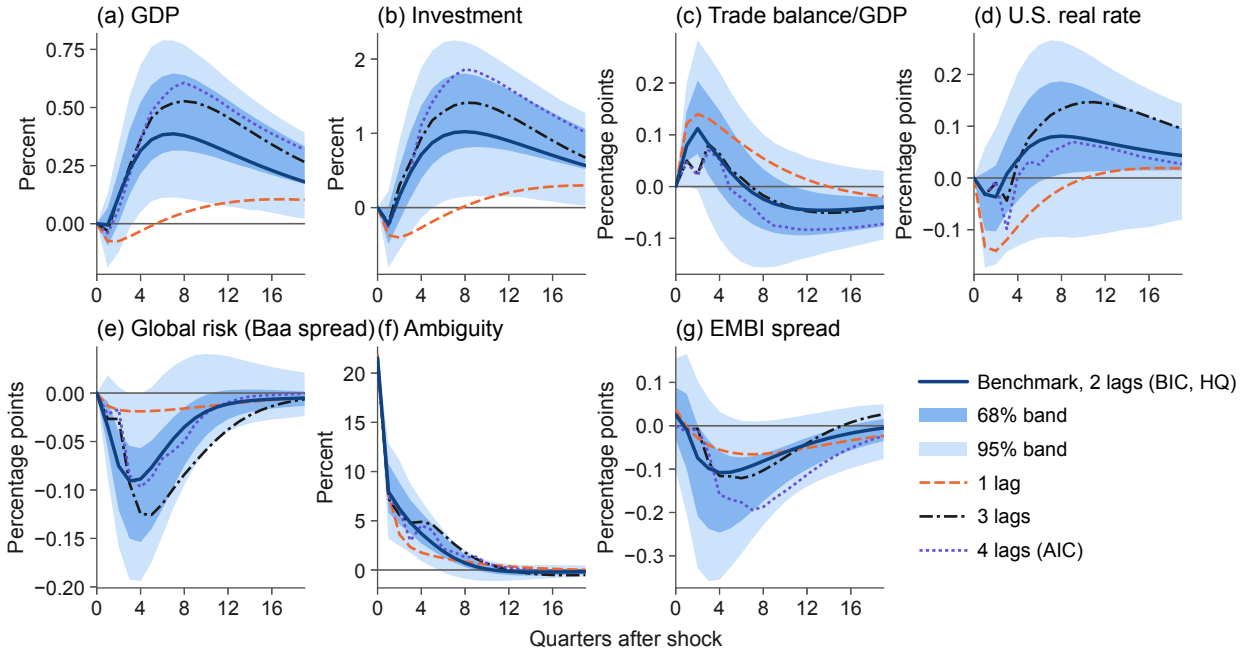


Figure C.42: IRFs to one std. dev. shock to ambiguity: alternative lag orders

Note: Point estimates with one, three and four lags against the benchmark with two lags (solid line, 68 and 95 percent bands).

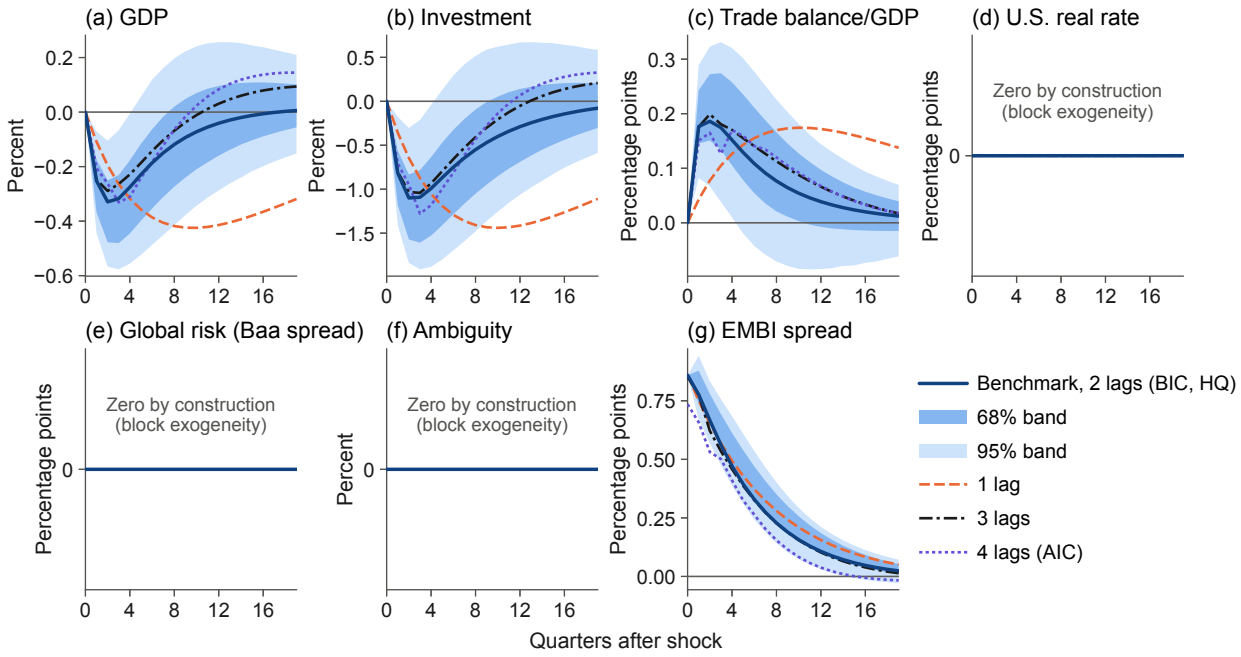


Figure C.43: IRFs to one std. dev. shock to country spread: alternative lag orders

Note: Point estimates with one, three and four lags against the benchmark with two lags (solid line, 68 and 95 percent bands).

C.11 Alternative specifications

The figures below show the results (i) when there are no spillovers between the variables in the shock block, (ii) when there are no spillovers from the shock block or from the real variables to the country spread, and (iii) across families of alternative specifications, including the case in which all variables depend on all other variables in the model and alternative orderings of the shock block variables. In Figures C.44 to C.48, the dashed line imposes the zero restrictions on the benchmark estimates without re-estimating the model, and the dash-dot line re-estimates the model under the restrictions. Tables C.14 to C.16 summarize every specification.

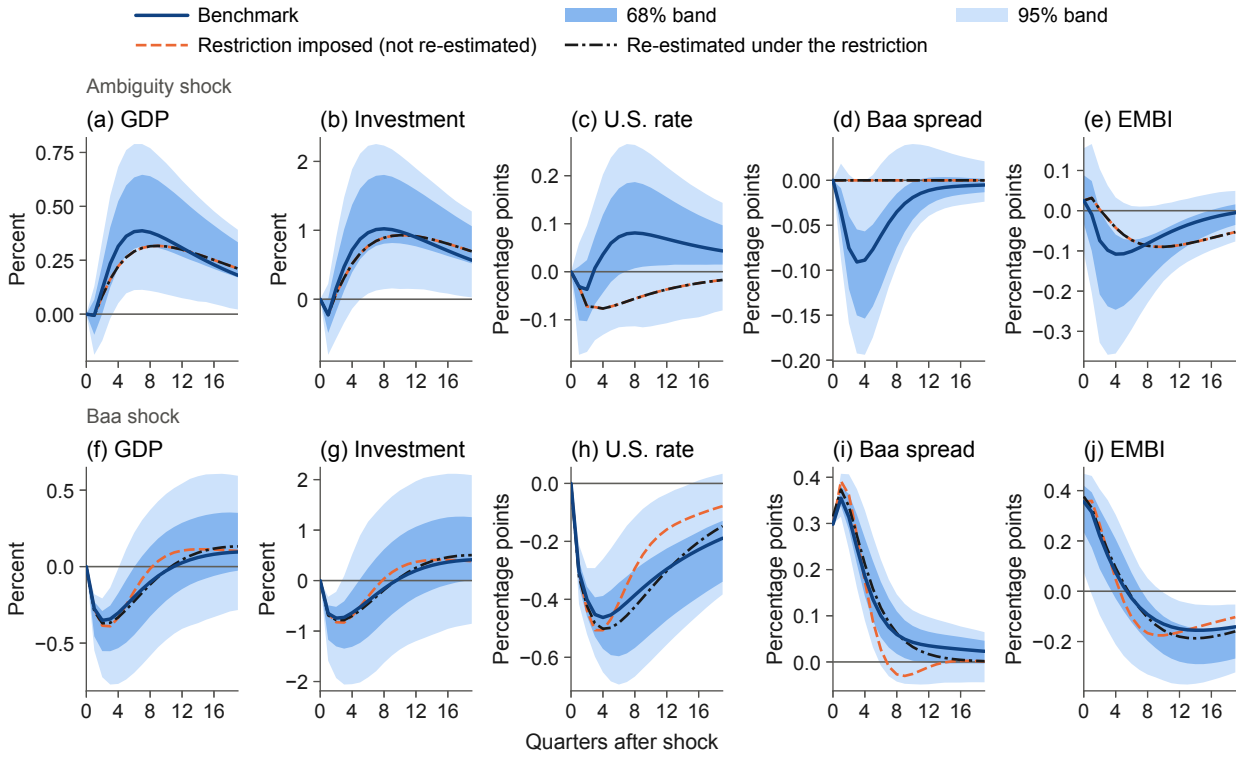


Figure C.44: IRFs when the BAA spread follows its own AR(2)

Note: Top row: responses to a one standard deviation ambiguity shock; bottom row: to a global financial risk shock. Solid line and shaded areas: benchmark with 68 and 95 percent bands. Dashed line: restrictions imposed on the benchmark estimates. Dash-dot line: model re-estimated under the restrictions.

Table C.14: Robustness: responses to an ambiguity shock (part 1: specification and countries)

	GDP		Investment		Baa	EMBI	FEVD
Specification	Peak (h)	68% / 95%	Peak (h)	68% / 95%	peak	peak	GDP, h=20
Benchmark							
Benchmark	0.387 (7)	2–19 / 4–19	1.023 (8)	3–19 / 6–19	−0.091**	−0.108*	6.1
Lag order							
One lag	0.105 (17)	14–19 / none	−0.404 (2)	1–2 / none	−0.019	−0.066**	0.5
Three lags	0.527 (8)	2–19 / 4–19	1.413 (8)	2–19 / 4–16	−0.126**	−0.121*	11.6
Four lags	0.606 (8)	3–19 / 4–19	1.864 (8)	3–19 / 4–19	−0.097*	−0.195**	14.3
Global-block ordering							
Ambiguity, Baa, R_{US}	0.374 (7)	3–19 / 5–19	0.991 (8)	3–19 / 8–19	−0.081*	−0.090*	5.8
Ambiguity, R_{US}, Baa	0.374 (7)	3–19 / 5–19	0.991 (8)	3–19 / 8–19	−0.081*	−0.090*	5.8
Baa, ambiguity, R_{US}	0.387 (7)	3–19 / 4–17	1.008 (8)	3–19 / 6–14	−0.098*	−0.096*	6.0
Baa, R_{US}, ambiguity	0.387 (7)	2–19 / 4–19	1.023 (8)	3–19 / 6–19	−0.091**	−0.108*	6.1
R_{US}, ambiguity, Baa	0.379 (7)	3–19 / 4–19	1.013 (8)	3–19 / 6–19	−0.079*	−0.106*	6.0
Exogeneity and zero restrictions							
All variables endogenous	0.374 (6)	2–19 / 3–19	0.964 (7)	3–19 / 5–19	−0.102**	−0.107**	5.7
Ambiguity exogenous (imposed)	0.353 (6)	2–19 / 3–19	0.916 (6)	3–19 / 5–19	−0.091**	−0.107**	4.6
Ambiguity exogenous (re-est.)	0.454 (6)	3–19 / 4–12	1.190 (7)	3–19 / 7	−0.109*	−0.132*	8.1
Baa AR(2) (imposed)	0.317 (9)	3–19 / 6–19	0.929 (10)	4–19 / 7–19	0.000	−0.090**	4.9
Baa AR(2) (re-est.)	0.317 (9)	3–19 / 6–19	0.929 (10)	4–19 / 7–19	0.000	−0.090**	4.9
EMBI without globals (imposed)	0.359 (8)	3–19 / 5–19	0.949 (9)	3–19 / 8–19	−0.091**	−0.059*	5.8
EMBI without globals (re-est.)	0.357 (7)	3–19 / 4–19	0.928 (9)	3–19 / 8–19	−0.091**	−0.040*	5.7
EMBI without real (imposed)	0.368 (6)	2–19 / 3–19	0.947 (7)	3–19 / 5–19	−0.091**	−0.089*	6.1
EMBI without real (re-est.)	0.379 (6)	2–19 / 4–19	0.984 (7)	3–19 / 6–17	−0.091**	−0.105*	6.3
R_{US} AR(2) (imposed)	0.403 (7)	3–19 / 4–14	1.087 (8)	3–19 / 5–13	−0.096*	−0.127*	6.3
R_{US} AR(2) (re-est.)	0.403 (7)	3–19 / 4–19	1.087 (8)	3–19 / 5–19	−0.096*	−0.127*	6.3
Countries (jackknife)							
Without Argentina (both units)	0.362 (6)	3–19 / 4–17	0.930 (8)	3–19 / 9	−0.091**	−0.124**	5.9
Without Argentina I	0.375 (6)	2–19 / 4–19	0.958 (7)	3–19 / 6–17	−0.091**	−0.128**	6.1
Without Argentina II	0.376 (7)	3–19 / 4–19	0.995 (8)	3–19 / 6–19	−0.091**	−0.105*	5.9
Without Brazil	0.449 (7)	2–19 / 3–19	1.453 (7)	2–19 / 4–19	−0.091**	−0.114*	7.9
Without Mexico	0.492 (7)	2–19 / 3–19	1.225 (8)	3–19 / 5–19	−0.091**	−0.148**	9.6
Without Peru	0.417 (6)	3–19 / 4–19	1.145 (6)	3–19 / 4–13	−0.091**	−0.100*	6.1
Without South Africa	0.360 (7)	3–19 / 4–17	0.940 (7)	3–19 / 6–16	−0.091**	−0.089*	5.1
Without Turkey	0.280 (11)	4–19 / 10–17	0.812 (14)	6–19 / 14–19	−0.091**	−0.083**	3.3

Note: Peak: largest absolute response of GDP or investment over h = 0-19 (percent) and its horizon; 68%/95%: horizons at which the band excludes zero with the sign of the peak (the investment dip at $h = 1$, which is significant at 68 percent in several specifications, is not shown). BAA and EMBI: largest absolute response (percentage points); * (**): the 68 (95) percent band excludes zero with the sign of the peak at some horizon. FEVD: ambiguity share of the 20-step-ahead GDP forecast error variance, percent. Imposed: zero restrictions imposed on the benchmark estimates; re-est.: model re-estimated under the restrictions. Bands: 10,000 replications of the residual bootstrap.

Table C.15: Robustness: responses to an ambiguity shock (part 2: samples and other economies)

	GDP		Investment		Baa	EMBI	FEVD
Specification	Peak (h)	68% / 95%	Peak (h)	68% / 95%	peak	peak	GDP, h=20
Benchmark							
Benchmark	0.387 (7)	2–19 / 4–19	1.023 (8)	3–19 / 6–19	−0.091**	−0.108*	6.1
Samples							
Ending 2007Q4	0.231 (15)	10–19 / 14–19	0.767 (15)	11–19 / 15–19	−0.032**	−0.153**	2.7
Ending 2011Q3	0.359 (9)	5–19 / 9–19†	1.066 (10)	7–19 / 11–19	−0.111**	−0.170**	6.5
Starting 2003Q1	0.469 (5)	2–19 / 3–8	1.044 (5)	3–19 / none	−0.131*	−0.115*	7.4
Starting 2010Q1	0.138 (3)	1, 3 / none†	−0.556 (11)	7–19 / 10–18	−0.092**	−0.061	1.2
Extended EME sample (master data)							
11 EMEs (12 units)	0.294 (8)	3–19 / 5–19	0.790 (9)	4–19 / 6–18	−0.091**	−0.151*	4.4
11 EMEs without Ecuador	0.316 (8)	3–19 / 4–19	0.884 (9)	3–19 / 5–19	−0.091**	−0.097*	5.0
11 EMEs, units with ≥ 40 quarters	0.279 (8)	3–19 / 5–18	0.740 (9)	4–19 / 9–13	−0.091**	−0.162**	4.3
Master data, benchmark sample							
Benchmark rebuilt from master data	0.388 (7)	3–19 / 4–19	1.031 (8)	3–19 / 6–19	−0.091**	−0.115*	6.1
Sample extended to 2021Q3 (master data)							
x_bench_end2022q4 data cut at 2019Q4 (de	0.343 (7)	3–19 / 5–14	0.966 (7)	3–19 / 6–13	−0.091**	−0.130**	4.8
Extended to 2021Q3 (master data)	−0.219 (2)	1, 9–14 / none	−0.563 (1)	1 / none	−0.039*	0.045*	0.5
Extended to 2021Q3, COVID dummy	−0.241 (2)	1–19 / 3, 5	−0.600 (1)	1–7 / none	−0.032	0.056*	1.0
x_bench_end2022q4 with 2020Q1-Q3 impulse	0.279 (7)	4–19 / 7–19	0.827 (8)	4–19 / 8–19	−0.032	−0.083*	3.6
Advanced small open economies (master data)							
12 advanced SOEs, 10-year yield	0.114 (11)	4–19 / none	0.298 (9)	5–19 / none	−0.091**	0.089**	1.8
12 advanced SOEs, spread vs U.S.	0.129 (12)	4–19 / none	0.370 (9)	4–19 / none	−0.091**	−0.045*	2.5
Pooled EMEs and advanced economies (master data)							
11 EMEs and 12 advanced SOEs	0.170 (8)	3–19 / none	0.443 (9)	4–19 / none	−0.091**	−0.065*	2.4

Note: Columns as in Table C.14. For advanced economies the EMBI column refers to the 10-year sovereign yield or its spread over the U.S. 10-year yield. Master data: the authors' extended 24-country dataset. † The 68 percent GDP band also excludes zero with the opposite sign: ending 2011:Q3 ($h = 1$); starting 2010:Q1 (h = 7-19).

Table C.16: Robustness: responses to an ambiguity shock (part 3: filters, measures and trends)

	GDP		Investment		Baa	EMBI	FEVD
Specification	Peak (h)	68% / 95%	Peak (h)	68% / 95%	peak	peak	GDP, h=20
Benchmark							
Benchmark	0.387 (7)	2–19 / 4–19	1.023 (8)	3–19 / 6–19	−0.091**	−0.108*	6.1
Filters							
BK (2, 250)	−0.122 (1)	1 / 1†	−0.396 (1)	1–2 / 1	−0.076*	−0.083*	1.2
BK (6, 32)	0.243 (7)	5–9 / 6–7†	0.636 (7)	5–9 / 6–8	−0.084**	−0.116	9.2
CF (2, 250)	0.373 (7)	3–19 / 4–19	0.948 (8)	3–19 / 6–16	−0.085*	−0.107*	5.8
CF (6, 32)	−0.200 (12)	11–14 / none†	−0.344 (12)	12–13 / none	−0.019*	0.045*	6.6
HP ($\lambda = 1600$)	0.093 (4)	3–7 / none	−0.335 (1)	1 / none	−0.046*	0.034	0.7
HP ($\lambda = 10^5$)	0.308 (6)	3–19 / 4–15	0.809 (7)	3–19 / 6–15	−0.090*	−0.106*	5.0
HP (10^5), all seven filtered	0.288 (6)	3–19 / 4–13	0.738 (7)	4–19 / 6–13	−0.111**	−0.080*	4.2
Ambiguity measure							
First difference (not detrended)	−0.096 (4)	none / none	−0.366 (1)	1 / none	0.050**	0.041	0.3
EWMA (0.35)	0.459 (7)	3–19 / 5–18	1.321 (8)	3–19 / 6–18	−0.126**	−0.156**	8.7
EWMA (0.55)	0.416 (6)	3–19 / 4–19	1.175 (8)	3–19 / 5–19	−0.114**	−0.125*	7.2
EWMA (default)	0.423 (7)	3–19 / 4–19	1.199 (8)	3–19 / 5–19	−0.117**	−0.130*	7.5
Only ambiguity HP (10^5)	0.394 (7)	2–19 / 4–19	1.038 (8)	3–19 / 5–19	−0.090*	−0.098*	6.3
Huber	0.364 (6)	2–19 / 4–19	0.919 (8)	3–19 / 6–19	−0.080*	−0.117*	5.4
Kalman smoothed	0.346 (9)	5–19 / 8–19	0.978 (10)	5–19 / 10–19	−0.093**	−0.115*	4.9
3-month moving average	0.288 (8)	5–19 / 9–19	0.808 (10)	5–19 / 12–19	−0.064*	−0.089*	3.5
Median	0.390 (6)	2–19 / 3–19	0.997 (8)	3–19 / 5–19	−0.085*	−0.116*	6.1
Mean (re-aggregated)	0.387 (7)	2–19 / 4–19	1.023 (8)	3–19 / 6–19	−0.091**	−0.108*	6.1
Not detrended	0.111 (13)	16–19 / none†	−0.481 (1)	1–2 / 1	−0.064*	−0.189**	0.6
Not detrended, trend in global eqs.	0.112 (7)	none / none†	−0.459 (1)	1–2 / 1	−0.090**	−0.163**	0.5
Not detrended, trends in all eqs.	0.294 (7)	3–19 / 4–19	0.727 (8)	4–19 / 7–19	−0.090**	−0.099**	3.8
Winsorized	0.408 (7)	2–19 / 4–19	1.073 (8)	3–19 / 5–19	−0.087*	−0.106*	6.9
Other data measures							
Baa minus 20-year Treasury	0.367 (7)	2–19 / 4–19	0.967 (9)	3–19 / 6–19	−0.075*	−0.112*	5.7
Real rate, 4-lag inflation	0.391 (7)	3–19 / 4–19	1.026 (8)	3–19 / 6–18	−0.092**	−0.106*	6.2
Trade balance not detrended	0.380 (7)	3–19 / 4–19	1.001 (8)	3–19 / 6–19	−0.091**	−0.104*	6.0
Trend terms							
Trend in global equations	0.386 (7)	2–19 / 3–19	1.021 (8)	3–19 / 5–19	−0.090**	−0.108**	6.1
Trends in all equations	0.294 (7)	3–19 / 4–19	0.727 (8)	4–19 / 7–19	−0.090**	−0.099**	3.8

Note: Columns as in Table C.14. Filters are applied to GDP, investment, trade balance/GDP and ambiguity. Ambiguity measures are alternative monthly-to-quarterly aggregations of the index, detrended unless stated. † The 68 percent GDP band also excludes zero with the opposite sign: BK (2, 250) (h = 5-11, 17-19); BK (6, 32) (h = 1-2); CF (6, 32) (h = 4-7, 18-19); not detrended (h = 1); not detrended, trend in global equations (h = 1-2).

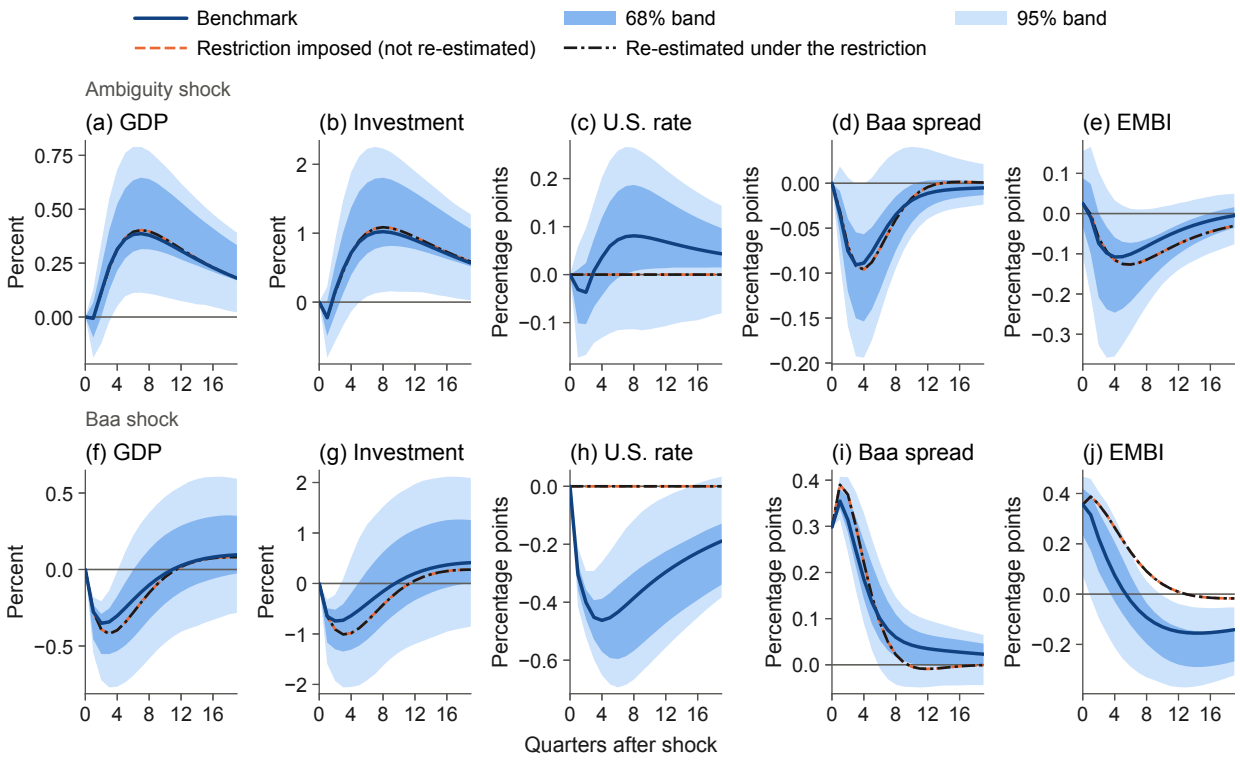


Figure C.45: IRFs when the U.S. interest rate follows its own AR(2)

Note: As Figure C.44.

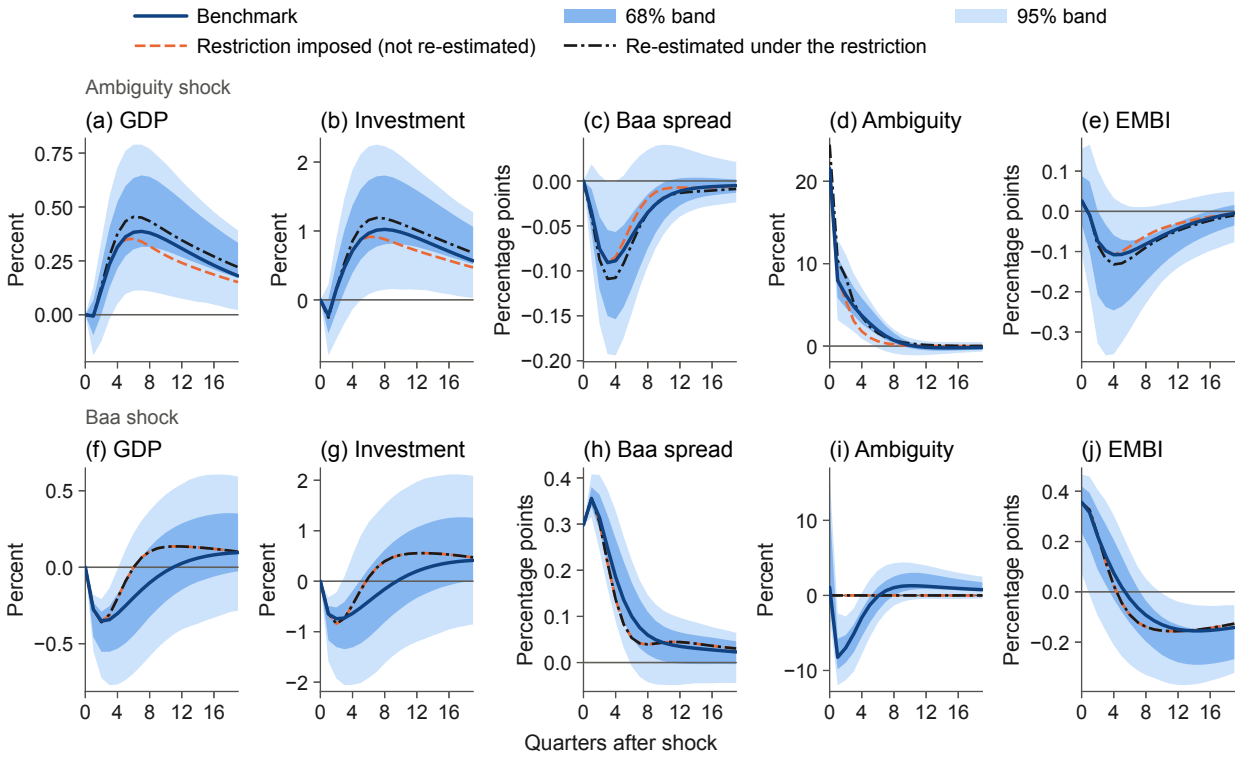


Figure C.46: IRFs when ambiguity is exogenous to the U.S. interest rate and the BAA spread

Note: As Figure C.44.

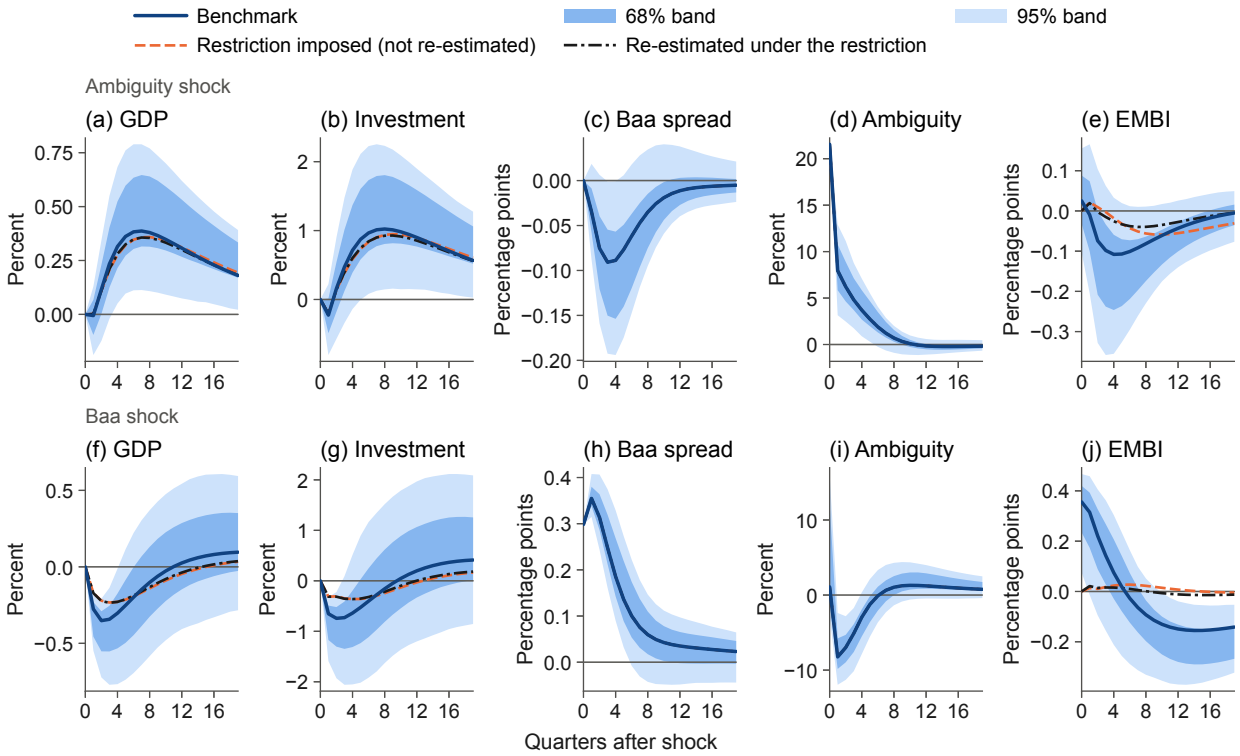


Figure C.47: IRFs when the country spread does not depend on shock block variables

Note: As Figure C.44.

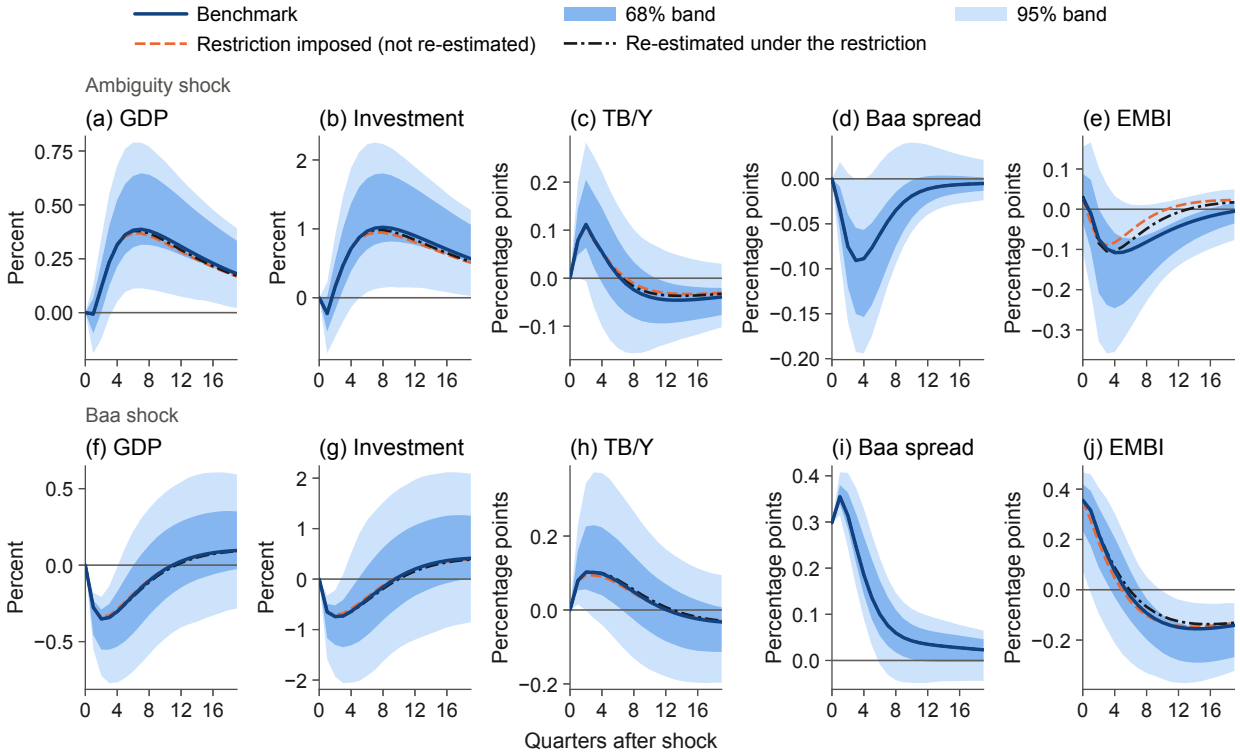


Figure C.48: IRFs when the country spread does not depend on real variables

Note: As Figure C.44.

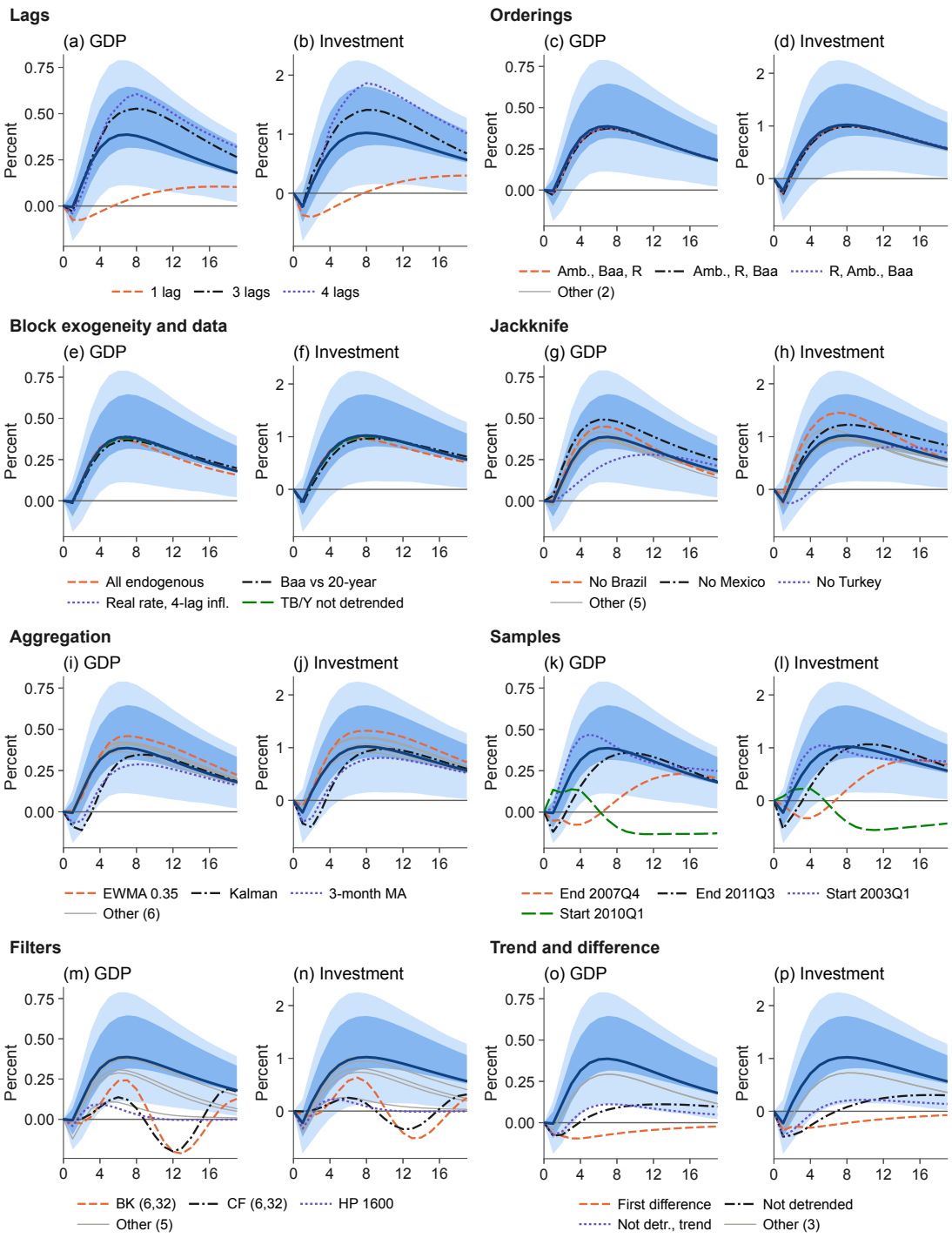


Figure C.49: Robustness of the responses to an ambiguity shock

Note: Responses of EME GDP and investment (percent) to a one standard deviation ambiguity shock. Solid line and shaded areas in every panel: benchmark with 68 and 95 percent bands. Lines: point estimates of the alternative specifications of each family: lag order; the five other orderings of the global block (R = U.S. real rate); no block exogeneity (all endogenous) and rebuilt data series; dropping one panel unit at a time; monthly-to-quarterly aggregations of the ambiguity index; sample periods; filters; ambiguity not detrended, with linear trends in the global (and country) equations, or in first differences. Where a family has more than four members, the three that deviate most from the benchmark are drawn in color and the others in gray.

C.12 Larger samples of countries, advanced economies and the pandemic

Figure C.53 shows the benchmark model estimated on eleven EMEs (Thailand is not included because a comparable EMBI series is unavailable), Figure C.54 on twelve small open advanced economies, and Figure C.55 the remaining variants: the eleven-country sample without Ecuador or with only the units observed for at least 40 quarters, the six countries rebuilt from the extended dataset, and the extension of the sample to 2021:Q3 with and without controls for the pandemic quarters. The pandemic-extended specification includes separate indicators for 2020:Q1, 2020:Q2, and 2020:Q3 to account for the unusually different collapse and rebound across those quarters; a single pooled pandemic indicator does not capture these quarterly movements adequately. Pooling the emerging and advanced economies mixes an EMBI spread with a domestic yield in the same equation and gives an intermediate response that we do not interpret.

Country-level responses. Figure C.50 and Table C.17 estimate the benchmark model unit by unit, with the global block common to all units, for the eleven EMEs. The GDP response to an ambiguity shock is positive over quarters 4 to 12 in ten of the eleven (Mexico is flat) and the EMBI spread falls in ten of eleven, while a global financial risk shock lowers GDP in all eleven and raises the spread in all eleven (Figure C.51). The mean-group estimator of Pesaran and Smith (1995), the simple average of the country responses, gives a GDP response of 0.30 percent at eight quarters against 0.29 percent for the pooled estimator, and the pooled response lies inside the 95 percent mean-group band at every horizon. The mean-group interval based on the cross-country dispersion of the estimates excludes zero at the 95 percent level from the sixth quarter; an interval that pools the bootstrap draws of the eleven country systems is wider, because Argentina's first sample has only 30 quarters, and excludes zero at the 68 percent level, and without that unit the mean group of the remaining ten EMEs is significant at the 95 percent level from the fifth quarter under the dispersion-based interval and from the fourth under the bootstrap interval. The responses differ in

size: Turkey (peak 1.0 percent), South Africa, Chile and Ecuador respond most, Brazil, Malaysia and the Philippines least, so that estimators that weight countries by precision (Swamy, 1970) or by GDP give 0.11 to 0.20 percent at eight quarters, and a hierarchical Bayesian panel VAR with an exchangeable prior (Jarociński, 2010) gives a common-mean response of 0.26 to 0.30 percent with a common shock variance and 0.14 to 0.22 percent with country-specific variances, which weight the low-volatility economies more. Scaled by each country's average spread, the EMBI falls by 2 to 3 percent of its level in the typical EME. The pooled result does not depend on Argentina, and Turkey is the most influential country in the six-country panel (without it the GDP peak is 0.28 percent, significant at the 68 percent level) but not in the eleven-country panel (0.22 percent, significant at the 95 percent level).

Table C.17: Country-level responses to an ambiguity shock and their averages

		GDP, $h = 7$			Investment, $h = 8$			EMBI spread, $h = 4$		
	T	Resp.	68%	95%	Resp.	68%	95%	Resp.	68%	95%
Benchmark countries										
Argentina I†	30	0.56	[−0.70,2.77]	[−4.45,5.30]	2.44	[−0.8,8.6]	[−12.1,15.7]	0.32	[−0.19,0.76]	[−0.80,1.48]
Brazil	100	0.04	[−0.18,0.27]	[−0.46,0.51]	−0.52	[−1.3,0.1]	[−2.1,0.9]	0.15	[−0.11,0.32]	[−0.39,0.50]
Mexico	103	−0.12	[−0.31,0.10]	[−0.54,0.33]	−0.10	[−0.6,0.3]	[−1.2,0.9]	0.11	[−0.06,0.17]	[−0.21,0.28]
Peru	91	0.23	[0.08,0.49]	[−0.19,0.72]	1.00	[0.4,2.3]	[−0.9,3.3]	−0.07	[−0.18,0.01]	[−0.29,0.11]
South Africa‡	71	0.42	[0.27,0.70]	[0.01,0.90]	1.31	[0.8,2.2]	[0.0,2.8]	−0.14	[−0.24,−0.11]	[−0.29,−0.04]
Turkey	82	0.67	[0.63,1.20]	[0.33,1.45]	1.94	[1.7,3.7]	[0.5,4.6]	−0.19	[−0.31,−0.11]	[−0.43,0.02]
Additional EMEs										
Chile	83	0.37	[0.20,0.76]	[−0.08,1.04]	1.20	[0.8,2.4]	[−0.1,3.2]	−0.07	[−0.12,−0.05]	[−0.15,−0.01]
Colombia	60	0.27	[0.24,0.66]	[−0.06,0.90]	0.72	[0.6,1.7]	[−0.2,2.3]	−0.08	[−0.17,−0.06]	[−0.22,0.00]
Ecuador	80	0.38	[0.13,0.61]	[−0.17,0.83]	0.83	[0.0,1.6]	[−1.2,2.3]	−0.35	[−0.98,0.27]	[−1.65,1.03]
Malaysia	40	0.03	[−0.02,0.16]	[−0.10,0.39]	−0.99	[−2.0,0.2]	[−2.8,3.6]	−0.04	[−0.11,0.02]	[−0.17,0.13]
Philippines	103	0.05	[−0.11,0.21]	[−0.28,0.39]	−0.21	[−1.0,0.3]	[−1.7,1.1]	−0.07	[−0.18,0.01]	[−0.29,0.13]
Summary estimates										
Mean group, 6	6 c.	0.30	[0.17,0.42]	[0.05,0.54]	1.01	[0.5,1.5]	[0.1,1.9]	0.03	[−0.05,0.11]	[−0.13,0.19]
Mean group, 11	11 c.	0.26	[0.19,0.34]	[0.12,0.41]	0.69	[0.4,1.0]	[0.1,1.3]	−0.04	[−0.09,0.01]	[−0.15,0.07]
Mean group, 10	10 c.	0.23	[0.16,0.31]	[0.09,0.38]	0.52	[0.2,0.8]	[−0.1,1.1]	−0.08	[−0.12,−0.03]	[−0.16,0.01]
MG, 11, block draws	11 c.	0.26	[0.20,0.55]	[−0.14,0.78]	0.69	[0.5,1.5]	[−0.5,2.2]	−0.04	[−0.14,0.02]	[−0.23,0.11]
Pooled, benchmark	7 u.	0.39	[0.32,0.65]	[0.11,0.79]	1.02	[0.8,1.8]	[0.1,2.2]	−0.11	[−0.25,−0.06]	[−0.36,0.03]
Pooled, 11 EMEs	12 u.	0.29	[0.22,0.50]	[0.06,0.61]	0.78	[0.6,1.4]	[0.1,1.7]	−0.15	[−0.32,−0.09]	[−0.43,0.04]
Swamy, 6	6 c.	0.20	[0.10,0.30]	[0.01,0.40]	0.40	[0.1,0.7]	[−0.2,1.0]	−0.09	[−0.13,−0.05]	[−0.17,−0.01]
Swamy, 11	11 c.	0.13	[0.08,0.19]	[0.03,0.24]	0.42	[0.2,0.6]	[0.0,0.8]	−0.07	[−0.10,−0.05]	[−0.12,−0.03]
GDP-wtd. MG, 6	6 c.	0.19	[0.09,0.46]	[−0.19,0.67]	0.44	[0.1,1.2]	[−0.7,1.8]	0.06	[−0.08,0.12]	[−0.19,0.21]
GDP-wtd. MG, 11	11 c.	0.18	[0.13,0.42]	[−0.09,0.58]	0.37	[0.2,1.0]	[−0.5,1.5]	0.03	[−0.08,0.07]	[−0.17,0.15]
HB, country var.	12 u.	0.17	[0.07,0.27]	[−0.03,0.39]	0.35	[0.1,0.7]	[−0.3,1.0]	−0.09	[−0.13,−0.06]	[−0.16,−0.03]
HB, common var.	12 u.	0.27	[0.13,0.41]	[−0.01,0.59]	0.70	[0.3,1.2]	[−0.2,1.8]	−0.08	[−0.19,0.05]	[−0.32,0.18]

Note: Unit-by-unit estimates of the benchmark specification: responses of GDP at $h = 7$ and investment at $h = 8$ (percent) and of the EMBI spread at $h = 4$ (percentage points) to a one standard deviation ambiguity shock, with 68 and 95 percent bands from 10,000 replications of the moving-block residual bootstrap applied unit by unit, Hall intervals. Summary rows: mean group (Pesaran and Smith, 1995), pooled LSDV, precision-weighted (Swamy, 1970) and GDP-weighted averages, and the hierarchical Bayesian common mean with a common or unit-specific shock variances (posterior medians and 95 percent credible sets).

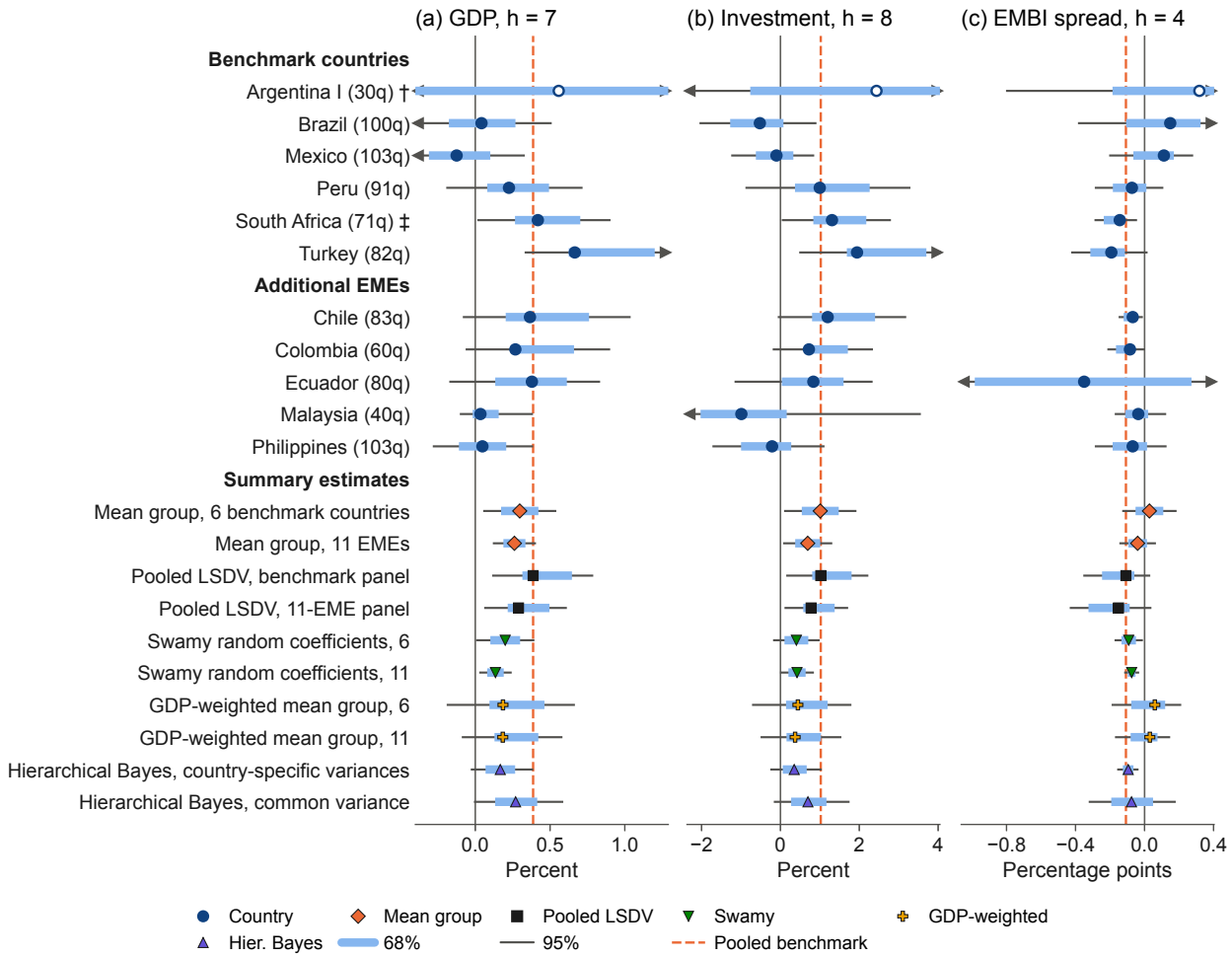


Figure C.50: Country-level responses to an ambiguity shock

Note: Responses of GDP at $h = 7$ and investment at $h = 8$ (percent) and of the EMBI spread at $h = 4$ (percentage points) to a one standard deviation ambiguity shock, estimated unit by unit with the benchmark specification (the global block common to all units), with 68 and 95 percent bands; sample length in quarters in parentheses. Summary rows: mean-group averages over the units shown, with intervals from the cross-country dispersion of the estimates (Pesaran and Smith, 1995), pooled LSDV estimates with block-bootstrap Hall bands, and the hierarchical Bayesian common mean with its credible set. The dashed line is the six-country pooled benchmark. †: 30 quarters; ‡: from 2002:Q2.

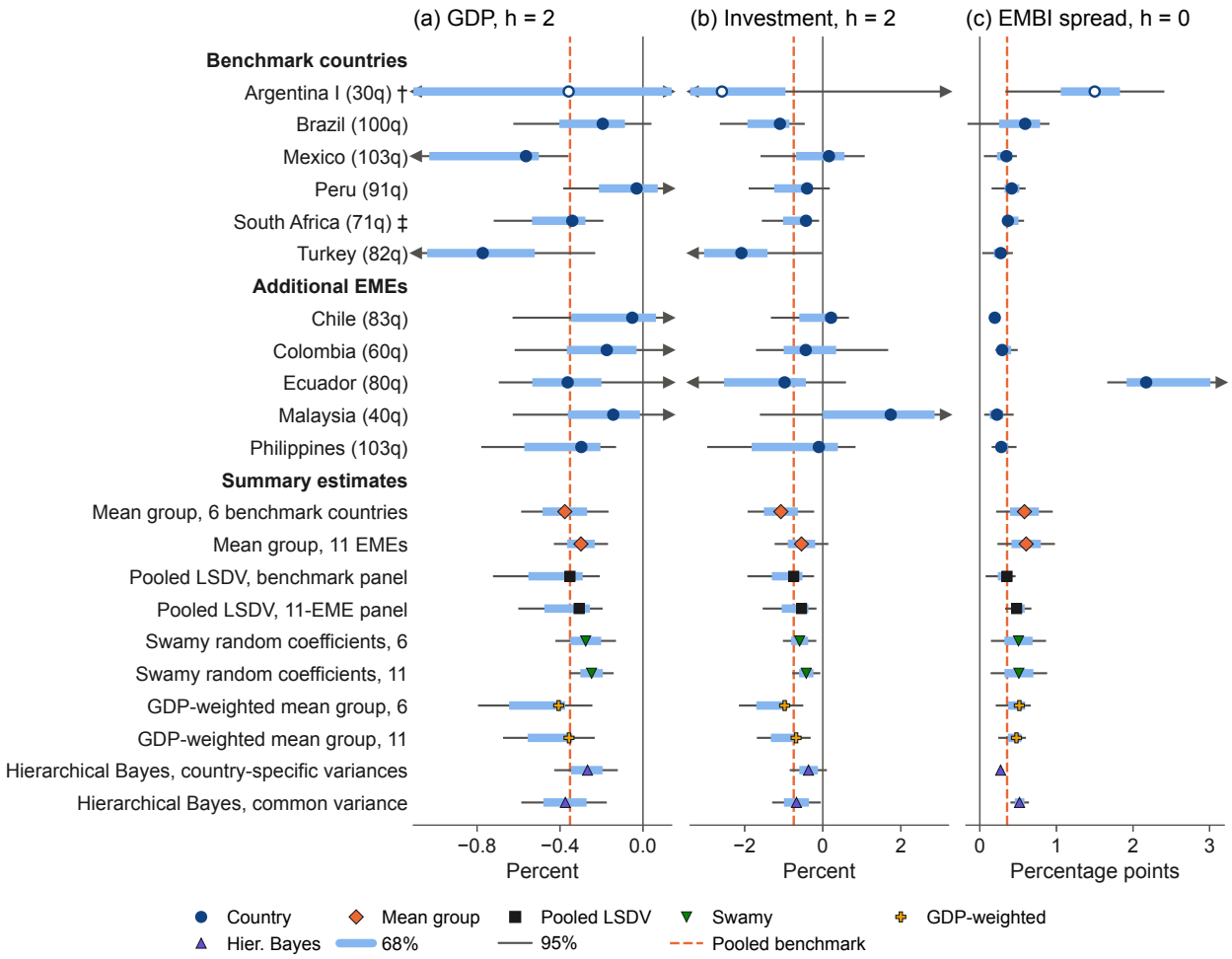


Figure C.51: Country-level responses to a global financial risk shock

Note: As Figure C.50, for a one standard deviation BAA spread shock.

C.13 Heterogeneity across EMEs

Figure C.52 and Table C.18 ask whether the responses to the two global shocks differ with a country's exposure to foreign investors, in panel local projections for the eleven EMEs that interact the shock with the lagged value of an exposure measure: portfolio debt and equity liabilities relative to GDP, U.S. holdings of the country's securities and the U.S. share among foreign holders, dollar-denominated cross-border credit, capital-account openness, the average level of the EMBI spread, the current account and reserves. No measure of exposure to foreign portfolio investors amplifies the GDP or investment response to the ambiguity shock in a way that survives dropping Ecuador or controlling for the sample trend (Spearman rank correlations between the country GDP responses and the exposure measures lie between -0.50 and 0.32). Spreads fall more where they are higher (rank correlation -0.65), but output and investment do not respond more once Ecuador's spreads and the trend are accounted for. Economies with larger current-account deficits respond more to both shocks, booming more after an ambiguity shock (rank correlation -0.63) and contracting more after a global financial risk shock (0.60 for GDP, 0.77 for investment), and the contraction after a global financial risk shock is smaller where reserves are higher and where foreign equity investors hold more of the country's equity. The null for exposure to U.S. investors is what the mechanism of Section 3.3.3 implies: the easing reaches EMEs through global financial conditions and bank lending rather than through bilateral U.S. purchases. With eleven countries these are cross-sectional patterns, and we do not build on them.

C.14 Monthly identification of the ambiguity shock

This section documents the monthly identification used in Section 3.3.5. We use the monthly ambiguity index of Izhakian (2020) for 1993:M2–2019:M12, in the same transformation as our quarterly variable (the log of the square root of the index, linearly detrended; the quarterly variable is the log of the square root of the quarterly average of the index). The monthly data set adds the U.S. real rate (the 3-month Treasury bill rate deflated with expected inflation), the BAA spread, the log VIX, the log of the Jurado et al. (2015) financial uncertainty index

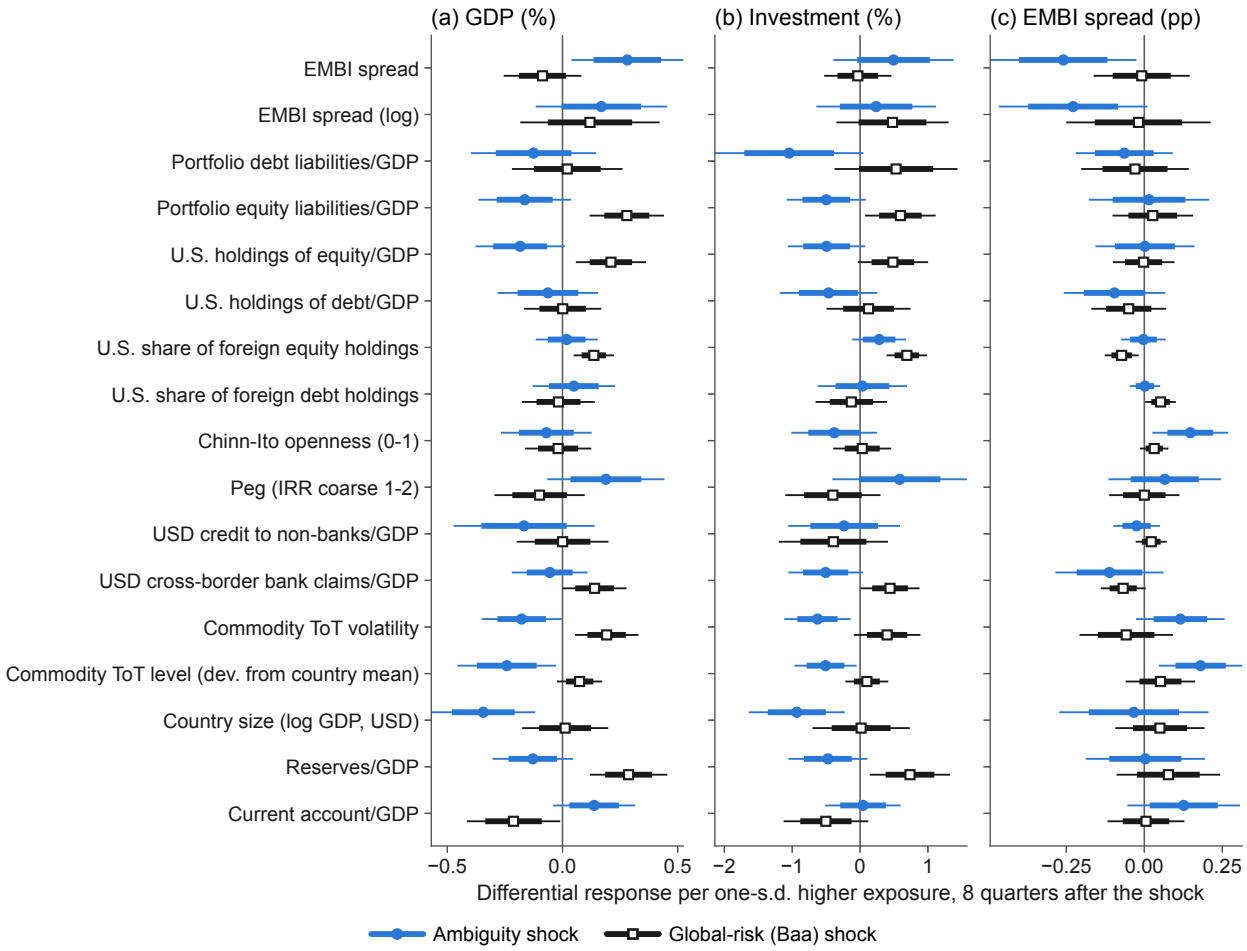


Figure C.52: Exposure interactions in the EME responses

Note: Coefficients on the interaction of the shock with the lagged, standardized exposure measure in panel local projections of GDP, investment and the EMBI spread on the benchmark ambiguity shock and on the global financial risk shock, eleven EMEs, unit fixed effects, the exposure level as an additional control; Driscoll and Kraay (1998) 68 and 90 percent bands.

Table C.18: Exposure interactions: summary

	Ambiguity shock			Global-risk (Baa) shock			
Exposure (one s.d.)	GDP	Investment	EMBI	GDP	Investment	EMBI	1 s.d.
Response at $h = 8$ without interaction							
No interaction	0.17	0.72*	−0.18*	−0.21*	−0.06	−0.27**	727 obs.
	(0.21)	(0.64)	(0.14)	(0.18)	(0.57)	(0.12)	
Differential response per one s.d. of lagged exposure, $h = 8$							
EMBI spread	0.28**	0.49	−0.26**	−0.09	−0.03	−0.01	4.34
	(0.15)	(0.54)	(0.14)	(0.10)	(0.30)	(0.09)	
EMBI spread (log)	0.17	0.24	−0.23*	0.12	0.48	−0.02	0.75
	(0.17)	(0.54)	(0.14)	(0.18)	(0.50)	(0.14)	
Portfolio debt liabilities/GDP	−0.13	−1.05*	−0.06	0.02	0.53	−0.03	6.85
	(0.17)	(0.67)	(0.09)	(0.15)	(0.55)	(0.10)	
Portfolio equity liabilities/GDP	−0.16*	−0.50*	0.02	0.28**	0.59**	0.03	8.78
	(0.12)	(0.35)	(0.12)	(0.10)	(0.32)	(0.08)	
U.S. equity holdings/GDP	−0.18*	−0.49*	0.00	0.21**	0.48*	0.00	4.10
	(0.12)	(0.35)	(0.10)	(0.09)	(0.32)	(0.06)	
U.S. debt holdings/GDP	−0.06	−0.46*	−0.10	0.00	0.13	−0.05	1.94
	(0.13)	(0.44)	(0.10)	(0.10)	(0.38)	(0.07)	
U.S. share of foreign equity holdings	0.02	0.28*	0.00	0.14**	0.69**	−0.07**	14.40
	(0.08)	(0.24)	(0.04)	(0.05)	(0.18)	(0.03)	
U.S. share of foreign debt holdings	0.05	0.03	0.00	−0.02	−0.13	0.05**	17.13
	(0.11)	(0.40)	(0.03)	(0.10)	(0.32)	(0.03)	
Chinn-Ito openness	−0.07	−0.38	0.15**	−0.02	0.03	0.03*	0.29
	(0.12)	(0.38)	(0.07)	(0.09)	(0.26)	(0.03)	
Peg (IRR coarse 1–2)	0.19*	0.58	0.07	−0.10	−0.40	0.00	0.46
	(0.15)	(0.60)	(0.11)	(0.12)	(0.43)	(0.07)	
USD credit to non-banks/GDP	−0.17	−0.24	−0.02	0.00	−0.39	0.02	8.60
	(0.19)	(0.50)	(0.05)	(0.12)	(0.49)	(0.03)	
USD cross-border bank claims/GDP	−0.06	−0.51*	−0.11*	0.14**	0.44**	−0.07*	4.22
	(0.10)	(0.33)	(0.11)	(0.08)	(0.27)	(0.04)	
Commodity ToT volatility	−0.18**	−0.63**	0.12*	0.19**	0.40*	−0.06	0.71
	(0.11)	(0.30)	(0.09)	(0.08)	(0.30)	(0.09)	
Commodity ToT gap	−0.24**	−0.51**	0.18**	0.07*	0.10	0.05	3.33
	(0.13)	(0.28)	(0.08)	(0.06)	(0.19)	(0.07)	
Country size (log GDP)	−0.34**	−0.93**	−0.03	0.01	0.02	0.05	1.12
	(0.14)	(0.43)	(0.15)	(0.11)	(0.44)	(0.09)	
Reserves/GDP	−0.13*	−0.47*	0.00	0.29**	0.74**	0.08	8.25
	(0.11)	(0.35)	(0.12)	(0.10)	(0.36)	(0.10)	
Current account/GDP	0.14*	0.04	0.13*	−0.21**	−0.50*	0.01	3.10
	(0.11)	(0.34)	(0.11)	(0.12)	(0.38)	(0.07)	

Note: Interaction coefficients at eight quarters (response per standard deviation of the exposure measure) from the local projections of Figure C.52, with standard errors, and Spearman rank correlations between the country-level responses of Table C.17 and the country averages of each exposure measure (exact p-values).

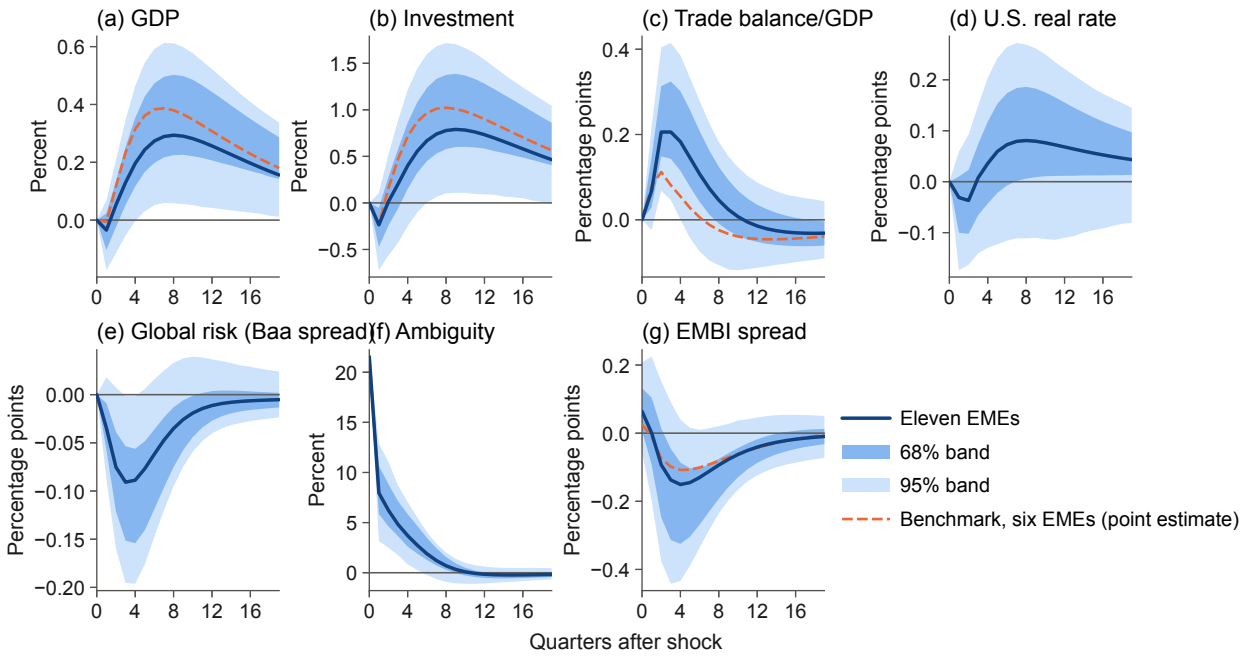


Figure C.53: IRFs to one std. dev. shock to ambiguity with larger sample

Note: Panel SVAR when the sample of countries expands to include Argentina, Brazil, Chile, Colombia, Ecuador, Malaysia, Mexico, Peru, Philippines, South Africa, and Turkey (12 panel units, 1994:Q2-2019:Q4). Solid line and shaded areas: point estimate with 68 and 95 percent bands; dashed line: six-country benchmark.

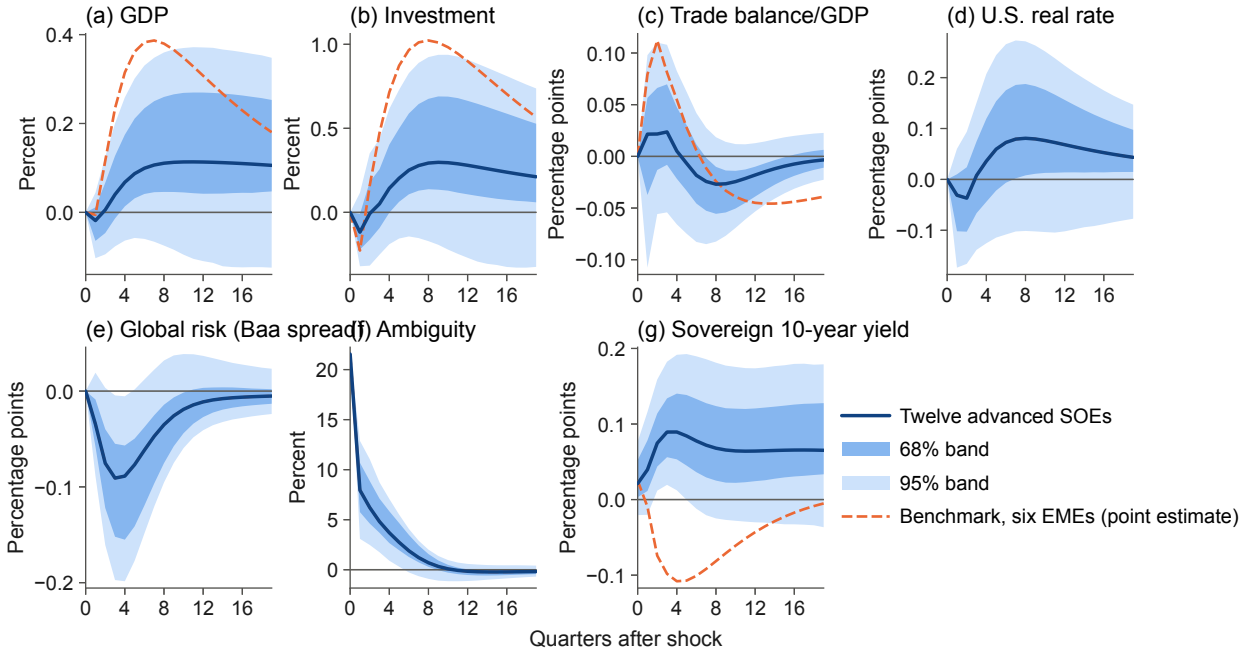


Figure C.54: IRFs to one std. dev. shock to ambiguity in twelve advanced small open economies

Note: Benchmark model estimated on Australia, Austria, Belgium, Canada, Denmark, Finland, Israel, the Netherlands, New Zealand, Norway, Sweden and Switzerland, with the 10-year sovereign yield in place of the EMBI spread and the real trade balance. Solid line and shaded areas: point estimate with 68 and 95 percent bands; dashed line: six-country benchmark.

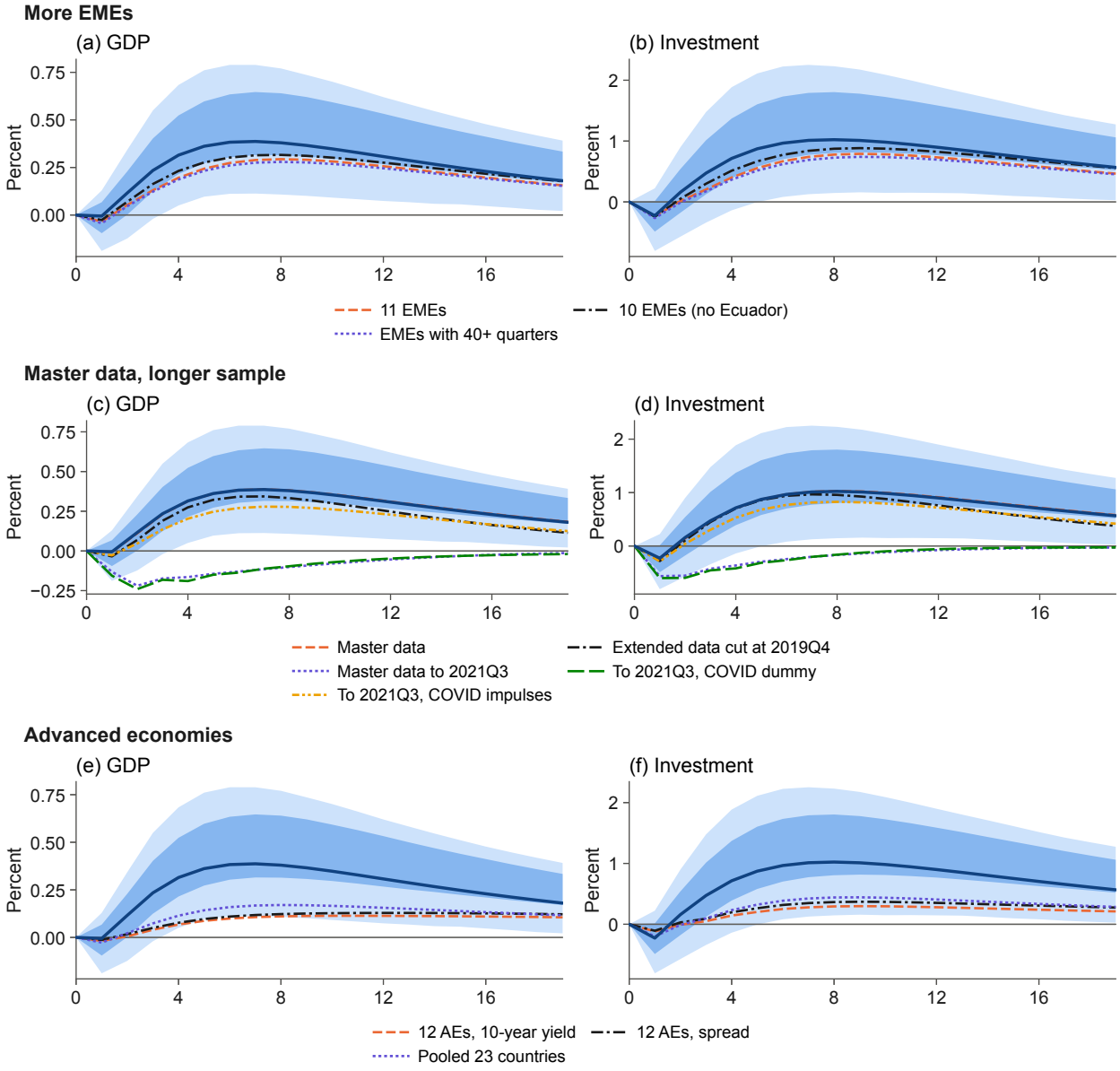


Figure C.55: Responses to an ambiguity shock in other samples

Note: Responses of GDP and investment (percent) to a one standard deviation ambiguity shock. Solid line and shaded areas: six-country benchmark with 68 and 95 percent bands. Lines: point estimates of the benchmark model on eleven EMEs, without Ecuador, or with only units observed for at least 40 quarters; the six countries rebuilt from the extended dataset, extended to 2021:Q3 without pandemic controls, with one dummy for 2020:Q1-Q3, or with one dummy per pandemic quarter, and the extended data cut back at 2019:Q4; twelve advanced small open economies with the 10-year sovereign yield or its spread over the U.S. 10-year yield in the country-spread slot; and all 23 countries pooled.

at the one-month horizon, the excess bond premium of Gilchrist and Zakrajšek (2012), the NASDAQ composite return and, in one system, the 10-year Treasury yield and the term premium of Adrian et al. (2013). The Bayesian and Hannan-Quinn criteria select two lags in every system. We consider four Cholesky orderings: M1 places ambiguity last, after the real rate, the BAA spread, the VIX, financial uncertainty, the excess bond premium and the stock return, so that the ambiguity shock is orthogonal within the month to all of them; M2 places ambiguity after the real rate and the BAA spread but before the VIX, financial uncertainty, the excess bond premium and the return, the analogue of our quarterly ordering; M3 is a minimal system with the real rate, the BAA spread, the VIX and ambiguity; M4 adds the 10-year yield and the term premium to M1, ordered before ambiguity. Bands come from 2,000 replications of a recursive-design wild bootstrap. Each monthly structural shock is summed within the quarter and standardized; Figure C.59 plots the M1 shock and its quarterly aggregate against the benchmark shock, with which it has a correlation of 0.69.

Figures C.60 to C.62 report the responses of every variable of the monthly systems to a one standard deviation ambiguity shock. In M1 the shock raises the VIX by 2.7 percent after two months (significant at the 90 percent level in months one to four) and lowers financial uncertainty, while the BAA spread and the excess bond premium do not move; in M2, where ambiguity precedes the VIX, the VIX falls by 6.5 percent on impact and the BAA spread and the excess bond premium rise moderately thereafter; in M4 the 10-year yield falls by 0.05 percentage points. Figures C.63 and C.64 report the panel and U.S. local projections of Section 3.3.5 for all outcomes and for the M1, M2 and M3 shocks alongside the benchmark quarterly shock, and Table C.22 summarizes them. Figures C.65 and C.66 report the external-instrument estimates: the aggregated monthly shock is regressed on the reduced-form residuals of the benchmark PSVAR (long-sample residuals for the global equations, panel residuals pooled by date for the country equations), the resulting impact vector is scaled to the benchmark impact on ambiguity (21.6 log points) and propagated with the benchmark companion matrix, and bands come from 2,000 replications of a moving-block bootstrap with blocks of eight quarters in which the first stage is re-estimated in every draw. The first-stage F statistics are 101 for M1 (70 with Newey-West standard errors), 226 for

M2 and 109 for M3, and the instruments explain 50, 70 and 52 percent of the variance of the ambiguity innovation.

The identified shocks. Figure C.56 plots the M1 shock with its largest realizations labelled, and Tables C.19 and C.20 list the ten largest positive and negative monthly and quarterly shocks with the period's stock return and VIX change. The four largest positive shocks are crash months with annualized realized volatility of 34 to 83 percent (October 2008, October 1997, July 2002, August 1998); December 2010 and January 2001 follow, and the remaining large positive shocks are quiet months. The largest negative shocks are moderate sell-offs (December 2018, January 2016, January 2009, January 2015, the taper tantrum of June 2013). Classifying months as risk-on (positive stock return and falling VIX) or risk-off, large positive shocks are risk-on months as often as the average month (47 against 46 percent; Fisher exact p-value 1.00), while large negative shocks are disproportionately risk-off (60 against 30 percent, p-value 0.02). The contemporaneous S&P 500 return and VIX change explain 2 percent of the variance of the monthly shock, and fifteen candidate drivers (news-based policy uncertainty, geopolitical risk, equity market volatility, realized variance, skewness and kurtosis, the VVIX, the variance risk premium, the global factor, two monetary policy surprise series and three survey sentiment measures) explain 14 percent jointly; the shock loads modestly on realized kurtosis, on falls in the variance risk premium and on the Jarociński and Karadi (2020) policy surprise. The quarterly benchmark shock, which is not orthogonalized to the VIX, loads in addition on falls in realized variance; about a third of its variance is explained by contemporaneous second- and higher-moment changes.

Controls for risk appetite, the global financial cycle and monetary policy. Table C.21 and Figures C.57 and C.58 add to M1, ordered before ambiguity, one control at a time: the risk aversion and economic uncertainty indices of Bekaert et al. (2022), the global factor in risky asset prices of Miranda-Agrippino and Rey (2022), the monetary policy surprises of Bauer and Swanson (2023) and the policy and central bank information shocks of Jarociński and Karadi (2020), the variance risk premium, realized variance and realized

skewness of daily S&P 500 returns, the news-based policy uncertainty and equity market volatility indices of Baker et al. (2016), the geopolitical risk index of Caldara and Iacoviello (2022) and IMF commodity prices. Each leaves the identified shock nearly unchanged (correlation 0.94 to 1.00 with the M1 shock; first-stage F statistics of 80 to 101 for the benchmark innovation) and the panel local projection of EME GDP at 0.45 to 0.59 percent, significant at the 90 percent level; the first principal component of the twelve controls, ordered first, gives 0.48 percent. After the shock, risk aversion rises by 1.6 percent and the global factor dips slightly. The monetary policy shocks are orthogonal to the ambiguity shock, lagged EME growth, investment, spreads and trade balances do not predict it (F-test p-values of 0.45 to 0.81), and the sample extended to 2021:M9 with indicators for the pandemic months gives the same shock (correlation 0.997) and a GDP response of 0.55 percent. Entering all twelve controls at once ordered before ambiguity lowers the local-projection response to 0.15 to 0.25 percent; the loss comes entirely from the risk aversion index and the variance risk premium, which are correlated 0.85 and 0.89 with the log VIX, being placed before ambiguity together with financial uncertainty, the excess bond premium and returns. The same two variables ordered after ambiguity leave the response at 0.41 to 0.44 percent, their lags added to the local projections at 0.43 percent, lags of all twelve controls at 0.69 percent, and the all-controls shock used as an external instrument in the benchmark PSVAR gives GDP responses of 0.30 to 0.36 percent. The question is therefore how the within-month component that ambiguity shares with near-volatility measures is attributed, not whether the expansion follows.

Table C.19: The largest monthly ambiguity shocks

Rank	Month	Shock (s.d.)	ΔVIX	S&P 500 (%)	Regime	Event (J6 reading of the public record)
Largest positive shocks						
1	2008M10	7.27	20.50	−18.6	risk-off	Post-Lehman crash: TARP enacted (3 Oct), coordinated central-bank rate cuts (8 Oct), VIX peaks near 80; S&P 500 -18.6%
2	1997M10	4.15	12.18	−3.5	risk-off	Asian crisis reaches Hong Kong (20-23 Oct); 27 Oct U.S. 'mini-crash' triggers NYSE circuit breakers for the first time
3	2002M7	3.44	6.63	−8.2	risk-off	WorldCom fraud and bankruptcy (21 Jul), S&P 500 bear-market low (23 Jul), Sarbanes-Oxley signed (30 Jul)
4	1998M8	3.28	19.48	−15.8	risk-off	Russian default and ruble devaluation (17 Aug), LTCM losses; S&P 500 -15.8%
5	2010M12	3.02	−5.79	6.3	risk-on	QE2 purchases under way, Bush tax-cut extension deal (17 Dec) after the Irish bailout (28 Nov); S&P 500 +6.3%, VIX falls
6	2001M1	2.17	−4.83	3.4	risk-on	Two 50 bp Fed cuts (3 Jan intermeeting, 31 Jan); NYSE and AMEX complete the switch to decimal pricing (29 Jan)
7	1993M10	1.77	−1.53	1.9	risk-on	No major market event; low-volatility advance (NAFTA debate)
8	2019M12	1.70	1.16	2.8	mixed	U.S.-China 'phase one' trade deal announced (13 Dec), UK election; Fed on hold
9	2009M12	1.69	−2.83	1.8	risk-on	Recovery rally after Dubai World standstill (late Nov); Greek downgrades (8 Dec)
10	2004M6	1.65	−1.16	1.8	risk-on	Fed begins tightening cycle (30 Jun, first hike since 2000); Iraq handover
Largest negative shocks						
1	2018M12	−2.75	7.35	−9.6	risk-off	Q4-2018 sell-off: Fed hike and balance-sheet 'autopilot' remark (19 Dec), government shutdown (22 Dec); S&P 500 -9.6%
2	2016M1	−2.72	1.99	−5.2	risk-off	China growth scare and renminbi weakness, oil below $30; S&P 500 -5.2%
3	2009M1	−2.42	4.84	−9.0	risk-off	Bank losses (Citi, Bank of America aid), recession data, stimulus debate; S&P 500 -9.0%
4	2015M1	−2.21	1.77	−3.2	risk-off	Oil collapse, SNB drops franc cap (15 Jan), ECB announces QE (22 Jan), Greek election (25 Jan)
5	2013M6	−1.77	0.56	−1.5	risk-off	Taper tantrum after Bernanke press conference (19 Jun); EM sell-off
6	2019M5	−1.67	5.59	−6.8	risk-off	U.S.-China escalation: tariffs raised to 25% (10 May), Huawei blacklisted; S&P 500 -6.8%
7	2012M5	−1.67	6.91	−6.5	risk-off	Euro crisis: Greek election and Grexit fears (6 May), Bankia rescue; S&P 500 -6.5%
8	2017M1	−1.65	−2.05	1.8	risk-on	Trump inauguration (20 Jan); low-volatility rally
9	2019M8	−1.57	2.86	−1.8	risk-off	New China tariffs announced (1 Aug), yuan breaks 7 and 'currency manipulator' label (5 Aug), 2s10s inversion (14 Aug)
10	2015M9	−1.56	−3.93	−2.7	mixed	After the August China devaluation and 24 Aug flash crash; Fed holds citing global risks (17 Sep), Glencore slump

Note: The ten largest positive and ten largest negative realizations of the M1 shock (standard deviations), with the month's S&P 500 return, the change in the VIX, the classification of the month as risk-on (positive return and falling VIX), risk-off (negative return and rising VIX) or mixed, and the event.

Table C.20: The largest quarterly ambiguity shocks

Rank	Quarter	Shock (s.d.)	ΔVIX	S&P 500 (%)	Regime	Event (J6 reading of the public record)
Largest positive shocks						
1	2008Q4	3.32	0.61	−25.6	risk-off	Lehman aftermath: TARP, coordinated rate cuts, VIX near 80, QE1 announced
2	2010Q4	1.56	−5.95	9.7	risk-on	QE2 announced (3 Nov), Irish bailout, Bush tax-cut extension; strong rally
3	2016Q2	1.47	1.68	1.9	mixed	Brexit referendum (23 Jun); Fed on hold
4	2012Q1	1.41	−7.90	11.3	risk-on	ECB second LTRO (29 Feb), Greek debt exchange (Mar); strong rally, VIX falls
5	2017Q3	1.35	−1.67	3.9	risk-on	Low-volatility rally; North Korea tensions (Aug), hurricanes Harvey and Irma
6	2019Q4	1.34	−2.46	8.2	risk-on	Fed third cut (30 Oct), repo-market operations, U.S.-China phase-one deal
7	2006Q4	1.32	−0.42	6.0	risk-on	Quiet rally, Fed on hold, VIX near 10
8	1995Q4	1.26	−0.22	5.3	risk-on	Bull market; federal government shutdowns (Nov, Dec)
9	1997Q4	1.26	1.10	2.4	mixed	Asian crisis: 27 Oct mini-crash, Korea IMF program (3 Dec)
10	2017Q4	1.21	1.53	5.9	mixed	Tax Cuts and Jobs Act passed (20 Dec); record-low VIX
Largest negative shocks						
1	2018Q4	−3.41	13.30	−15.0	risk-off	Q4-2018 sell-off: rates, Fed tightening, trade war, government shutdown
2	2018Q1	−2.48	8.93	−1.2	risk-off	VIX spike and short-volatility ETP collapse (5 Feb), steel tariffs (1 Mar)
3	2011Q3	−2.43	26.44	−15.5	risk-off	U.S. downgrade (5 Aug), euro-area crisis; VIX +26 points
4	2000Q4	−1.69	6.28	−8.4	risk-off	Dot-com bust continues, contested presidential election
5	2008Q1	−1.68	3.11	−10.4	risk-off	Credit crisis; Bear Stearns rescue (16 Mar)
6	2018Q2	−1.53	−3.88	2.9	risk-on	Trade tensions begin, Italian political crisis (29 May)
7	2016Q1	−1.41	−4.26	0.8	risk-on	China and oil sell-off (Jan-Feb), then recovery after mid-February
8	2010Q2	−1.34	16.95	−12.6	risk-off	Greek bailout (2 May), flash crash (6 May), euro-area crisis
9	2000Q1	−1.31	−0.65	2.3	risk-on	Nasdaq peak (10 Mar), end of the technology boom
10	1997Q3	−1.23	1.38	6.8	mixed	Thai baht floated (2 Jul): start of the Asian crisis; U.S. equities strong

Note: As Table C.19, for the benchmark quarterly ambiguity shock.

Table C.21: The monthly identification with additional controls

			EME GDP, panel LP (six EMEs)			Proxy SVAR GDP	
Variant	Corr. with M1	F (HAC)	Peak (h)	90% band	90% horizons	Peak (h)	68% horizons
Baseline							
M1 (T3.3)	1.00	101 (70)	0.53 (11)	[0.13, 0.93]	9–12	0.26 (10)	9–20
Benchmark quarterly shock	–	–	0.53 (6)	[0.05, 1.02]	6	0.32 (8)	5–20
One control ordered before ambiguity							
BEX risk aversion	0.96	83 (54)	0.36 (10)	[−0.10, 0.81]	none	0.27 (9)	9–20
BEX uncertainty	1.00	100 (67)	0.53 (11)	[0.13, 0.93]	10–12	0.27 (10)	9–20
MAR global factor	0.99	88 (62)	0.59 (11)	[0.21, 0.98]	6, 9–12	0.23 (10)	11–20
Variance risk premium	0.94	80 (31)	0.45 (6)	[0.05, 0.85]	4–6, 11	0.32 (9)	8–20
Realized variance	0.99	93 (68)	0.55 (11)	[0.08, 1.02]	10–12	0.27 (10)	9–20
Realized skewness	0.98	89 (58)	0.54 (11)	[0.13, 0.95]	10–12	0.27 (10)	10–20
EPU (news)	0.99	96 (67)	0.49 (12)	[0.10, 0.88]	10–12	0.24 (10)	10–20
Equity market volatility tracker	1.00	101 (72)	0.50 (11)	[0.11, 0.90]	10–12	0.26 (10)	10–20
Commodity prices	0.99	95 (70)	0.49 (11)	[0.08, 0.89]	6, 9–12	0.25 (10)	10–20
Geopolitical risk	0.99	100 (81)	0.53 (11)	[0.17, 0.89]	6, 9–12	0.28 (10)	9–20
Bauer-Swanson (first or exogenous)	0.99	92 (54)	0.51 (11)	[0.14, 0.88]	6, 9–12	0.27 (10)	10–20
Jarocinski-Karadi MP and information (first or exogenous)	0.97	82 (56)	0.56 (11)	[0.17, 0.95]	6, 9–12	0.25 (10)	10–20
Principal components of the 12 controls							
PC1, before ambiguity	0.99	99 (74)	0.48 (11)	[0.09, 0.87]	10–12	–	–
PC1 and PC2, before	0.98	90 (62)	0.49 (11)	[0.11, 0.86]	10–12	–	–
PC1, after ambiguity	1.00	96 (71)	0.46 (11)	[0.09, 0.84]	10–12	–	–
PC1 and PC2, after	0.99	91 (60)	0.45 (11)	[0.10, 0.80]	10–12	–	–
Risk aversion and VRP							
Before ambiguity	0.86	61 (32)	0.24 (4)	[−0.07, 0.55]	none	–	–
After ambiguity	0.97	85 (45)	0.41 (10)	[0.00, 0.82]	11–12	–	–
M1; lags 1–4 in the LP	–	–	0.43 (6)	[0.03, 0.83]	6	–	–
All 12 controls							
Before ambiguity (VAR, $p = 2$)	0.72	24 (16)	0.25 (5)	[−0.07, 0.57]	none	0.30 (9)	8–20
Before ambiguity (VAR, BIC p)	0.75	23 (15)	0.23 (10)	[−0.27, 0.73]	none	0.34 (9)	7–20
Two-step	0.91	50 (28)	0.46 (6)	[0.07, 0.85]	5–6	0.35 (8)	7–20
Contemporaneous only (lag 0)	0.77	29 (17)	0.19 (5)	[−0.07, 0.46]	none	0.36 (9)	6–20
After ambiguity (lags only)	0.86	50 (20)	0.33 (6)	[0.02, 0.64]	6, 11	0.20 (12)	14–20
M1; lags 1–2 in the LP	–	–	0.74 (12)	[0.46, 1.02]	4–5, 9–12	–	–
M1; lags 1–4 in the LP	–	–	0.69 (9)	[0.28, 1.10]	4–12	–	–
Sample to 2021M9, COVID dummies (LP to 2021Q3 with 2020 target-quarter dummies)							
M1	1.00	111 (88)	0.55 (11)	[0.19, 0.90]	9–12	0.25 (10)	10–20
All controls except Bauer-Swanson	0.73	25 (17)	0.17 (5)	[−0.14, 0.48]	none	0.28 (10)	8–20

Note: Monthly VAR M1 with one control added before ambiguity (rows), the first principal component of the twelve controls, all controls ordered before or after ambiguity, and the extended sample. Corr.: correlation of the identified shock with the M1 shock. F: first-stage F statistic of the quarterly aggregate for the benchmark ambiguity innovation. GDP: peak of the panel local projection of EME GDP on the shock (percent), with the 90 percent Driscoll and Kraay (1998) band and the horizons significant at the 90 percent level; six benchmark countries. Proxy-SVAR: GDP peak when the shock is used as an external instrument in the benchmark PSVAR.

Table C.22: The ambiguity shock identified at monthly frequency

Shock	Corr. with benchmark	GDP Peak (h)	GDP 90%	Investment Peak (h)	Investment 90%	EMBI spread Trough (h)	EMBI spread 90%
A. Panel local projections on the shock, six countries, $h = 0, \ldots, 12$							
Benchmark quarterly shock	1.00	0.53 (6)	6	1.45 (6)	–	−0.30 (10)	5–10
M1: ambiguity last	0.69	0.53 (11)	9–12	1.93 (11)	6, 9–12	−0.34 (10)	8–11
M2: before VIX, FU1, EBP	0.81	0.46 (11)	11–12	1.65 (11)	11–12	−0.30 (10)	6–11
M3: R, Baa, VIX, ambiguity	0.70	0.64 (11)	5–6, 9–12	2.25 (11)	4–12	−0.29 (10)	8–11
M4: M1 + yields	0.66	0.30 (6)	–	0.88 (6)	–	−0.25 (10)	8–10
M1, not detrended	0.67	0.21 (6)	–	1.01 (12)	12	−0.28 (10)	8–11
VIX-orthogonal (T3.2)	–	0.53 (6)	4–6, 9–12	1.71 (6)	4–6, 9–12	−0.29 (10)	8–11
GDP with two lags of the quarterly EBP, VIX and FU1 added to the controls							
Benchmark quarterly shock		0.55 (11)	6, 10–12				
M1: ambiguity last		0.57 (11)	5–7, 9–12				
M2: before VIX, FU1, EBP		0.61 (11)	10–12				
M3: R, Baa, VIX, ambiguity		0.62 (11)	4–12				

Instrument	First stage F	First stage R^2	GDP Peak (h)	GDP 68%	GDP 95%	Investment Peak (h)	Investment 68%	Investment 95%
B. Proxy SVAR: the shock as an external instrument for the benchmark model's ambiguity innovation								
All impact responses estimated from the instrument								
Benchmark quarterly shock	841	0.98	0.32 (8)	5–20	–	0.80 (9)	7–20	–
M1	70	0.50	0.26 (10)	10–20	–	0.71 (12)	12–20	–
M2	261	0.70	0.20 (10)	12–20	–	0.52 (12)	17	–
M3	99	0.52	0.28 (9)	9–20	–	0.78 (11)	11–20	–
M4	51	0.47	0.24 (10)	11–20	–	0.64 (12)	13–20	–
M1, not detrended	66	0.48	0.22 (11)	12–20	–	0.60 (13)	14–20	–
Country block recursive (GDP, investment, TB/Y zero on impact)								
Benchmark quarterly shock	841	0.98	0.39 (7)	3–20	6–10	1.02 (8)	4–20	–
M1	70	0.50	0.34 (8)	5–20	–	0.95 (10)	5–20	–
M2	261	0.70	0.34 (8)	4–20	–	0.95 (9)	5–20	–
M3	99	0.52	0.33 (8)	5–20	11–13	0.93 (10)	5–20	–
M4	51	0.47	0.34 (8)	4–20	11	0.94 (9)	5–20	–
M1, not detrended	66	0.48	0.33 (8)	5–20	–	0.94 (10)	5–20	–
Benchmark Cholesky, block-8 Hall bands			0.39 (7)	2–19	4–19	1.02 (8)	3–19	6–19

Note: Monthly shocks: structural residual of the ambiguity equation of monthly VARs, 1993:M2–2019:M12, two lags, in the orderings M1 to M4 described in the text, summed within the quarter and standardized. Corr.: correlation with the benchmark ambiguity shock, 1994:Q4–2019:Q4. Panel A: panel local projections of GDP, investment (percent) and the EMBI spread (percentage points) on the shock, six benchmark countries, unit fixed effects, controls two lags of the outcome, of the country variables and of the U.S. real rate, BAA spread and ambiguity; Driscoll and Kraay (1998) standard errors. Peak: largest coefficient over $h = 0$–12 (trough for the EMBI) with its horizon; 90%: horizons at which the 90 percent interval excludes zero. Panel B: the shock instruments the reduced-form ambiguity innovation of the benchmark PSVAR; F: first-stage F statistic (Newey-West); R^2: share of the innovation's variance explained; responses scaled to the benchmark impact on ambiguity and propagated with the benchmark companion matrix; "country block recursive": GDP, investment and the trade balance do not respond within the quarter, as in the benchmark. Bands: 2,000 replications of a moving-block bootstrap (blocks of eight quarters). The last row is the benchmark Cholesky response.

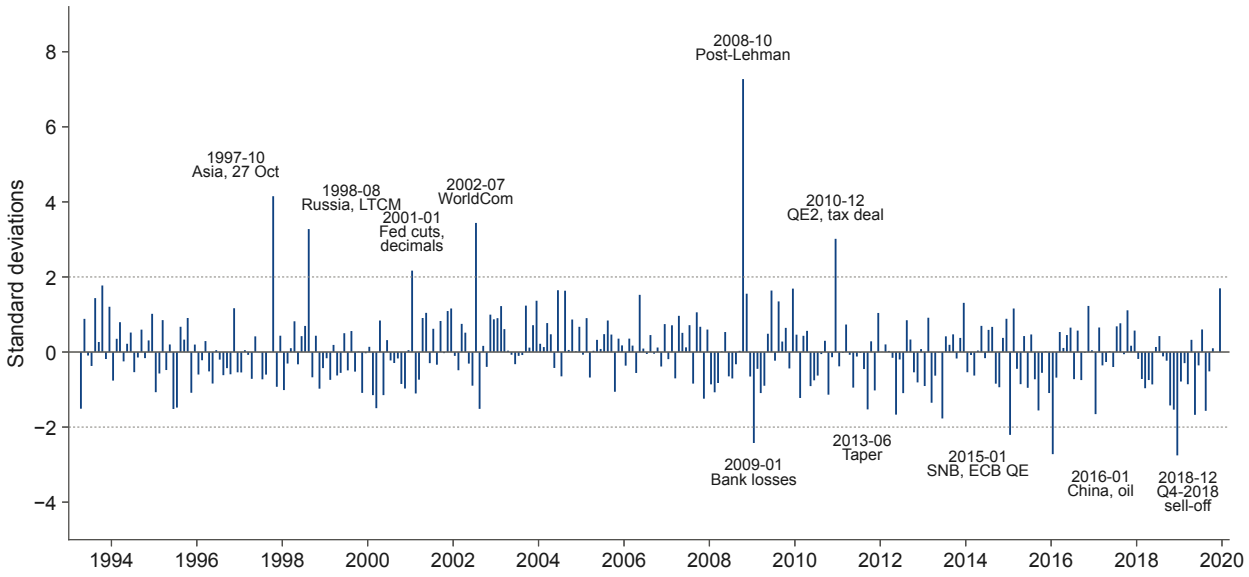


Figure C.56: The monthly ambiguity shock

Note: The M1 ambiguity shock, 1993:M4-2019:M12, in standard deviations, with the largest realizations labelled. Dotted lines at ±2 standard deviations.

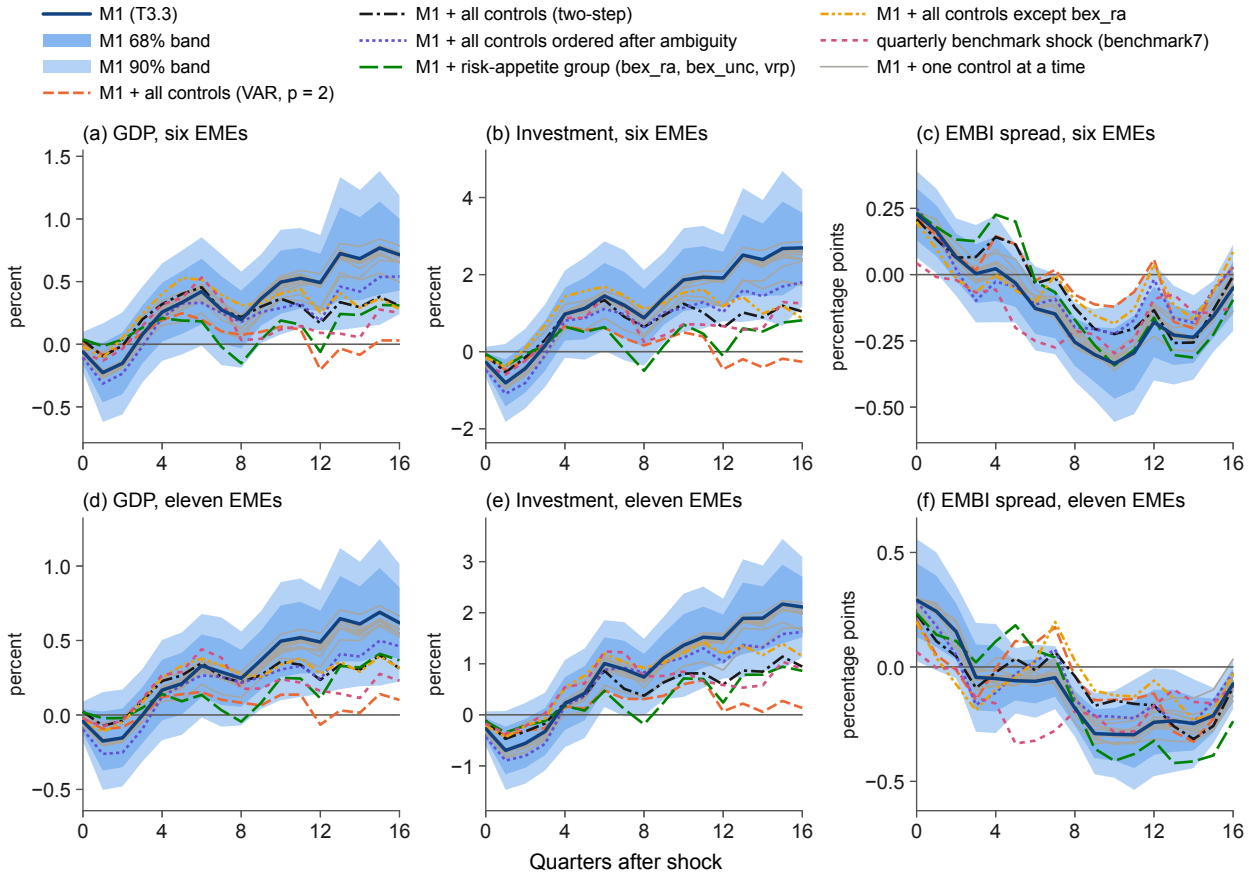


Figure C.57: EME responses to the monthly ambiguity shock with additional controls

Note: Panel local projections of EME GDP, investment (percent) and the EMBI spread (percentage points) on the monthly ambiguity shock, six benchmark countries (top row) and eleven EMEs (bottom row). Solid line and shaded areas: the M1 shock with 68 and 90 percent Driscoll and Kraay (1998) bands. Thin gray lines: M1 with one control at a time ordered before ambiguity. Other lines: all controls in the VAR; all controls ordered after ambiguity; the risk-appetite group (risk aversion, economic uncertainty and the variance risk premium) ordered before ambiguity; all controls except risk aversion; the two-step variant; and the quarterly benchmark shock.

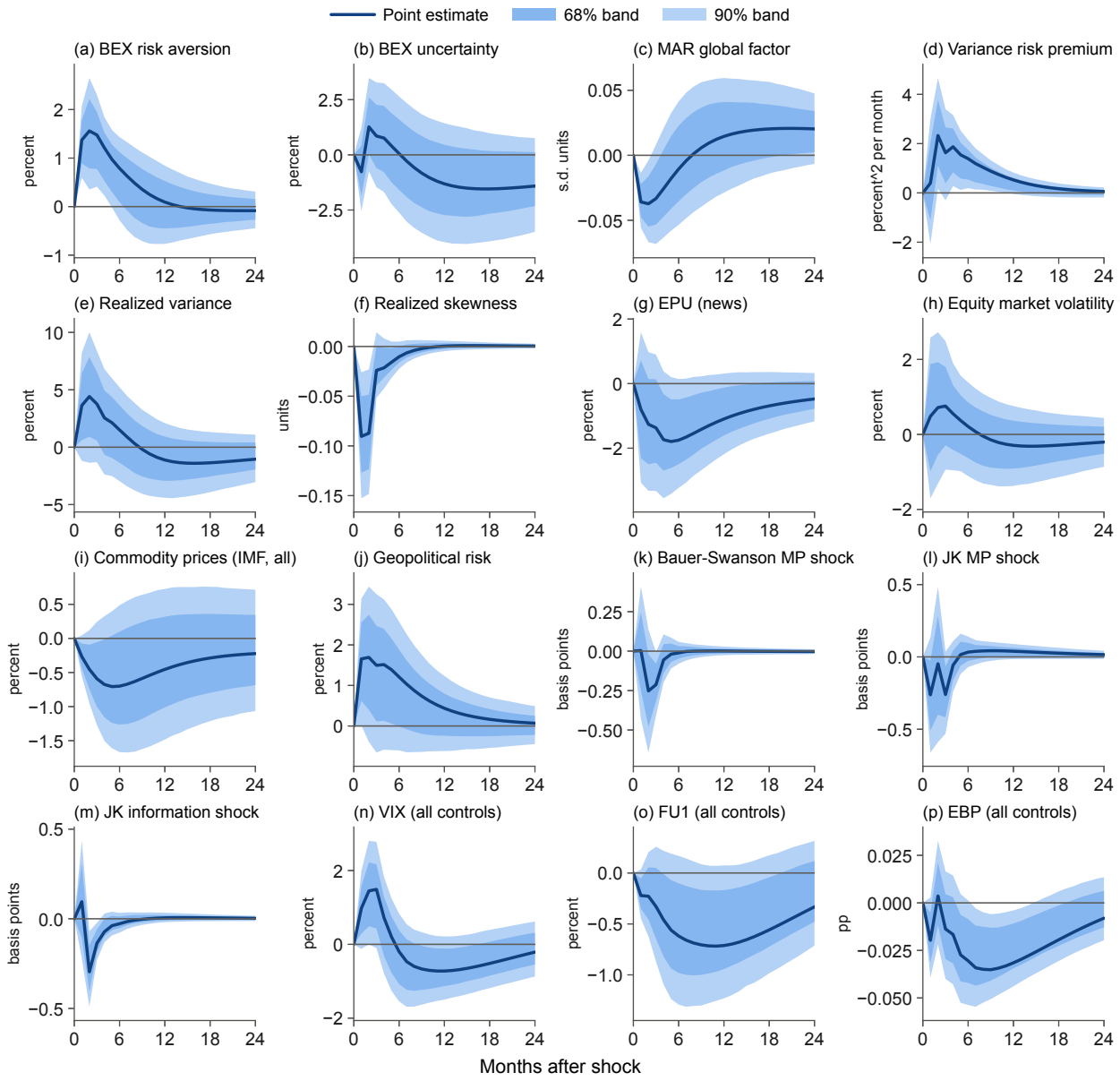


Figure C.58: Responses of the controls to the monthly ambiguity shock

Note: Responses of each control to a one standard deviation ambiguity shock in the monthly VAR M1 with the control added before ambiguity; 68 and 90 percent wild-bootstrap bands.

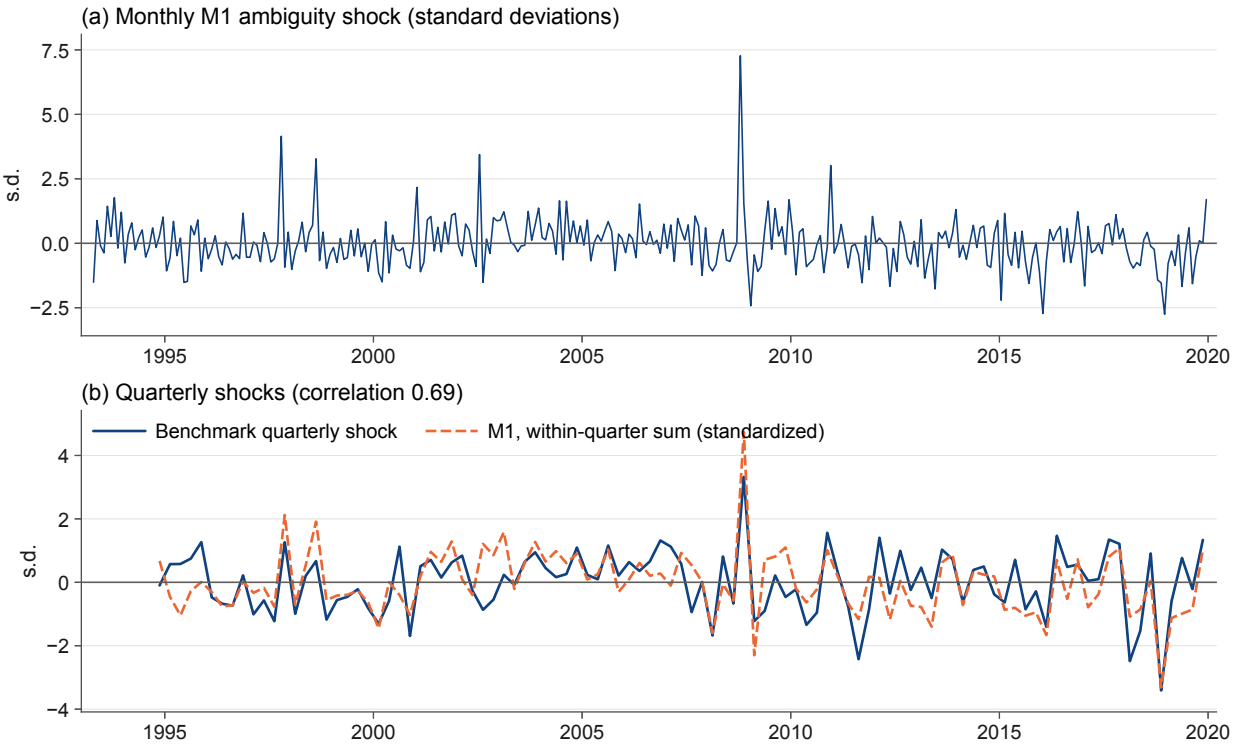


Figure C.59: Monthly and quarterly ambiguity shocks

Note: Panel (a): the monthly M1 ambiguity shock (unit standard deviation), 1993:M4–2019:M12. Panel (b): the benchmark quarterly ambiguity shock and the M1 shock summed within the quarter, both standardized, 1994:Q4–2019:Q4.

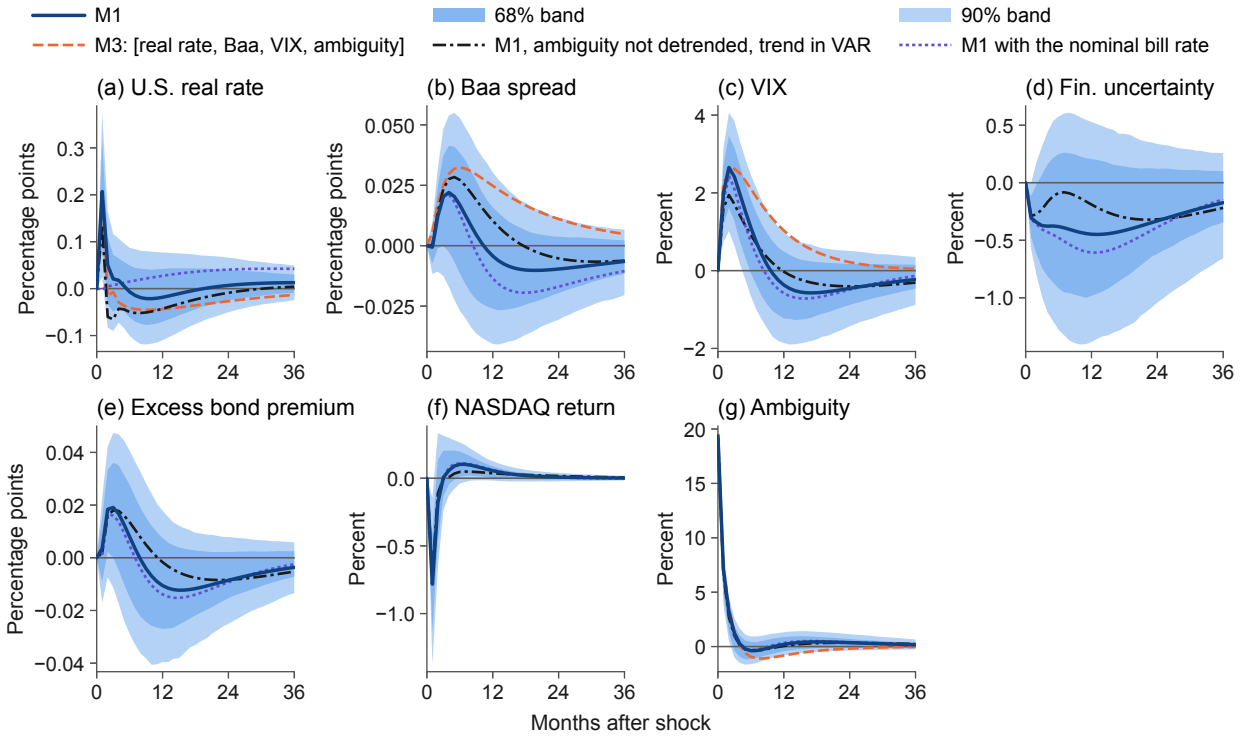


Figure C.60: Monthly VAR M1: responses to an ambiguity shock

Note: Responses of every variable of the monthly VAR M1, ordered [U.S. real rate, BAA spread, VIX, financial uncertainty (1-month horizon), excess bond premium, NASDAQ return, ambiguity], to a one standard deviation ambiguity shock; 1993:M2–2019:M12, two lags. Logs and returns in percent; rates, spreads and the term premium in percentage points. 68 and 90 percent bands from 2,000 replications of a recursive-design wild bootstrap. Overlays: point responses of the minimal system M3, of M1 with non-detrended ambiguity and a linear trend in every equation, and of M1 with the nominal 3-month bill rate in place of the real rate.

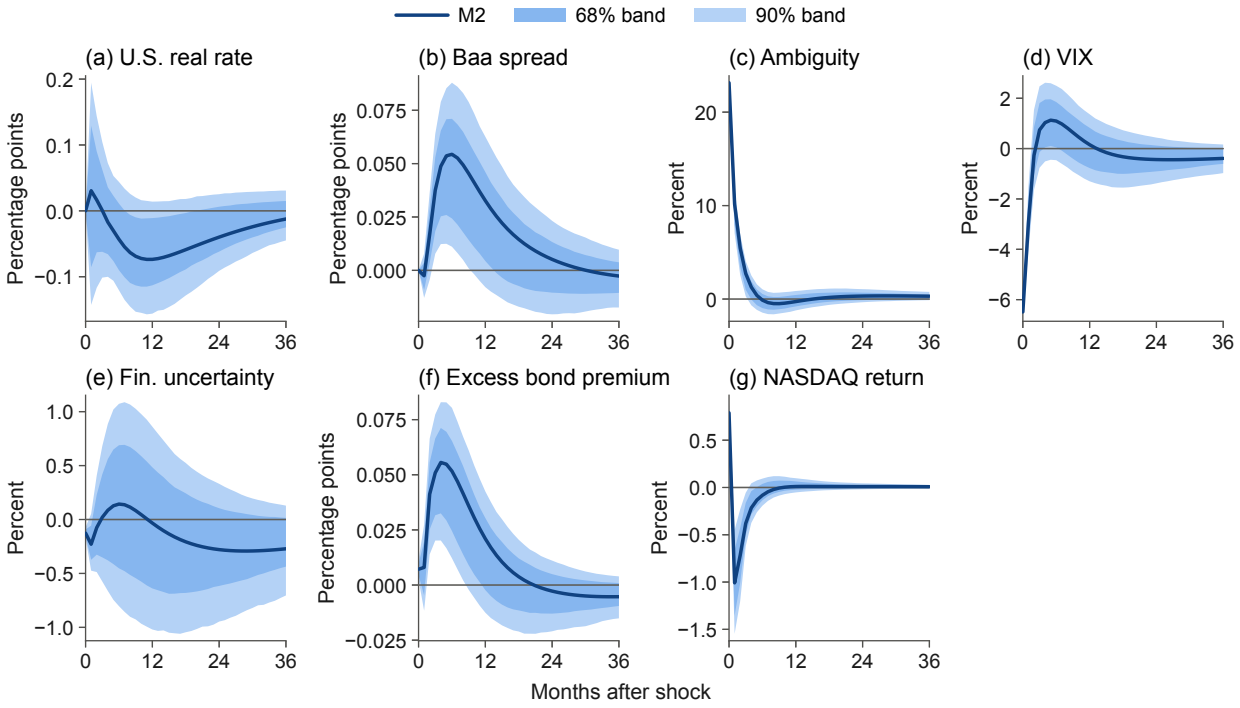


Figure C.61: Monthly VAR M2: responses to an ambiguity shock

Note: As Figure C.60, for the ordering [U.S. real rate, BAA spread, ambiguity, VIX, financial uncertainty, excess bond premium, NASDAQ return].

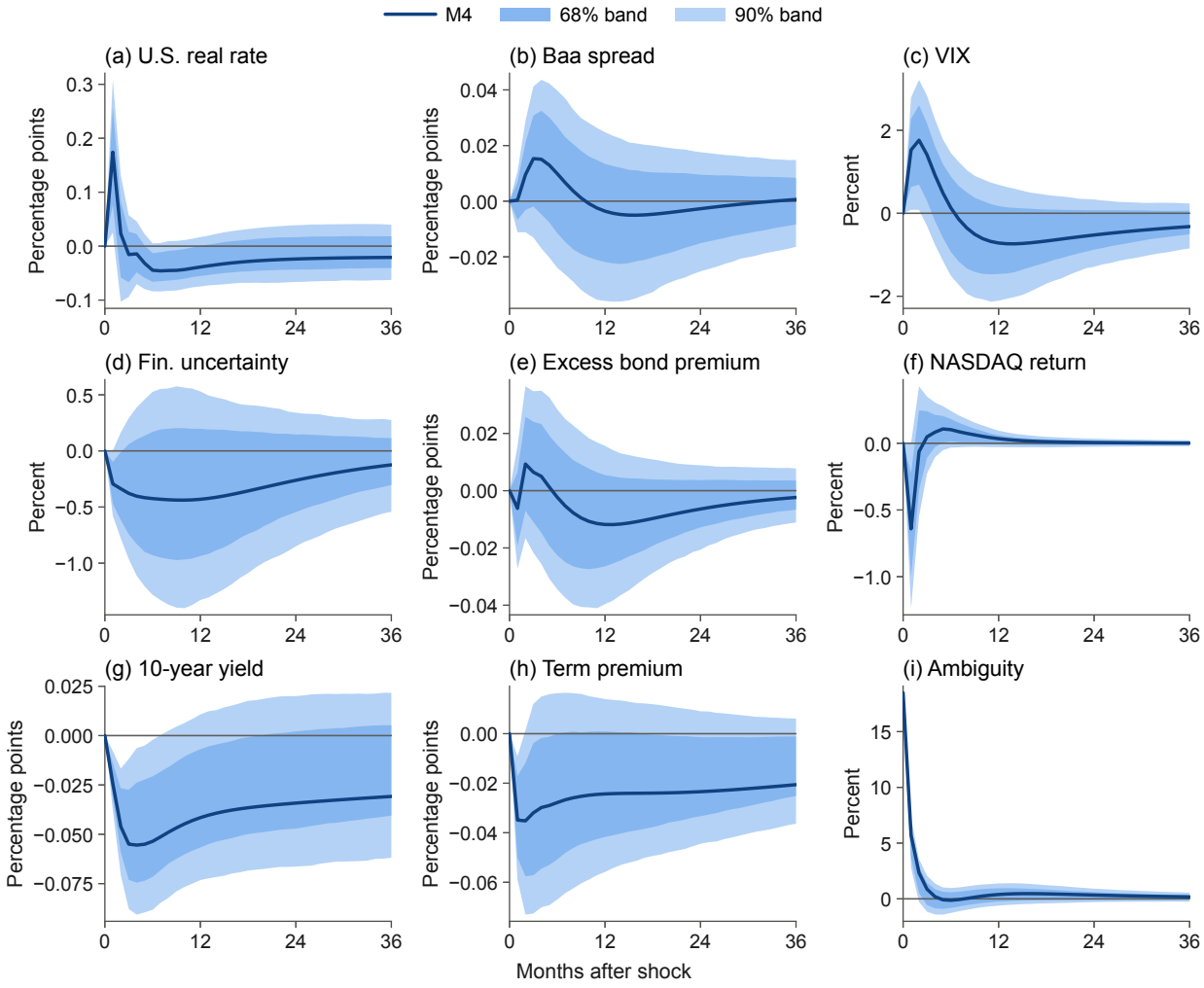


Figure C.62: Monthly VAR M4: responses to an ambiguity shock

Note: As Figure C.60, for the ordering [U.S. real rate, BAA spread, VIX, financial uncertainty, excess bond premium, NASDAQ return, 10-year Treasury yield, term premium, ambiguity].

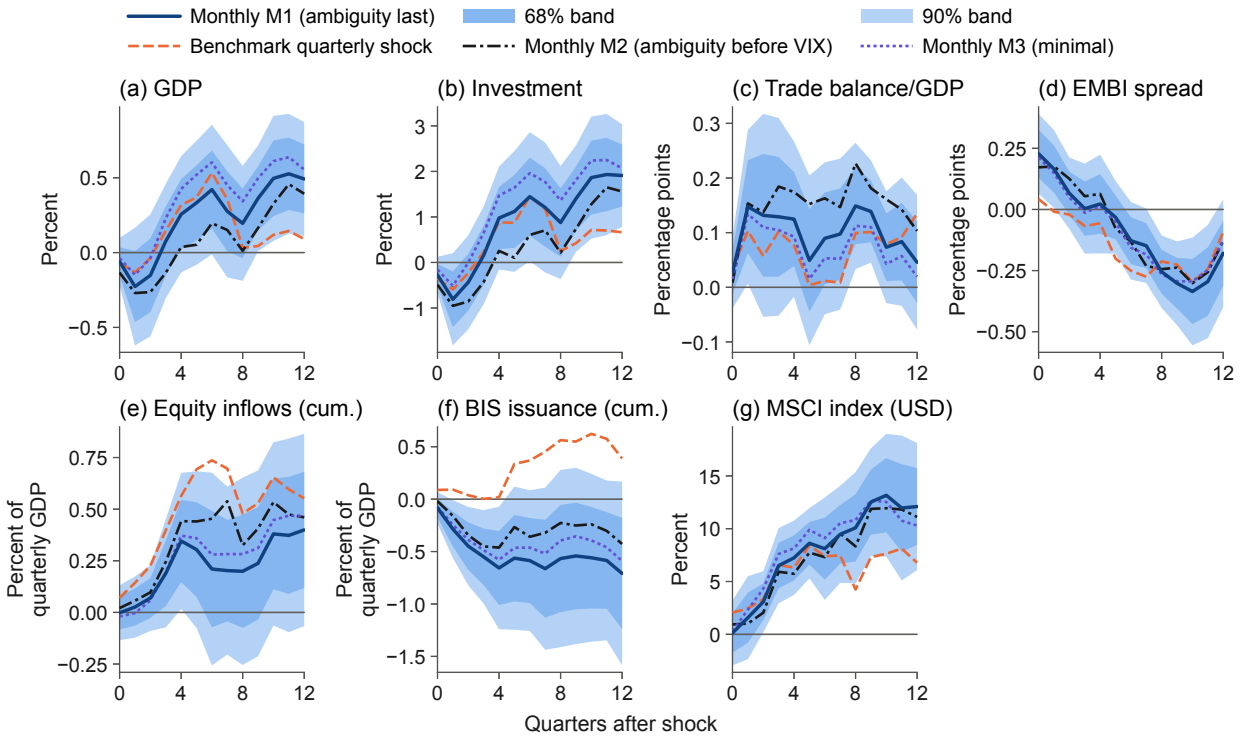


Figure C.63: EME responses to the quarterly-aggregated monthly ambiguity shock

Note: Panel local projections on the M1 shock summed within the quarter and standardized; six benchmark countries, unit fixed effects, controls two lags of the outcome, of GDP, investment, the trade balance and the EMBI spread and of the U.S. real rate, BAA spread and ambiguity; 68 and 90 percent bands from Driscoll and Kraay (1998) standard errors. Lines: the same projections on the benchmark quarterly shock and on the M2 and M3 shocks. Flows are cumulative, in percent of quarterly GDP; the MSCI index is $100 \times$ log, in U.S. dollars.

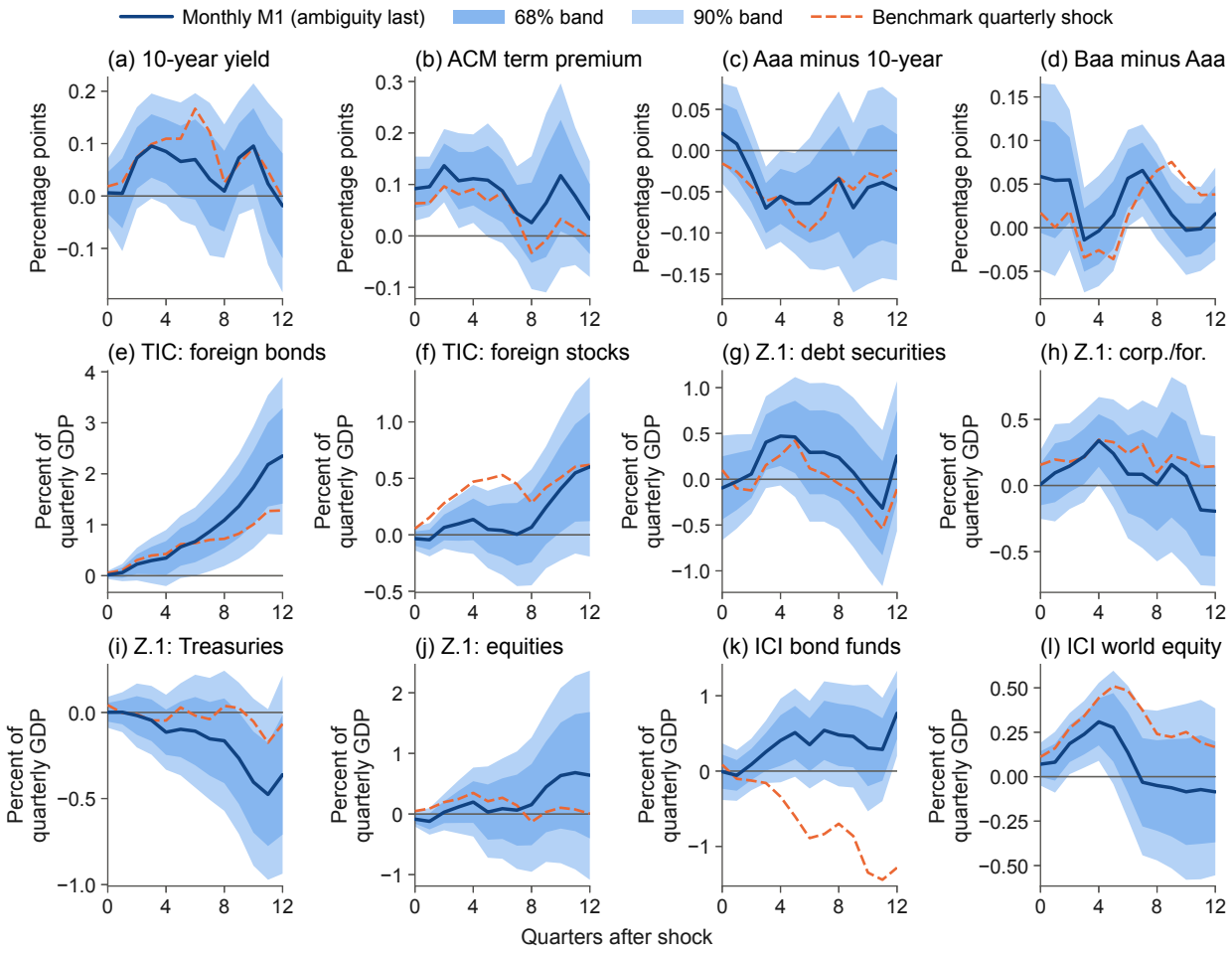


Figure C.64: U.S. prices and flows after the quarterly-aggregated monthly ambiguity shock

Note: U.S. time-series local projections on the M1 shock summed within the quarter and standardized; controls two lags of the outcome and of the U.S. real rate, BAA spread and ambiguity; 68 and 90 percent bands from Newey and West (1987) standard errors. Prices in percentage points; flows cumulative, in percent of quarterly U.S. GDP (TIC: U.S. residents' net purchases of foreign bonds and stocks; Z.1: mutual funds' net purchases; ICI: net new cash flow, 2007–2019). Dashed: the same projection on the benchmark quarterly shock.

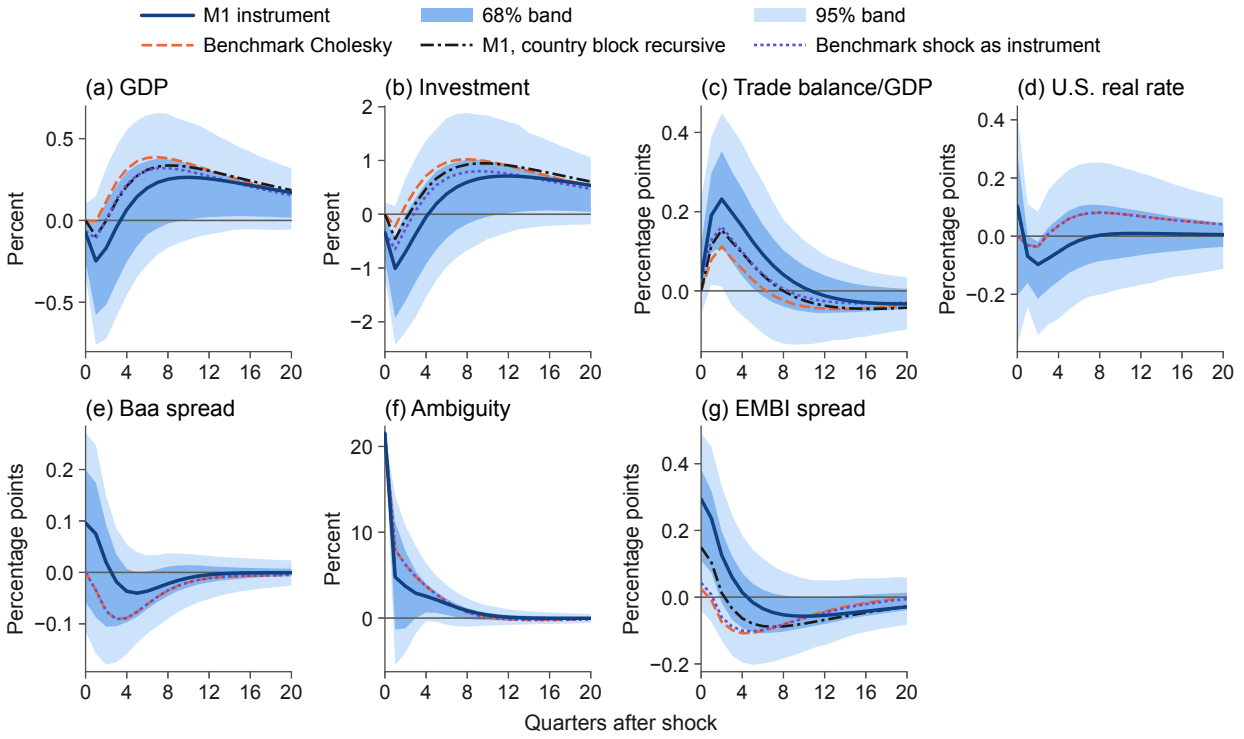


Figure C.65: External-instrument estimates with the monthly M1 shock

Note: Responses of the benchmark PSVAR to an ambiguity shock identified with the quarterly-aggregated M1 shock as an external instrument, scaled so that ambiguity rises by 21.6 log points on impact and propagated with the benchmark companion matrix. "Country block recursive": GDP, investment and the trade balance do not respond within the quarter, as in the benchmark. 68 and 95 percent bands from 2,000 replications of a moving-block bootstrap (blocks of eight quarters, first stage re-estimated in each draw). The benchmark Cholesky responses are shown for comparison.

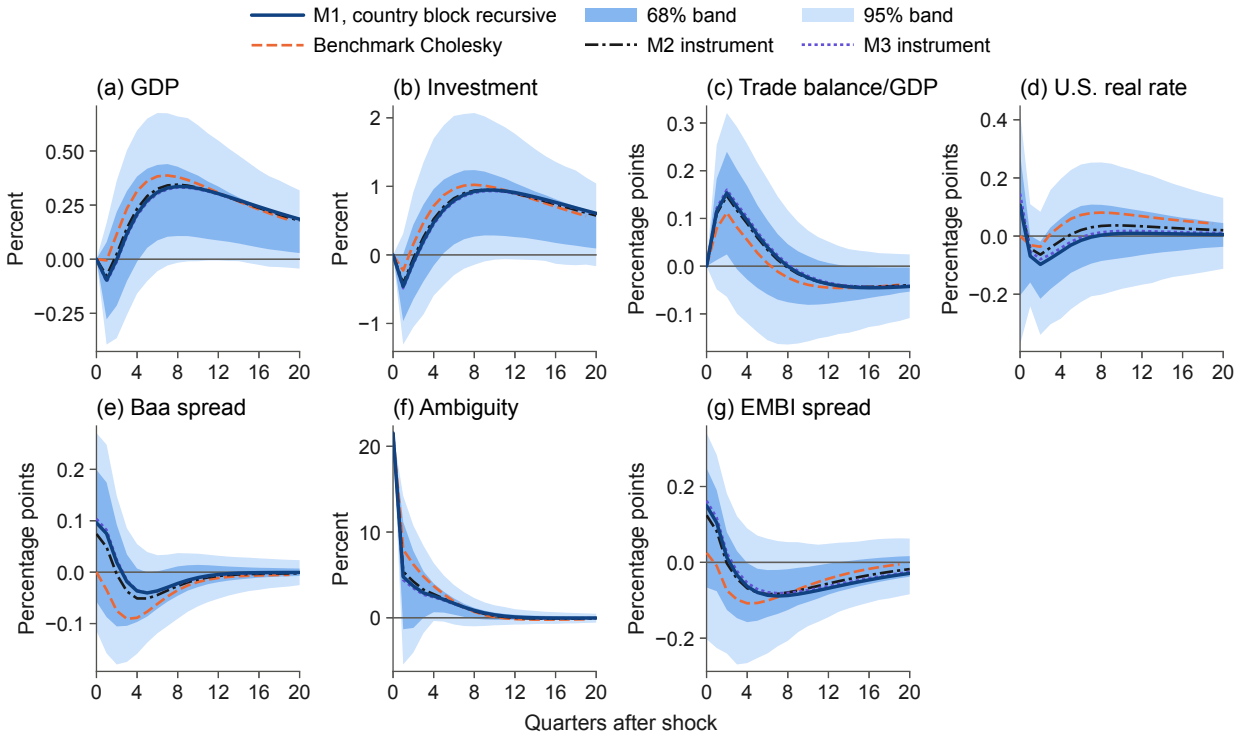


Figure C.66: External-instrument estimates: M1, M2 and M3 instruments

Note: As Figure C.65, country block recursive, for the M1, M2 and M3 instruments.

C.15 Historical decomposition

The historical decomposition of Section 3.4 writes each variable of the panel, from 1995:Q1 (after the two initial lags) onwards, as the sum of a deterministic component, which collects the unit effects and the propagation of the initial conditions, and the cumulated contributions of the seven identified structural shocks, computed from the estimated moving-average representation; the global shocks are propagated from the long global sample. The decomposition is exact. Table C.23 reports the contributions over the main episodes for the cross-country average of EME GDP and for investment, with bands from the moving-block bootstrap. Ambiguity shocks account for 3.4 of the 9.4 percentage-point rise of average GDP relative to trend from 2003:Q2 to 2008:Q1 and for 9.4 of the 31.1 points for investment. During 2008:Q3-2009:Q4 their cumulative contribution stays at about 2 percent of trend GDP; 1.9 points of this are the legacy of the 2003-07 shocks and 0.4 points the contribution of the 2008:Q4 shock (3.3 standard deviations), the largest of the sample, which is partly offset by negative shocks in the first half of 2009. The 2008-09 collapse is driven by global financial risk shocks, whose contribution to the EMBI spread also dominates its history (22 against 3.5 percent of its in-sample variation). Figure C.67 shows that the ambiguity contribution is the same under the five alternative orderings of the global block and when the shock is identified at monthly frequency and used as an external instrument. With indicators for 2008:Q4 and 2009:Q1 in every equation, the ambiguity response and contribution are unchanged (GDP peak 0.27 to 0.35 percent), whereas the output contraction after a global financial risk shock loses most of its size, so that the global-risk effect on EME output is identified largely from the crisis quarters. Figure C.68 repeats the decomposition for the eleven-country panel, in which ambiguity shocks account for 2.6 of the 8.3 points of the pre-crisis rise.

Table C.23: Historical decomposition: contributions by episode

A. Benchmark: contributions to average EME GDP

Episode	Average EME GDP	Ambiguity shocks			Global risk	U.S. real rate	Country spread	Domestic shocks
		Contrib.	68%	95%				
Rise, 2003Q2–2008Q1 (change)	9.37	3.39	[2.72, 5.57]	[0.84, 6.82]	0.91	0.18	1.59	3.29
Peak, 2008Q1 (level)	6.75	2.75	[2.16, 4.60]	[0.48, 5.63]	−0.39	0.12	0.59	3.67
Collapse, 2008Q1–2009Q2 (change)	−8.21	−0.74	[−1.76, −0.30]	[−2.77, 0.57]	−2.83	−0.40	−0.14	−4.09
2008Q3–2009Q4 (mean)	1.03	2.05	[1.43, 3.61]	[−0.17, 4.30]	−2.19	−0.08	0.39	0.85
Recovery, 2009Q2–2013Q2 (change)	6.62	−2.68	[−4.73, −1.77]	[−5.67, 0.46]	3.64	0.19	−0.30	5.78
2003Q1–2007Q4 (mean)	1.32	0.68	[0.11, 1.44]	[−0.85, 2.24]	0.15	−0.15	0.00	0.63
2010Q1–2013Q2 (mean)	3.78	0.27	[−0.40, 1.31]	[−2.24, 2.48]	−0.12	−0.06	0.22	3.47

B. Robustness of the attribution (point contributions)

Identification	Ambiguity shocks					Global-risk shocks		
	Rise	Peak	Collapse	08Q3–09Q4	Recovery	Rise	Collapse	08Q3–09Q4
Orderings of the global block								
Benchmark: R_{US}, Baa, ambiguity	3.39	2.75	−0.74	2.05	−2.68	0.91	−2.83	−2.19
Ambiguity, R_{US}, Baa	3.37	2.83	−0.83	2.10	−2.59	1.05	−2.98	−2.25
R_{US}, ambiguity, Baa	3.25	2.66	−0.59	2.11	−2.64	1.05	−2.98	−2.25
Ambiguity, Baa, R_{US}	3.37	2.83	−0.83	2.10	−2.59	1.00	−3.10	−2.13
Baa, ambiguity, R_{US}	3.58	2.95	−1.03	2.00	−2.66	0.80	−2.90	−2.04
Baa, R_{US}, ambiguity	3.39	2.75	−0.74	2.05	−2.68	0.80	−2.90	−2.04
Crisis date dummies								
Dummy 2008Q4	2.39	2.58	−0.52	2.06	−2.47	2.11	−1.02	−0.58
Dummies 2008Q3–2009Q1	2.86	2.78	−0.86	1.95	−1.97	0.61	−0.10	0.14
Proxy SVAR (monthly shock as instrument)								
M1 shock, global impact	2.61	2.24	−0.37	2.08	−1.96	–	–	–
M1 shock, all impacts	2.07	1.71	−0.64	1.46	−1.12	–	–	–
M2 shock, global impact	2.88	2.47	−0.52	2.12	−2.21	–	–	–
M2 shock, all impacts	1.83	1.55	−0.82	1.04	−0.92	–	–	–
M3 shock, global impact	2.59	2.26	−0.48	2.02	−1.86	–	–	–
M3 shock, all impacts	2.27	1.94	−0.67	1.64	−1.32	–	–	–
M4 shock, global impact	2.83	2.46	−0.60	2.08	−2.09	–	–	–
M4 shock, all impacts	2.15	1.82	−0.90	1.34	−1.09	–	–	–

Note: Panel A: change in the cross-country average of EME GDP relative to trend over each episode (percentage points) and the contributions of ambiguity shocks (with 68 and 95 percent bands from 10,000 replications of the moving-block bootstrap), global financial risk shocks, country spread shocks, the U.S. real rate shock and the three domestic shocks; the 2008:Q1 row gives the level of the cumulated ambiguity contribution at its peak. Panel B: the ambiguity contributions under the five alternative orderings of the global block, with indicators for 2008:Q4 and 2009:Q1 in every equation, and with the monthly identified shock as an external instrument (point estimates).

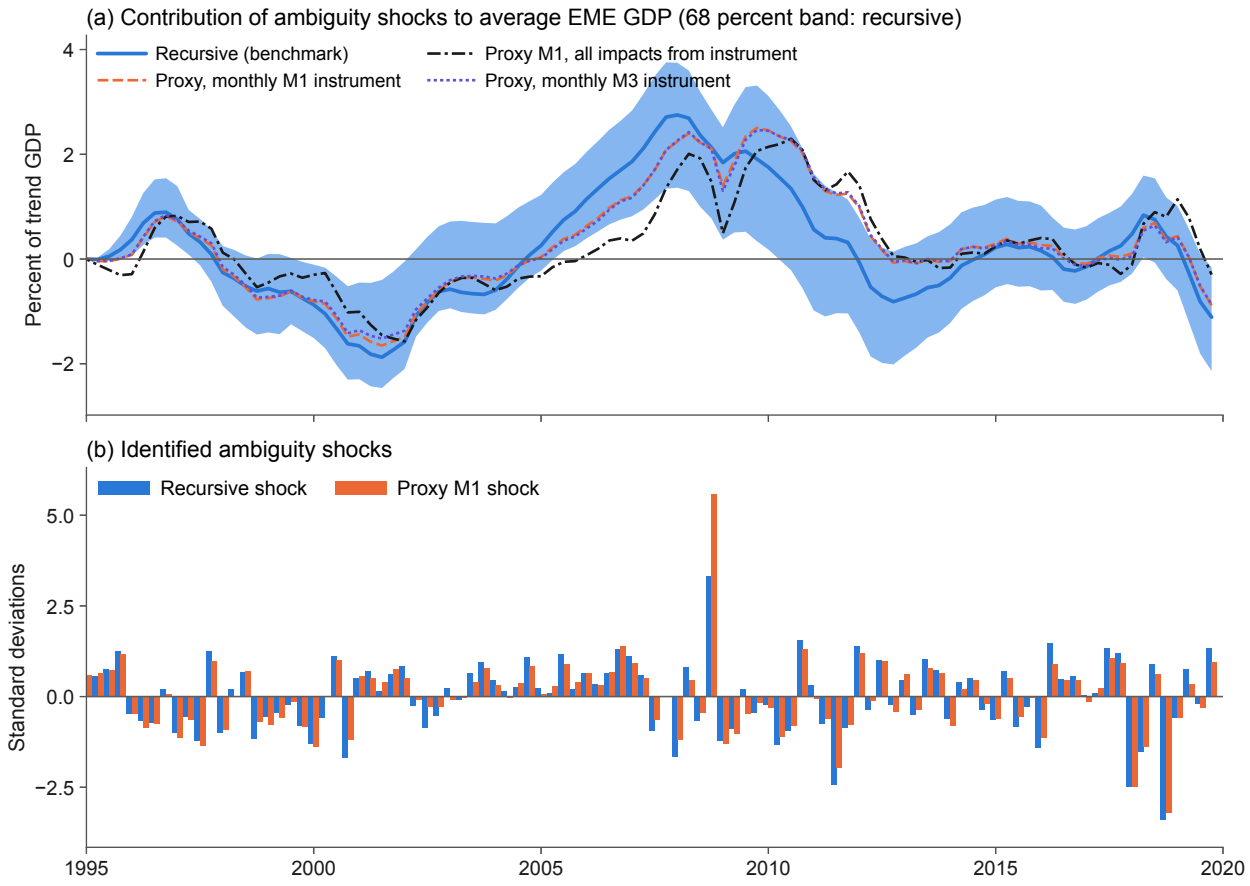


Figure C.67: The contribution of ambiguity shocks under alternative identifications

Note: Panel (a): cumulated contribution of ambiguity shocks to average EME GDP (percent of trend) in the benchmark recursive identification (with its 68 percent band) and with the monthly M1 or M3 shock used as an external instrument for the ambiguity innovation, either for the ambiguity column only or for all impact responses. Panel (b): the benchmark and proxy-identified quarterly ambiguity shocks (standard deviations).

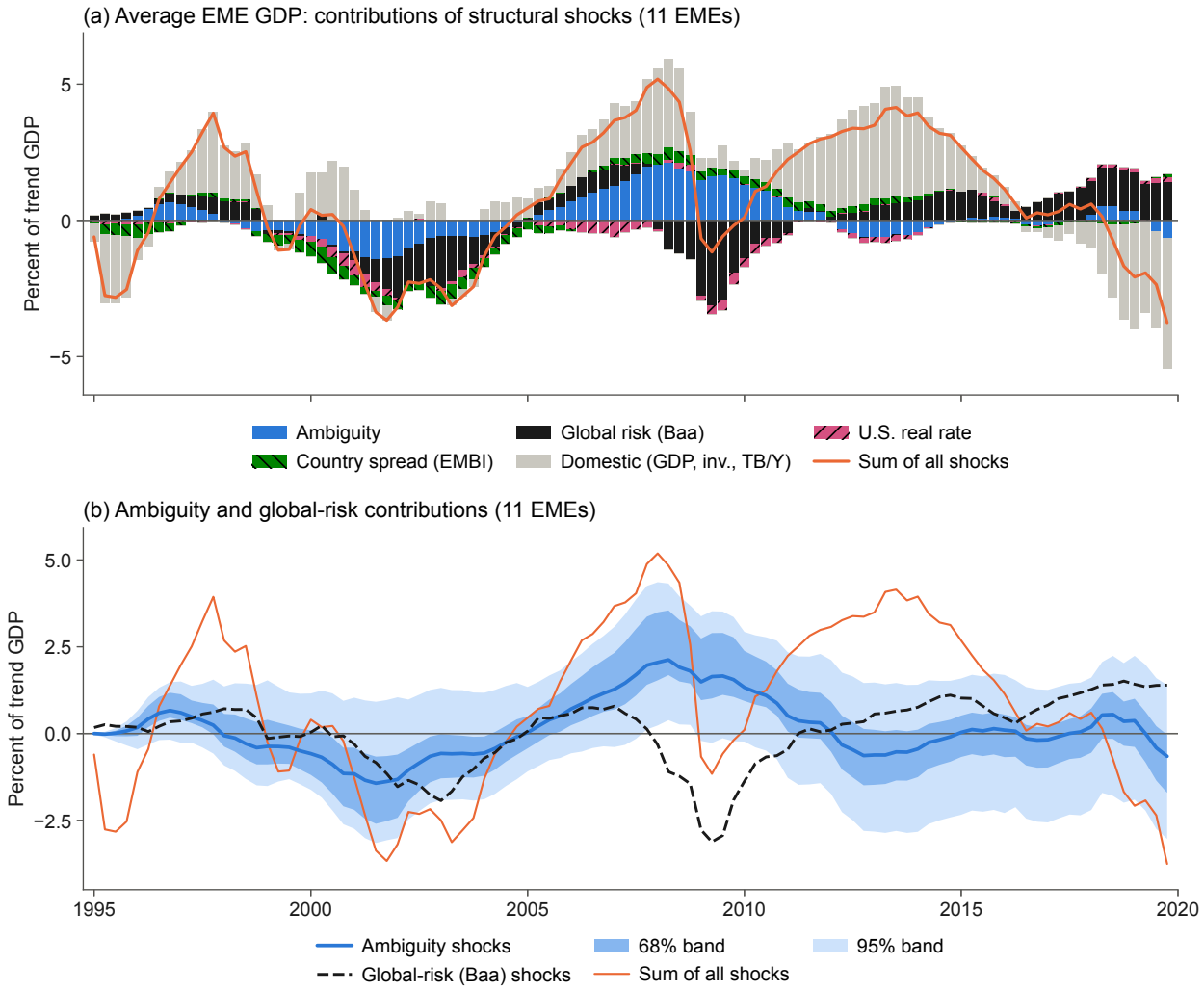


Figure C.68: Historical decomposition of average EME GDP, eleven EMEs

Note: As Figure 3.8, for the eleven-country panel.

C.16 A portfolio interpretation: derivations

This section derives the results of Section 3.3.1, states the condition on foreign ambiguity, and sets out the extensions mentioned there: a long Treasury that hedges equity, market clearing, and the dollar. Every closed-form expression has been checked against numerical optimization of the portfolio problem on a grid of calibrations. Throughout, Σ is the covariance matrix of the risky excess returns, m the vector of reference means, γ risk aversion, and $w_0 = 1 - \mathbf{1}'w$ the riskless share.

The criterion. Under the α-maxmin criterion the investor evaluates w by $\alpha \min_\mu U(w;\mu) + (1-\alpha)\max_\mu U(w;\mu)$ with $U(w;\mu) = w'\mu - \frac{\gamma}{2}w'\Sigma w$ and $\mu_U \in [m_U - a, m_U + a]$. Because U is linear in μ, the extremes are attained at the endpoints, $\min_\mu w'\mu = w'm - a|w_U|$ and $\max_\mu w'\mu = w'm + a|w_U|$, which gives the objective in (3.1) with $\kappa = (2\alpha - 1)a$. For $\alpha > 1/2$ the objective is strictly concave with a kink at $w_U = 0$, so the solution is unique.

Proof of Proposition 1. Define $\beta \equiv \rho\sigma_U/\sigma_F$ and the hedged U.S. premium $h \equiv m_U - \beta m_F$. If $w_U > 0$ the first-order conditions give $w = \gamma^{-1}\Sigma^{-1}(m_U - \kappa,\, m_F)'$, and with

$$\Sigma^{-1} = \frac{1}{\sigma_U^2\sigma_F^2(1-\rho^2)}\begin{pmatrix} \sigma_F^2 & -\rho\sigma_U\sigma_F \\ -\rho\sigma_U\sigma_F & \sigma_U^2 \end{pmatrix}$$

the solution is

$$w_U = \frac{h - \kappa}{\gamma\sigma_U^2(1-\rho^2)}, \qquad w_F = \frac{m_F - \rho\,(\sigma_F/\sigma_U)\,(m_U - \kappa)}{\gamma\sigma_F^2(1-\rho^2)},$$

so that $w_U > 0$ requires $\kappa < h$. Differentiating,

$$\frac{\partial w_U}{\partial a} = -\frac{2\alpha - 1}{\gamma\sigma_U^2(1-\rho^2)}, \qquad \frac{\partial w_F}{\partial a} = \frac{(2\alpha-1)\,\rho}{\gamma\sigma_U\sigma_F(1-\rho^2)}, \qquad \frac{\partial w_0}{\partial a} = \frac{(2\alpha-1)(1-\beta)}{\gamma\sigma_U^2(1-\rho^2)},$$

and the ratio $-(\partial w_F/\partial a)/(\partial w_U/\partial a)$ equals β. The foreign share rises if and only if $(2\alpha - 1)\rho > 0$, and the riskless share rises if and only if $\beta < 1$. If $\kappa \geq h$, the Kuhn-Tucker condition

at $w_U = 0$ holds, $w_F = m_F/(\gamma\sigma_F^2)$, and all derivatives are zero. For $\alpha < 1/2$ the kink is convex and the signs reverse. At the calibration of Section 3.3.1 with $\alpha = 1$ and $\gamma = 3$, one percentage point of ambiguity moves 18 percentage points of wealth out of U.S. equity, 6.6 into the foreign asset and 11.5 into the riskless asset, and the investor exits U.S. equity at $a = h = 2.6$ percentage points (Figure C.69, panel a).

Ambiguity about the foreign asset. If the foreign mean is also ambiguous, with $a_F = \delta_r a$, the foreign share rises with a if and only if $\delta_s \equiv \delta_r\sigma_U/\sigma_F < \rho$: co-movement of foreign ambiguity with U.S. ambiguity, measured in Sharpe-ratio units, must be smaller than the return correlation. A constant level of foreign ambiguity shifts the shares but not the comparative statics, and U.S. ambiguity can bring an excluded foreign asset into the portfolio, since the condition for holding it, $m_F - \kappa_F > \rho(\sigma_F/\sigma_U)(m_U - \kappa)$, is relaxed as κ rises.

Smooth ambiguity. Under the smooth ambiguity preferences of Klibanoff et al. (2005) with exponential utility, a power second-order function with ambiguity aversion $\theta \geq 0$ and a normal second-order prior $\mu \sim N(m, \Omega)$, the certainty equivalent is $w'm - \frac{\gamma}{2}w'\Sigma w - \frac{\lambda}{2}w'\Omega w$ with $\lambda = \gamma + \theta$, the robust mean-variance form of Maccheroni et al. (2013). With $\Omega = \mathrm{diag}(s^2, 0)$ and $\Gamma \equiv \gamma\Sigma + \lambda\Omega$, the solution is $w = \Gamma^{-1}m$ and $\partial w/\partial s^2 = -\lambda w_U\Gamma^{-1}e_U$, so that $\partial w_F/\partial s^2 = \lambda\Gamma_{12}w_U/\det\Gamma > 0$ if and only if $\rho w_U > 0$, with the same reallocation ratio $\Gamma_{12}/\Gamma_{22} = \beta$. There is no exit region: $w_U > 0$ for every s when $h > 0$ (Figure C.69, panel b). Own-asset ambiguity thus acts as U.S.-specific variance $\lambda s^2/\gamma$ that is uncorrelated with the foreign asset.

Proof of Proposition 2. With $w = \gamma^{-1}\Sigma^{-1}m$ and $d\Sigma = bb'\,d\sigma_G^2$, $dw = -\Sigma^{-1}(d\Sigma)w = -(b'w)\,\Sigma^{-1}b\,d\sigma_G^2$, and by the Sherman-Morrison formula $\Sigma^{-1}b = V^{-1}b/(1+\sigma_G^2\,b'V^{-1}b)$, which is positive componentwise when $b > 0$. If $b'w > 0$ every risky share falls and the riskless share rises (Figure C.69, panel d). For comparison, a rise in σ_U with ρ fixed gives $\partial w_F/\partial\sigma_U = \rho\tilde{S}_U/(\gamma\sigma_U\sigma_F(1-\rho^2)) > 0$, where $\tilde{S}_U = (m_U - \kappa)/\sigma_U$, with a reallocation ratio $\beta\tilde{S}_U/(2\tilde{S}_U - \rho S_F)$ between zero and β (panel c): U.S.-specific volatility lies between ambiguity and common risk.

Transmission to EMEs. Let the EME spread s, investment i and output y satisfy $s-\bar{s} = -\zeta(D-\bar{D}) - \eta(y-\bar{y})$, $i-\bar{i} = -\iota(s-\bar{s})$ and $y-\bar{y} = \chi(i-\bar{i})$, with $D = W\, w_F(m_F, a)$ the foreign demand for EME claims and $m_F = \bar{m}_F + (s-\bar{s})$. With $\kappa_y \equiv \chi\iota$ and $\Delta \equiv 1 + \zeta W\, \partial w_F/\partial m_F - \eta\kappa_y > 0$,

$$\frac{ds}{da} = -\frac{\zeta W}{\Delta}\frac{\partial w_F}{\partial a}, \qquad \frac{dy}{da} = \frac{\kappa_y\, \zeta W}{\Delta}\frac{\partial w_F}{\partial a}.$$

The sign of the output response is the sign of $\partial w_F/\partial a$, and without the demand response the spread-output feedback multiplies the effect by $1/(1-\eta\kappa_y) > 1$. A common volatility shock lowers w_F, raises the spread and lowers output. With EME firms supplying claims elastically, $S_F = \bar{S}_F - \varepsilon_F(m_F - \bar{m}_F)$, market clearing gives $dm_F = -W(\partial w_F/\partial a)\, da/(W(\Gamma^{-1})_{FF} + \varepsilon_F)$, so that the EME expected return falls, its asset price rises and investment rises by $-\varepsilon_F\, dm_F$.

Market clearing in the U.S. Let demand for the n risky assets be $x = W\Gamma^{-1}(m - \kappa e_U)$ and supplies $S_j = \bar{S}_j - \varepsilon_j(m_j - \bar{m}_j)$, with $E = \mathrm{diag}(\varepsilon_j)$ and $M = W\Gamma^{-1}$. Then $dm/d\kappa = (M+E)^{-1}Me_U$. With all supplies fixed ($E = 0$), $dm/d\kappa = e_U$: the expected return on U.S. equity rises one for one with the worst-case shift, nothing else moves and there is no reallocation. With an elastic supply of U.S. capital ($\varepsilon_U \to \infty$), $dm_U/d\kappa \to 0$, the stock of U.S. capital falls, and the other markets absorb the partial-equilibrium shift of Proposition 1. The evidence of Section 3.3.2, lower U.S. investment and no significant rise in expected U.S. equity returns, is the second case.

Long Treasuries as an equity hedge. With n risky assets the derivative of the shares is $\partial w/\partial\kappa = -\gamma^{-1}\Sigma^{-1}e_U$, and since $(\Sigma^{-1})_{jU} = -\rho_{jU\cdot\mathrm{rest}}\sqrt{(\Sigma^{-1})_{jj}(\Sigma^{-1})_{UU}}$, the share of asset j rises with U.S. ambiguity if and only if its partial correlation with U.S. equity, given the other risky returns, is positive: an asset gains from U.S. ambiguity if it is a partial substitute for U.S. equity. Let the long Treasury be a risky asset with a negative loading on the common factor, as the stock-bond correlation has been since about 2000 (Campbell et al., 2017), and the Treasury bill the riskless asset. Then the Treasury share falls after an ambiguity shock, because lower equity exposure lowers the demand for the hedge, and

rises after a common volatility shock; with a fixed Treasury supply the term premium rises after the former and falls after the latter. At an illustrative calibration (Treasury volatility 8 percent, correlations of -0.3 with U.S. equity and -0.23 with the foreign asset), one percentage point of ambiguity moves the shares of U.S. equity, the Treasury and the foreign asset by -18.8, -6.8 and +6.4 percentage points, and with supply elasticities of 100, 0 and 4 the U.S. premium and the term premium rise by 0.12 percentage points and the foreign premium falls by 0.54 per percentage point of κ, U.S. capital falls and foreign capital rises. The same partial-correlation rule says that the cross-country incidence of the reallocation should follow a country's return correlation with U.S. equity rather than the size of U.S. holdings in it.

The dollar and reserves. With a second currency and intermediaries of limited risk-bearing capacity (Gabaix and Maggiori, 2015), the change in the exchange rate is proportional to net cross-border purchases. The ambiguity shock raises net U.S. purchases of foreign assets, so the dollar depreciates; a common volatility shock reverses the flows and the dollar appreciates. If EME central banks absorb part of the inflow by buying dollars, EME reserves rise and the bilateral EME exchange rates move little, which is the pattern of Figure C.31.

Measured ambiguity and volatility. Under the normal approximation used to construct the index, a given dispersion of the daily means of returns produces less measured ambiguity when volatility is higher. A common volatility shock with an unchanged dispersion of beliefs therefore lowers measured ambiguity, a mechanical complement to the belief-coordination channel of Section 3.3.4; conversely, a rise in the index at constant volatility maps into a rise in the ambiguity parameter a of the model.

Mapping to the evidence. Table C.24 lists the predictions of the model and its extensions next to the evidence.

Table C.24: The portfolio model and the evidence

Evidence	Model prediction	Where
EME GDP and investment rise after a U.S. ambiguity shock	Proposition 1 with the spread schedule: output rises if $\rho > 0$ and the investor is ambiguity averse	Sections 3.1, 4
EME equity prices rise; spreads fall with a lag	Lower foreign expected return in the EME asset market	Sections 3.1, 3.3.3
U.S. residents buy foreign bonds and stocks	Proposition 1; the partial-correlation rule (any partial substitute for U.S. equity)	Section 3.3.3
Treasury yields and term premia rise; no rise in safe-asset demand	Treasury-hedge extension	Section 3.3.3
U.S. investment falls; U.S. expected equity returns do not rise significantly	Market clearing with elastic U.S. capital	Section 3.3.2
Dollar weakens; EME currencies flat and reserves rise	Currency extension with central-bank absorption	Appendix C.8
Global financial risk shock: flight to safety, EME contraction, stronger dollar	Proposition 2	Sections 3.2, 3.3.3
No amplification by the size of U.S. holdings	Incidence follows partial correlations, not holdings	Appendix C.13
Expansion builds over two to three years	Not in the static model; spread-output feedback and investment adjustment	Section 3.1
Measured ambiguity falls after a global financial risk shock	Belief coordination; volatility lowers measured ambiguity for a given dispersion of beliefs	Sections 3.2, 3.3.4

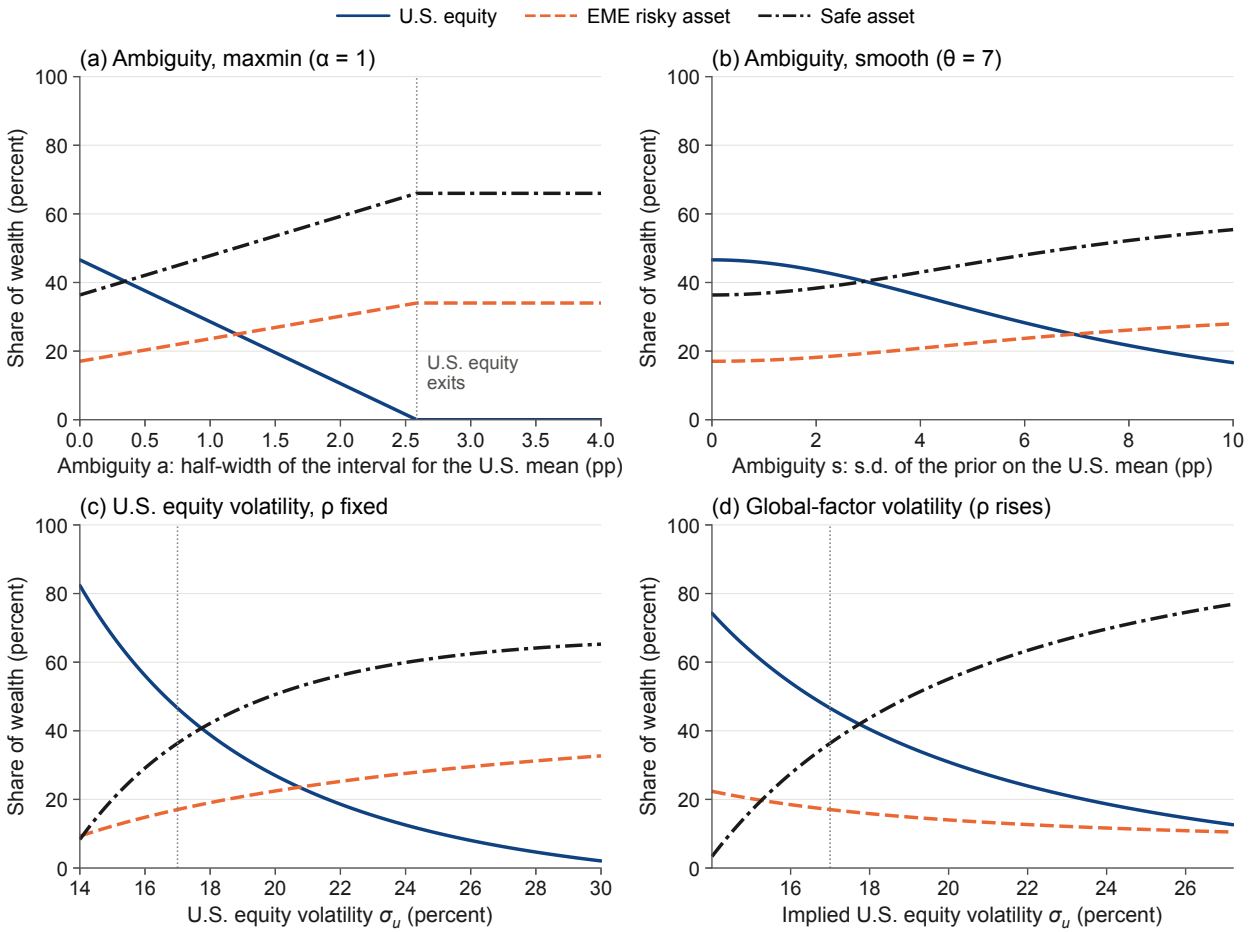


Figure C.69: Portfolio shares in the two-asset model

Note: Shares of wealth in U.S. equity, the foreign risky asset and the riskless asset. Panel (a): α-maxmin with $\alpha = 1$ as the half-width a of the interval for the U.S. mean rises; U.S. equity exits at $a = 2.6$ percentage points. Panel (b): smooth ambiguity with $\theta = 7$ as the standard deviation s of the prior on the U.S. mean rises. Panel (c): U.S. equity volatility with the correlation held fixed. Panel (d): volatility of a common factor. Calibration: $m_U = 5.5$, $\sigma_U = 17$, $m_F = 8$, $\sigma_F = 28$ percent, $\rho = 0.6$, $\gamma = 3$; dotted lines mark the baseline.